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Measurements & Instrumentation

(Theory with worked out Examples, Self and Classroom Practice Questions)

PSUs – Electrical Engineering

ACE is the leading institute for coaching in IES, GATE & PSUs

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**CONSISTENTLY TOP RANKS IN IES
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Foreword

Measurements & Instrumentation for PSUs

The Study material for Measurements & Instrumentation is thoroughly revised this year to meet the requirements of all categories of students. It consists of

- Brief synopsis covering the entire syllabus
- Worked out examples, concept-wise
- **Level 1 Problems** : For self practice by the students after attending the Lectures. However, the faculty concerned will clear the doubts, if any.
- **Level 2 Problems** : The concerned faculty will analyze and solve these questions in the class.

The students are advised to go through the theory duly understanding the worked out examples to the extent possible and bring the booklet to the class. By doing so, the students will get familiarized with some of the concepts and can really enjoy the lecture in the class. Also, the students can interact with the faculty in a better way.

Thanks to All the Professors who extended their valuable services in the preparation of this Booklet.

It is believed that this volume is also a valuable aid to the students appearing for competitive exams like IES, DRDO, ISRO and other PSUs.

With best wishes to all the Students

Y.V. Gopala Krishna Murthy,
M Tech. MIE,
Managing Director,
ACE Engineering Academy

Syllabus

SI units, measurement of current, voltage, power, power-factor and energy. Measurement of resistance, inductance capacitance and frequency-bridge methods, transducers and their applications to the measurement of non-electrical quantities like temperature, pressure, strain, displacement etc., cathode ray oscilloscope.

Main Index

<i>S. No</i>	<i>Chapter Name</i>	<i>Page No.</i>
01	Electrical Measurements	1 - 86
02	Electronic Measurements	87 - 100
03	Transducers	109 - 172

Electrical Measurements

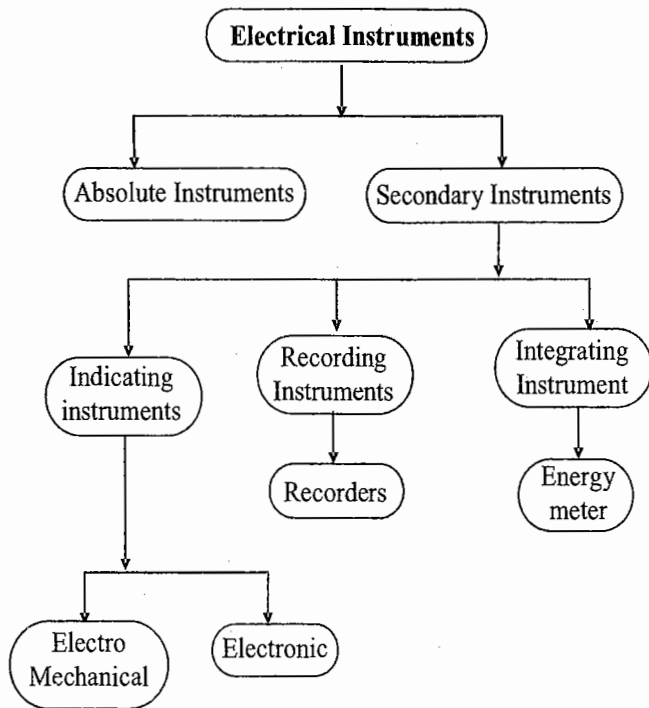
Contents

<i>S. No</i>	<i>Chapter Name</i>	<i>Page No.</i>
01	Basics of Electrical Instruments	1 - 4
02	Electromechanical Indicating Instruments	5 - 28
03	Measurement of Power & Energy	29 - 42
04	Bridge Measurement of R, L & C	43 - 66
05	Error Analysis	67 - 80
06	Potentiometers & Instrument Transformers	81 - 86

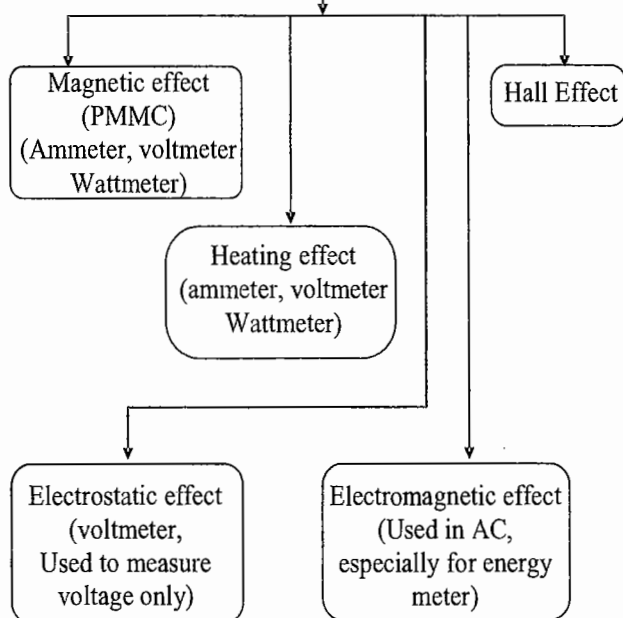
1

Basics of Electrical Instruments

ELECTRICAL INSTRUMENTS:



Principle of operation of indicating instruments



Absolute instrument:

These instruments doesn't give direct readings but gives in terms of instrumental constant. But these are accurate meters. Hence these instruments are used in researching laboratories.

Eg: Tangent galvanometer, Religh's current balance, Absolute electrometer

Secondary instruments: These instruments give direct readings.

Indicating instruments: These instruments will give instantaneous values of the quantity to be measured. Scale and Pointer mechanism are invalid

Eg: PMMC, MI, Dynamometer wattmeter, frequency meter, power factor meter

Recording instrument:

These instruments records the value to be measured (or) observed. Recorded over a graph paper by a light weight pen

- These are used to observe load variation continuously.

Eg: Recording voltmeter, Recording wattmeter, storage oscilloscope

Integrating instruments:

These instruments add the measured value to the existing value. These instruments will give the total electricity consumed over a period of time

Eg: Energy meter (or) kWhr, kVARh meter and ampere hour (Ah) meter

Indicating instruments:

Range: the meter is calibrated to read a value from minimum to maximum value



Span: Its length of the scale. It can be calculated by the algebraic difference of maximum and minimum value

Deflecting torque (T_d):

- Electromagnetic torque (T_{em})
- For the produced deflecting torque the pointer starts from zero position
- If the meter is provided with only deflecting torque, then pointer indicates full value

Controlling torque (T_c):

It is to stop the pointer at the required final value, controlling torque acts exactly equal and in opposite direction to deflecting torque.

Damping torque: The pointer makes oscillations with respect to final value, to eliminate these oscillations damping torque is required.

NOTE: In meter, first produces ' T_d ' proportional to quantity to be measured and then T_c starts producing and raises until it balances the T_d . At final steady state position $T_c = T_d$. So no damping torque (Because speed of the pointer is zero)

NOTE: T_c depends on T_d but damping torque independent of T_c and T_d .

Deflecting torque can be produced by:

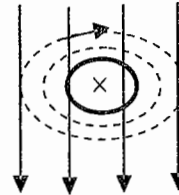
1. Magnetic effect
2. Electrostatic effect
3. Induction effect
4. Thermal effect
5. Chemical effect
6. Hall effect

Magnetic effect:

Whenever a current carrying conductor placed in magnetic field it experiences force by **Lorentz's** principle

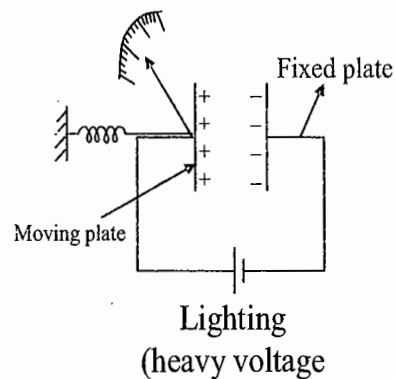
$$F = BIL\sin\theta$$

That force direction is found by **fleming left hand rule**



Here conductor moves "right to left"
(By fleming's left hand rule)

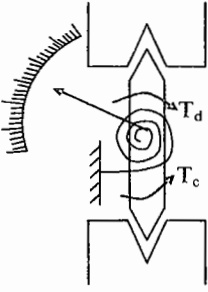
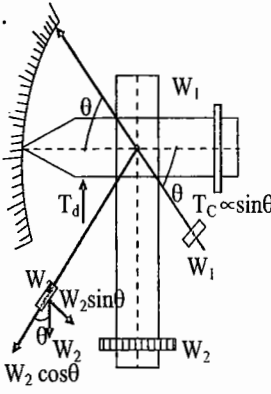
Electrostatic effect:



- When a high voltage is applied to capacitor then capacitor plates are charged with opposite polarities electro static force is produced, because of this the plates will move together.

If one plate is fixed, pointer is connected to movable plate Then we can see the voltage measured by pointer. These are specially used to measure very high dc voltages.

Controlling torque:

Spring control	Gravity control
1. Hair spring is used to provide the control	1. Weights are used to balance the motion of pointer
2. 	2. 
3. Springs are connected, in series with moving coil i.e. Springs are carrying current into the coil and out from the coil	3. These must be placed in vertical direction only
4. In MI instrument there is no physical contact between coil and spring i.e spring is not carrying the current	4. This method will give non linear scales
5. Spring is made up of phosphorous bronze	5. Controlling torque can be only upto 90°. After this controlling torque will decreases.

Note: In PMMC spring snaps means it's broken so no current it's carrying

Note: Spring control instruments can placed in any direction and gives linear scales. Hence it is used in all instruments

Example 1:

A weight of a 3.5g is used as the controlling weight in a gravity controlled instrument. Find its distance from spindle if the deflecting torque corresponding to a deflection of 58° is $2.51 \times 10^{-3} \text{ Nm}$

- (a) 45.2mm (b) 78.6 mm
(c) 66.3 mm (d) 86.2mm

Ans:(d)

Sol: Deflecting torque due to a weight is

$$T_d = mg \sin \theta \times d$$

Where,

d = distance of the weight from spindle

$$d = \frac{T_d}{mg \sin \theta}$$

$$= \frac{2.51 \times 10^{-3}}{3.5 \times 10^{-3} \times 9.81 \times \sin 58^\circ}$$

$$= 0.0862 \text{ m} = 86.2 \text{ mm}$$

Eddy current damping:

- In this method Al vane and permanent magnet are used
- Most effective damping method
- This method is suitable for the instruments having strong magnetic field (i.e. permanent magnet)

Eg: PMMC

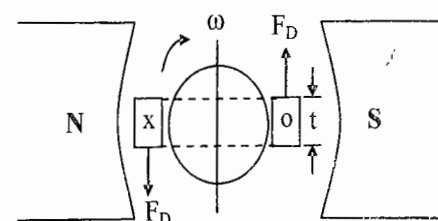


Fig: Eddy current damping by metal former

Air friction damping:

The gap between Al piston and cylinder hole must be very small and should be uniform.

- This method is suitable for the instrument having weak magnetic field

Ex: MI Instrument, Electro dynamo meter, Watt meter.

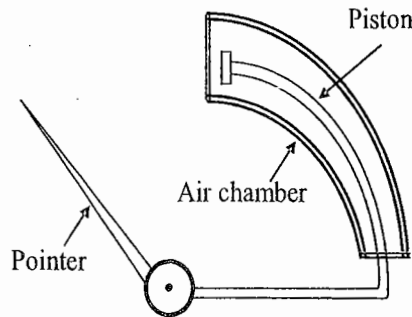


Fig: Air friction damping

Fluid friction damping:

Here high viscous fluid is used (mineral oil) used for instrument having low deflecting torque

Eg: Electro static voltmeter

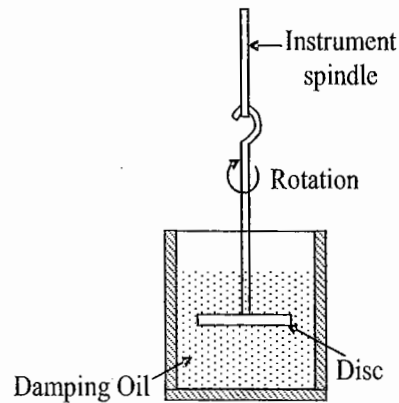


Fig: Fluid Friction Damping

These instruments must be placed in vertical direction only

- These are not portable instruments

Effectiveness order:

Eddy current > fluid friction > Air friction

- Under damping oscillations are present and pointer takes more time to give the reading
- For over damping no oscillations but pointer moves very slowly to the setting.
- For critical damping pointer reaches final value in quickest possible time without any oscillations called "dead beat"
- All indicating instruments are 2nd order instruments i.e. $0.6 < \xi < 0.8$. Here time taken by pointer depends only on "damping torque."

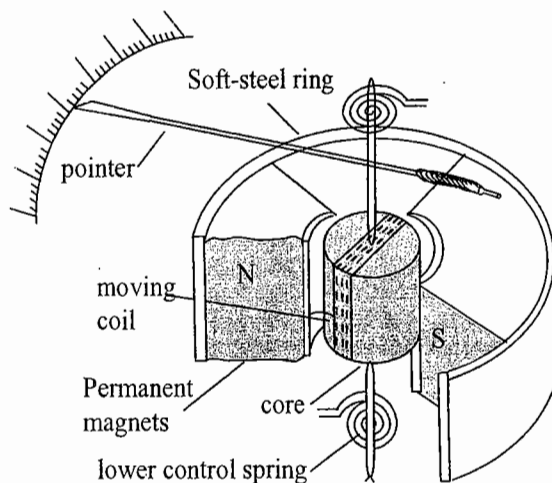
Electromechanical Indicating Instruments

Analog ammeter, voltmeter and ohm meter

Types of instruments used for measurements of current and voltage

- PMMC (Permanent magnet moving coil)
- Moving iron type (AC & DC)
- Electrodynamometer type (AC & DC)
- Electro thermic instrument (AC & DC)
Eg: (Hot wire, thermocouple)
- Electrostatic type (AC & DC)

PMMC: (Permanent magnet moving coil/ D'Arsonal instrument:



- It is similar to a permanent magnet DC motor with a limited rotation
- It works whenever a current carrying conductor is placed in a magnetic field. Experiences a mechanical force according to Lorentz's principle
 $T_d = BINA \text{ N-m}$ ($\theta = 90^\circ$ for max)
- Springs will carry the current

- Spring control is used to produce controlling torque springs are made up of bronze
Note: Radial magnetic field concept is used
- Eddy current damping is used (Al frame produce damping torque) for voltmeter
- In ammeter moving coil is wound on aluminium former, aluminium former produces damping torque.

Torque produced in PMMC:

$$T_d = BINA$$

$$T_c = k_c \theta$$

At steady state position ($T_c = T_d$) not ($T_d = T_c$)

$$T_c = T_d$$

$$k_c \theta = BINA$$

$$\theta = \frac{BINA}{k_c} \quad (\because B, N, A, k_c = \text{constant})$$

$\theta \propto I \rightarrow$ Hence scale in PMMC is linear.

- Hence with ageing effect of spring the reading will be more

$$\theta \propto A$$

$$\theta \propto B$$

$$\theta \propto N$$

Hence with the decreasing field of magnets, reading will be less.

Note: M.C instrument works only for DC

Example 1:

A 1 - ϕ , 230V 50Hz supply is given to a (0-300V) m.c instrument then it reads '0 V'.

Sol: If high frequency ac supply is given, for positive half cycle pointer moves zero to final value and final value to zero. And for negative half cycle current drawn is reversed and then pointer moves below zero. And

shows the '0' reading because of stoppers like this is to be happen '50' times in one second. So because of inertia the pointer stays '0' position only and vibrates there.

Note: If low frequency ac supply (say 1 Hz) then the pointer oscillates between '0' and final reading continuously

- When mc instrument connected in series with the load then deflection (θ) is proportional to the current passing through the load. Hence it works as a ammeter
- When mc instrument connected across the load then the current passing through meter proportional to potential difference across two terminals of meter

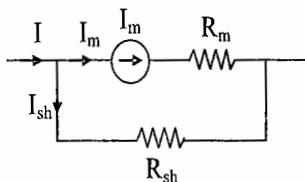
$$\theta \propto V$$

Moving coil instruments average values of current and voltage

Extension range of PMMC:

- Measuring the higher values of dc currents voltages by a small circuit modifications in the existing instruments is called extension of range of an instrument

1. dc ammeter:



$$I_{sh} = (I - I_m)$$

$$I_m R_m = I_{sh} R_{sh}$$

$$R_{sh} = \frac{R_m}{m-1} \Rightarrow m = \frac{I}{I_m}$$

'm' is called multiplication factor

- To increase range of an ammeter, A low value of Resistance must be connected in parallel with the meter. It is called as shunt resistance (R_{sh})
- The internal resistance (R_m) of an ammeter ideally zero ohm (sc).

Note: The ammeter must be connected in series with the load. If it is connected in parallel with the load the supply voltage appear across the ammeter and also because of its low internal resistance meter draws more current and meter will be damaged

Example 2:

(0 – 1 mA) ammeter has its interval resistance of 4Ω . It's range is to be extended to 100 A. Then the value of shunt resistance required is

$$\text{Sol: } I = 100\text{A} \quad m = \frac{I}{I_m} = 100 \text{ k}$$

$$\begin{aligned} I_m &= 1 \text{ mA} & R_{sh} &= \frac{R_m}{m-1} \\ & & &= \frac{4}{100\text{k}-1} \\ & & &= 40 \mu\Omega \end{aligned}$$

Example 3:

If an analog PMMC (0–10) A ammeter is provided with no controlling mechanism and the moving parts are free to rotate, the reading of the instrument if a current of 1A DC Passed through the moving coil.

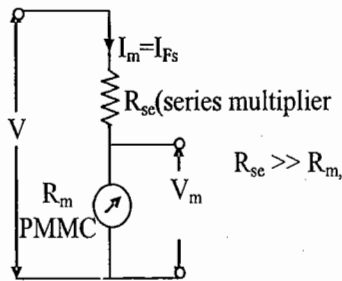
- (a) The pointer will remain stationary
(b) The pointer will continuously rotate
(c) 1A
(d) 10 A

Ans: (b)



Sol: Since there is no control mechanism, T_C will not develop and T_d will develop because coil is carrying current.

Voltmeter Series Multipliers



$$V_m = \frac{V R_m}{R_{sc} + R_m}$$

$$m = \frac{V}{V_m} = \text{multiplying factor}$$

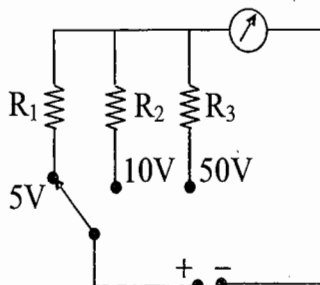
$$m = \frac{V}{V_m} = \frac{\text{Fullscale of voltage}}{\text{measure value of voltage}}$$

$$\text{Voltmeter sensitivity} = S_v = \frac{1}{I_{fs}} = \frac{R_{sc} + R_m}{V} \Omega$$

- Our human body resistance is 600 - 1000Ω

Example 4:

A multi-range voltmeter is shown in the figure with full scale deflection current of $50\mu A$ and meter resistance of $1K\Omega$. What are the values of resistances R_1 , R_2 & R_3 respectively?



- (a) 100 KΩ, 200 KΩ, 1000 kΩ
- (b) 101 KΩ, 201 KΩ, 1001 kΩ
- (c) 99 KΩ, 199 KΩ, 999 kΩ
- (d) 5 KΩ, 10 KΩ, 50 kΩ

Ans: (c)

$$\text{Sol: } R_1 = R_m \left[\frac{V_1}{V_m} - 1 \right] = R_m \left[\frac{V_1}{I_m \times R_m} - 1 \right]$$

$$R_1 = 1K \left(\frac{5}{(50 \mu \times 1K)} - 1 \right) = 99 K\Omega$$

$$R_2 = 1K \left(\frac{10}{(50 \mu \times 1K)} - 1 \right) = 199 K\Omega$$

$$R_3 = 1K \left(\frac{50}{(50 \mu \times 1K)} - 1 \right) = 999 K\Omega$$

Note: In case of ammeter full scale current is minimum then ammeter is more sensitive

Ex: An ammeter with 1 mA of full scale current is more sensitive than an ammeter with 2 mA full scale current.

- The internal resistance of voltmeter is very high and ideally it is infinite
- In all the electrical instruments the responsible quantity for measurement is current
- In all electronic instruments the responsible quantity is voltage

Note: If ammeter is connected in parallel it will damage. If voltmeter is connected in series it won't damage because of its high internal resistance

- If voltmeter connected in series with load, it shows almost supply volts (full deflection) when load resistance is less and ammeter reads currents through voltmeter as shown fig. 1.

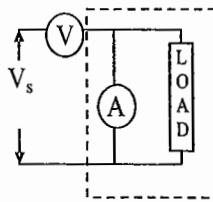


Fig (1)

Here ammeter and load in parallel $m\Omega \parallel M\Omega = m\Omega$ only. So total parallel combination is order of ' $m\Omega$ '.

So voltmeter is almost full scale

$$V \approx V_s$$

$$A \approx 0$$

For the below connection

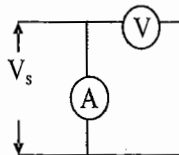


Fig (2)

Ammeter will damage and It becomes short circuit. So no voltage will read across (V)

So A = damaged

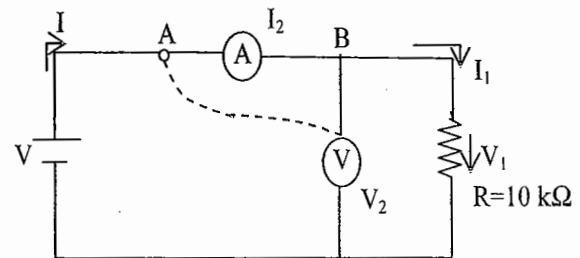
$$V \approx \text{zero}$$

Note: Basically there is no constructional difference between dc ammeter and dc voltmeter. If internal resistance of any ammeter is known ammeter can be converted as voltmeter (or) voltmeter can be converted as ammeter.

Note: PMMC measures average value and MI instrument will measure effective value (or) rms value.

Example 5:

Which of the following statements to determine the value of resistance R in the circuit shown is true?



- The voltage V_2 indicated by the voltmeter is higher than the voltage V_1 across the consumer.
- The Indicated current I_1 is smaller than I
- The voltage V_2 indicated by the voltmeter is less than the voltage V_1 across the consumer
- For a more accurate result, the voltmeter should be connected in front of the ammeter (Point A)

Ans: (d)

Tips to Solve a Voltmeter with different Ratings

- If voltmeter reading is given in 'V and S_v

$$R_s + R_m = S_v \times V$$

$$\frac{R_s + R_m}{V} = S_v$$

- If ammeter ratings are given in 'R' I_{fs}

$$R_s + R_m = R$$

- V, I_{fs} are given then

$$R_s + R_m = \frac{V}{I_{fs}}$$

$R_s \rightarrow R_{se}$ series multiplier resistance

R_m meter resistance

I_{fs} full scale current

From the above relations find individual meters resistor values 'R' then apply voltage division rule for finding the voltage measured by each meter

Loading effect: If a "Low sensitivity" voltmeter is used to measure the voltage across resistance. Then some of current will be passing through the voltmeter. So that meter reads wrong value this is called loading effect.

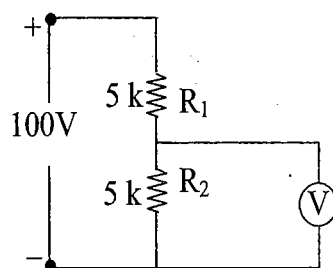
To reduce the loading effect high sensitivity voltmeters should be used

If the output of half wave rectifier is passing through PMMC then it will measure $I_{avg} = \frac{I_m}{\pi}$

If the output full wave rectifier is passing through PMMC. Then it will measure $I_{avg} = \frac{2I_m}{\pi}$

Example 6:

In the circuit shown in figure, voltage across R_2 is measured by two different voltmeters P and Q having sensitivity of $1K\Omega/V$ and $20K\Omega/V$. Both the meter are used on 50V scale, which of the following statement is true



- Voltmeter P will read more accurate than voltmeter Q
- Both the voltmeters read same voltage
- Voltmeter Q will read more accurate than voltmeter P.
- None of above

Ans: (c)

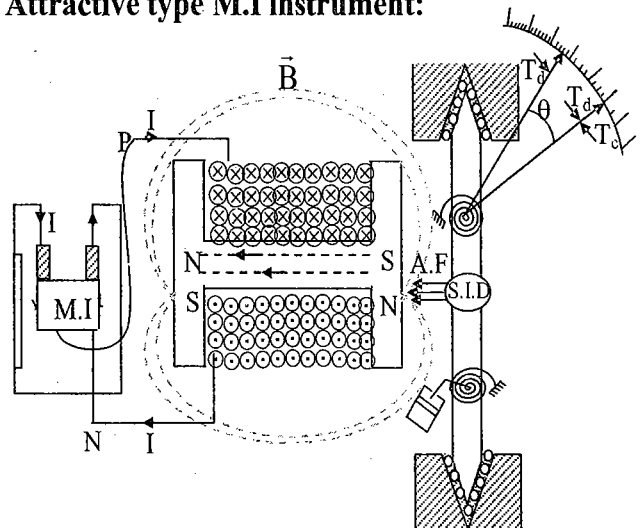
Sol: Voltmeter with high sensitivity will give more accurate results.

Moving Iron Instruments

These are two types:

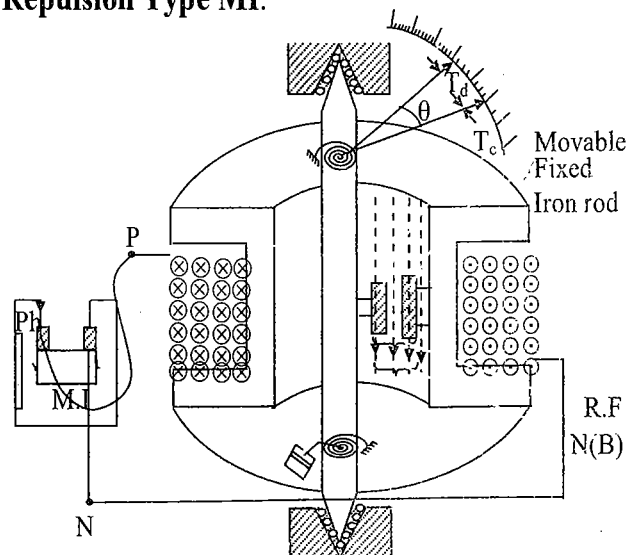
- Attractive type MI instrument
- Repulsion type MI instrument

Attractive type M.I instrument:



- When a current is passed through the coil, coil will setup a magnetic field, which attracts the iron disc, then pointer gives the deflections. This is attracted type M.I.

Repulsion Type MI:





- When the moment switch is closed, the current will enter and magnetic field is formed in hollow cylinder handle. Due to the presence of magnetic field, the repulsion force, developed due to the force spindle will rotate.

Note: The deflecting torque is derived in transient period only i.e. in 'dt' sec.

$$L = \frac{N\phi}{I}$$

$$LI = N\phi$$

$$\frac{d}{dt}(LI) = \frac{d}{dt}(N\phi)$$

$$I \cdot \frac{dL}{dt} + L \frac{dI}{dt} = N \frac{d\phi}{dt} = V$$

Electrical energy supplied to coil in dt sec.

$$\begin{aligned} VI dt &= \left(I \cdot \frac{dL}{dt} + L \frac{dI}{dt} \right) I dt \\ &= I^2 dL + LI dI \text{ -----(1)} \end{aligned}$$

$$\text{Initial energy stored} = \frac{1}{2} L \cdot I^2$$

Change in stored energy in 'dt' sec

$$\begin{aligned} &= \frac{1}{2} [L \cdot 2I dI + I^2 dL] \\ &= LI dL + \frac{1}{2} I^2 dL \text{ -----(2)} \end{aligned}$$

Mechanical work done in 'dt' sec

$$= T_d \cdot d\theta \text{ ---(3)}$$

Energy conservation law

$$(1) = (2) + (3)$$

$$I^2 dL + LI dL = LI dL + \frac{1}{2} I^2 dL + T_d \cdot d\theta$$

$$\frac{1}{2} I^2 dL = T_d \cdot d\theta$$

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

$$\text{If } \frac{dL}{d\theta} \text{ is constant then } T_d \propto I^2$$

Spring control is used to produce controlling torque i.e. $T_c = K_c \theta$

Air friction damping is used to produce Damping torque.

Note: Eddy current damping is not possible because of presence of Iron disc and poor magnetic field.

At final steady state position $T_c = T_d$

$$\theta = \frac{1}{2} \cdot \frac{I^2}{K_c} \cdot \frac{dL}{d\theta}$$

$$\text{If } \frac{dL}{d\theta} \text{ is constant}$$

$$\theta \propto I_{rms}^2$$

M.I instrument has non-linear sequence (or) sequence law scale.

Cramped (very close) at beginning & no number of division at the full scale.

M.I instruments work for A.C and D.C.

Note: For AC even though current direction reversed magnetic field direction reversed but always attractive force is present between coil magnetic field & iron disc. i.e. unidirectional torque.

Note: MI will give wrong values for both AC & DC. For DC MI instrument will give accurate values, hence MI is not used for DC.

Note: There is no alternate instrument for AC.

To measure correctly, Hence MI is used only measure AC & scale is calibrated in terms of R.M.S value.

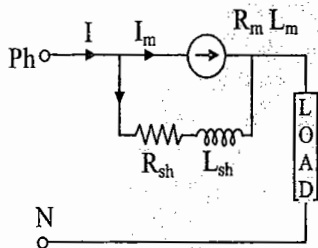


MI instrument can be used as Ammeter up to 50A measurements only & as a voltmeter up to 750V only. Because of the magnetic saturation of coil. If the O/P of H.W.R is passing through M.I. It

measures $\frac{I_m}{2}$.

If the O/P of F.W.R is passing through M.I. It measures $\frac{I_m}{\sqrt{2}}$.

To increase range of ammeter, extra number of turns are connected in parallel with the meter



If, $x_L = 2\pi fL$, $Z_m = R_m + jx_m$

$$Z = \sqrt{R^2 + x^2}$$

$$I_{sh} Z_{sh} = I_m Z_m$$

$$\frac{I_{sh}}{I_m} = \frac{Z_m}{Z_{sh}} = \frac{\sqrt{R_m^2 + (2\pi fL_m)^2}}{\sqrt{R_{sh}^2 + (2\pi fL_{sh})^2}}$$

$$\frac{\sqrt{R_m^2 + (\omega L_m)^2}}{\sqrt{R_{sh}^2 + (\omega L_{sh})^2}} = \frac{R_m \sqrt{1 + \left(\frac{\omega L_m}{R_m}\right)^2}}{R_{sh} \sqrt{1 + \left(\frac{\omega L_{sh}}{R_{sh}}\right)^2}}$$

$$\text{If } \frac{L_{sh}}{R_{sh}} = \frac{L_m}{R_m}$$

$$\frac{I_{sh}}{I_m} = \frac{R_m}{R_{sh}}$$

Frequency Independent. Error due to frequency is eliminated.

If shunt time constant = meter time constant. Then division of current in meter and shunt is independent of frequency and depends only on resistance. It is very difficult practically

Example 7:

A moving iron voltmeter calibrated for 50 Hz supply and having an inductance of the coil as 0.8 H draws 0.1 A when connected to 500 V a.c. What will be its reading when connected to 300 V dc _____ (in V)

Ans: 300.4

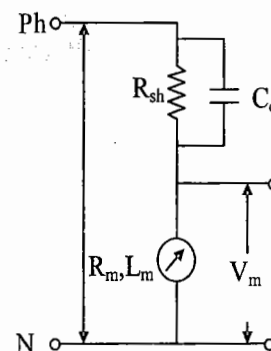
$$\text{Sol: } Z = \frac{V_{ac}}{I_{ac}} = \frac{500}{0.1} = 5000 \Omega$$

$$R = \sqrt{(5000)^2 - (2\pi \times 50 \times 0.8)^2} \\ = \sqrt{(5000)^2 - (251.327)^2} = 4993.7 \Omega$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{300}{4993.7} = 0.06008 \text{ A}$$

$$\therefore \text{Reading of voltmeter when connected for 300V dc} = \frac{500}{0.1} \times 0.06008 = 300.4 \text{ V}$$

Error in voltmeter:



To reduce the frequency error a compensating capacitor is connected across multiple resistance (R_{se})



$$\text{Compensating capacitor } (c_c) = \frac{0.41 L m}{R_{se}^2}$$

Note: There is a physical constructional difference between AC ammeter and AC voltmeter.

AC ammeter has thick cross-section and few number of turns (low resistance)

$$\downarrow R_a = \frac{\rho \cdot \ell (\downarrow) \rightarrow \text{few turns}}{a (\uparrow) \text{ more area}}$$

Note: And AC voltmeter has more number of turns with thin cross section (high resistance)

$$\uparrow R_v = \frac{\rho \cdot \ell \rightarrow \text{more turns}}{a \rightarrow \text{Less area}}$$

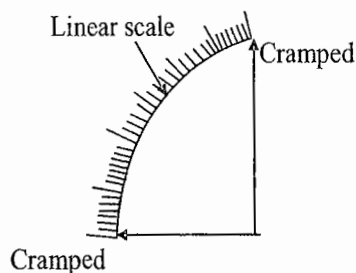
$$\text{If } f \uparrow = X_L \uparrow$$

To reduce the Increased reactance of L we have to connect a capacitor for decrease the increased reactance

$$\text{If } f \uparrow = (X_L \uparrow - X_C \downarrow)$$

Note: We cant connect capacitor in series, it forms series RLC as 'f' increases X_L & X_C gets compensated X cancelled. The CKT (meter) then becomes only connected with R. & It will damage for high currents. So we have to connect capacitor in parallel to R_{sh}

Condition for linear scale:



$$T_d = T_c$$

$$\frac{1}{2} I^2 \cdot \frac{dL}{d\theta} = k\theta$$

$$\text{Linear scale } I \propto \theta \quad I = K_1 \theta$$

$$\frac{1}{2} (K_1 \theta)^2 \frac{dL}{d\theta} = k\theta$$

$$\theta^2 \cdot \frac{dL}{d\theta} = 2 \frac{k}{K_1} \theta$$

$$\theta \frac{dL}{d\theta} = 2 \frac{k}{K_1} = \text{constant}$$

→ Condition for linear scale

Note: To measure very high values of AC currents and A.C voltages, Instrument Transformers are used i.e C.T (Current Transformers) and P.T (Potential Transformer) Because of cost point of view of potential transformers, we are using capacitive voltage transformers are used in electrical substations

To convert M.I ammeter into voltmeter is same as in converting PMMC instruments.

Example 8:

A 0 – 200mA moving iron ammeter is converted to a 0-600V, 50Hz voltmeter by adding a series resistance with the coil. The coil has negligible resistance and an inductance $L = (0.1 + 0.2\theta) / 4\pi$ henry, where θ is the deflection in radian. The total angular span of the meter is 1000. Find the series resistance?

- (a) 11.22 Ω (b) 0.6 k Ω
(c) 1.2k Ω (d) 3 k Ω

Ans: (d)

Sol:

The impedance required to convert the meter to 600V voltmeter.



$$Z = \frac{600}{200 \times 10^{-3}} \Rightarrow 3k\Omega$$

Inductance of meter coil for full scale deflection

$L =$

$$\frac{(0.1 + 0.2 \times 100 \times \pi / 180)}{4\pi} \Rightarrow 0.0357$$

Reactance of meter coil at 50Hz

$$= 2\pi \times 50 \times 0.0357 \Rightarrow 11.22 \Omega$$

$\therefore$ Series resistance required

$$\begin{aligned} R_s &= \sqrt{(3000)^2 - (11.22)^2} \\ &= 2999.9 \\ &\approx 3k\Omega \end{aligned}$$

Example 9:

Which type of the instrument has the high power consumption?

- (a) Moving iron
- (b) PMMC
- (c) Hotwire
- (d) All instruments have identical power consumption

Ans: (c)

Errors in M.C & M.I Instruments

1. Frictional Error

To reduce frictional error, so for the same developed torque, weight of the system should be

less than torque i.e., $\left(\frac{\text{torque}}{\text{weight}}\right)$ ratio has to be high

In MC Instrument coil is moving hence weight of moving system is more compared to MI

Note: If weight decreases, Torque/ weight increases sensitivity increases, error ↓, Accuracy ↑

Frictional error is moderately more in M.C compared to M.I

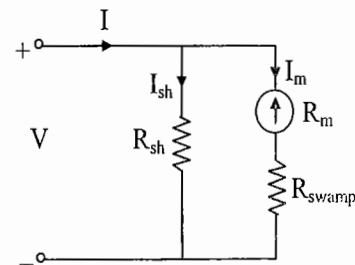
2. Temperature Error.

- Because of temperature spring is expand and indicates less than the actual value and coil resistance also changes
- D.C Current is present in MC and RMS current is present in M2, Hence the heat developed is more for RMS current compared to D C Current
- Temperature error is more in MI than MC

In MC instrument

The temperature effect is more in Ammeters than voltmeters because ammeter resistance is very less.

To reduce temperature effect in Ammeter a 'swamping resistor' is connected in series with meter.



$$I_m = \frac{I R_{sh}}{R_{sh} + R_m + R_{swamp}}$$

$I/I_m = m = \text{multiplying factor}$

$$\frac{R_{sh} + R_m + R_{swamp}}{R_{sh}} = \frac{I}{I_m} = m$$



$$1 + \frac{R_m + R_{\text{swamp}}}{R_{\text{sh}}} = m$$

$$R_{\text{sh}} = \frac{(R_{\text{swamp}} + R_m)}{m - 1}$$

R_{swamp} is made of manganin for D.C

R_{swamp} is made of constantan for A.C

Its Value is 30-40 times the ammeter resistance.

The value of shunt resistance (with swamping resistor) is more than the value of shunt resistance (without swamping resistor)

Conclusion:

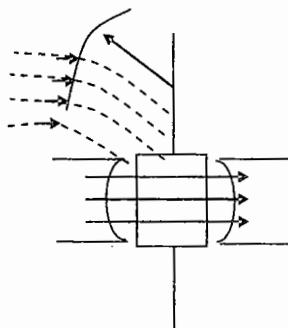
To reduce temperature effect swamping resistor is used

3. Frequency error

If same Input is measured at different frequencies then the meter gives different values, then it is called frequency error.

- MC used for DC so no frequency error in MC
- MI measures AC hence frequency error is present

Stray magnetic Field error



MC → Paramagnetic is used (strong magnetic field)

M.I → electromagnetic (weak magnetic field)

- Due to external magnetic field Inside meter operating field is disturbed. It is called stray magnetic field error.
- Stray magnetic field depends on magnitude of external magnetic field and its direction
- when magnitude of external magnetic field is in phase with main field, stray magnetic field error will be more
- To reduce this error Iron shielding is used
- The stray magnetic field error is more in MI compared to MC

Hysteresis & Eddy current Error in M.I

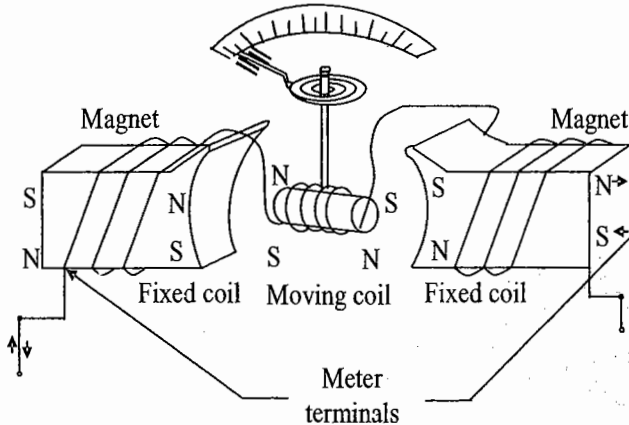
Hysteresis Error:

- Because the use of high permeability soft iron has large hysteresis loss than this error will be more compare to MC
- To reduce the hysteresis loss Nickel-Iron alloy is used In the manufacturing of M.I which has low value of hysteresis area & hence minimum error

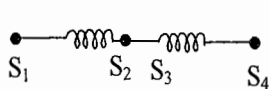
Eddy Current Error :

- In MI Measurement eddy current error is $\propto f^2$
- Eddy current error is constant for high frequency. Hence because of this reason M.I Instrument is suitable up to 125 Hz only
- Compared to M.C temperature $\uparrow$, stray load losses $\uparrow$, Hysteresis frictional loss $\downarrow$.
- To measure very low currents & voltage at very high frequency (50 MHz) thermocouple ammeter and thermocouple voltmeters are used

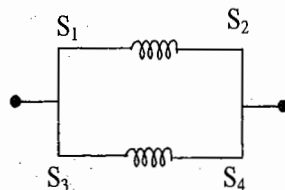
Electro Dynamometer



- In dynamometer type instruments air core is used for both fixed coil and moving coil. Hence this instrument is almost free from hysteresis and eddy current error. Hence it is used as transfer type instrument for power frequency
- Thermocouple instrument can be used as transfer type instrument for high frequency.



Series for low currents



Parallel for high currents

Fixed coil connections for low and high currents

Operation:

It works on principle of dc motor ie when a current carrying conductor placed in a magnetic field experiences a mechanical force.

$$T_d = I_1 I_2 \cos \phi \frac{dM}{d\theta}$$

Note: $\cos \phi$ here is not a power factor

I_1 = fixed/current coil current

I_2 = moving/potential coil current

θ = angle between I_1 and I_2

M = Mutual inductance

- Spring control is used to produce controlling torque (springs are carrying current)

$$T_c = k_c \theta$$

- Air friction damping is used also called **pneumatic damping**

At final steady state position $T_c = T_d$

$$\theta = \frac{I_1 I_2 \cos \phi}{k_c} \frac{dM}{d\theta}$$

For dc: $T_d = I_1 I_2 \frac{dM}{d\theta}$

If I_1 & I_2 are same then $T_d = I^2 \frac{dM}{d\theta}$

EMMC used as

Voltmeter, Ammeter and wattmeter for ac and dc.

Wattmeter is more popularly used.

Application

To convert as dynamometer ammeter connect both fixed coil and moving coil in series. This combination in series with load

	PMMC	MI	EMMC/Dynamometer
T_d	BINA	$\frac{1}{2} I^2 \frac{dL}{d\theta}$	$I_1 I_2 \cos \phi \frac{dM}{d\theta}$
θ	$\frac{BINA}{k_c}$	$\frac{1}{2} \frac{I^2}{k_c} \frac{dL}{d\theta}$	$\frac{I_1 I_2 \cos \phi}{k_c} \frac{dM}{d\theta}$



Example 10:

The mutual inductance of a dynamometer ammeter varies with deflection θ (degree) as $M=6\cos(\theta+30^\circ)$ mH.

The deflection torque produced by a current of 50 mA corresponding to a deflection of 60° is

- (a) $10 \mu\text{N-m}$ (b) $15 \mu\text{N-m}$
(c) $20 \mu\text{N-m}$ (d) $25 \mu\text{N-m}$

Ans: (b)

Sol: $T_d = I^2 \frac{dM}{d\theta};$

$$M = 6 \cos(\theta+30^\circ) \times 10^{-3}$$

$$\left. \frac{dM}{d\theta} \right|_{60^\circ} = 6 \sin(\theta+30^\circ) \times 10^{-3}$$

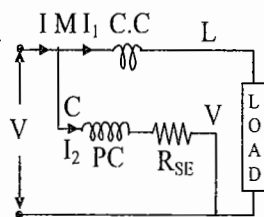
$$= 6 \times 1 \times 10^{-3} = 6 \times 10^{-3}$$

$$T_d = (50 \times 10^{-3})^2 \times 6 \times 10^{-3}$$

$$= 15000 \times 10^{-9} = 15 \mu\text{N-m}$$

• **To convert as Dynamometer wattmeter:**

Connect current coil in series with the load and potential coil in parallel with load



M = Main $I_1 = I$

L - Load $I_2 = \frac{V}{R_s}$

C - common

V - voltage

$$T_d = I_1 I_2 \cos\phi \frac{dM}{d\theta}$$

$$= I \frac{V}{R_s} \cos\phi \frac{dM}{d\theta}$$

$$= VI \cos\phi \left(\frac{1}{R_s} \frac{dM}{d\theta} \right)$$

$$T_d = P \left(\frac{1}{R_s} \frac{dM}{d\phi} \right)$$

$$\text{Power} = VI \cos\phi$$

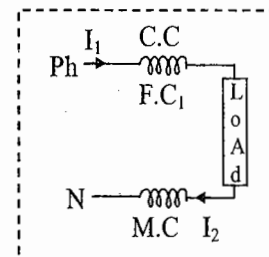
At balance

$$T_c = T_d$$

$$\theta = \frac{P}{R_s} \frac{dM}{d\theta}$$

$$\theta \propto P \quad \text{It is average active power.}$$

• **To convert as dynamometer ammeter.**



FC - current coil

MC - voltage coil (or) potential coil

$$I_d = I_1 = I_2$$

$$T_d = I^2 \cos\phi \frac{dM}{d\theta}$$

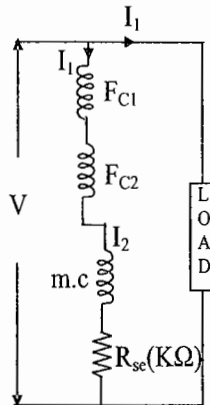
$$T_c = k_c \theta$$

$$k_c \theta = I^2 \cos\phi \frac{dM}{d\theta}$$

$$\theta \propto I^2$$

Non linear scale

- This type ammeter when used for ac reads rms value.
- To convert as dynamometer voltmeter**
Fixed coil and moving coil connected in series and this combination is connected in parallel with the load.



$$I_1 = I_2 = \frac{V}{R_{sc}}$$

$$T_d \propto I^2$$

$$T_d \propto \frac{V^2}{R_{sc}^2}$$

$$T_c = k_c \theta$$

$$T_c = T_d$$

$$\theta = \frac{V^2}{R_{sc}^2 k_c} \frac{dM}{d\theta}$$

$$\theta \propto V^2$$

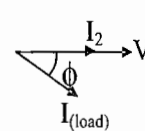
- Non – linear scale used for ac and dc measurements
- Moving coil measures potential hence it's called potential coil
- Fixed coil carries the load current and it is connected in series with the load hence it's called current coil.

Error due to type of wattmeter connection:

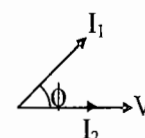
MC connection	LC connection
(MC short dia)	(LC Short dia)
Measured total power = $P_m = P_{load} + I^2 r$ $P_m = P_T + I^2 r$ Error = $\left(\frac{P_m - P_T}{P_T} \right) \times 100$ Error = $P_m - P_T = I_1^2 r$ $r \rightarrow$ Resistance of current coil	Assume CC resistance is small Measured power $= P_m = P_{load} + \frac{V^2}{R_s}$ Error = $P_m - P_{load}$ $= \frac{V^2}{R_s} = I_2^2 R_s$ $I_2 = I - I_1$
Error is minimum if load current 'I' is small	For large load current $I_1 \uparrow, I_2 \downarrow = I - I_1 \uparrow$ I_2 is small, so error $I_2^2 R_s$ decreases Error is minimum if load current is high

$$I^2 r = \frac{V^2}{R_s}$$

- A potential coil is connected with a series resistance (R_s) which is in the order of $K\Omega$ hence the inductive reactance of the PC is small and neglected.
- Current flowing through potential coil is in phase with voltage 'V'



for lag load



for lead load

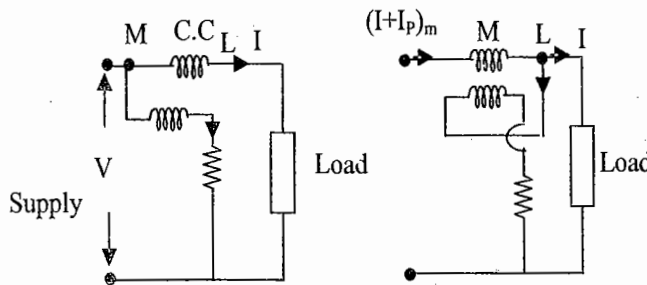


Note: Thickness of the current coil conductor is higher to withstand the higher load currents

- Standard conductors are used to reduce the eddy current losses.

Example 11:

A 250V, 10A electrodynamic wattmeter has a resistance of current and potential coil of 0.5Ω and 12500Ω respectively. Find the percentage error due to each of the two methods of connections shown in figure below with unity power factor loads at 250V. With currents 12A neglect error due to pressure coil inductance.



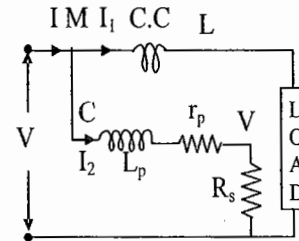
- (a) 2.54% ; 0.146% (b) 2.34% ; 0.166%
(c) 2.43% ; 0.616% (d) 2.15% ; 0.158%

Ans: (b)

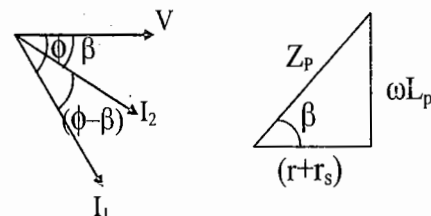
$$\begin{aligned} \text{Sol: } P_t &= V_L I_L \cos \phi & P_t &= V_L I_L \cos \phi \\ &= 250 \times 12 \times 1 & &= 250 \times 12 \times 1 \\ &= 3000 \text{ W} & &= 3000 \text{ W} \\ (P_m)_{MC} &= P_t + I_L^2 R_c & (P_m)_{LC} &= P_t + \frac{V^2}{R_p} \\ &= 3000 + (12)^2 \times 0.5 & & \\ &= 3000 + \frac{(250)^2}{12500} & & \\ \% \text{ error} &= \frac{3072 - 3000}{3000} \times 100 & & \\ &= \frac{3005 - 3000}{3000} \times 100 & & \\ &= 2.4\% & & = 0.166\% \end{aligned}$$

Error in Watt Meters

Error due to potential coil inductance:



L_p = inductance of PC
 r_p = resistance of PC ($r_p \ll R_s$)
 R_s = series multiplier resistance ($K\Omega$)
 Z_p = impedance of PC
 $= ((r_p + r_s) + j\omega L_p)$
 β = PC phase angle



$$\begin{aligned} \cos \beta &= \left(\frac{R_s}{Z_p} \right) \\ \beta &= \tan^{-1} \left(\frac{\omega L_p}{r_p + r_s} \right) \end{aligned}$$

$$T_d \propto I_1 I_2 \cos(\phi - \beta) \frac{dM}{d\theta}$$

$$\begin{aligned} W_M (\text{measured power}) &\propto \theta \\ &= \frac{V}{Z_p} \frac{I}{K_c} \cos(\phi - \beta) \frac{dM}{d\theta} \end{aligned}$$

$$W_M = \frac{VI \cos \beta \cos(\phi - \beta)}{R_s k_c} \frac{dM}{d\theta}$$

W_M (True power) without ' L_p '



$$T_d \propto I_1 I_2 \cos \phi \frac{dM}{d\theta}$$

$$T_d \propto \frac{VI \cos \phi}{R_s} \cdot \frac{dM}{d\theta}$$

$$W_T = \frac{VI}{R_s} \cos \phi \frac{dM}{d\theta}$$

$$\frac{W_T}{W_m} = \frac{\cos \phi}{\cos \beta \cos(\phi - \beta)}$$

$$\text{Correction factor (c.f)} = \frac{\cos \phi}{\cos \beta \cdot \cos(\phi \mp \beta)}$$

'-' for lag

'+' for lead

$$W_T = \frac{\cos \phi}{\cos \beta (\cos \phi \cdot \cos \beta + \sin \phi \cdot \sin \beta)} \cdot W_m$$

$$\frac{W_T}{W_m} = \frac{\cos \phi}{\cos \beta \cos \phi \cdot \cos \beta (1 + \tan \phi \cdot \tan \beta)}$$

$$W_T = \frac{1 + \tan^2 \beta}{1 + \tan \phi \tan \beta} \cdot W_m$$

$$W_T = \frac{1}{1 + \tan \phi \tan \beta} \cdot W_m$$

$$W_m - W_T = W_T (\tan \phi \tan \beta)$$

$$W_m - W_T = VI \cos \phi \cdot \frac{\sin \phi}{\cos \phi} \cdot \sin \beta$$

$$\text{Error} = W_m - W_T = VI \sin \phi \cdot \tan \beta$$

$$\text{error} = \frac{W_m - W_T}{W_T} \times 100$$

$$= \frac{VI \sin \phi \cdot \tan \beta}{VI \cos \phi} \times 100$$

$$\% \text{ error} = \pm (\tan \phi \tan \beta \times 100)$$

Example 12:

A voltage $100 \sin \omega t + 40 \cos (3 \omega t - 30^\circ) + 50 \sin (5 \omega t + 45^\circ)$ V is applied to the pressure coil circuit of a wattmeter and through the current coil is passed a current of $8 \sin \omega t + 6 \cos (5 \omega t - 120^\circ)$ A. What percentage of reading on the wattmeter is due to fundamental?

- (a) 31.9% (b) 50%
(c) 91.2% (d) 156.7%

Ans:(c)

Sol: The wattmeter reading is

$$W = \frac{1}{2} [100 \times 8 \cos 0 + (6 \times 50) \sin (45 - (-30))] \\ = 438.82 \text{ W}$$

The fundamental power is

$$= \frac{1}{2} \times 100 \times 8 \Rightarrow 400 \text{ W}$$

The wattmeter reading w.r.t fundamental

$$\text{power} = \frac{400}{438.82} \times 100 \Rightarrow 91.2 \%$$

Example 13:

If the Reactance of the pressure coil circuit of a wattmeter is 1% of it's resistance, calculate the % error due to this cause at power factor of 0.8 respectively.

- (a) 0.72% (b) 0.75%
(c) 0.71% (d) 0.78%

Ans: (b)

Sol: From Δ le $\tan \beta = \frac{X_p}{r_p}$

$$= \frac{0.01 \times r_p}{r_p} = 0.01$$

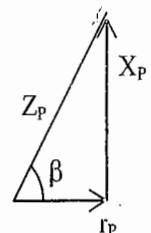
$$\beta = \tan^{-1}(0.01)$$

$$\cos \phi = 0.8$$

$$\phi = \cos^{-1}(0.8)$$

$$\tan \phi = \tan [\cos^{-1}(0.8)] = 0.75$$

$$\therefore \% \text{ error} = + (\tan \phi \times \tan \beta \times 100) \\ = + (0.75 \times 0.01 \times 100) \\ = 0.75\%$$





⇒ Because of potential coil self inductance wattmeter reads more than the actual power for 'Lag' power factor error is +ve (for lag p.f)

⇒ Wattmeter reads less than the actual power and error is -ve for lead p.f

$$\text{Error} \propto \frac{1}{\cos \phi} \text{ (Lead/lag)}$$

Error Due to different load p.f

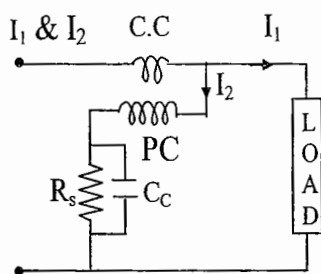
ϕ	$\cos \phi$	$\sin \phi$	$(W_m - W_T) = V.I. \sin \phi \cdot \tan \beta$
0	1	0	0 (min error)
30	0.866	0.5	$0.5 V.I \tan \beta$
90	0(low)	1	$VI \tan \beta$ (max error)

Note: Under low p.f load conditions the error in the measurement of the power is maximum in a normal (or) unity p.f. meter.

To reduce this error under low p.f load conditions. A specially designed low p.f watt meter is used.

Note: In case of O.C test of a transformer the p.f of transformer is between (0.2 to 0.4) under this low p.f condition to measure the power accurately L.P.F. watt meters are used.

Low power factor wattmeter: (Diagram of LPF)



In designing of the L.P.F. wattmeter

L.P.F. → $\cos \phi$ → low

$$T_d \propto I_1 I_2 \cos \phi$$

$$T_d \propto \frac{V}{R_s} I_1 \cdot \cos \phi$$

For L.P.F ' $\cos \phi$ ' is low. so that " T_d " is low to increase the value of " T_d " A series multiplier resistance ' R_s ' is kept low Value $R_s \downarrow$, $T_d \uparrow$

- To compensate Inductance of the 'PC' a Compensating capacitor is connected in parallel is

$$C_c \approx \frac{L_p}{R_s^2}$$

Note: In dynamometer watt meter the P.C should be highly resistive nature.

Example 14:

An electrodynamic wattmeter is used for measurement of power in a single phase circuit. The load voltage is 100V and the load current is 9A at a lagging power factor of 0.1. The wattmeter voltage circuit has a resistance of 3000Ω and inductance of 30mH Estimate the percent error in wattmeter reading when the pressure coil is connected on the supply side. The frequency is 60Hz.

- (a) 6.82% (b) 5.485%
(c) 9.82% (d) 7.44%

Ans: (d)

Sol: Load power = $100 \times 9 \times 0.1 = 90W$

Load power factor $\cos \phi = 0.1$

∴ $\phi = 84.26^\circ$ and $\sin \phi = 0.995$ and $\tan \phi = 9.95$

Resistance of pressure coil circuit $R_p =$

3000Ω

Reactance of pressure coil circuit =

$$2\pi \times 60 \times 30 \times 10^{-3} = 11.304\Omega$$



As the phase angle of the pressure angle of the pressure coil circuit is small

$$\beta = \tan \beta \approx \frac{11.304}{3000} = 0.00376 \text{ rad}$$

When the pressure coil is connected on the load side, the wattmeter measures power loss in pressure coil circuit in addition to load power.

$$\begin{aligned} \text{True power} &= VI \cos \phi \\ &= 100 \times 9 \times 0.1 \\ &= 90 \text{ W} \end{aligned}$$

Let us consider only the effect of Inductance
Reading of watt meter

$$\begin{aligned} &= \text{true power} (1 + \tan \phi \tan \beta) \\ &= 90 (1 + 9.95 \times 0.00376) \\ &= 93.36 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Power loss in pressure coil circuit} &= \frac{V^2}{R_p} \\ &= \frac{(100)^2}{3000} \\ &= 3.33 \text{ W} \end{aligned}$$

∴ Reading of wattmeter considering the power loss in pressure coil circuit.

$$= 93.36 \text{ W} + 3.33 = 96.697$$

$$\% \text{error} = \frac{96.69 - 90}{90} \times 100 = 7.44\%$$

Example 15:

The torque produced in an indicating instrument by utilizing magnetic, electro dynamic, thermal, chemical and electro static effect is known as.

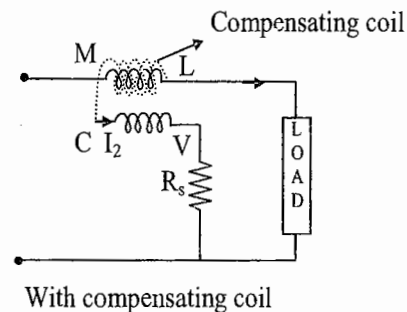
- Controlling torque
- Deflecting torque
- Damping torque
- Restoring torque

Ans:(b)

Compensation Coil (LC connection)

- In L.C Connection Current Coil carrying additionally potential coil current, because of this additional flux is produced in current coil & it links with P.C induces an EMF & causes current in potential coil because of this power loss is present in P.C
- To compensate the power loss in pressure coil compensating coil with equal number of turns of current coil is connected in series opposition to potential coil.

Compensating coil requires only for LC connection not for MC connection.



Errors in Dynamometer Type Instruments

1. Frictional Error (Torque/weight):

- Potential coil has more number of turns then weight of moving coil is increased.
- Then It has more Frictional error compared to M.C and M.I

2 Temperature Error:

- Because presence of two coils net heat developed is more. Then temperature error is more compare to other Instruments.



3 Frequency error:

- When it is used as a voltmeter (or) ammeter Net Inductance is more then the frequency error is more
- But when used as a wattmeter Inductance is very less, then frequency error is low.

Hysteresis & Eddy current error:

- Because of the use of Air core, The hysteresis and eddy current errors are almost zero.
- Both Fixed coil & moving coils are electro magnet then poor magnetic field is present

- The external magnetic field can easily distort the inside operating field. Hence stray magnetic field is more compared to other

Note: For portable wattmeter to reduce the stray magnetic field error; Iron shielding is used.

- For precision watt meters to reduce stray magnetic field error, static system is used, in this two set of fixed coil and moving coils are used.

Class Room Practice Questions

LEVEL-1 Questions

01. Consider the following statements regarding the use of electro-dynamometer type indicating instruments:

It is used for measuring

1. direct current
2. alternating current
3. direct voltage
4. alternating voltage

Of these statements

- (a) 1 and 2 are correct
- (b) 3 and 4 are correct
- (c) 1 and 3 are correct
- (d) 1, 2, 3 and 4 are correct

02. Which one of the following measuring instruments would consume the LOWEST power from the source during measurement?

- (a) Permanent magnet moving coil
- (b) Electronic multimeter
- (c) Electrostatic instrument
- (d) Moving iron instrument

03. **Assertion (A):** The needle of an indicating instrument attains a position where deflecting and control torques acting on the moving system are equal and opposite.

Reason(R): The oscillations of the needles are suppressed by the damping mechanism

- (a) Both A and R are individually true and R is the correct explanation of A
- (b) Both A and R are individually true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

04. Match List I (Instrument) with List II (Property/use) and select the correct answer using the codes given below the Lists:

List I

- A. PMMC
- B. Moving iron
- C. Thermocouple
- D. Electrostatic type

List II

1. Square law type scale
2. Very good high frequency response
3. Linear scale over the entire range
4. Voltmeter

Codes:

	A	B	C	D
(a)	4	1	2	3
(b)	3	2	1	4
(c)	4	2	1	3
(d)	3	1	2	4

05. **Assertion(A):** Any indicating instrument should have a damping torque.

Reason(R): Without a damping torque, the pointer takes more settling time

06. In an electro-dynamometer, a moving coil has an area A , turn N and carries a current I producing a magnetic flux B . The torque on the moving coil is proportional to

- (a) I
- (b) I^2
- (c) $B \cdot I^2$
- (d) $A \cdot N \cdot B \cdot I^2$

07. **Assertion(A):** The electro-dynamo-meter type instruments have high power consumption.

Reason(R): The current under measurement has to provide the field flux.

08. Which one of the following measuring devices has minimum loading effect on the quantity under measurement?

- (a) PMMC
- (b) CRO
- (c) Hot wire
- (d) Electro-dynamometer



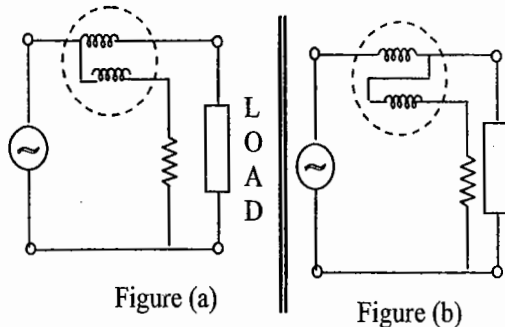
09. Which of the following meters requires an external power source for its operation?
(a) PMMC meter
(b) Hot wire ammeter
(c) Electronic voltmeter
(d) Electrodynamicometer
10. Consider the following statements regarding the controlling torque:
1. It is not present in power factor meter.
2. It opposes the deflecting torque
3. It is provided by air friction or by fluid friction
Which of these statements are correct?
(a) 1, 2 and 3 (b) 1 and 3 only
(c) 2 and 3 only (d) 1 and 2 only
11. A voltmeter with an internal resistance of $200\text{k}\Omega$ when connected across an unknown resistance reads 250V . The milliammeter internal resistance = 0 connected in series with the above combination reads 10mA . The actual value of the unknown resistance is
(a) $25\text{k}\Omega$ (b) $200\text{k}\Omega$
(c) $28.56\text{k}\Omega$ (d) $20\text{k}\Omega$
12. An unshielded moving iron voltmeter is used to measure the voltage in an a.c. circuit. If a stray d.c. magnetic field having a component along the axis of the meter coil appears, the meter reading would be
(a) unaffected
(b) decreased
(c) increased
(d) either decreased or increased depending on the direction of the d.c. field
13. An advantage of a permanent magnet moving coil instrument is that it is
(a) free from friction error
(b) has high (torque/weight of the moving parts) ratio
(c) has low (torque/weight of the moving parts) ratio
(d) can be used on both a.c. and d.c.
14. The scale of a voltmeter is uniform. Its type is
(a) moving iron
(b) induction
(c) moving coil permanent magnet
(d) moving coil dynamometer
15. The effect of stray magnetic fields on the actuating torque of a portable instrument is maximum when the operating field of the instrument and the stray fields are
(a) perpendicular (b) parallel
(c) inclined at 60° (d) inclined at 30°
16. An indicating instrument is more sensitive if its torque to weight ratio is
(a) much larger than unity
(b) of the order of unity
(c) much less than unity
(d) made deflection-dependent
17. Which one of the following materials is used in the fabrication of swamping resistance of a PMMC instruments?
(a) Copper (b) Aluminium
(c) Manganin (d) Tungsten
18. Which one of the following types of instruments can be used to determine the r.m.s value of ac voltage of high magnitude (10kV) and of any wave shape?
(a) Moving iron instruments
(b) Dynamometer type instruments
(c) Induction instruments
(d) Electrostatic instruments
19. A compensated wattmeter has its reading corrected for error due to
(a) the frequency



- (b) friction
(c) power consumed in current coil
(d) power consumed in potential coil
20. **Assertion (A) :** General purpose dynamometer type wattmeter cannot indicate the correct value of power at low power factors.
Reason (R) : The presence of self-inductance in the pressure coil circuit introduces an error in the indicated value which increases appreciably with decrease in power factor of the load.
21. A wattmeter has a range of 1000 W with an error of $\pm 1\%$ of full scale deflection. If the true power passed through it is 100W, then the relative error would be
(a) $\pm 10\%$ (b) $\pm 5\%$
(c) $\pm 1\%$ (d) $\pm 0.5\%$
22. What is the effect of inductance in the pressure coil on performance of a dynamometer type wattmeter?
(a) It reads low on lagging power factor and high on leading power factor
(b) It reads high on lagging power factor and low on leading power factor
(c) Its reading is not affected to all
(d) It always reads low
23. The current and potential coil of a dynamometer type wattmeter were accidentally interchanged while connecting. After energizing the circuit, it was observed that the wattmeter did not show the reading. This could be due to the
(a) Damage to potential coil
(b) Damage to current coil
(c) Damage to both the potential and current coil
(d) Loose contacts
24. The magnetic field responsible for the production of the deflecting torque in an accurate dynamometer type wattmeter, being very weak, the accuracy of the measurement can be increased by providing a
(a) Magnetic shield around the instrument
(b) Compensating winding along with the pressure coil
(c) Astatic arrangement to the moving system of the instrument
(d) Capacitance shunt across a portion of the pressure coil
25. The moving coil in a dynamometer wattmeter is connected
(a) in series with the fixed coil
(b) across the supply
(c) in series with the load
(d) across the load
26. A dynamometer type wattmeter responds to the
(a) average value of active power
(b) average value of reactive power
(c) peak value of active power
(d) peak value of reactive power
27. Consider the following statements in respect of two types of a dynamometer wattmeter connections are shown above in figure (a) and figure (b).
1. Figure(a) is used when applied voltage is high and load current flowing is low.
 2. Figure (b) is used when applied voltage is low and load current flowing is high
 3. If the order of connections in Statements 1 and 2 are in opposite order, gross error will be increased.
 4. If the order of connections in Statements 1 and 2 are in opposite order, gross error will be reduced.



Which of the statements given above are correct?



- (a) Only 1 and 2 (b) Only 1 and 3
(c) 1, 2 and 3 (d) 1, 2 and 4

28. A compensated wattmeter has its reading corrected for error due to which one of the following parameter?

- (a) Frequency
(b) Friction
(c) Power consumed in current coil
(d) Power consumed in pressure coil

LEVEL-2 Questions

01. The current passing through a 10Ω resistance is given by $i = 3 + 4\sqrt{2} \sin 314t$ this current is measure by a PMMC meter. What is the measured value?

- (a) 3A (b) 4A
(c) 5A (d) $4\sqrt{2}$ A

02. The current following in a 20Ω resistor is given by $i = 2 + 4\sin 314t$. This current is measured by a hot wire ammeter. What is the measured value?

- (a) 2A (b) 3.46A
(c) 4A (d) 2.83A

03. Which one of the following meter has maximum loading effect an the circuit under measurement?

- (a) $1000\Omega/\text{volt}$ (b) $100\Omega/\text{volt}$
(c) $1\text{m}\Omega/\text{volt}$ (d) $10\text{m}\Omega/\text{volt}$

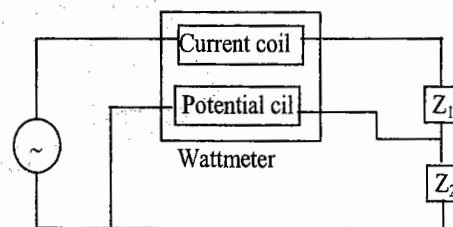
04. A current of $-8 + 6\sqrt{2} (\sin \omega t + 30^\circ)$ A is passed though three meters. They are a center zero PMMC meter, true r.m.s meter and moving Iron instrument. The respective reading (in A) will be

- (a) 8, 6, 10 (b) 8, 6, 8
(c) -8, 10, 10 (d) -8, 2, 2

05. A capacitor is connected across a portion of resistance of the multiplier in order to make the pressure coil circuit of the wattmeter non-inductive. The value of this resistance is r while the total resistance and inductance of the pressure circuit are respectively R_p and L . The value of the capacitance C is

- (a) $\frac{L}{R_p^2}$ (b) $\frac{0.41L}{r^2}$
(c) $\frac{L}{r^2}$ (d) $\frac{0.41L}{R_p^2}$

06. A wattmeter is connected as shown in the fig. the wattmeter reads



- (a) zero always
(b) Total power consumed by Z_1 & Z_2
(c) power consumed by Z_1
(d) power consumed by Z_2

07. A d.c. voltmeter has a sensitivity of $1000\Omega/\text{volt}$. When it measures half full scale in 100 V range, the current through the voltmeter is



- (a) 100 mA (b) 1 mA
(c) 0.5 mA (d) 50 mA

08. In a PMMC instrument, the central spring stiffness and the strength of the magnet decrease by 0.04% and 0.02% respectively due to a rise in temperature by 1°C . With a rise in temperature of 10°C , the instrument reading will

- (a) increase by 0.2%
(b) decrease by 0.2%
(c) increase by 0.6%
(d) decrease by 0.6%

Solutions

LEVEL-1

01. Ans: (d)

02. Ans: (b)

Sol: Electronic instruments have very high input impedance. So, amount of current taken, and hence power consumed will be very less for them.

03. Ans: (b)

04. Ans: (d)

Sol: Torque equation for PMMC:

$$K\theta = BINA$$

$$\theta \propto I, \text{ Hence, linear scale}$$

Torque equation for MI:

$$K\theta = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

$$\theta \propto I^2$$

Hence, square law type scale

Thermocouple instruments have very good high frequency response.

05. Ans: (a)

06. Ans: (b)

07. Ans: (a)

08. Ans: (b)

Sol: CRO has input impedance in the range of $\text{M}\Omega$. So it has least loading effect among given.

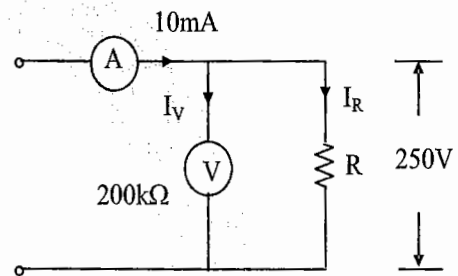
09. Ans: (c)

Sol: The Electronic instrument needs an external source of supply.

10. Ans: (d)

11. Ans: (c)

Sol:



$$V = 250 \text{ V}, R_V = 200 \text{ k}\Omega, I = 10 \text{ mA}$$

$$I_V = \frac{250}{200 \text{ k}\Omega} = 1.25 \text{ mA}$$

$$I_R = 10 \text{ mA} - 1.25 \text{ mA} = 8.75 \text{ mA}$$

$$R = \frac{250 \text{ V}}{8.75 \text{ mA}} = 28.56 \text{ k}\Omega$$

12. Ans: (d)

13. Ans: (b)

14. Ans: (c)

15. Ans: (b)

16. Ans: (a)

Sol: Torque to weight ratio determines the sensitivity of meter.

It's value should be much higher than unity for better sensitivity.

17. Ans: (c)

Sol: Manganin has negligible temperature coefficient of resistance. So, it is used to overcome temperature errors in instrument.



18. Ans: (d)

Sol: For measurement of very high voltage, Electrostatic voltmeters are used due to their low current.

19. Ans: (d)

20. Ans: (a)

21. Ans: (a)

Sol: $W = (0 - 1000) \text{ W}$

Error = $\pm 1\% \rightarrow \text{fsd}$

$W_T = 100 \text{ W}$

Error = $\pm \frac{1}{100} \times 1000 \text{ W} = \pm 10 \text{ W}$

%Error w.r.t true value

$= \pm \frac{10 \text{ W}}{100 \text{ W}} \times 100 = \pm 10\%$

22. Ans: (b)

23. Ans: (b)

Sol: Current coil has very small resistance. If it is connected in parallel to supply then very high current will flow through it causing damage to it.

24. Ans: (c)

Sol: Accuracy is improved by providing a static arrangement to the moving system of the instrument.

25. Ans: (b)

26. Ans: (a)

27. Ans: (c)

28. Ans: (d)

Level-2 Key

01. (a) 02. (b) 03. (c) 04. (c)

05. (b) 06. (d) 07. (c) 08. (a)

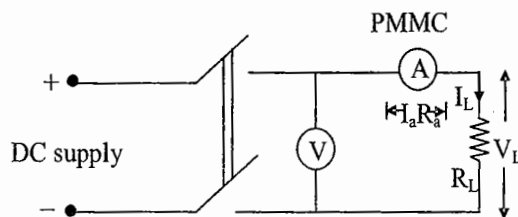
3

Measurement of Power & Energy

- DC power $\rightarrow$ (Only for R-Load) $\Rightarrow P = VI$
 - Voltmeter – Ammeter method
 - Ammeter – voltmeter method
- AC power $\rightarrow R, L$ & C loads $\Rightarrow P = VI \cos\phi$
Measured by Dynamometer type wattmeter

Measurements of DC power

1. V – A method:



Conclusions

- $P_m > P_t$ (Measured power greater than true power)
- Error is due to ammeter only
- Suitable for measurement of high resistance and low load current

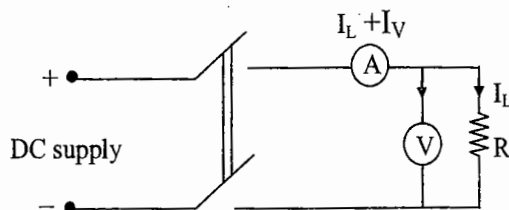
$$P_m = \text{voltmeter reading} \times \text{Ammeter reading}$$

$$= (V_L + V_a) \times I_L$$

$$= V_L I_L + V_a I_L$$

$$P_m = P_{\text{true}} + I_L^2 R_L \rightarrow \text{Power loss}$$

2. A – V method:



$$(i) P_m > P_t$$

(ii) Error is due to voltmeter

(iii) Suitable for measurement of low resistance and high load currents

$$P_m = V_L \times (I_L + I_v)$$

$$P_m = V_L I_L + V_L I_v$$

$$P_m = P_t + V_L I_v$$

Note:

In both the methods ($P_m > P_t$)

Note:

In both the methods the error due to always “Load side” of instruments

- AC power can be measured by using wattmeter as discussed previous lesson

Measurement of poly phase power:

3 - ϕ

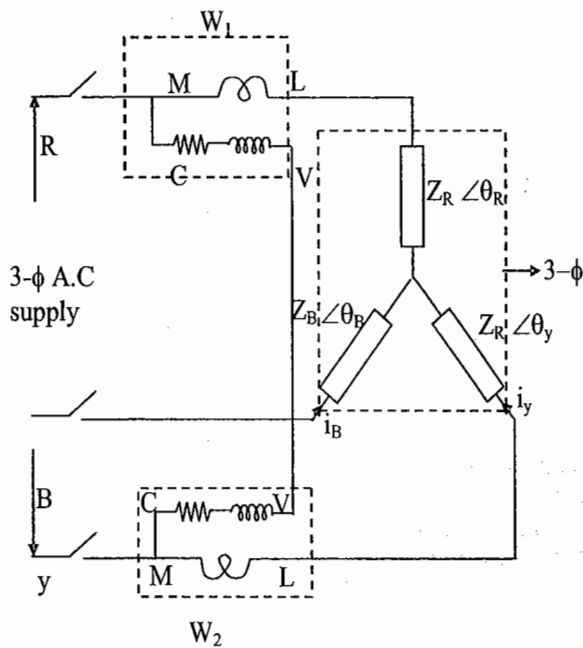
Balanced ($I_n = 0$) $Z_1 = Z_2 = Z_3$ & $\theta_1 = \theta_2 = \theta_3$

Unbalanced ($I_n \neq 0$)

3 - ϕ 4 wire			3 - ϕ 3 wire		
	Min	Max		Min	Max
Balanced	1	3	Balanced	2	2
unbalanced	2	3	Unbalanced	2	2

- For 3 ϕ 4- wire $\rightarrow$ neutral is present
- As per blondel's theorem, if neutral is not available for 'n' wire system then 'n-1' no. of wattmeter's are required for 'n' wire system.

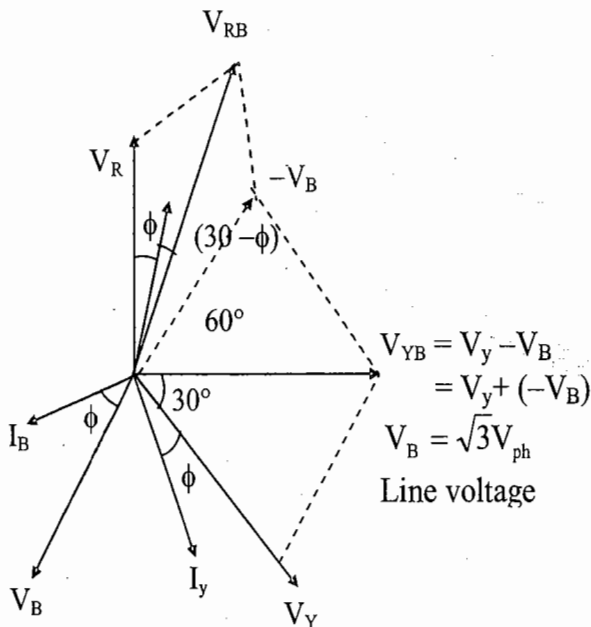
Consider 3- ϕ 'Y' connected lagging power factor balanced load with “R-Y-B” phase sequence ('B' phase made common)



3- ϕ Y- connected "lagging loads

$$W_1 = \vec{V}_{RB} I_R \cos \angle V_{RB} \times I_R$$

$$W_2 = \vec{V}_{YB} I_Y \cos \angle V_{YB} \times I_Y$$



$$W_1 = \vec{V}_{RB} \vec{I}_R \cos(30 - \phi)$$

$$W_2 = \vec{V}_{YR} \vec{I}_Y \cos(30 + \phi)$$

For balanced condition

$$V_{RB} = V_{YB} = V_L$$

$$I_R = I_Y = I_L$$

$$W_1 = V_L I_L \cos(30 - \phi)$$

$$W_2 = V_L I_L \cos(30 + \phi)$$

$$P = W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi$$

In star connection

$$V_L = \sqrt{3} V_{ph}; I_L = I_{ph}$$

$$P = 3 V_{ph} I_{ph} \cos \phi \rightarrow \text{Total } 3\phi \text{ power}$$

$$W_1 - W_2 = V_L I_L \sin \phi$$

$$W_1 - W_2 = \sqrt{3} V_{ph} I_{ph} \sin \phi$$

$$V_{ph} I_{ph} \sin \phi = \frac{W_1 - W_2}{\sqrt{3}}$$

$$Q_1 - \text{phase} = \frac{W_1 - W_2}{\sqrt{3}}$$

Reactive power for 3 - ϕ

$$Q_T = 3 \left(\frac{W_1 - W_2}{\sqrt{3}} \right) = \sqrt{3} (W_1 - W_2)$$

Power factor angle

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{V_L I_L \sin \phi}{\sqrt{3} V_L I_L \cos \phi}$$

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{\tan \phi}{\sqrt{3}}$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{3} (W_1 - W_2)}{(W_1 + W_2)} \right)$$

$$\text{p.f} = \cos \phi$$

Advantages of 2 - wattmeter method:

Note:

- (1) It can be used for balanced and unbalanced, whether neutral is available (or) not



Example 1:

The power flowing in a 3phase 3wire balanced load system is measured by two wattmeter method. The reading of wattmeter A is 7500W and of wattmeter B is -1500W ; then find what is the power factor of the system.

Sol: $P_1 = 7500W$

$$P_2 = -1500W$$

$$P = P_1 + P_2 = 7500 - 1500 = 6000W$$

$$P_1 - P_2 = 7500 + 1500 = 9000W$$

$$\phi = \tan^{-1} \left\{ \sqrt{3} \left[\frac{P_1 - P_2}{P_1 + P_2} \right] \right\}$$

$$= \tan^{-1} \sqrt{3} \left(\frac{9000}{6000} \right) = 69^\circ$$

$$p.f = \cos \phi = \cos 69^\circ = 0.358$$

Example 2:

A 3 ϕ , 500 V motor has a power factor of 0.4. Two wattmeter's connected to measure the input to be 30kW, Then the reading of each wattmeter.

(a) $W_1 = 34.85 \text{ kW}, W_2 = -4.85 \text{ kW}$

(b) $W_1 = 38.26 \text{ kW}, W_2 = -8.26 \text{ kW}$

(c) $W_1 = 69.68 \text{ kW}, W_2 = -39.681 \text{ kW}$

(d) $W_1 = 0 \text{ kW}, W_2 = 30 \text{ kW}$

Ans: (a)

Sol: $W_1 + W_2 = 30 \text{ kW} \text{ ----- (1)}$

$$\cos \phi = 0.4$$

$$\tan \phi = 2.291$$

$$\tan \phi = \frac{\sqrt{3}(W_1 - W_2)}{(W_1 + W_2)}$$

$$2.291 = \frac{\sqrt{3}(W_1 - W_2)}{30 \text{ kW}}$$

$$W_1 - W_2 = 39.68 \text{ kW} \text{ ----- (2)}$$

Solving equation (1) & (2)

$$W_1 = 34.84 \text{ kW}$$

$$W_2 = -4.85 \text{ kW}$$

Example 3:

The result of ratio of the readings of two watt meters connected to measure power in a balanced 3 - ϕ , 3 - wire load is 'r'. The load is known to be inductive with a lagging power factor. Find the power factor of the load?

(a) $\sqrt{0.2 + 3 \left(\frac{r-1}{r+1} \right)^2}$ (b) $\frac{1}{\sqrt{1 + 3 \left(\frac{r-1}{r+1} \right)^2}}$

(c) $\frac{r-1}{r+1}$ (d) $\frac{1}{\sqrt{2 + 3 \left(\frac{r-1}{r+1} \right)^2}}$

Ans: (b)

Sol: $\tan \phi = \sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \Rightarrow \sqrt{3} \left(\frac{\frac{W_1}{W_2} - 1}{\frac{W_1}{W_2} + 1} \right)$

$$\cos \phi = \frac{1}{\sec \phi}$$

$$= \frac{1}{\sqrt{1 + \tan^2 \phi}} \Rightarrow \frac{1}{\sqrt{1 + 3 \left(\frac{r-1}{r+1} \right)^2}} \because r = \frac{W_1}{W_2}$$



Readings of W_1, W_2 for different load p.f:

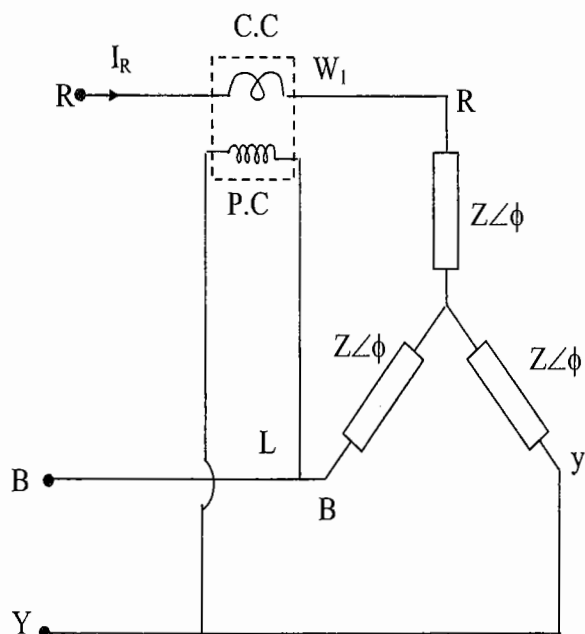
ϕ	$\cos\phi$	$W_1 = V_L I_L \cos(30 - \phi)$	$W_2 = V_L I_L \cos(30 + \phi)$	$W_1 + W_2$	Observations
0	1	$\frac{\sqrt{3}}{2} V_L I_L$	$\frac{\sqrt{3}}{2} V_L I_L$	$\sqrt{3} V_L I_L$	$W_1 = W_2$
30	0.866	$V_L I_L$	$\frac{V_L I_L}{2}$	$1.5 V_L I_L$	$W_2 = 2W_1$
60	0.5	$\frac{\sqrt{3}}{2} V_L I_L$	0	$\frac{\sqrt{3}}{2} V_L I_L$	$W_1 = 0$
90	0	$\frac{V_L I_L}{2}$	$-\frac{V_L I_L}{2}$	0	$W_1 = -ve$ $W_2 = +ve$

Note:

If one of the wattmeter shows negative then reverse either potential coils (or) current coil terminals and record the wattmeter reading with (-ve sign.)

Eg: Wattmeter reading is 20W after reversing current coil terminals then take the reading as 'W = -20W'.

Measurement of Reactive power:



$$W_1 = V_{YB} I_R \cos \angle V_{YB} - I_R$$

$$W_1 = V_{YB} I_R \cos(90 - \phi)$$

$$W = V_L I_L \sin \phi$$

$$W = \sqrt{3} V_{ph} I_{ph} \sin \phi$$

$$Q_T = \sqrt{3} (W) = 3 V_{ph} I_{ph} \sin \phi \rightarrow 3 - \phi \text{ reactive power}$$

NOTE:

- For measurement of reactive power the current coil is connected to one of the phases of a 3 - ϕ network
- Then the potential coil is to be connected between the remains two phases.

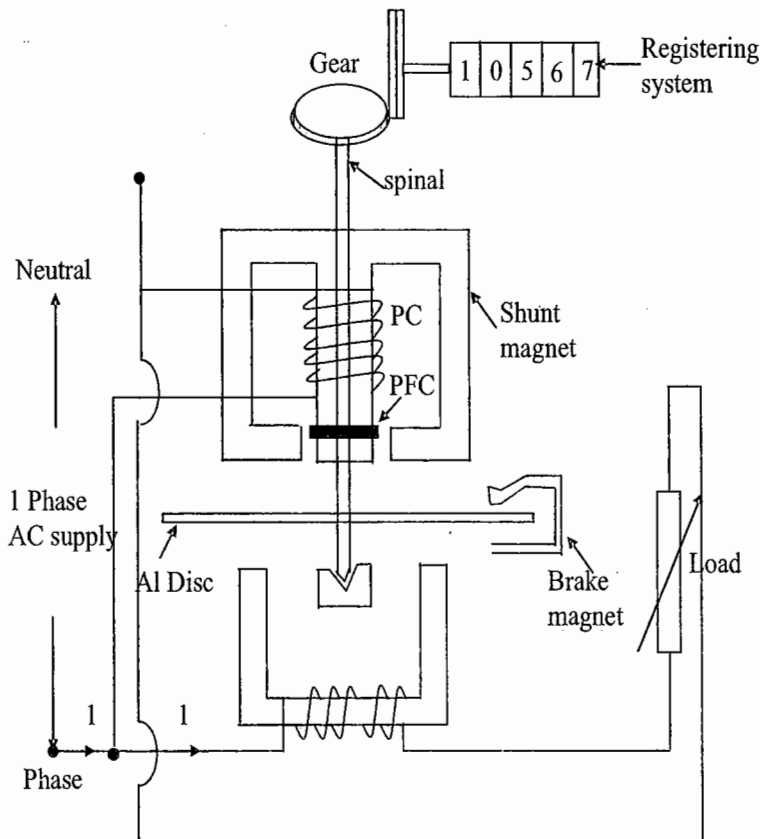
Measurement of Energy

The power delivered over a period of time is called energy

$$\text{Energy} = \int_0^t V I \cos \phi dt$$

1- ϕ energy meter induction type:

Energy meter is an integrating instrument which works on the principle of induction.



Energy meter consists of

- (i) Driving mechanism
- (ii) Rotating mechanism
- (iii) Breaking mechanism
- (iv) Recording (or) registering mechanism

Driving mechanism:

- (a) shunt magnet
- (b) series magnet

Shunt magnet:

Shunt magnet is "highly inductive and it has large no. of turns

- On shunt magnet potential coil is placed and it is directly connected to the supply
- P.C carries a current of I_p . It is proportional to supply voltage only and it is independent of load.

Series magnet:

It is wound with a coil called as "current coil" (CC) is connected in series with load and carries the current ' I_c ' proportional to load I .

Rotating mechanism:

A light weight aluminum disc is used and it is placed on spindle. The speed of aluminum disc is proportional to power consumed by the load

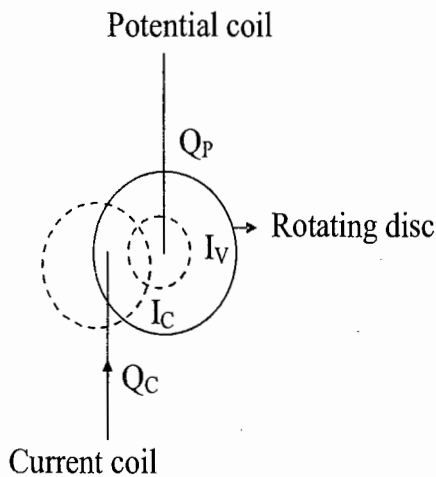
Operation:

Potential coil produces a flux (ϕ_p) and this flux is passing through the rotating disc. Which induces an eddy voltage (e_p) and an eddy current (I_{ep}). When current coil produces flux ϕ_c , which induces an eddy current I_{ec} in the disc. The interaction of (I_{ec}) with potential coil flux (ϕ_p) and

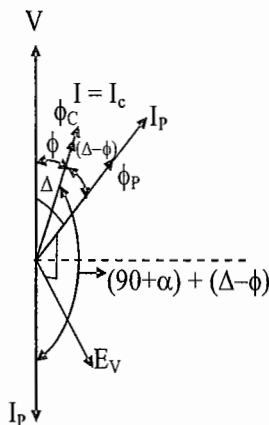
interaction of I_{ep} with current coil flux ' ϕ_c ' produces a torque in the rotating disc. This torque is proportional to the power consumed by the load.

Note:

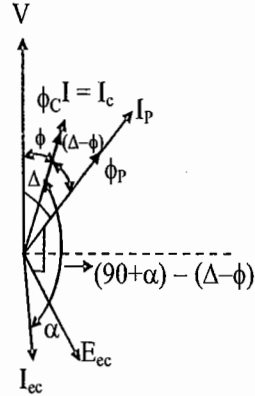
If this power consumption is measured over a time period (dt), then the no. of revolutions of rotating disc is proportional to the energy consumed by the load. But between two torques the angle is 90° . The resultant of T_{d1} and T_{d2} net driving torque (T_d) is produced, with this disc starts rotating



Phasor for potential coil emf:



Phasor for current coil emf:



$$T_{d1} = \phi_C I_{ep} \cos \angle \phi_c \text{ \& } I_{ep}$$

$$T_{d2} = \phi_P I_{ec} \cos \angle \phi_c \text{ \& } I_{ec}$$

$$T_{d1} = VI \cos(90 + \alpha) - (\Delta - \phi)$$

$$T_{d2} = VI \cos(90 + \alpha) + (\Delta - \phi)$$

$$T_d = T_{d1} - T_{d2}$$

$$= VI((\cos(90 + \alpha) - (\Delta - \phi)) - \cos((90 + \alpha) + (\Delta - \theta)))$$

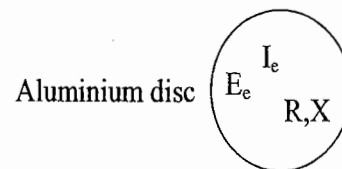
$$= VI(2 \sin(90 + \alpha) \sin(\Delta - \phi))$$

$$T_d = VI(2 \cos \alpha \sin(\Delta - \phi))$$

$$2 \cos \alpha \rightarrow \text{Al disc material}$$

$$T_d \propto VI \sin(\Delta - \phi)$$

- If all disc reactance $X = 0$, then driving torque will be maximum



$$T_d \propto VI \sin(\Delta - \phi) \rightarrow W_n(\text{measure value}) X = 0$$

$$T_d \propto VI \cos \phi \rightarrow W_T = (\text{True value}) \alpha = 0$$

$$\text{Error} = W_m - W_T = VI \sin(\Delta - \phi) - VI \cos \phi$$

$$\text{Error} = VI(\sin(\Delta - \phi) - \cos \phi)$$

$$\text{Error} = \frac{\sin(\Delta - \phi) - \cos \phi}{\cos \phi} \times 100$$

Example 4:

A correctly adjusted single phase 240V induction watt hour meter has a meter constant of 600 revolutions per kwh. Determine the speed of the disc, for a current of 10A at a power factor of 0.8 lagging. If the lag adjustment is altered so that phase angle between voltage coil flux and applied voltage is 86° calculate the error introduced at 0.5 pf lagging

- (a) 19.8rpm; -10% (b) 19.8rpm; +10%
(c) 19.2rpm; -11.8% (d) 19.2rpm; -12.3%

Ans: (d)

Sol: Energy consumed in one minute with rated current and 0.8pf lagging

$$240 \times 10 \times 0.8 \times \left(\frac{1}{60}\right) \times 10^{-3} = 0.032 \text{ kwh}$$

$$\therefore \text{Revolutions made in one minute} = 0.032 \times 600$$

$$= 19.2 \text{ rpm}$$

$$\therefore \text{speed of the DISC} = 19.2 \text{ rpm}$$

$$\text{Error introduced} = \frac{\sin(\Delta - \phi) - \cos \phi}{\cos \phi} \times 100$$

$$\text{We have } \Delta = 86^\circ$$

$$\text{p.f} = 0.5 \text{ lagging}$$

$$\therefore \text{error} = \frac{\sin(86^\circ - 60^\circ) - \cos 60^\circ}{\cos 60^\circ} \times 100$$

$$= -12.3\%$$

- Energy meter works on 1- ϕ I.M principle i.e. the angle between potential coil and cc flux is maintained 90° . Then two driving torque T_{d1} and T_{d2}

At steady state of a/d disc

$$\frac{k_3 \phi_m^2 N R}{Z} = T_d$$

$$N = T_d \frac{Z}{k_3 \phi_m^2 R}$$

$$N = V I \cos \phi$$

$$\int_0^t N dt \propto \int_0^t V I \cos \phi$$

↓ ↓
Mech. Electrical
Energy energy

No. of revolutions $\propto$ Electrical energy

Note:

- Because of braking torque, it runs at steady state

$$T_B \propto \phi_m \cdot d$$

$$N \propto \frac{T_B}{d \phi_m} \Rightarrow N \propto \frac{1}{d}$$

d = distance between spindle and permanent magnet

- To increase the speed permanent magnet is kept nearer to the spindle
- To decrease the speed magnet is kept far away from the spindle
- If angle (Δ) between "PC voltage (V) and PC coil flux (Q_p)" (or) I_p is 90° . Then energy meter records true power and error is zero.

Note:

In energy meter the nature of potential coil should be highly inductive and has large no. of turns, then error is zero.

Breaking torque:

$$E_e \propto \phi_m N$$

$$= k_1 \phi_m N$$

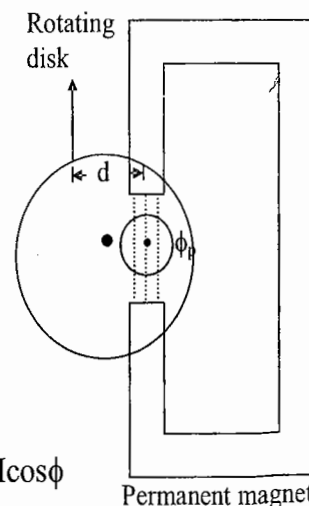
$$T_e \propto \phi_m I_e R$$

$$T_B = k_2 \phi_m I_e R$$

$$T_B = \frac{k_3 \phi_m^2 N R}{Z}$$

$$T_B \propto \phi_m^2$$

$$\therefore T_B \propto N, T_d \propto V I \cos \phi$$





N = speed

R = Radius of spindle and permanent magnet

Formulae in energy meter:

Energy = power $\times$ time

$$= VI \cos \phi \times t$$

$$= \frac{VI \cos \phi}{1000} \times \frac{t}{3600} \text{ kWhr}$$

($\because$ t in sec, 1 kWhr = 1 unit At full load $I = i$,

at half load $I = \frac{i}{2}$)

Energy meter constant:

$$K = \frac{\text{No. of revolutions}}{\text{kWhr}}$$

$$K = \frac{N}{\text{pt}}$$

P = power kW

t = time in hr

N = speed of disc = revolutions

$$\text{Creeping error} = \frac{\text{Total no. of revolutions made}}{\text{Energy meter const } t(K)}$$

% Creeping error

$$= \frac{\text{No. of revolutions due to creeping}}{\text{Revolutions due to load}} \times 100$$

% Energy meter constant

$$= \frac{\text{Energy recorded}}{\text{True energy consumed}} \times 100$$

= +ve – fast

= -ve – slow

Example 5:

A single phase kwh meter makes 500 revolutions per kwh. It is found on testing as making 40 revolutions in 58.1 seconds at 5 kw full load. The percentage error in the meter is _____ (in %)

Ans: 0.86

Hint: Energy consumed $(E.C)_t = \text{kwh}$

$$= 5 \times \frac{58.1}{3600} = 0.0807 \text{ kwh}$$

$$(E.C)_m = \frac{\text{rev}}{M.C} = \frac{40}{500} = 0.08 \text{ kwh}$$

$$\% \text{ error} = \frac{0.08 - 0.0807}{0.0807} \times 100 = -0.86\%$$

Example 6:

Statement for Linked Answer Questions

A single phase energy meter is tested for half an hour run at a supply voltage of 230V and a load current of 10A at 0.8 lag power factor. The dial reading at start was 57.35 and at the end of the test was 58.25. The meter constant is 1200 revolutions per kwh. The meter made 890 revolutions in 30 minute run

(i) The error in meter registration is

- (a) 0.02 kwh more (b) 0.04 kwh less
(c) 0.02 kwh less (d) 0.2 kwh more

Ans: (c)

$$\text{Sol: (Energy consumption)}_t = \frac{V_L I_L \cos \phi}{1000} \times 0.5$$

$$= \frac{230 \times 10 \times 0.8}{1000} \times 0.5$$

$$= 0.92 \text{ kwh}$$

$$(E.C)_m = 58.25 - 57.35 = 0.90 \text{ kwh}$$

$$\therefore \text{error} = A_m - A_t = 0.90 - 0.92 = -0.02 \text{ kwh}$$

(ii) The percentage error in meter revolutions

- (a) 19.38% fast (b) 19.38% slow
(c) 17.21% fast (d) 1.28% slow

Ans: (b)

$$\text{Sol: (Rev)}_t = M.C \times E.C = 1200 \times 0.92 = 1104 \text{ rev}$$

$$(\text{Rev})_m = 890 \text{ rev}$$

$$\% \text{ error} = \frac{890 - 1104}{1104} \times 100 = -19.38\%$$

Creeping error:

When PC is energized and there is no current in current coil, then the disc revolves slowly and continuously. This effect is called creeping error.

Causes:

- Main reason is over static friction compensation
- Over voltages
- Excessive lubrications, vibrations, stray magnetic field of Instrument.
- To eliminate the creeping error, two holes are drilled on opposite sides of the disc diameter.
- A small Iron piece is attached to the edge of an disc, under creeping. Thin Iron piece is attracted by permanent magnet so disc stops creeping.

Example 7:

A 230V, 1 – ϕ watt hour meter has a constant load of 4A passing through it for 6 hours at unity p.f if the meter disc makes 2208 revolutions during this period. calculate the power factor of the load if the number of revolutions made by the meter are 1472 when operating at 230V and 5A for 4 hours.

- (a) 0.8 (b) 0.6
(c) 0.75 (d) 0.92

Ans:(a)

Sol: No. of revolutions = $kVI \cos \phi \times t \rightarrow (1)$

$$\therefore k = \frac{2208}{230 \times 4 \times 1 \times 6} = 400 \text{ rev/kwh}$$

$$\therefore \text{for 1472 revolutions } \cos \phi = ?$$

$$(1) \Rightarrow 1472 = (400 \times 230 \times 5 \times \cos \phi \times 4) \times 10^{-3}$$

$$\Rightarrow \cos \phi = 0.8$$

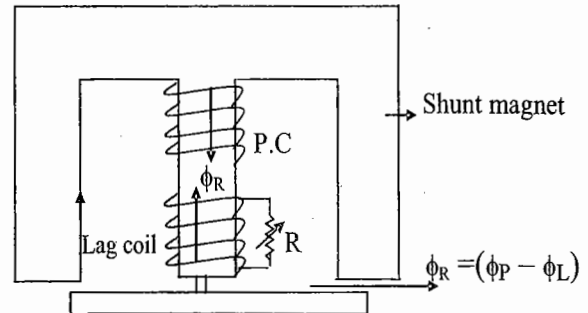
Phase lag/ Quadrature / Inductive /Lag coil

For the measurement of true energy by the energy meter potential coil current (I_P) lags behind the

voltage (V) by an angle 90° . But in practical it is Δ (You can see this Δ from phasor diagram of energy meter)

To make this angle 90° a Lag coil is connected below the potential coil which will produces flux ϕ_L .

As shown in below figure



By adjusting the lag coil (means by moving it up or down) we can make ϕ_R exactly in quadrature with the voltage. So that Iron losses in the core is small.

Example 8:

A single phase energy meter of the induction type is rated 230 V, 10A, 50 Hz and has a meter constant of 600 rev/Kwh when correctly adjusted. If quadrature adjustment is slightly disturbed so that the lag is 85° , calculate the percentage error at full load, 0.8 pf lag.

- (a) -5.5% (b) -6.9%
(c) -9.3% (d) -1.3%

Ans: (b)

Sol: $T_d \propto \sin \alpha$

Where α = angle between two fluxes.

$$\text{Given } \phi = \cos^{-1} (0.8) \Rightarrow 36.8$$

$\therefore$ The value of angle under measurement is

$$\alpha^1 = 85 - 36.8 \Rightarrow 48.2^\circ$$

$$\text{But true value is } \alpha = 90^\circ - 48.2$$

$$= 53.2$$

$$\therefore \text{error} = \frac{\sin \alpha^1 - \sin \alpha}{\sin \alpha} \times 100$$



$$= \frac{\sin 48.2^\circ - \sin 53.2^\circ}{\sin 53.2^\circ} \times 100$$

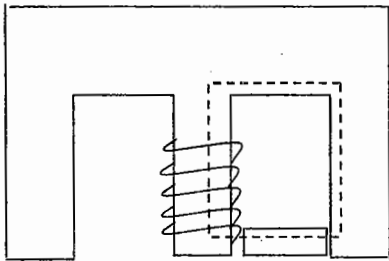
$$= -6.9\%$$

SHADING BANDS

Which are similar to the “lag coil” which are made of Cu. By adjusting the position of Shading Band ϕ_L is adjusted such that ϕ_R is kept at 90° w.r.t voltage.

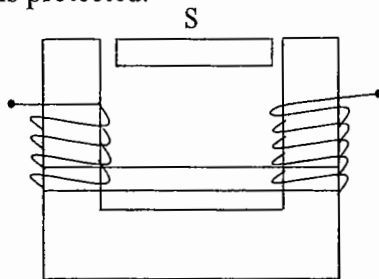
“OVER VOLTAGE” Compensation :-

Saturable shunt magnets are kept between side limb of the shunt magnet. Under normal air gap between saturable shunt magnet and the side limbs produces Maximum reluctance but under high voltage reluctance breaks so that flux is diverted hence meter is protected.



Over current protection

A saturable series magnet is kept between the side limbs of series magnet under heavy load currents flux diverts through this magnet hence meter is protected.



Temperature compensation:

The permanent magnet is made up of “Mu temperature” which is less affected by the surrounding temperature.

Testing of Energy Meter

Phantom/fictitious Loading :

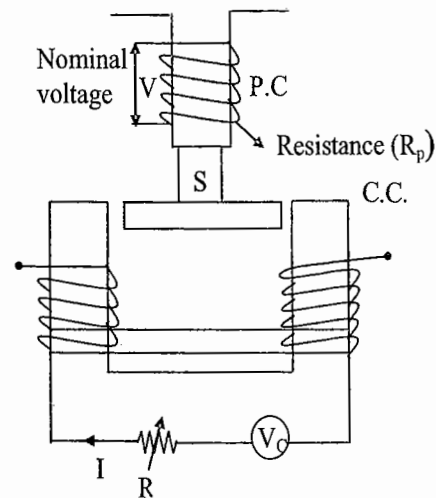
Phantom (or) fictitious loading is used to reduce the energy loss due to actual loading. So that the energy meter is tested at the desired load current and nominal supply voltage.

Procedure:

The potential coil of the energy meter is applied with normal supply voltage.

The current coil is supplied with a low voltage supply variable resistance is connected in series with it.

By varying resistance (R) we can make the current through current coil equal to the load current called fictitious loading.



Under Normal loading :

$$\text{Total power loss} = \frac{V^2}{R_p} + VI \quad \text{“I” is load current}$$

so due to normal loading energy loss is high.

Note: In energy meter potential coil (or) current coil terminals are reversed then disc rotates in the opposite direction.

Class Room Practice Questions

LEVEL-1 Questions

01. Which of the following statement is true about two wattmeter method for power measurement in three phase current.

- (a) power can be measured using two wattmeter method only for star connected three phase circuits
- (b) when two meters show identical readings, the power factor is 0.5
- (c) when power factor is unity, one of the wattmeter reads zero
- (d) when the readings of the two wattmeters are equal but of opposite sign, the power factor is zero.

02. In the measurement of power on balanced load by two-Wattmeter method in a 3-phase circuit, the readings of the Wattmeters are 3kw and 1kW respectively, the latter being obtained after reversing the connections of the current coil. The power factor of the load is

- (a) 0.277
- (b) 0.554
- (c) 0.625
- (d) 0.866

03. In calibration of a dynamometer Wattmeter by potentiometer, phantom loading arrangement is used because

- (a) the arrangement gives accurate results
- (b) the power consumed in calibration work is minimum
- (c) the method gives quick results
- (d) the onsite calibration is possible

04. Which of the following statements are correct in case of a power factor meter?

- 1. The deflection is proportional to the phase angle between field coil and crossed coil
- 2. The restoring torque is provided by a controlling torque

3. It consists of two coils mounted at right angles to each other.

Select the correct answer using the code given below:

- (a) 1 and 2
- (b) 2 and 3
- (c) 1 and 3
- (d) 1, 2, and 3

05. In a single-phase induction type energy meter, the lag adjustment is done to ensure that

- (a) current coil flux lags the applied voltage by 90°
- (b) pressure coil flux lags the applied voltage by 90°
- (c) pressure coil flux is in phase with the applied voltage
- (d) current coil flux lags the pressure coil flux by 90°

06. If an inductance type energy meter runs fast, it can be slowed down by

- (a) lag adjustment
- (b) light load adjustment
- (c) adjusting the position of braking magnet and moving it closer to the centre of the disc
- (d) adjusting the position of braking magnet and moving it away from the centre of the disc

07. **Assertion (A):** The rotating disc in an energy meter is made up of a magnetic material

Reason (R): Braking takes place due to eddy current generated by the braking magnet

08. Consider the following statements associated with an energy meter:

- 1. It is an integrating type instrument
- 2. It is an induction type instrument



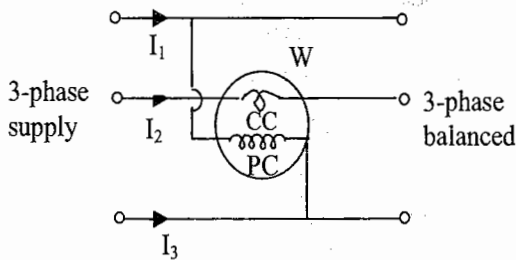
3. It uses a permanent magnet for rotation of aluminium disc
4. It employs a high control torque
Which of these statements are correct
(a) 1, 2, 3 and 4 (b) 1 and 2 only
(c) 2 and 3 only (d) 3 and 4 only
09. A single-phase energy meter having meter constant of 200 rev/kWh is operating on 230V, 50 Hz supply with a load of 10A and at unity power factor for three hours continuously. The number of revolutions shown by the meter during this period is
(a) 13800 (b) 1380
(c) 276 (d) 138
10. The meter constant of a single-phase energy meter is 500rev/kWh. It is found that with a load of 5kW, it makes 40 revolutions in 50 sec. The percentage error is
(a) 5.25% (b) 10.5%
(c) 15.25% (d) 20%
11. A water boiler at home is switched on to the a.c. mains supplying power at 230v/50 Hz. The frequency of instantaneous power consumed by the boiler is
(a) 0 Hz (b) 50 Hz
(c) 100 Hz (d) 150 Hz
12. The two wattmeter method is used to measure active power on a three phase, three wire system. If the phase voltage is unbalanced, then the power reading is
(a) affected by both negative sequence and zero sequence voltages
(b) affected by negative sequence voltage but not by zero sequence voltages
(c) affected by zero sequence voltages but not by negative sequence voltages
(d) not affected by negative or zero sequence voltages
13. Damping torque in the disc of an a.c. energy meter is provided by which one of the following?
(a) Electrostatic effect
(b) Magneto static effect
(c) Eddy current effect
(d) Chemical effect
14. What is the major cause of creeping in an energy meter?
(a) Over compensation for friction
(b) Mechanical vibrations
(c) Excessive voltage across the potential coil
(d) Stray magnetic fields
15. The brake magnet of an error free electro mechanical Induction type energy meter is shifted from its position and moved a small distances towards the edge of the disc. Then the meter will
(a) continue to work correctly without any change in its characteristics
(b) continue to work, but registers a lower kWhr
(c) continue to work, but registers a higher kWhr
(d) stop working
16. Disc in an induction type energy meter rotates in the opposite direction when
(a) a large capacitor is connected on the load side
(b) a large inductor is connected on the load side
(c) current coil and voltage coil are wrongly connected
(d) either current coil or voltage coil is wrongly connected
17. The two-wattmeter method is used to measure power in a balanced 3-phase



circuit, drawing lagging current. The power factor if one of the wattmeter reads zero is
(a) 0.5 (b) 0.8 (c) 1.0 (d) 0.0

LEVEL-2 Questions

01. In two-wattmeter method of power measurement, one of the wattmeter will show negative reading when the load pf angle is strictly
(a) $< 30^\circ$ (b) $< 60^\circ$
(c) $> 30^\circ$ (d) $> 60^\circ$
02. An electro-dynamometer type wattmeter is connected (as shown in the given figure) in a 3-phase supply and having a 3-phase balanced load. E and I are the values of phase voltage and current and ϕ is the phase angle between them. The wattmeter reading will be



- (a) Proportional to $EI \sin \phi$
(b) Proportional to $EI \cos \phi$
(c) proportional to $EI \tan \phi$
(d) zero
03. The power of a three-phase three-wire balanced system was measured by two Wattmeter method. The reading of one of the Wattmeter was found to be double that of the other. What is the power factor of the system?
(a) 1 (b) 0.866 (c) 0.707 (d) 0.5
04. A single – phase energy meter is operating on 230V, 50Hz supply with a load of 20A

for two hours at upf. The meter makes 1380 revolutions in that period. The meter constant is

- (a) 695 rev/kWh (b) 150 rev/kWh
(c) 0.15 rev/kWh (d) 1/150 rev/kWh

05. The pressure coil of an induction type energy meter is
(a) Highly resistance (b) Highly inductive
(c) purely resistive (d) Purely inductive
06. The minimum number of wattmeter(s) required to measure 3 – phase, 3 – wire balanced or unbalanced power is
(a) 1 (b) 2 (c) 3 (d) 4

Solutions

LEVEL-1

01. Ans: (d)

02. Ans: (a)

Sol: $W_1 = 3 \text{ kW}$,

$W_2 = -1 \text{ kW}$ (after reversing connections of CC)

$$\phi = \tan^{-1} \left[\sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \right]$$

$$= \tan^{-1} \left[\sqrt{3} \left(\frac{3 \text{ kW} - (-1 \text{ kW})}{3 \text{ kW} + (-1 \text{ kW})} \right) \right]$$

$$= 73.89^\circ$$

$$\text{Load power factor} = \cos \phi$$

$$= \cos(73.89) = 0.277$$

03. Ans: (b)

04. Ans: (c)

05. Ans: (b)

06. Ans: (d)

07. Ans: (c)

08. Ans: (b)

09. Ans: (b)



Sol: Energy = $VI \cos \phi \times \text{time}$

$$= \frac{230 \times 10 \times 1 \times 3}{1000} = 6.9 \text{ kWhr}$$

No. of revolutions (N)
 $= \text{meter constant (k)} \times \text{Energy in kWhr}$
 $= 200 \times 6.9 = 1380$

10. Ans: (c)

Sol: Energy consumed in kWhr (True value)

$$= \frac{5 \text{ kw} \times 50}{3600} = 0.06944 \text{ kw hr}$$

Energy recorded in kw hr (measured value)

$$= \frac{40 \text{ rev}}{500 \text{ rev/kWhr}} = 0.08 \text{ kWhr}$$

$$\% \text{Error} = \frac{0.08 - 0.0694}{0.06944} \times 100$$

$$= 15.273\%$$

11. Ans:(c)

Sol: Instantaneous power = VI

$$= [V_m \sin \omega t] [I_m \sin (\omega t)]$$

$$= V_m I_m \sin^2 \omega t$$

$$= \frac{V_m I_m}{Z} (1 - \cos 2\omega t)$$

$$= V_{\text{rms}} I_{\text{rms}} (1 - \cos 2\omega t)$$

So, frequency of the instantaneous power will be twice that of supply frequency.

12. Ans: (b)

13. Ans: (c)

14. Ans: (a)

15. Ans: (b)

16. Ans: (d)

17. Ans: (a)

Level-2 Key

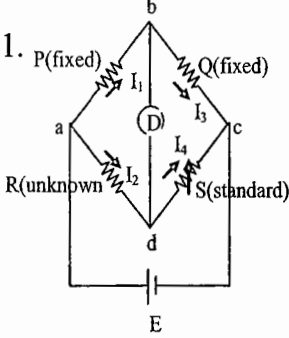
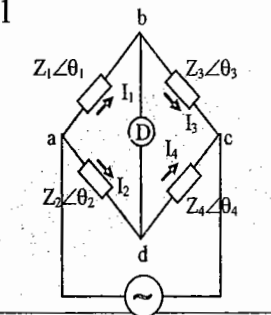
01. (d)	02. (a)	03. (b)	04. (b)	05. (b)
06. (b)				

4

Bridge Measurement of R, L & C

There are two types of bridges

1. DC Bridges
2. AC Bridges

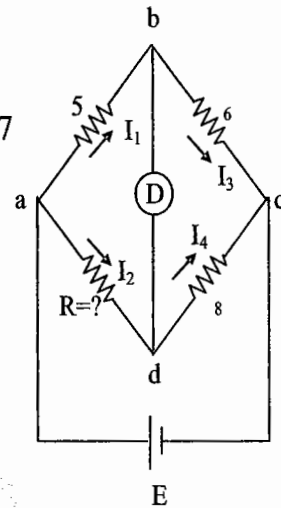
DC Bridge	AC Bridge
1. 	1. 
2. source: Battery	2. AC source, oscillator
3. Detector: DC Galvanometer (0Hz)	3. Vibrational galvanometer (5-20Hz) • Telephone detectors in head phones (200-5kHz) • Tuned amplifier (>5kHz)
4. Balanced PS = QR	4. $Z_1 Z_4 = Z_2 Z_3$ and $\theta_1 + \theta_4 = \theta_2 + \theta_3$
5. Application: only 'R' measurement	5. L, C, f, Q -factor, D-factor, & ϵ_r

Note: Don't connect Galvanometer and source to the same points. It may be unbalance conditions.

- If the below Bridge is balanced then the value of R is
PS = QR

$$5 \times 8 = 6 \times R$$

$$R = \frac{5 \times 8}{6} = 6.67$$



Example 1:

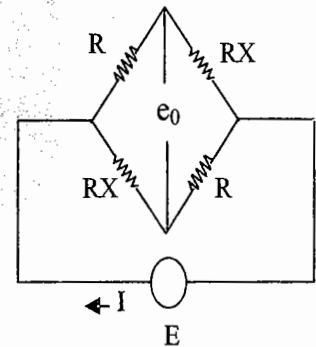
For the circuit given in figure the unbalance voltage e_0 is

(A) $e_0 = \frac{IRX}{2}$

(B) $e_0 = 2IRX$

(C) $e_0 = IR(1+X)$

(D) $e_0 = \frac{IR(X-1)}{2}$



Ans: (D)

Sol: equivalent resistance = $(R + RX) // (R + RX)$

$$= \frac{(R + RX)}{2}$$

$$\therefore I = \frac{E}{(R + RX)/2}$$

$$\Rightarrow E = I \frac{(R + RX)}{2}$$

$$e_0 = E \left[\frac{RX}{R + RX} \right] - E \left[\frac{R}{R + RX} \right]$$

$$= E \left[\frac{RX - R}{R + RX} \right] = \frac{IR(X-1)}{2}$$



When bridge is balanced, adjacent branch voltages are equal.

- To reduce the sliding balance, select uncommon elements as variables always (R_4 & C_4).
- If the selected uncommon elements are present in same arm. Then time taken for bridge balance is much reduced then it is called converges to sliding balance.

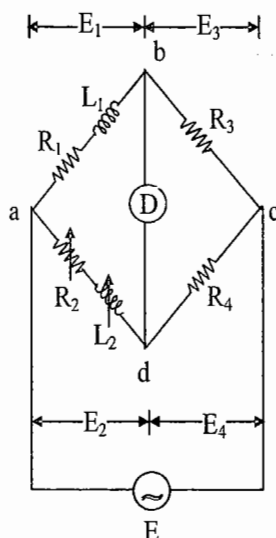
• **Measurement of Inductance :**

- 1) Maxwell – Inductance Bridge
- 2) Maxwell – Inductance –capacitance bridge (medium Q coils $1 < Q < 10$)
- 3) Hays Bridge – (High Q coils $Q > 10$)
- 4) Owen's Bridge – (Incremental Inductance (μ_0))
- 5) Anderson's bridge – ("low Q coils" ($Q < 1$))

(accurate measurement of Inductor from milli Henry to wide range)

⇒ out of above 5 bridges Anderson bridge is most accurate bridge.

Maxwell –Inductance Bridge:



$$(R_1 + j\omega L_1) \times R_4 = (R_2 + j\omega L_2) \times R_3$$

Under balanced conditions

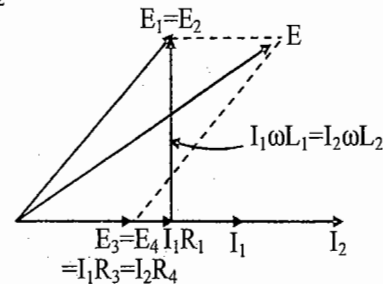
$$R_1 R_4 = R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4}$$

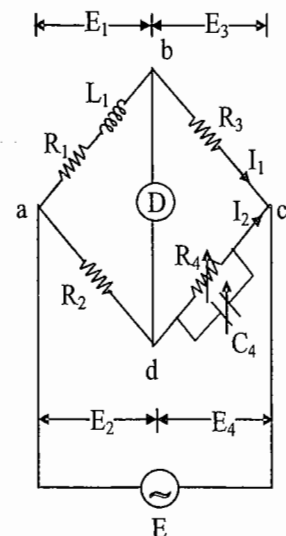
$$L_1 R_4 = R_2 R_3$$

$$L_1 = \frac{L_2 R_3}{R_4}$$

To make Bridge balance select uncommon elements from the above two equations those are " R_2 & L_2 "



Maxwell –Inductance –Capacitance Bridge:



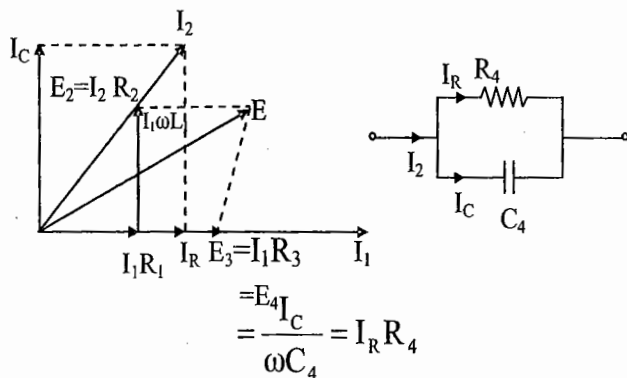
$$R_1 R_4 = R_2 R_3$$

$$R_1 = \frac{R_2 R_3}{R_4}$$

$$L_1 = R_2 R_3 R_4$$



$$Q \text{ factor} = \frac{\omega L_1}{R_1} = \omega R_4 C_4$$



$$|V_4| = I_R R_4 \rightarrow I_R \text{ in phase } |V_4|$$

$$= \frac{I_C}{\omega C_4} \rightarrow I_C \text{ leads } V_4 \text{ by } 90^\circ$$

$$I_2 = I_R + I_C$$

- It's not suitable for high & low 'Q' coils it's used for measurement of a medium Q coils Inductance & resistance from a known value of capacitance

$$Q = \frac{\omega L_1}{R_1} = \frac{X_{L1}}{R_1} \rightarrow \text{high} \approx \infty$$

$$Z_1 = R_1 + j X_1$$

$$\theta_1 = \tan^{-1} \left(\frac{X_{L1}}{R_1} \right)$$

$$\theta_1 = 90^\circ$$

$$\theta_1 + \theta_4 = \theta_2 + \theta_3$$

$$\theta_4 = -90^\circ$$

For high 'Q' coil

$$\tan^{-1}(\omega C_4 R_4) = 90^\circ$$

$$(\omega C_4 R_4) \leq \infty \rightarrow (\text{Not practical})$$

Hence this bridge is not used for High Q factor measurement

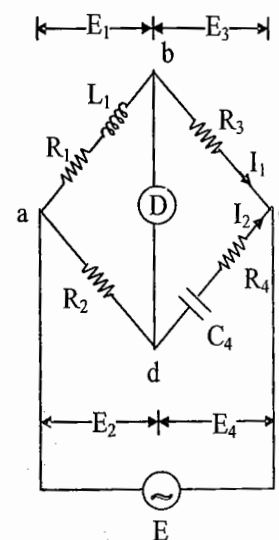
If low Q coil:

$$\theta_1 = \tan^{-1}(0) = 0$$

$$\text{So } \omega C_4 R_4 = 0$$

So also we can't change R_2 & R_3 values to this low values. So bridge is not suitable for low Q_1 factor measurement.

Hays Bridge:



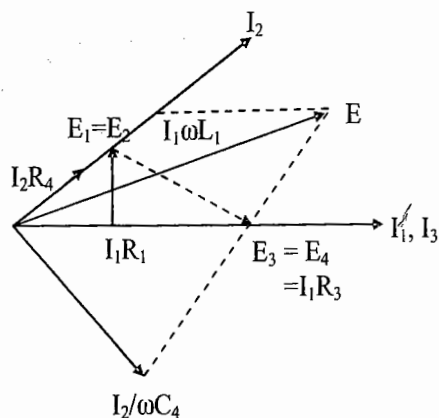
Under bridge balance condition

$$Z_1 Z_4 = Z_2 Z_3$$

$$L_1 = \frac{R_2 R_3 C_4}{(1 + \omega^2 R_4^2 C_4^2)}$$

$$R_1 = \frac{\omega^2 R_2 R_3 R_4 C_4^2}{(1 + \omega^2 R_4^2 C_4^2)}$$

$$Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega R_4 C_4}$$



Phasor diagram

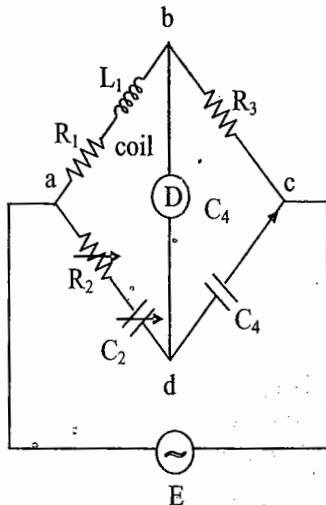
This is the "slowest bridge"

- used only for "high Q-factor measurement"



Owen's bridge:

This is the bridge which is having two capacitances in the bridge.



Under balanced conditions

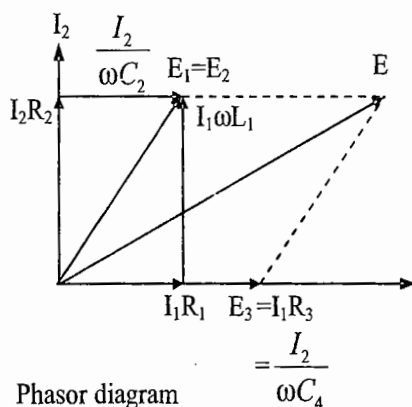
$$Z_1 Z_4 = Z_2 Z_3$$

$$(R_1 + j\omega L_1) \frac{1}{j\omega C_4} = \left(R_2 + \frac{1}{j\omega C_2} \right) R_3$$

$$R_1 = \frac{C_4}{C_2} R_3$$

$$Q = \frac{\omega L_1}{R_1} = \omega R_2 C_2$$

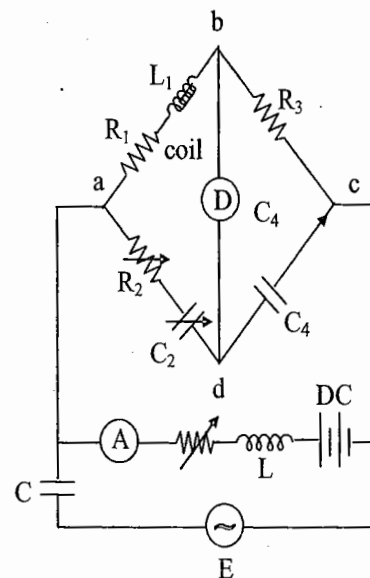
- This Bridge is suitable for medium Q-factor measurement.
- Owens bridge is used for incremental inductance measurement from known values of the fixed capacitance (C_4)



AIR CORE	IRON CORE
$L_{air} = \frac{N^2 \mu_0 A}{l}$	$L_{Iron} = \frac{\mu_0 \mu_r N^2 A}{l}$
	Physical Properties
	$L_{Iron} = L_{air} \times \mu_r$
	$L = \frac{N^2 \mu_0 A}{l} \times \mu_r$
	$\mu_r > 1$

$\mu_r \rightarrow$ Incremental Inductance

Modified Bridge for measurement of incremental Inductance



Note: The purpose of using the D.C. supply is to compensate the saturation effect of coil by supplying D.C. current

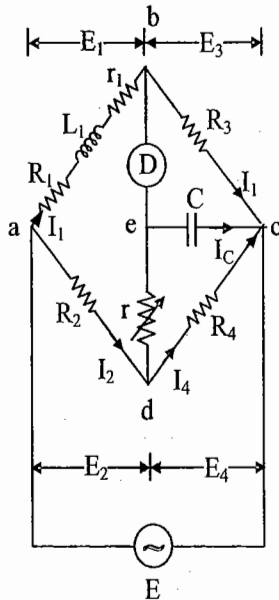
- Capacitor "C" blocks DC current. So It's used to block the DC current entering into AC source.



- Inductor "L" blocks AC. It is used to block AC current entering into the DC source.

ANDERSON'S BRIDGE:

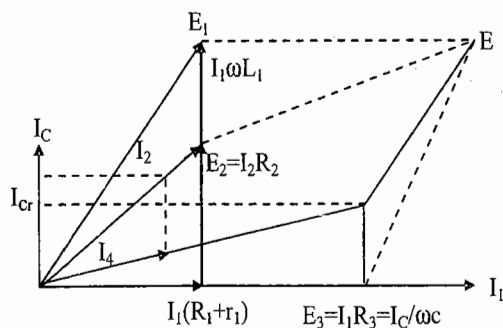
It is called as "5-point Bridge"



Under balanced condition

$$I_1(r + R_1 + j\omega L_1) = I_2 R_2 + I_C r \text{ and}$$

$$I_C \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_C) R_4$$



$$I_1(r + R_1 + j\omega L_1) = I_2 R_2 + I_1 j\omega C R_3 r \text{ (or)}$$

$$I_1(r + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2$$

$$j\omega C R_3 I_1 \left(r + \frac{1}{j\omega C} \right) = (I_2 - I_1 j\omega C R_3 r) R_4$$

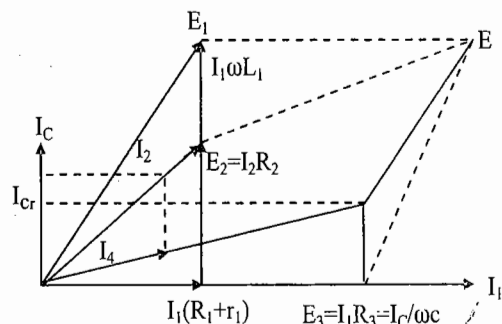
$$I_1(r + R_1 + j\omega L_1 - j\omega C R_3 r)$$

$$= I_1 \left(\frac{R_2 R_3}{R_4} + \frac{j\omega C R_2 R_3 r}{R_4} + j\omega C R_3 R_2 r \right)$$

$$R_1 = \frac{R_2 R_3}{R_4} - r_1$$

$$L_1 = C \frac{R_3}{R_4} (r(R_4 + R_2) + R_2 R_4)$$

- Anderson bridge uses a fixed capacitance 'C' and for the bridge balance, the resistances r & r₁ are selected as variables.
- These are used for measurement of Low Q coil
- This bridge used for accurate measurement of Inductance over 'mH to wide range'
- It is the modification of Maxwell inductance capacitance bridge.
- Bridge will give two balance equations because of 'fixed point'



$$V_1 = V_2 + I_C r$$

$$V_{bc} = I_1 r_3 = V_3$$

$$V_{bc} = \frac{I_C}{j\omega C} = \frac{I_c}{\omega_c} \angle -90$$

$$V_{dc} = I_4 R_4 = V_{de} + V_{ec}$$

$$= I_C r + V_3$$

$$I_2 = I_C + I_4$$



Disadvantages

- (1) more complex bridge
- (2) shielding of the bridge is complicated

To measure the mutual – Inductance:

- (1) Heavy side campbell bridge
- (2) Carey foster bridges are used.

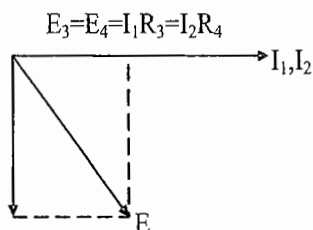
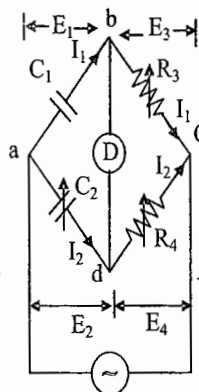
Capacitance Measurement

Capacitance can be measured from the following three bridges.

- (1) De-sauty's Bridge
- (2) Modified De-sauty's Bridge
- (3) Schering Bridge

De-sauty's Bridge

This bridge is suitable for perfect capacitor value measurement only (No-dielectric losses)

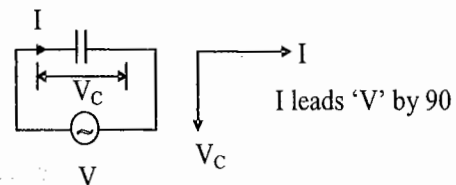


Under bridge balance condition

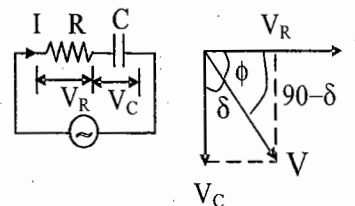
$$\frac{1}{j\omega C_1} R_4 = \frac{1}{j\omega C_2} R_3$$

$$C_1 = C_2 \frac{R_4}{R_3}$$

Pure capacitor (No-dielectric loss)



Practical capacitor (with dielectric loss)



$\sin \delta = \text{p.f of capacitor}$

$$\tan \delta = \frac{V_R}{V_C} = \frac{IR}{IX_C} = \omega RC$$

D-factor = $\tan \delta = \omega RC$

(or)

Derating factor (or) Dissipation factor

Example 2:

The four arms of a bridge are :

Arm ab: an imperfect capacitor C_1 with an equivalent series resistance of r_1

Arm bc: a non- inductive Resistance R_3

Arm cd; a non – inductive resistance R_4

Arm da; an imperfect capacitor C_2 with an equivalent series resistance of r_2 series with a Resistance R_2

A supply of 450Hz is given between terminals "a" and "c" and the detector is connected between



“b” and “d”. At balance: $R_2 = 4.8\Omega$; $R_3 = 2000\Omega$; $R_4 = 2850\Omega$ and $C_2 = 0.5\mu\text{F}$ and $r_2 = 0.4\Omega$ calculate the value of C_1 and r_1 and also find the dissipating factor for this capacitor.

(a) $0.712\mu\text{F}$; 3.65Ω ; 0.000734

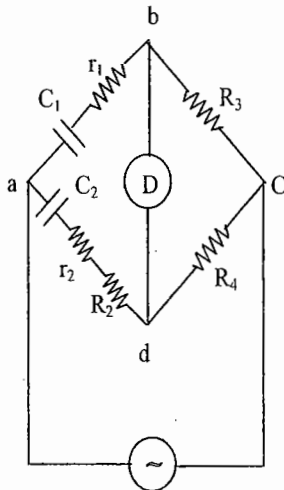
(b) $0.712\mu\text{F}$; 0.365Ω ; 0.00734

(c) $0.0712\mu\text{F}$; 3.65Ω ; 0.00734

(d) $0.712\mu\text{F}$; 3.65Ω ; 0.00734

Ans: (d)

Sol:



at balance

$$\left(r_1 + \frac{1}{j\omega c_1}\right)R_4 = \left(r_2 + R_2 + \frac{1}{j\omega c_2}\right)R_3$$

Separating the real and imaginary terms

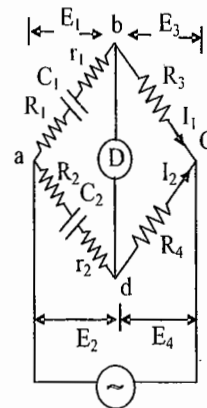
$$r_1 = (r_2 + R_2) \frac{R_3}{R_4} = (0.4 + 4.8) \times \frac{2000}{2850} = 3.65\Omega$$

And
$$C_1 = C_2 \frac{R_4}{R_3}$$

$$= (0.5 \times 10^{-6}) \times \frac{2850}{2000} \text{ F} = 0.712\mu\text{F}$$

$$\begin{aligned} \text{Dissipating factor } D_1 &= \tan\delta_1 = \omega C_1 r_1 \\ &= 2\pi \times 450 \times 0.712 \times 10^{-6} \times 3.65 \\ &= 0.00734 \end{aligned}$$

Modified De sauty's Bridge :



Used for measurement of C,D (dissipation factor)

$$C_1 = \frac{R_4}{R_3} C_2$$

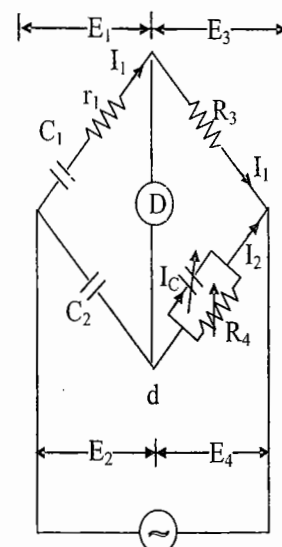
$$r_1 = \frac{R_3}{R_4} (R_2 + r_2) - R_1$$

Dissipation factor $\tan \delta = \omega C_1 r_1$

- this bridge is used for measurement of D factor practical capacitor

Schering Bridge:

This bridge is used for measurement of capacitance, D-factor (accurately), Dielectric loss, relative permittivity and properties of insulating materials.





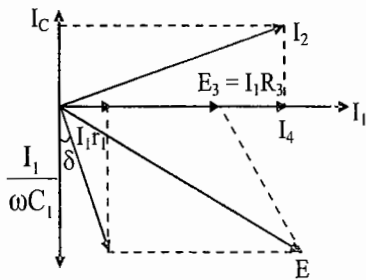
From the Bridge balance equations

$$C_1 = \frac{R_4}{R_3} C_2$$

$$r_1 = \frac{C_4}{C_2} R_3$$

$$D\text{-factor} = \omega C_1 r_1 = \omega \frac{R_4}{R_3} C_2 \cdot \frac{C_4}{C_2} R_3$$

$$D = \omega R_4 C_4$$



Example 3:

Common – data question Q(i) & Q(ii)

A dielectric sheet of thickness 1mm is tested at 50 Hz between two electrodes of 10 cm diameter. The Schering bridge employed has a standard compressed air capacitor C_2 of 100pF; a non – inductive resistor R_4 of 350Ω in parallel with a variable capacitor C_4 and a non – inductive variable resistor R_3 . At balance $C_4 = 0.4\mu\text{F}$, $R_3 = 250\Omega$.

(i) Find the power factor of the specimen

- (A) 0.04 (B) 0.18
(C) 0.23 (D) 0.12

Ans: (A)

Sol: Power factor $\sin \delta = \tan \delta = \omega C_4 R_4$

$$= 2\pi \times 50 \times 0.4 \times 10^{-6} \times 350$$

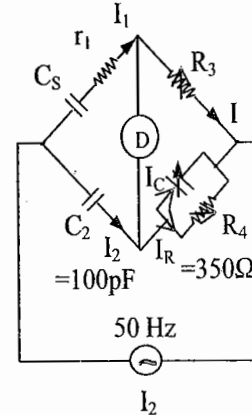
$$= 0.043$$

(ii) Find the permittivity of the sheet.

- (A) 1 (B) 2
(C) 3 (D) 4

Ans: (B)

Sol:



under balance

$$Z_s Z_4 = Z_2 Z_3$$

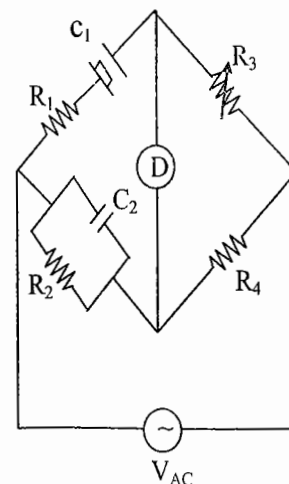
$$C_s = \frac{R_4}{R_3} C_2$$

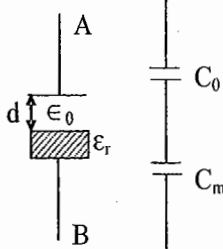
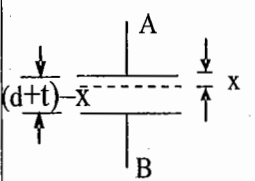
$$= \frac{350}{250} \times 100 \times 10^{-12} = 140 \times 10^{-12} \text{ F}$$

$$\text{Permittivity } \epsilon_r = \frac{C_s \times d}{\epsilon_0 \times A}$$

$$= \frac{140 \times 10^{-12} \times 0.001}{8.854 \times 10^{-12} \times \frac{\pi}{4} (0.1)^2} = 2$$

Measurement of (ϵ_r) relative permittivity:



 $d \rightarrow$ Distance between electrode specimen $t \rightarrow$ thickness of specimen $C_{AB} \rightarrow$ equalant capacitance	 Electrode is moved a distance of x (micrometer) to get same capacitance of equal to C_{AB}
$C_{AB} = \frac{C_1 C_2}{C_1 + C_2}$ $C_1 = \frac{\epsilon_0 A}{d}; C_2 = \frac{\epsilon_0 \epsilon_r A}{t}$ $C_{AB} = \frac{\epsilon_0 \epsilon_r A}{(t + d\epsilon_r)} \rightarrow (a)$	$C_{AB} = \frac{\epsilon_0 A}{(d+t)-x} \rightarrow (b)$

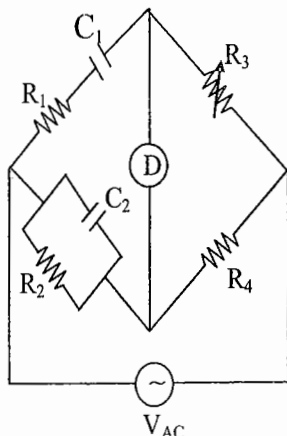
By equating equation "a & b"

$$\frac{\epsilon_0 \epsilon_r A}{(t + d\epsilon_r)} = \frac{\epsilon_0 A}{(d+t)-x}$$

$$\epsilon_r = \frac{t}{t-x}$$

Measurement of Frequency

Wien's Bridge:



from bridge balance

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad \frac{R_1}{R_2} + \frac{C_2}{C_1} = \frac{R_3}{R_4}$$

If $C_1 = C_2, R_1 = R_2$

$$\text{Then } \frac{R_3}{R_4} = 2; f = \frac{1}{2\pi R_1 C_1}$$

Applications:

- Used in distortion analyzer
- this bridge is not suitable for measurement of frequency of signals having harmonics, because bridge balance is difficult
- It's used for measurement of frequency from 100Hz to 100kHz

Causes of errors in AC bridges:

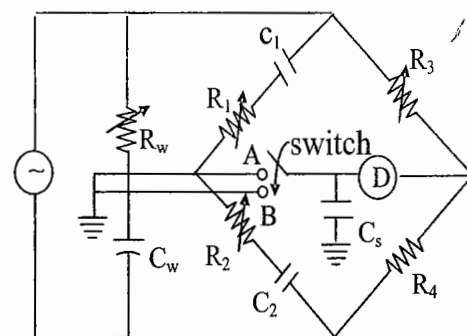
Stray conductance effect: Because of improper insulation leakage currents are present

Mutual inductance effect: The magnetic field coupling between the components

Stray capacitance effect: The capacitance forming between arms

Eddy current Error

Wagner Earth / Ground connections:





This connection is used to eliminate the stray capacitance produced between "Bridge arms & arms to the ground"

Procedure:

- Switch is kept on position A, make bridge balanced during this process C_s (stray capacitance) is charges to the certain potential.
- Now switch is moved to position 'B' by adjusting the Wagner resistance R_w . Bridge is balanced
- This indicates potential of C_s is equal to ground potential
- This process will continue either position of the switch i.e at A (or) B. the galvanometer shows zero deflection
- This corresponding to potential of C_s is equal to ground potential so that stray capacitance effect is reduced.

Resistance Measurements:

Resistance materials:

Manganian $\rightarrow$ Cu $\rightarrow$ 84% , Ni $\rightarrow$ 12%, mg $\rightarrow$ 4% and $\alpha = 0.00015/^\circ\text{C}$.

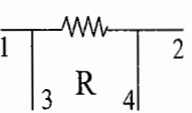
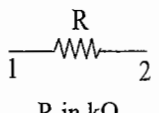
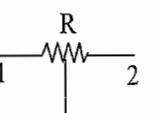
Constantan $\rightarrow$ Cu $\rightarrow$ 88%, Ni $\rightarrow$ 12% and $\alpha = 0.0017/^\circ\text{C}$.

Both have low value of ' α ' so it won't change with temperature

Cu $\alpha = 0.004/^\circ\text{C} \rightarrow$ high value it's changes with temperature

Classification of resistance:

Low Resistance	Medium Resistance	High Resistance
Resistance $\leq 1\Omega$	Resistance $R(1\Omega \text{ to } 100k\Omega)$	Resistance $R > 100k\Omega$
Ex: Ammeter internal resistance, all	Ex: DC machine shunt field	Ex: cable insulation resistance, all

motors, generators and transformer's winding resistances, Earthing resistance, all semi-conductor devices when forward biased	winding resistance, all heater coil resistances, human body resistance, voltmeter resistance (100Ω)	motors, generators and transformers winding's insulation resistance, all semi conductor devices when reverse biased
To measure low resistance 4 terminal required	To measure medium resistance only two terminals are required	To measure high resistance 3 terminals are required
		
1 & 2 are current terminals, 3 & 4 voltage terminals	1 & 2 are terminals	Guard terminal (3) is used to eliminate leakage currents.
Methods to measure it 1. Potentiometer method 2. Kelvin double bridge	Methods to measure it 1. voltmeter, ammeter method 2. sub situation method 3. wheat stone bridge 4. ohm meter Method	Methods to measure it 1. loss of charge method 2. meggar 3. mega ohm bridge 4. direct deflection



Example 4:

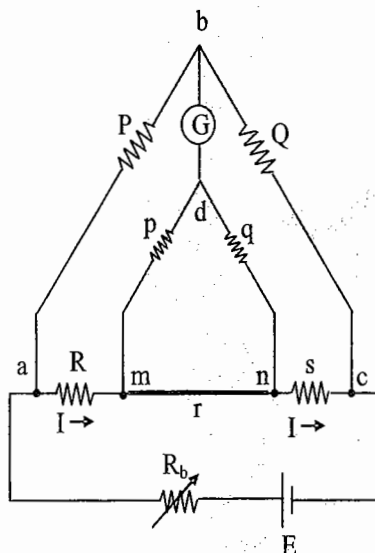
The methods for measurement of low and medium resistance cannot be used for measurement of high resistance as the

- (A) effect of thermo-electric e.m.f is considerable
- (B) effect of contact resistance is considerable
- (C) effect of connecting leads is considerable
- (D) resistance offered to the flow of current

Ans: (D)

Low Resistance Measurement

Kelvin's double bridge:



DC Bridge: source is DC so it is called DC Bridge

Under bridge balance conditions

$$R = \frac{P}{Q}(S) + \frac{qr}{p+q+r} \left(\frac{P}{Q} - \frac{p}{q} \right)$$

Including lead resistance

P, Q → External arm resistances

p, q → Internal arm resistances

s → standard variable resistance

r → terminals / contacts / leads resistance

R → unknown resistance

m → reversible switch to eliminate thermo electric emf

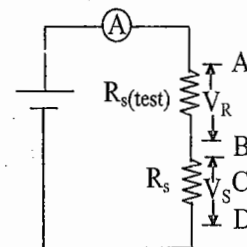
when lead resistance eliminated

$$R = \frac{P}{Q}S \quad \left(\because \frac{P}{Q} = \frac{p}{q} \right)$$

Advantage: It's used to eliminate the lead resistance effect so that suitable for measurement upto "micro ohms range"

Applications: Transformer winding.

Potentiometer method:



V_R = unknown resistance

V_S = known resistance

R_S = standard resistance

R_{Test} = test resistance

$$\therefore R = \frac{V_R}{V_S} R_S$$

Example 5:

Measurement of resistance by using potentiometer 4V is observed across the test resistance and 6V is observed across the standard resistance of 2mΩ. find test resistance?

Sol: $V_R = 4V$, $V_S = 6V$, $R_S = 2m\Omega$

$$R_T = \frac{V_R}{V_S} R_S \Rightarrow \frac{4}{6} \times 2m\Omega = \frac{4}{3} m\Omega$$



Note: difficulties in measurement of low resistance

- 1) terminals resistance
- 2) thermo electric emf's

Bifilar winding is used to reduce the inductance effect and aryton -perry winding is used to reduce capacitance effect.

Medium Resistance Measurement

V-A method	A-V method
$R_m = \frac{V}{A}$ V = Voltmeter reading A = Ammeter reading $= \frac{V_a + V_R}{I_R}$ $= \frac{I_R R + I_R R_a}{I_R}$ $R_m = R_a + R$ $R = R_m - R_a$ $\% \text{ error} = \frac{R_m - R}{R} \times 100$ $\% \text{ error} = \frac{R_a}{R} \times 100$	$R_m = \frac{V}{A}$ V = Voltmeter reading A = Ammeter reading $R_m = \frac{V}{I}$ By nodal $R = \frac{R_v R_m}{R_v + R_m}$ From above equation $\frac{R_m - R}{R} = \frac{-R_m}{R_v}$ Assume $R_m \approx R_T$ $\% \text{ Error} = \frac{R_m - R}{R} = \frac{-R_T}{R_v} \times 100$
The measured resistance is more than the actual value due to ammeter resistance	The measured resistance is less than the actual value due to voltmeter resistance
To reduce % error, V-A connection is	To reduce % error A-V connection is suitable

suitable for high resistance measurement in medium scale range

for low resistance measurement in medium scale range

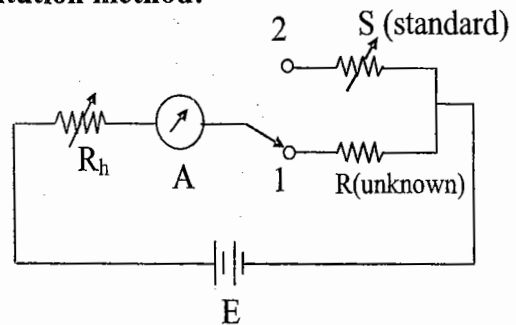
- To get same error on both V-A & A-V methods the value of unknown resistance "R" is equate both % errors

$$\frac{R_a}{R} = \frac{R}{R_v}$$

$$R = \sqrt{R_a R_v}$$

Note: in any method, the error due to only load side instrument

Substitution method:



Switch at position (1): vary the value of Rheostat 'R_h', such that the ammeter pointer reaches, the

chosen scale value. Then $I_1 = \frac{E}{R_h + R_a + R}$

Switch at position (2): Then current flowing through ammeter is I₂.

Equation $I_1 (R_h + R_a + R) = I_2 (R_h + R_a + S)$

$$R = \frac{I_2}{I_1} [R_h + R_a + S] - (R_h + R_a)$$

If $I_1 = I_2$ (by $R_T = S$ ue)

If instead of ammeter, galvanometer is used then

$$R = \frac{\theta_2}{\theta_1} [R_h + R_a + S] - [R_h + R_a]$$



Example 6:

In a measurement of resistance by substitutions method a standard $0.5\text{M}\Omega$ resistor is used. The galvanometer has a resistance of $10\text{K}\Omega$ and gives deflection as follows

- (i) With standard resistor, 41 divisions
- (ii) With unknown Resistance 51 divisions

Find the unknown resistance?

- (a) $0.2\text{M}\Omega$ (b) $0.4\text{M}\Omega$
- (c) $0.5\text{M}\Omega$ (d) $0.8\text{M}\Omega$

Ans: (b)

Sol: let S = Standard Resistance

R = unknown Resistance

G = Galvanometer resistance

θ_1 = deflection with standard resistor

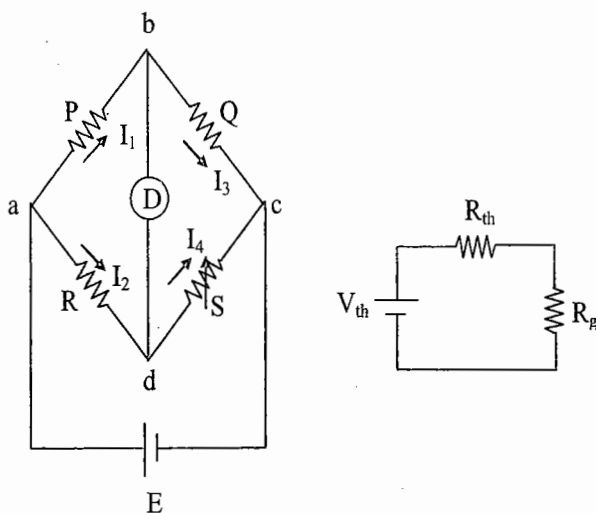
θ_2 = deflection with unknown resistor

$$\therefore \frac{\theta_1}{\theta_2} = \frac{R + G}{S + G}$$

Hence, unknown Resistance

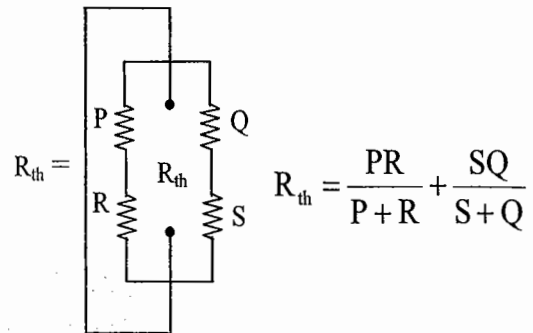
$$\begin{aligned} R &= (S + G) \frac{\theta_1}{\theta_2} - G \\ &= (0.5 \times 10^6 + 10 \times 10^3) \left[\frac{41}{51} \right] - 10 \times 10^3 \\ &= 0.4 \times 10^6 \Omega = 0.4\text{M}\Omega \end{aligned}$$

Wheat stone Bridge:



$$V_{th} = V_{bd} = V_b - V_d$$

$$= V \left[\frac{S}{S+Q} - \frac{R}{P+R} \right] = V \left[\frac{SP - RQ}{(S+Q)(P+R)} \right]$$



$$R_{th} = \frac{PR}{P+R} + \frac{SQ}{S+Q}$$

$$I_{th} = I_g = \frac{V_{th}}{R_{th} + R_g}$$

Under balanced condition

$$I_g = 0$$

$$SP - RQ = 0$$

$$SP = RQ$$

$$R = \frac{P}{Q} S$$

- Measured resistance 'R' is independent of galvanometer internal resistance and source internal resistance
- Hence the measured value is more accurate
- **Current sensitivity (S_i)** : It's the ratio of deflection of the galvanometer to that of the current flowing through the galvanometer

$$S_i = \frac{\theta}{I_g} \text{ mm}/\mu\text{A}$$

- **Voltage sensitivity:** It's ratio of deflection of galvanometer to that of the voltage across the galvanometer

$$S_v = \frac{\theta}{V_{th}} \text{ mm}/\text{V}$$



- **Bridge sensitivity:** $S_B = \frac{\theta}{\left(\frac{\Delta R}{R}\right)} \text{ mm}$

Note: In case of Bridge circuits sensitivity is an important parameter compared to accuracy resolution, precession etc

For maximum bridge sensitivity $S_B = \frac{V.S_V}{4}$

Example 7:

In wheat stone Bridge, with the added advantage of doubling the sensitivity of the bridge, which of the following effects on the measurement is eliminated

- (1) Resistance of connecting leads
- (2) Parasite emf's
- (3) temperature effect
- (4) Thermo electric emf

- (a) 1 and 4 (b) 1 and 3
(c) 2 and 4 (d) All the above

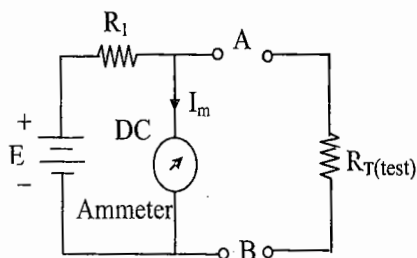
Ans: (c)

Ohm meter method:

Two types of methods

- (1) series ohm meter
- (2) shunt ohm meter

Shunt ohm meter:

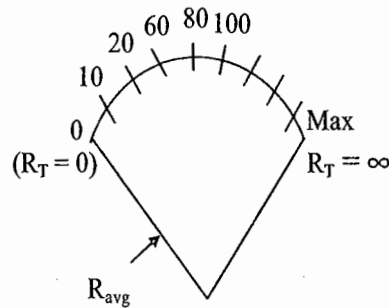


- (i) $R = \infty$ (O.C)

$$I_m \rightarrow \text{max}$$

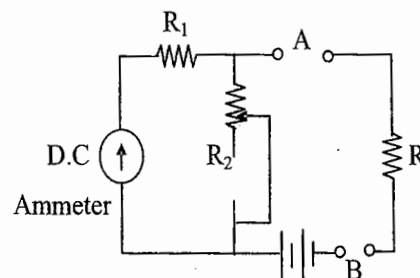
- (ii) $R = 0$ (S.C) $\Rightarrow I = 0$

$$I \propto R$$



- It is basically a **PMMC** instrument
- It works only for current
- The current scale is calibrated in terms of unknown resistance (R)
- It is used in continuity checking of electro static components

Series ohm meter:



$R_2 \rightarrow$ zero adjusting resistor

- (i) $R = \infty$ (O.C), $I = 0$

$$I_m \rightarrow \text{max}$$

- (ii) $R = 0$ (S.C) $\Rightarrow I = \infty$

(full scale deflection)

$$I \propto \frac{1}{R}$$

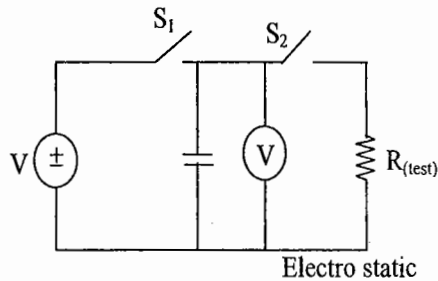
- In series ohm meter, the left extreme end is marked as ' $\infty\Omega$ ' and right extreme end is marked as ' 0Ω '.

Note: In series ohm meter, if battery is replaced with P.M.D.C generator then it's called as "meggar"



High resistance Measurement

(1) loss of charge method



'V' is initial voltage across capacitor

→ voltage across capacitor of any time

$$V_c = V \cdot e^{-t/RC}$$

$$R = \frac{0.4343t}{C \log_{10}(V/V_c)}$$

t = seconds

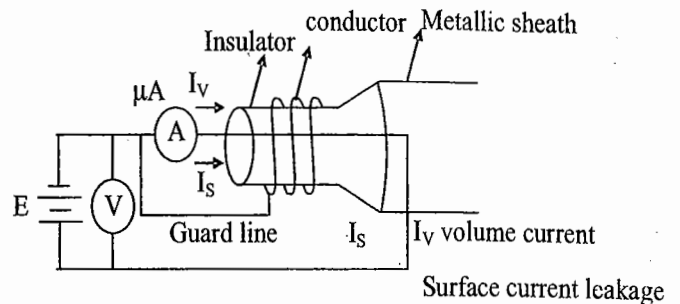
- It is used to find cable insulation resistance in researching laboratories

Example 8:

In loss of charge method for measuring insulation resistance of cable, voltage across capacitance of $0.4343 \mu\text{F}$ is dropped from 200V to 20V in (one minute). Find the insulation resistance value?

Sol: $R = \frac{0.4343 \times 1 \times 60}{0.4343 \times 10^{-6} \log_{10}\left(\frac{200}{20}\right)} = 60 \text{ M}\Omega$

Direct –deflection method



I_v → volume current, passing through the conductor through the insulation and through metallic sheath

I_s → surface leakage current passing through the conductor, over the insulation under the metallic sheath.

Ammeter is reading both I_v & I_s . hence the measured resistance has an error.

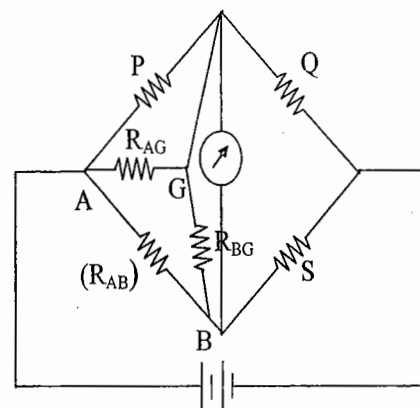
$$I = I_v + I_s$$

Guard wire: It's a conductor with small number of turns, tightly connected over the insulation and maintained at supply terminals.

$$R = \frac{\text{voltmeter reading}}{\text{Ammeter reading}}$$

Ammeter reading is only I_v

Mega Ohm Bridge:



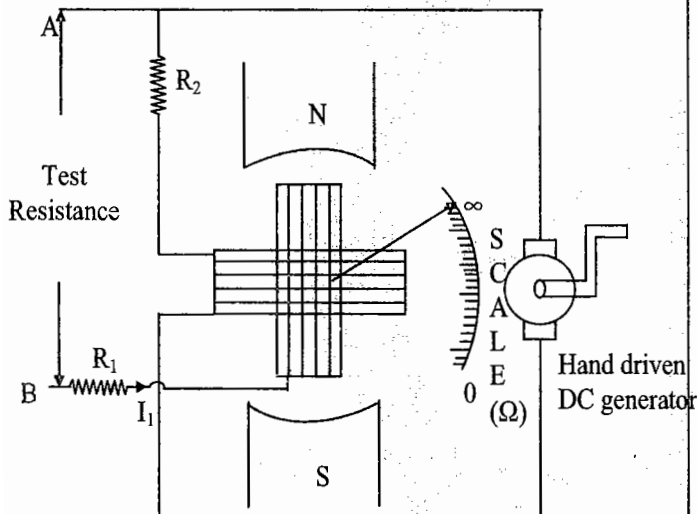


$$R_{AG} || P = P_1$$

- R_{BG} Galvanometer resistance $\rightarrow$ resultant resistance is very less then only R_{AB} is present
- Then $R_{AB} = \frac{P_1}{Q} S$
- Guard terminal of source to eliminate resistances produced by the leakage currents inside the cable insulation



Electro dynamo-meter meggar



- Potential coil (P.C) and current coil (C.C) are moving and these coils are connected such that their corresponding torques acts in opposite direction i.e both P.C and C.C are rigidly connected 90° .
- Resistors R_1 & R_2 are connected in series with P.C & C.C
- Here left extreme end ' ∞ ' is marked on a scale and '0' is marked on right extreme end of scale.

P.C

$$F_1 = BI_1 l \sin \theta$$

C.C

$$F_1 = BI_1 l \sin(90 + \theta)$$

$$= BI_1 l \cos \theta$$

When pointer gives reading

$$F_1 = F_2$$

$$BI_2 l \sin \theta = BI_1 l \cos \theta$$

$$\tan \theta = \frac{R_2}{R_1 + R}$$

$$\theta \propto \frac{1}{R}$$

$$\theta = 0 ; R = \infty$$

$$\theta = \infty ; R = 0$$

- The meggar voltage ratings are 500V, 1000V, 2 kV, 2-5kV,
- The minimum application voltage is 100V

Applications:

- Used for measurement of Insulation resistance of cables
- Winding insulation resistance of motors, generator, Transformers bushings

Note:

- In meggar spring control is not required

Example 9:

The controlling torque in a meggar is provided by

- (a) springs
- (b) Weights attached to the moving system
- (c) It does not need any controlling torque
- (d) None of the above

Ans: (c)

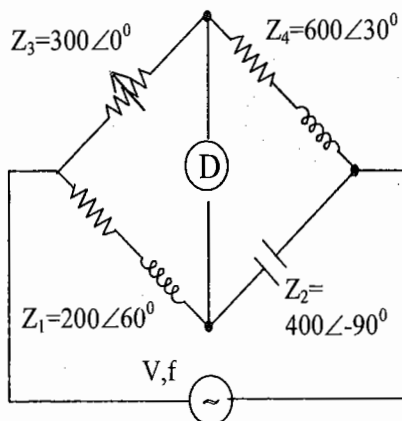
Class Room Practice Questions

LEVEL-1 Questions

01. The bridges suitable for the measurement of an unknown inductance in terms of a known capacitance would include
- Maxwell and Hay
 - Maxwell and Schering
 - Hay and Schering
 - Maxwell, Hay and Schering

02. The Wheatstone bridge method of resistance measurement is ideally suitable for the measurement of resistance values in the range of
- $0.001\ \Omega$ to $1\ \Omega$
 - $0.1\ \Omega$ to $100\ \Omega$
 - $100\ \Omega$ to $10\text{k}\Omega$
 - $100\text{k}\Omega$ to $10\text{M}\Omega$

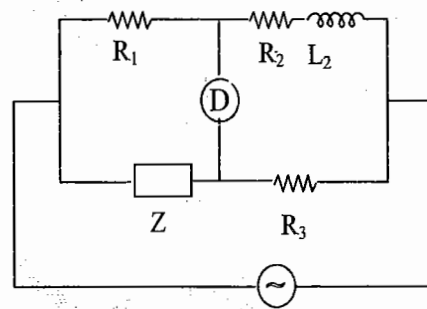
03. In the A.C bridge shown in the given figure, the impedances Z_1 , Z_2 , Z_3 and Z_4 in ohms at the supply frequency are as indicated. Z_3 is variable impedance. The bridge



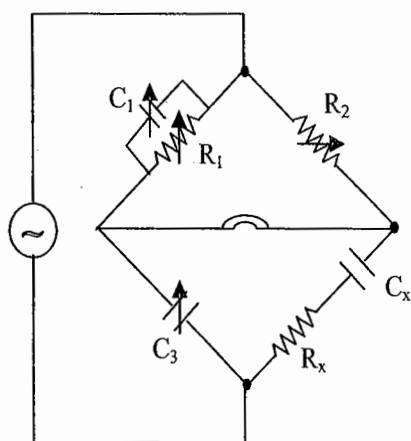
- is balanced with the indicated values of impedances
- can be balanced if $Z_3 = 400\angle 0^\circ$
- can be balanced if $Z_4 = 600\angle 60^\circ$
- cannot be balanced with the given configuration

04. In a balanced Wheatstone bridge, if the position of detector and source are interchanged, the bridge will still remain balanced. This inference can be drawn from
- reciprocity theorem
 - duality principle
 - compensation theorem
 - equivalence theorem

05. The ac bridge shown in the given figure will remain balanced if impedance Z consists of



- resistance and inductance in series
 - resistance and capacitance in parallel
 - capacitance only
 - inductance only
06. The accuracy of Kelvin's double bridge for the measurement of low resistance is high because the bridge
- uses two pairs of resistance arms
 - has medium value resistances in the ratio arms
 - uses a low resistance link between standard and test resistance
 - uses a null indicating galvanometer.
07. For the bridge shown in the given figure, at balance the values of R_x , C_x and Q_x will be



- (a) $R_x = \frac{C_1 R_2}{C_3}$, $C_x = \frac{R_1 C_3}{R_2}$, $Q_x = \omega C_1 R_1$
- (b) $R_x = \frac{C_1 R_2}{C_3}$, $C_x = \frac{R_1 C_3}{R_2}$, $Q_x = \frac{1}{\omega C_1 R_1}$
- (c) $R_x = \frac{C_1 R_1}{C_3}$, $C_x = \frac{R_2 C_3}{R_1}$, $Q_x = \frac{1}{\omega C_1 R_1}$
- (d) $R_x = \frac{C_1 R_1}{C_3}$, $C_x = \frac{R_2 C_3}{R_1}$, $Q_x = \omega C_1 R_1$

08. Vibration galvanometers, tunable amplifiers and head phones are used in
(a) d.c. bridges
(b) a.c. bridges
(c) Both d.c and a.c bridges
(d) Kelvin double bridge
09. The capacitance and loss angle of a given capacitor specimen are best measured by
(a) Wheatstone bridge
(b) Maxwell bridge
(c) Anderson bridge
(d) Schering bridge
10. Which one of the following is measured by the loss of charge method?
(a) Low R
(b) High R
(c) Low L
(d) High L
11. What should be the main characteristic(s) of the null detector in a bridge measurement?

1. Accuracy
2. Precision
3. Sensitivity
4. Resolution

Select the correct answer using the code given below:

Code:

- (a) Only 1 and 2
(b) Only 2 and 3
(c) Only 3 and 4
(d) Only 3

12. Which one of the following bridge is used for measurement of dielectric loss and power factor of a capacitor?

- (a) Maxwell's bridge
(b) Anderson bridge
(c) De Sauty's bridge
(d) Schering bridge

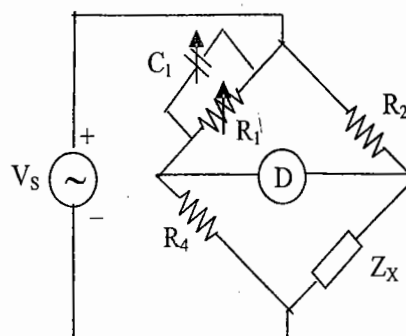
13. **Assertion (A):** The accuracy of the null detector does not play any role in a bridge measurement.

Reason (R): Under balance condition of the bridge, no current flows through the null detector.

14. Which one of the following is a frequency sensitive bridge?

- (a) De-Sauty bridge
(b) Schering bridge
(c) Wien's bridge
(d) Maxwell's bridge

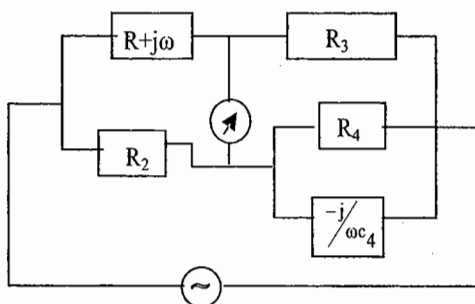
15. The bridge circuit shown in the fig below is used for the measurement of an unknown element Z_x . The bridge circuit is best suited when Z_x is a





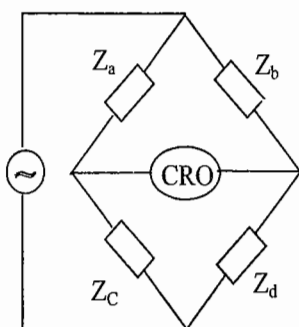
- (a) Low resistance (b) High resistance
(c) Low Q Inductor (d) lossy Capacitor

16. The maxwell's bridge shown in the fig. is at balance the parameters of the inductive coil are



- (a) $R = \frac{R_2 R_3}{R_4}$ $L = C_4 R_2 R_3$
(b) $L = \frac{R_2 R_3}{R_4}$ $R = C_4 R_2 R_3$
(c) $R = \frac{R_4}{R_2 R_3}$ $L = \frac{1}{C_4 R_2 R_3}$
(d) $L = \frac{R_4}{R_2 R_3}$ $R = \frac{1}{C_4 R_2 R_3}$

17. Given, $Z_a = 100 \angle 50^\circ$, $Z_b = 300 \angle 90^\circ$, $Z_c = 200 \angle 0^\circ$ the value of Z_d for the bridge shown in the figure to be balanced is

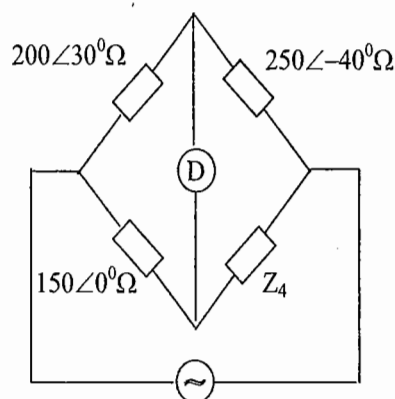


- (a) $600 \angle -40^\circ$ (b) $600 \angle 140^\circ$
(c) $600 \angle -140^\circ$ (d) $150 \angle 40^\circ$

18. While using Maxwell bridge, the Q factor of a coil is obtained as

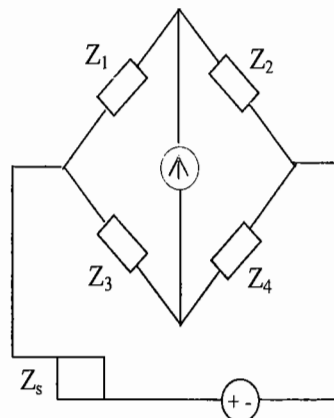
- (a) $\frac{1}{\omega CR}$ (b) ωCR
(c) $\frac{\omega C}{R}$ (d) $\frac{R}{\omega C}$

19. At the balance condition of the ac bridge shown in the above figure, the value of Z_4 would be



- (a) $120 \angle 70^\circ \Omega$ (b) $187.5 \angle -10^\circ \Omega$
(c) $187.5 \angle -70^\circ \Omega$ (d) $3333.3 \angle 70^\circ \Omega$

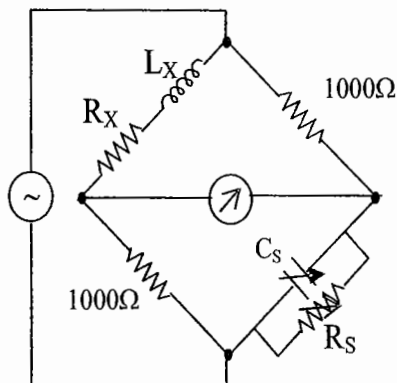
20. Consider the circuit as shown above, Z_1 is an unknown impedance and measured as $Z_1 = (Z_2 Z_3)/Z_4$. The uncertainties in the value of Z_2 , Z_3 and Z_4 are $\pm 1\%$, $\pm 1\%$ and $\pm 3\%$ respectively. The overall uncertainty in the measured value of Z_1 is



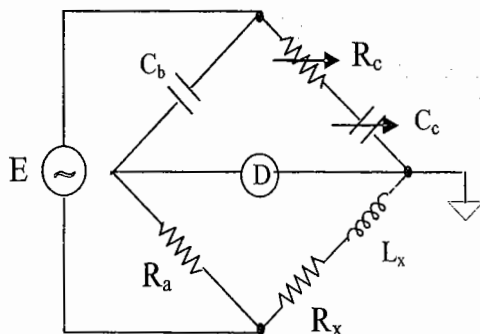


- (a) $\sqrt{11}\%$ (b) $\pm 4\%$
(c) $\pm 5\%$ (d) $\sqrt{5}\%$

21. In the bridge circuit shown above, at balance condition, the value of $C_s = 0.5\mu\text{F}$ and $R_s = 1000\Omega$. The values of inductance L_x and resistance R_x are



- (a) $L_x = 0.5\text{H}$, $R_x = 1000\Omega$
(b) $L_x = 0.25\text{H}$, $R_x = 2000\Omega$
(c) $L_x = 0.5\text{H}$, $R_x = 3000\Omega$
(d) $L_x = 0.25\text{H}$, $R_x = 500\Omega$
22. The figure shows 'Owen bridge' arranged to measure incremental inductance of the unknown inductance L_x , R_x . At balance, what are the values of L_x and R_x ?



- (a) $L_x = \frac{R_a R_c}{C_b}$, $R_x = \frac{R_a C_b}{C_c}$
(b) $L_x = R_a R_c C_b$, $R_x = R_x C_b C_c$

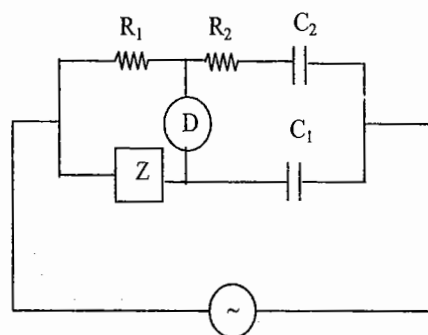
(c) $L_x = \frac{R_a R_c}{C_b}$, $R_x = \frac{R_a C_c}{C_b}$

(d) $L_x = R_a R_c C_b$, $R_x = \frac{R_a C_b}{C_c}$

23. Which of the following bridges is also used in an oscillator?

- (a) Maxwell (b) Schering
(c) Hay (d) Wien

24. The a.c bridge in fig remains balance if Z comprises of



- (a) Resistance only
(b) Capacitance only
(c) Resistance and capacitance in parallel
(d) Resistance and inductor in series

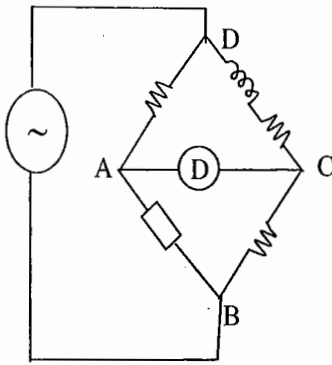
25. Kelvin double bridge is best suited for the measurement of

- (a) Resistances of very low value
(b) low value capacitance
(c) Resistance of very high value
(d) high value capacitance

26. Wien bridge is best suited for the measurement of

- (a) frequency (b) capacitor
(c) inductor (d) Resistor

27. Figure shows a balanced ac bridge excited by a voltage source of fixed frequency



- (a) an R and L in series
- (b) an R and C in series
- (c) an R and C in parallel
- (d) an R and L in parallel

28. Match the parameter with the type of bridge used for measurement.

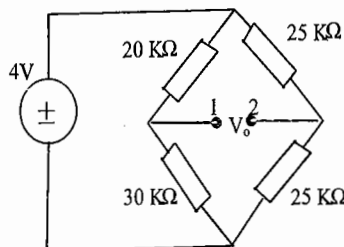
LIST - I

- (a) low value of R
- (b) High value of Q
- (c) Inductance L
- (d) capacitance C

LIST - II

- (e) Schering bridge
- (f) maxwell-wien bridge
- (g) Kelvin double bridge
- (h) Hay bridge

29. The output resistance across the terminals 1 and 2 of the DC bridge in fig is

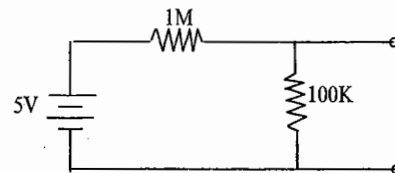


- (a) 12.5 kΩ
- (b) 24.5 kΩ
- (c) 25.0 kΩ
- (d) 100 kΩ

30. If the supply voltage to the bridge decreased by 5%, the sensitivity of the measurement system.

- (a) decreases by 5%
- (b) does not change
- (c) increases by 5%
- (d) increases by 20%

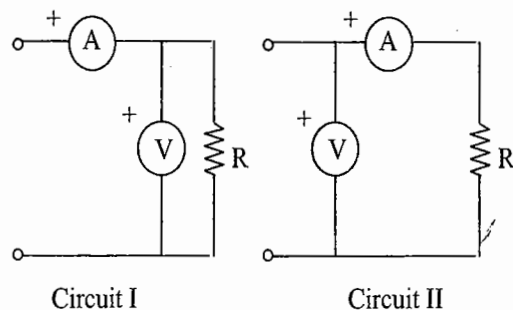
31. A 20,000 ohms/V meter is used to measure the voltage across 100k on 0-1V scale as in the circuit. The voltage measured is



- (a) 5V
- (b) 0.45V
- (c) 4.45V
- (d) 0.082V

LEVEL-2 Questions

01. The power in a resistor R is estimated by measuring the voltage and current using the voltmeter-ammeter method. Two different arrangements can be used as shown in circuits I and II. Less erroneous results are obtained by adopting.

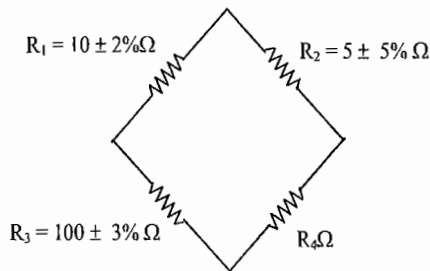


- (a) circuit I for low values of R
- (b) circuit II for low values of R
- (c) circuit I for high values of R
- (d) circuit II for low and high values of R

02. The arms of a Wheatstone bridge are shown in the given figure. For the balanced



condition, the least tolerance value of R_4 will be



- (a) $50 \pm 2\% \Omega$ (b) $50 \pm 3\% \Omega$
(c) $50 \pm 5\% \Omega$ (d) $50 \pm 10\% \Omega$

03. Match List I (Bridges) with List-II (Quantity measured) and select the correct answer using the *codes* given below the lists:

List I

- A. Anderson Bridge
B. Kelvin Double Bridge
C. Schering Bridge
D. Wien Bridge

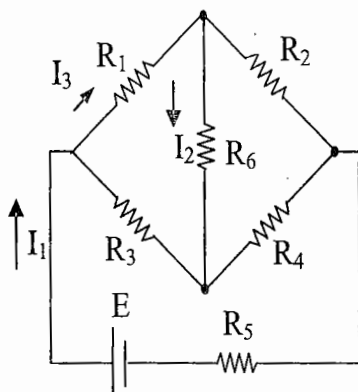
List II

1. Frequency
2. Resistance
3. Inductance
4. Capacitance

Codes:

	A	B	C	D
(a)	3	2	1	4
(b)	3	2	4	1
(c)	2	1	3	4
(d)	2	3	4	1

04. In the balanced Wheatstone bridge shown in the figure, if the value of R_6 is increased, the current I_2



- (a) will increase
(b) will decrease
(c) will remain unchanged
(d) None

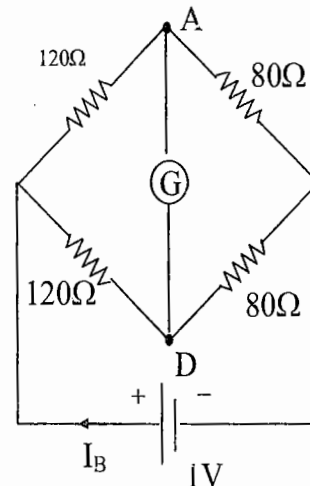
05. Hay's bridge is suitable for the measurement of which one of the following?

- (a) Inductance with $Q < 10$
(b) Inductance with $Q > 10$
(c) Capacitance with high dissipation
(d) Capacitance with low dissipating factors

06. Which bridge is used to determine frequency?

- (a) Anderson bridge
(b) De Sauty bridge
(c) Wien bridge
(d) Campbell bridge

07. In the circuit shown, when the current through the branch AD is zero, the battery current I_B is



- (a) 1 mA (b) 2 mA
(c) 10 mA (d) 20 mA

08. The equations under balance condition for a bridge are $R_1 = \frac{R_2 R_3}{R_4}$ and $L_1 = R_2 R_3 C_4$

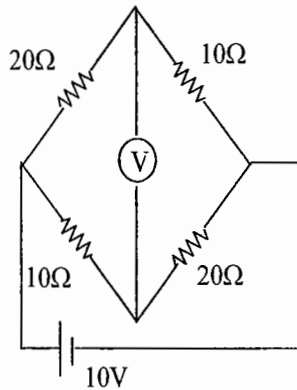
Where R_1 and L_1 are unknown quantities. Where one of the following sets of



parameters should be chosen as variables in order to achieve converging balance?

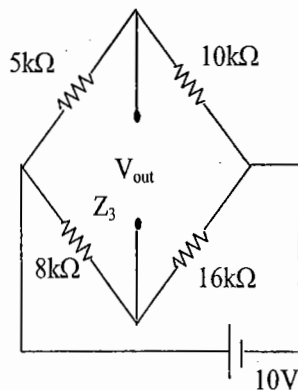
- (a) R_1 and R_3 (b) R_2 and C_4
(c) R_4 and C_4 (d) R_3 and C_4

09. The reading of high impedance voltmeter V in the bridge circuit shown in the given figure is



- (a) zero (b) 3.33V
(c) 4.20V (d) 6.66V

10. In the Wheatstone bridge shown in the given figure, if the resistance in each arm is increased by 0.05% then the value of V_{out} will be



- (a) 50mV (b) 5mV
(c) 0.1V (d) zero

Solutions

LEVEL-1

01. Ans: (a) 02. Ans: (c)

03. Ans: (d) 04. Ans: (a)

05. Ans: (b)

Sol: It is Maxwell-Inductance capacitance bridge.

06. Ans: (c)

07. Ans: (a)

Sol: It is Schering bridge

08. Ans: (b)

09. Ans: (d)

10. Ans: (b)

11. Ans: (d)

12. Ans: (d)

13. Ans: (a)

14. Ans: (c)

15. Ans: (c)

16. Ans: (a)

Sol: It is Maxwell-Inductance Capacitance bridge.

17. Ans: (b)

Sol: $Z_a Z_d = Z_c Z_b$

$$100 \times Z_d = 200 \times 300$$

$$Z_d = 600 \Omega$$

$$\theta_a + \theta_d = \theta_c + \theta_b$$

$$50^\circ + \theta_d = 0^\circ + 90^\circ$$

$$\theta_d = 40^\circ$$

$$Z_d \angle \theta_d = 600 \Omega \angle 40^\circ$$

18. Ans: (b)

Sol: Maxwell-Inductance bridge, Q-factor can't be found because absence of capacitor but in Maxwell-Inductance capacitance bridge Q-factor is found. $Q\text{-factor} = \omega RC$



19. Ans: (c)

Sol: $Z_4 \times 200 = 150 \times 250$

$$Z_4 = \frac{150 \times 250}{200} = 187.5 \Omega$$

$$30^\circ + \theta_4 = 0^\circ + (-40^\circ) \Rightarrow \theta_4 = -70^\circ$$

$$Z_4 \angle \theta_4 = 187.5 \angle -70^\circ$$

20. Ans: (c)

$$\text{Sol: } Z_1 = \frac{Z_2 Z_3}{Z_4}$$

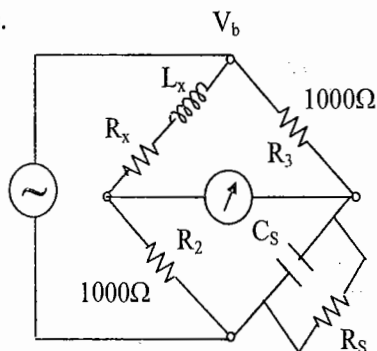
Uncertainty (or) % limiting error, use same method, add them.

$$Z_1 = \frac{Z_2 Z_3}{Z_4}$$

$$= \pm (1\% + 1\% + 3\%) = \pm 5\%$$

21. Ans: (a)

Sol: It is Maxwell-Inductance Capacitance bridge.



$$\frac{L_x}{C_s} = R_2 R_3$$

$$L_x = R_2 R_3 C_s$$

$$= 1000 \times 1000 \times 0.5 \mu = 0.5 \text{ H}$$

$$R_x R_s = R_2 R_3$$

$$R_x = \frac{R_2 R_3}{R_s}$$

$$R_x = \frac{1000 \times 1000}{1000} = 1000 \Omega$$

22. Ans: (d)

$$\text{Sol: } \frac{L_x}{C_b} = R_a R_c$$

$$L_x = R_a R_c C_b$$

$$\frac{R_x}{C_b} = \frac{R_a}{C_c}$$

$$R_x = R_a \frac{C_b}{C_c}$$

23. Ans: (d)

24. Ans: (c)

Sol: The given bridge is Schering bridge (it has 3 capacitors). So 'z' comprises of resistance and capacitance in parallel.

25. Ans: (a)

26. Ans: (a)

27. Ans: (c)

Sol: The given bridge is Maxwell Inductance Capacitance bridge. Therefore unknown element is R and C in parallel.

28. Ans: a-g, b-h, c-f, d-e

29. Ans: (b) 30. Ans: (b) 31. Ans: (d)

Level-2 Key

01. (a)	02. (d)	03. (b)	04. (c)	05. (b)
06. (c)	07. (c)	08. (c)	09. (b)	10. (d)

Error: The difference of magnitude between measured value (A_m) and true value (A_t) is called Error.

- In Engineering, Error is represented with $\pm$
- In general, Error means (Absolute error) i.e.,
 1. No plus, No minus
 2. Only magnitude
- Error represented by ' ϵ '

$$\epsilon = A_m - A_t$$

($\therefore A_m$ is measured value, A_t is true value)

These errors are broadly classified into two types:

1. Static error (error is constant w.r.t time)
2. Dynamic error (error is changing w.r.t time)

Static error: The error which is independent of time is called Static error.

Dynamic error : The error which depends on time is called dynamic error.

+ve $\Rightarrow A_m > A_t$

-ve $\Rightarrow A_m < A_t$

Error may be +ve (or) -ve

Correction Factor (CF) :

The value which we are added (or) subtracted from the measured value in order to get the true value (or) actual value is called Correction factor.

It may be +ve (or) -ve

C.F = -(ϵ)

For ex: Error (ϵ) = ($A_m - A_t$) = 0.2

Correction factor = -(ϵ) = -0.2

So the measured value decreases by '0.2'

Relative Static Error (R.S.E):

The static error taken over the true value is called relative static error.

$$R.S.E = \frac{E}{A_t}$$

$$R.S.E = \frac{A_m - A_t}{A_t}$$

$$\%R.S.E = \frac{A_m - A_t}{A_t} \times 100$$

Basic characteristics of an Instrument:

1. Accuracy
2. Precision
3. Repeatability
4. Reproducibility
5. Sensitivity
6. Resolution
7. Hysteresis
8. Dead time and Dead zone
9. Drift

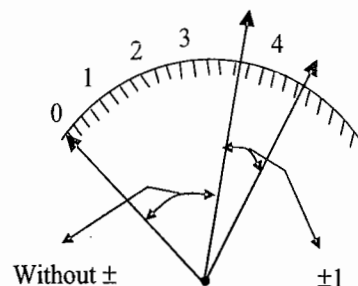
1. Accuracy: It is the degree of closeness of measured quantity to true quantity. (i.e., conformity to truth)

2. %GAE (Guaranteed Accuracy Error) :

When accuracy is mentioned in terms of error it's called "Guaranteed Accuracy Error"

If this error expressed w.r.t full scale deflection it's called %GAE.

It is expressed by Manufacturer, it is constant for meter.



Error is represented by ' $\pm$ '

Deflection is represented without ' $\pm$ '



Instrument	%GAE	
A	$\pm 1\%$	Error
B	$\pm 99\%$	
C	98%	Deflection
D	1%	

So while choosing meter error should be as minimum as possible & Deflection should be as high as possible.

3. Limiting error (or) Guaranteed error:

It's calculated from %GAE

L.E is constant for a meter and it's also expressed by manufacturer

L.E give the range of output for a given input reading.

% Limiting error: If the error is expressed w.r.t true value is called %L.E

$$\% \text{ Limiting error} = \frac{A_m - A_t}{A_t} \times 100$$

It's a variable error and it is calculated by observer.

[In solving problems, %GAE = Accuracy]

$$\% \text{ Accuracy} = (1 - |\epsilon_r|) \times 100$$

Note: For a good instrument %GAE, limiting error and %L.E must be minimum and accuracy (or) %accuracy must be high.

Ex: $\% \epsilon_r = \pm 2\% \Rightarrow \text{Accuracy} = 98\%$

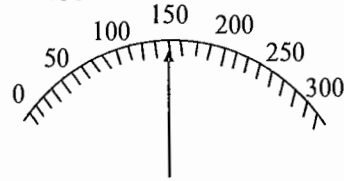
When $\epsilon_r = 0.02$ (when measuring 1V with 100 V full scale) then $\% \text{ Accuracy} = (1 - |\epsilon_r|) \times 100$
 $= (1 - 0.02) \times 100 = 98\%$

Example 1:

If (0 – 300 V) voltmeter has an accuracy of $\pm 1\%$, then find %LE, when measuring

- (a) 200 V (b) 150 V
(c) 100 V (d) 50 V

$$\text{Sol: L.E} = \frac{1}{100} \times 300 = 3V$$



$$(a) \% \text{LE of } 200 \text{ V} = \pm \frac{3}{200} \times 100 = \pm 1.5\%$$

$$(b) \% \text{LE of } 150 \text{ V} = \pm \frac{3}{150} \times 100 = \pm 2\%$$

$$(c) \% \text{LE of } 100 \text{ V} = \pm \frac{3}{100} \times 100 = \pm 3\%$$

$$(d) \% \text{LE of } 50 \text{ V} = \pm \frac{3}{50} \times 100 = \pm 6\%$$

From the above result we can conclude that :
 As the value to be measured increased, then %L.E decreases.

All indicating instruments give more %LE below half of f.s.d of scale.

Example 2:

To measure 10Volts, if one selects a (0–100)volts range voltmeter whose accuracy is within $\pm 1\%$, then the error in this measurement is _____

- (a) $\pm 7.5\%$ (b) $\pm 5\%$
(c) $\pm 10\%$ (d) $\pm 20\%$

Ans: (c)

Sol: (0–100)V

$$\% \text{ G.A.E} = \pm 1\%$$

$$100 \times \frac{\pm 1}{100} = \pm 1$$

$$\text{Error} = 100 \times \frac{\pm 1}{10} = \pm 10\%$$

Example 3:

Three DC currents I_1 , I_2 and I_3 meet at a node with I_1 entering and I_2 and I_3 leaving the node. I_1 and I_2 are measured as 100mA and 99mA with a $\pm 1\%$ accuracy. Then the value of I_3 and the accuracy of I_3 are.

- (a) $1A \pm 2$ (b) $199mA \pm 2$
(c) $1mA \pm 2$ (d) $1mA \pm 1.99$



Ans: (d)

Sol: $I_1 = 100\text{mA}$

$I_2 = 99\text{mA}$

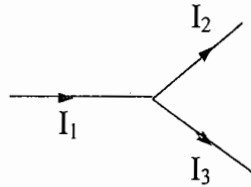
Accuracy = 1%

By KCL

$I_1 + (-I_2) = I_3$

$100\text{mA} \pm 1\% - 99\text{mA} \pm 1\% = I_3$

$I_3 = 1\text{mA} \pm 1.99$



Example 4:

An ammeter has a range of 0 to 30A. The Instrument gave the following readings.

Current flow (A) 0 5 10 15 20 25

Ammeter reading (A) 1 4 12 14 22 28

The non-linearity of the instrument in terms of full scale reading

(a) 20% (b) 10% (c) 25% (d) 15%

Ans: (b)

Sol: Full scale reading = 30V

$E_{\text{error}} = 28 - 25 = 3\text{V}$

$\therefore 3\text{V}$ is a 10% of full scale Reading

Precision:

It indicates the degree of closeness between the measured values.

For a good instrument, high precision is preferred when the measured values are very close.

Ex: While measuring '10A', precision instrument measures 9.9, 9.8, 9.7, 9.6

- An high precision instrument may be accurate (or) may not be accurate i.e., precision doesn't give the guarantee about accuracy.
- But an high accuracy instrument is a precision instrument (i.e.,) accuracy guarantees the precision

Note:

For a good instrument, accuracy is at most important condition but precision is of second order preferences.

Repeatability:

It indicates the degree of closeness among measure values, when same input value

measurand is measured number of times by same instrument with observer under same conditions.

Reproducibility :

It also indicates the degree of closeness, when the same input value is measured number of times by one instrument with many number of observers, under different conditions.

- Precision is the measurement of repeatability (or) reproducibility.
- Precision also depends on number of significant figures in digital instruments.
- As the number of significant figures increases then actual information conveyed also increases i.e. precision increases.

Ex: 130 (3 significant) = here chance of lead (129 or 131)

If 130.00 V (5 significant figures) = (129.9 or 130.1) Here precision increases.

Example 5:

Two Resistors R_1 and R_2 are connected in series with $R_1 = 28.7\Omega$ and $R_2 = 3.624\Omega$.

Calculate the total resistance to the appropriate number of significant figures.

- (a) 32.32 (b) 32.324
(c) 32.3 (d) 32.3240

Ans: (c)

Sol: $R_1 = 28.7\Omega$ (Three significant figure)

$R_2 = 3.624\Omega$ (Four significant figure)

Total Resistance $R_1 = 28.7 + 3.624$

$= 32.324\Omega$

(Five significant figures)

$= 32.3$

(Three significant figures)

This is done because one of the resistances is accurate only to three significant figures and therefore the result should also be reduced to three significant figures and hence the value 32.3Ω

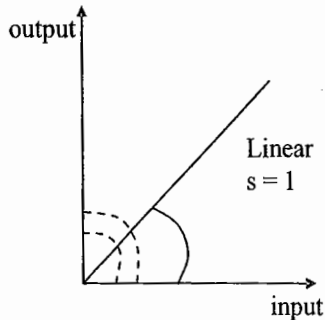
Sensitivity (or) Figure of merit (or) Scale factor

(S): It indicates the relation between change in output readings for change in input reading for an instrument.

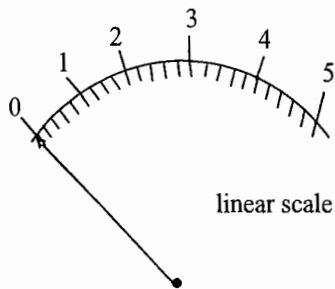


This relation may be linear (or) Non-linear.

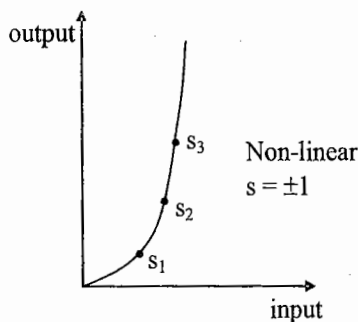
$$s = \frac{\text{change in output}}{\text{change in input}} = \tan \theta$$



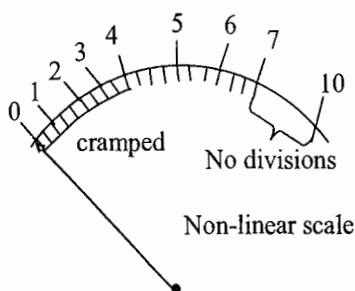
$$\Rightarrow \theta \propto I$$



Ex: PMMC, Dynamometer



$$\Rightarrow \theta \propto I^2$$



Ex: MI

- For an instrument $s = 1$ preferred. It indicates the change in output is proportional to change in input.
- $s < 1$ (or) $s > 1$ indicates non-linear reading, then the scale is cramped at the beginning and almost no divisions at the end.

Note: Sensitivity playing an important role in designing of type of scale at the time of designing

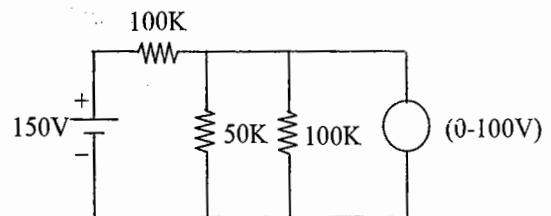
It's the reciprocal of full scale deflection current.

In order to reduce the loading error, high sensitive instruments are preferred.

Example 6:

Two voltmeters with the ranges of (0 – 100 V) and has the sensitivities of 1000 Ω/V , 2000 Ω/V . If these voltmeters are connected individually to measure the voltages across 50 K Ω resistors then find the corresponding readings and % error.

Sol:



$$V_{50K\Omega} = 150 \times \frac{50}{150} = 50V$$

(i) V_A

$R_A = 1000 \times 100 = 100 \text{ k}\Omega$ (resistance of voltmeter which is across it)

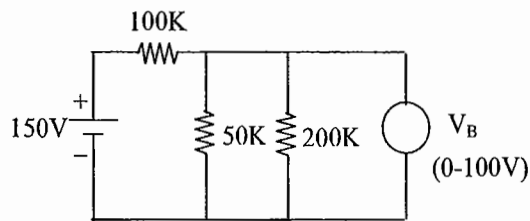
$$50 \text{ K} \parallel 100 \text{ K} = 33.3 \text{ k}\Omega$$

$$V_A = V_{33.3K\Omega} = 150 \times \frac{33.3k\Omega}{100K + 33.3K}$$

$$\% \text{error} = \frac{37.5 - 50}{50} \times 100 = -25\%$$



(ii)



$$R_B = 20000 \times 100 = 200 \text{ K}\Omega$$

$$50 \text{ K}\Omega \parallel 200 \text{ K}\Omega = 40 \text{ K}\Omega$$

$$V_B = V_{48.78} = \frac{150 \times 40}{(40 + 100)} = 42.85 \text{ V}$$

$$\% \text{error} = \frac{42.85 - 50}{50} \times 100 = -14.3\%$$

Loading effect:

When a voltmeter is connected across a resistor to measure voltage, voltmeter draws current for its working, then voltmeter internal resistance and the resistance of voltage to be measured, resultant resistance decreases, then voltmeter reads less than the actual value. It's called loading effect.

Note :

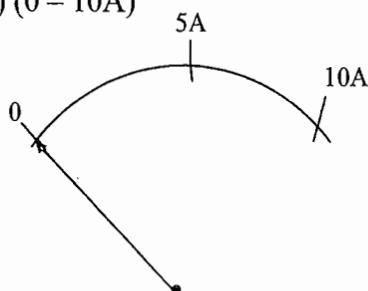
To reduce loading effect, high sensitivity voltmeters to be selected for the same range.

- Compared to electrical instruments, electronic instruments have high sensitivity and more accurate.

Resolution (or) Discrimination

Even a smallest change in input, for which there is a definite numerical change in output, this numerical change is called resolution.

(a) (0 – 10A)



$$\text{Resolution} = \frac{10}{2} = 5 \text{ A / division}$$

$$\text{Resolution} = \frac{\text{full scale deflection (fsd)}}{\text{No. of divisions}}$$

- So as number of divisions increases, resolution increases for the same range.

Note: For non-linear scale we can't calculate resolution.

Hysteresis Error:

If the meter gives less than the actual value for ascending values and shows more than the actual values for descending values.

If the meter gives two different readings for same input value is called hysteresis error.

Causes:

1. Electric Hysteresis:

The meter has two different flux densities while increasing and decreasing time of the same input value.

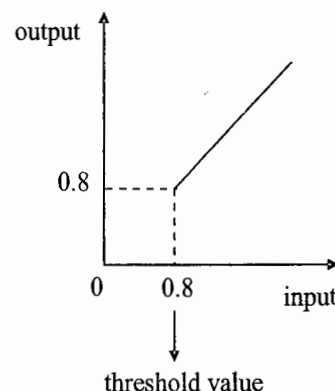
2. Mechanical Hysteresis:

Whatever the stress is developed while loading the meter, the same stress can't be removed.

Threshold:

If the input to the instruments is increased gradually from zero value, then a minimum value is required on a input to get a detachable change in output.

The minimum value is called threshold.





Dead time:

The pointer takes some minimum time to start from its zero position, because of inertia of moving system. The minimum time is called Dead time.

Dead zone: For a large change in input for which there is no output for some portion of the scale is called Dead zone.

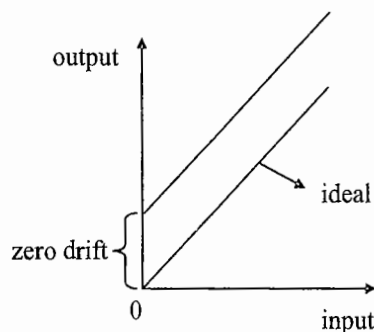
Drift:

It's an error; but constant throughout the range of scale.

Zero drift:

For an instrument, there is variation in output, which is not caused by change in input.

- It's due to instability of components and internal temperature change.



- For a good instrument, accuracy, precision, repeatability, reproducibility, sensitivity, resolution, linearity, fidelity (dynamic error is zero) are desirable characteristics.
- For a good instrument, limiting error, hysteresis error, dead time, dead zone, drift and non-linearity are undesirable characteristics.

Example 7:

The following readings are taken by one voltmeter to the same input values are 102.1 V, 102.4V, 102.3 V, 102.6 V, 102.7 V, 102.9V, then find the range of errors.

Sol: $V_{avg} = \frac{102.1 + 102.4 + 102.3 + 102.6 + 102.7 + 102.9}{6}$

$= 102.5 \text{ V} \rightarrow \text{True value } (A_t)$

From question

$V_{min} = 102.1$

$V_{max} = 102.9$

$V_{max} - V_{avg} = 102.9 - 102.5 = 0.4$

$V_{avg} - V_{min} = 102.5 - 102.1 = 0.4$

Range of error $= \pm \frac{(V_{avg} - V_{min}) + (V_{max} - V_{avg})}{2}$

$= \pm \frac{0.4 + 0.4}{2} = \pm 0.4$

Note: To obtain true value, measure the input value number of times and take the average. If it is infinite times, 100% true value is obtained

Types of Static Errors

- Gross errors:** These are completely due to human mistakes i.e., doing loose connections, reading the values, taking the values, memorizing and doing calculations. Gross errors are not common for all observers. These are variable errors.

Note: The reading can be taken correctly from the meter when meter is in vertical position and eye contact should be over the pointer (i.e., 90° to the meter)

- Systematic error:**
 - Instrumental
 - Environmental
 - Observational

(i) Instrumental :

- Due to inherent short coming of instrument, ageing effect, frictional error, Hysteresis error.
- Misuse of an instrument : choosing wrong range of an instrument.



(c) loading effect

(ii) Environmental error:

Due to temperature, pressure, humidity, dust etc.

(iii) Observational error: or (parallax error)

Note: To eliminate the shadow of the pointer a mirror strip is placed in between scale and pointer (parallel error is eliminated)

Systematic errors are constant errors these are common for all observers.

3. Random errors or accidental errors:

These are due to the un identified sources These errors comes suddenly and eliminate suddenly but reason is not known.

These are more but effect is very small to eliminate the random error statistical analysis is carried out i.e. multi sample data in which the same input value is measured no. of times which are instrument by one observer or many observers.

$x_1, x_1, x_1, \dots, x_n \Rightarrow 'n' \text{ times}$

Average value $(\bar{x}) = \frac{x_1 + x_2 + \dots + x_n}{n}$

2. Deviation $(d) \Rightarrow d_1 = x_1 - \bar{x}$

$$d_2 = x_2 - \bar{x}$$

$$\therefore d_n = x_n - \bar{x}$$

3. Average deviation $= \frac{|d_1| + |d_2| + \dots + |d_n|}{n}$

To know precision of an instrument.

4. Standard deviation: Deviation from average value

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + \dots + d_n^2}{n}} \quad (n > 20)$$

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2}{n-1}} \quad (n \leq 20)$$

5. Probable error (r) = $\pm 0.6745\sigma$

6. Probable error of mean = $(r_m) = \frac{r}{\sqrt{n-1}}$

7. Uncertainty Error

$$X = f(x_1, x_2, x_3, \dots, x_n)$$

$W_{x1}, W_{x2}, \dots, W_{xn}$ are the uncertainties of

$x_1, x_2, \dots, x_n$

Uncertainty of $x = w_x =$

$$\pm \sqrt{\left(\frac{\partial x}{\partial x_1}\right)^2 \times w_{x1}^2 + \left(\frac{\partial x}{\partial x_2}\right)^2 \times w_{x2}^2 + \dots + \left(\frac{\partial x}{\partial x_n}\right)^2 \times w_{xn}^2}$$

8. Standard Error

$$X = f(x_1, x_2, x_3, \dots, x_n)$$

$\sigma_{x1}, \sigma_{x2}, \dots, \sigma_{xn}$ are the standard deviation error of $x_1, x_2, \dots, x_n$

Standard deviation error of ' x ' = $\sigma_x =$

$$\pm \sqrt{\left(\frac{\partial x}{\partial x_1}\right)^2 \times \sigma_{x1}^2 + \left(\frac{\partial x}{\partial x_2}\right)^2 \times \sigma_{x2}^2 + \dots + \left(\frac{\partial x}{\partial x_n}\right)^2 \times \sigma_{xn}^2}$$

Note:

For the calculation of uncertainty and standard deviations convert percentage values into real values and substitute in formulae.

Example 8:

Uncertainty distribution is used for

(a) analysis of multi-sample data

(b) analysis of single sample data

(c) analysis of both single and multiple sample data

(d) none of the above

Ans: (b)



Example 9:

A set of independent 5 measurements were made to determine the weight of a lead shot. The weights in gramme were

1.570, 1.597, 1.591, 1.562, 1.577

Find the probable error of the mean

- (a) 4.9×10^{-3} (b) 11.6×10^{-3}
(c) 14.5×10^{-3} (d) 1.57

Ans: (a)

Sol: Find arithmetic mean

$$= \frac{1.570 + 1.597 + 1.591 + 1.562 + 1.577}{5}$$

= 1.5794 gramme

Deviation from the mean

$$d_1 = 1.570 - 1.5794 = -0.0094$$

$$d_2 = 1.597 - 1.5794 = 0.0176$$

$$d_3 = 1.591 - 1.5794 = 0.0116$$

$$d_4 = 1.562 - 1.5794 = -0.0174$$

$$d_5 = 1.577 - 1.5794 = -0.0024$$

Stranded deviation

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2}{5-1}} = 0.0145.$$

$$\text{Problem error of the mean} = \frac{0.6745 \sigma}{\sqrt{5-1}} = 4.9 \text{m gram}$$

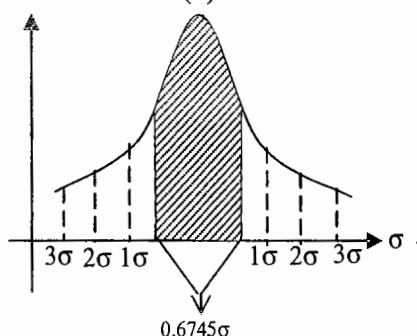
Example 10:

In which of following deviation's has the highest probability range in specifying the measured data.

- (a) $\pm 3\sigma$ limit (b) $\pm 2\sigma$ limit
(c) Probable error (d) Standard deviation

Ans: (a)

Sol:



From graph we can approximated as
Deviation Probability

$$0.6745 \sigma \rightarrow \approx 50\% \text{ (Probable error)}$$

$$\pm \sigma \rightarrow \approx 68.28\% \text{ (Standard deviation)}$$

$$\pm 2\sigma \rightarrow \approx 95.46\%$$

$$\pm 3\sigma \rightarrow \approx 99.74\%$$

Example 11:

The manufactures providing four meter with labeling their accuracies given below

(1) Meter "A" with Accuracy $\pm 1\%$

(2) Meter "B" with Accuracy $\pm 98\%$

(3) Meter "C" with Accuracy 99%

(4) Meter "D" with Accuracy 1%

Among the given above meters, the high accurate meters are

- (a) only BD (b) only BC
(c) only AC (d) only AD

Ans: (c)

Sol: For example let us consider we are measuring 5A current by using above given meters then

Meter A gives $\Rightarrow 5 \pm 1\% \Rightarrow 5 \pm 0.01 \Rightarrow 4.99$
(or) 5.01

Meter B gives $\Rightarrow 5 \pm 98\% \Rightarrow 5 \pm 0.98 \Rightarrow 4.02$
(or) 5.98

Meter C gives $\Rightarrow 99\% \text{ of } 5A \Rightarrow 4.95A$

Meter D gives $\Rightarrow 1\% \text{ of } 5A \Rightarrow 0.05A$

Conclusion: from above we can concluded meter "A" and meter C gives the highest accuracy readings among the given meters.

Example 12:

Power factor of a circuit is determined by :

$$\cos \phi = \frac{P}{VI}$$

Where P = Power in watt

V = Voltage in volt

I = current in Ampere

The relative error in power, current and voltage are



respectively $\pm 0.5\%$; $\pm 1\%$; $\pm 1\%$ calculate the relative error in power factor. Also calculate the uncertainty in the power factor if the errors are specified as uncertainties

- (a) 2.5%;1.8 (b) 2.3%; 1.5
(c) 2.5%;1.89 (d) 2.5%; 1.5

Ans: (d)

Sol: Given data

Relative error in power $\Rightarrow \pm 0.5\%$

Relative error in voltage $\Rightarrow \pm 1\%$

Relative error in current $\Rightarrow \pm 1\%$

$$\text{Given that } \cos \phi = \frac{P}{VI}$$

$$\begin{aligned} \therefore \text{Relative error in power factor} \\ &= 0.5\% + 1\% + 1\% \\ &= 2.5\% \end{aligned}$$

Class Room Practice Questions

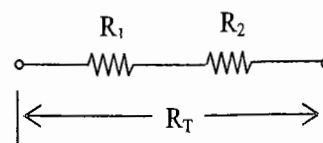
LEVEL-1 Questions

01. The term 'precision' used in instrumentation means
- gradual departure of the measured value from the calibrated value
 - smallest increment in the measurand that can be detected by the instrument.
 - maximum distance or angle through which any part of a mechanical system may be moved in one direction without causing motion of the next part.
 - the ability of the instrument to give output readings close to each other, when the input is constant.
02. Threshold of a measurement system is
- the smallest change in input which can be detected
 - a measure of linearity of the system
 - the smallest input which can be detected
 - a measure of precision of the system
03. A large no. of 230Ω resistors are obtained by combining 120Ω resistors with a standard deviation of 4.0Ω and 110Ω resistors with a standard deviation of 3.0Ω . The standard deviation of the 230Ω resistors thus formed will be
- 3.5Ω
 - 5.0Ω
 - 7.0Ω
 - 12.0Ω
04. The following terms used in the context of an instrument are numbered as shown
- | | |
|--------------|----------------|
| 1. accuracy | 2. sensitivity |
| 3. precision | 4. resolution |
- Match these with their possible definitions listed below
- P. repeatability of readings on successive observation

- Q. smallest perceptible change in the output
 R. Deviation of the output from the true value
 S. minimum value of the input from the true value
 T. Ration of the change in the instrument readings to the change in the measured variable
- 1-P, 2-Q, 3-R, 4-S
 - 1-S, 2-Q, 3-P, 4-T
 - 1-R, 2-T, 3-P, 4-Q
 - 1-T, 2-Q, 3-p, 4-R

05. For a proving ring used in a press, the value of force and its uncertainty are given as $12355 \pm 2354\text{N}$ without any rounding off. If the error should be rounded off at the maximum to two significant figures only. The value of the reading is rounded off as
- $12300 \pm 2300\text{N}$
 - $12360 \pm 2350\text{N}$
 - $12400 \pm 2300\text{N}$
 - $2400 \pm 2400\text{N}$
06. The percentage limiting error, in the case of an instrument reading of 8.3V with a 0 to 150 voltmeter having a guaranteed accuracy of 1% full-scale reading is
- 1.810%
 - 0.181%
 - 18.10%
 - 0.0018%

07. Two resistances R_1 and R_2 are connected as shown in the given figure. If $R_1 = 528 \pm 5\Omega$ and $R_2 = 325 \pm 3\Omega$, then the total resistance R_T will be



- $853 \pm 2\Omega$
- $853 \pm 5\Omega$
- $853 \pm 3\Omega$
- $853 \pm 8\Omega$



08. With reference to 'random errors' in measurement, the standard deviation σ can be expressed, in terms of deviation of any individual observation from the mean of the group ' d_m ' and the number of observations in the group ' n ' as

$$(a) \sigma = \frac{\sum d_m}{n} \quad (b) \sigma = \sqrt{\frac{\sum d_m^2}{n}}$$

$$(c) \sigma = \sqrt{\frac{\sum d_m^2}{n-1}} \quad (d) \sigma = 0.6745 \sqrt{\frac{\sum d_m^2}{n-1}}$$

09. The "accuracy" of a measuring instrument is determined by the

- (a) closeness of the value indicated by it to the correct value of the measured
- (b) repeatability of the measured value
- (c) speed with which the instruments reading approaches the final value
- (d) least change in the value of the measured that could be detected by the instrument

10. The difference between the measured value and the true value is called

- (a) gross error (b) relative error
- (c) probable error (d) absolute error

11. A 300 V full scale deflection voltmeter has an accuracy of $\pm 2\%$, when it reads 222V. The actual voltage

- (a) lies between 217.56 V and 226.44V
- (b) lies between 217.4v and 220.3V
- (c) lies between 216V and 228V
- (d) is exactly 222V

12. A voltmeter has a range 0-20 V and manufacturer rates its accuracy as $\pm 1\%$ fsd. Match List I (voltage values) with List II (Error as percentage of True value) and select the correct answer using the codes given below

List-I

- A. 2V
- B. 5 V
- C. 10 V
- D. 20 V

List-II

- 1. 4%
- 2. 10%
- 3. 2%
- 4. 1%

Codes:

	A	B	C	D
(a)	3	1	2	4
(b)	2	1	3	4
(c)	2	4	3	1
(d)	3	4	2	1

13. The voltage across an impedance is measured by a voltmeter having input impedance comparable with the impedance causing an error in the reading. What is this error called?

- (a) random error (b) gross error
- (c) systematic error (d) loading effect error

14. Match List - I (Term) with List II (Statement) and select the correct answer using the code.

List - I

- A. Relative error
- B. Precision
- C. Calibration
- D. Resolution

List - II

- 1. The ability of the device to give identical output when repeat measurement are made with the same input signal
- 2. The ratio of difference between the true value of the measurand
- 3. The smallest increment in measurand that can be detected with certainty by the instrument
- 4. The process of making adjustments on the scale so that the instrument readings conform to an accepted standard



Codes:

	A	B	C	D
(a)	2	3	4	1
(b)	4	1	2	3
(c)	4	3	2	1
(d)	2	1	4	3

15. A resistance is measured by a voltmeter-ammeter method using DC excitation and a voltmeter of very high resistance connected directly across the unknown resistance. If the voltmeter and ammeter are subject to maximum error of $\pm 2.4\%$ and $\pm 1.0\%$ respectively, then the magnitude of maximum error in the value of resistance obtained from the measurement is nearly
- (a) 1.4% (b) 1.7%
(c) 2.4% (d) 3.4%

16. **Assertion (A):** Precision is a necessary prerequisite of accuracy.

Reason(R): Precision guarantees accuracy.

- (a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

17. Consider the following statements:

In a measuring instrument,

1. linearity is more important than sensitivity.
2. high precision indicates high accuracy
3. accuracy cannot be better than resolution.

Of these statements

- (a) 1, 2 and 3 are correct
(b) 1 and 2 are correct
(c) 2 and 3 are correct
(d) 1 and 3 are correct

18. Which one of the following has the highest accuracy

- (a) Standard resistance
(b) Standard inductance
(c) Standard capacitance
(d) Standard mutual inductance

19. The total current $I = I_1 + I_2$ in a circuit is measured as $I_1 = 150 \pm 1A$, $I_2 = 250 \pm 2A$, where the limits of error are given as standard deviations. I is measured as

- (a) $(400 \pm 3)A$ (b) $(400 \pm 2.24)A$
(c) $(400 \pm 1.5)A$ (d) $(400 \pm 1)A$

20. Match **List-I (Accuracy)** with **List-II (Type of the standard)** and select the correct answer using the codes given below the lists:

List - I (Accuracy)

- A. Least accurate
B. More accurate
C. Much more accurate
D. Highest possible accurate

List - II (Type of the standard)

1. Primary
2. Secondary
3. Working
4. International

Codes:

	A	B	C	D
(a)	3	4	1	2
(b)	1	4	3	2
(c)	3	2	1	4
(d)	1	2	3	4

21. Which one of the following statements is correct?

The application of the instrument in wrong manner in the procedure of measurement results in a/an

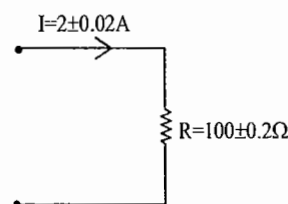
- (a) systematic error (b) random error
(c) gross error (d) instrument error



22. Which one of the following statements correctly represents the systematic errors?
- These errors can be calculated from the details of the instruments
 - These are the residual errors
 - These errors may occur under controlled conditions
 - These are the errors committed by the experimenters
23. When reading is taken at half scale in the instrument, the error is
- exactly equal to half of full-scale error
 - equal to full-scale error
 - less than full-scale error
 - more than full-scale error
24. The errors introduced by an instrument fall in which category?
- Systematic error
 - Random errors
 - Gross errors
 - Environmental errors

LEVEL-2 Questions

01. A 10V full scale voltmeter having $100\text{k}\Omega$ sensitivity is used to measure the output of a photovoltaic cell having an internal resistance of $1\text{M}\Omega$. The voltmeter reads 5V. The voltage generated by the photovoltaic cell is
- 5 V
 - 10 V
 - greater than 5V but less than 10 V
 - greater than 10 V
02. The current flowing through the resistor R is as indicated in the given figure. The computed value of power is



- $400 \pm 0.42 \text{ W}$
 - $400 \pm 4.60 \text{ W}$
 - $400 \pm 8.85 \Omega$
 - $400 \pm 1065 \text{ W}$
03. Two resistors $R_1 = 36 \text{ ohms}$ and $R_2 = 75 \text{ ohms}$, each having tolerance of $\pm 5\%$ are connected in series. The value of the resultant resistance will be
- $111 \pm 0 \text{ ohm}$
 - $111 \pm 2.778 \text{ ohm}$
 - $111 \pm 5.55 \text{ ohm}$
 - $111 \pm 7.23 \text{ ohm}$
04. A galvanometer with a full scale current of 10 mA has a resistance of 1000Ω . The multiplying power (the ratio of measured current to galvanometer current) of a 100Ω shunt with this galvanometer is
- 110
 - 100
 - 11
 - 10
05. Two $100 \mu\text{A}$ full scale PMMC meters are employed to construct a 10 V and a 100 V full scale voltmeter. These meters will have figures of merit (sensitivities) as
- $10 \text{ k}\Omega/\text{V}$ and $10 \text{ k}\Omega/\text{V}$
 - $100 \text{ k}\Omega/\text{V}$ and $10 \text{ k}\Omega/\text{V}$
 - $10 \text{ k}\Omega/\text{V}$ and $100 \text{ k}\Omega/\text{V}$
 - $10 \text{ k}\Omega/\text{V}$ and $1 \text{ k}\Omega/\text{V}$
06. Three voltmeters are connected in series across 120V DC supply as
- 100V and 5mA
 - 100V and $250\Omega/\text{V}$
 - 100mA and $5 \text{ k}\Omega$
- Then estimate the reading of each voltmeter.



07. A moving coil voltmeter has uniform scale with 100 divisions; the full scale reading is 200V and $\frac{1}{10}^{\text{th}}$ of a scale division can be estimated with a fair degree of accuracy. Resolution of the instrument is
(a) 2V (b) 0.2 V
(c) 20 V (d) 200V

Solutions

LEVEL-1

01. Ans: (d)

02. Ans: (a)

03. Ans: (b)

Sol: $R_{eq} = R_1 + R_2$

$$\frac{\partial R_{eq}}{\partial R_1} = \frac{\partial R_{eq}}{\partial R_2} = 1$$

$$\text{S.D. of } R_{eq} = \sqrt{\left(\frac{\partial R_{eq}}{\partial R_1}\right)^2 (\partial R_1)^2 + \left(\frac{\partial R_{eq}}{\partial R_2}\right)^2 (\partial R_2)^2}$$

$$= \sqrt{(3)^2 + (4)^2} = 5$$

$$R_{eq} = 230 \pm 5 \Omega$$

04. Ans: (c)

05. Ans: (d)

06. Ans: (c)

Sol: Limiting error = $\frac{150}{8.3} \times 1\% = 18.1\%$

07. Ans: (d)

Sol: $R_T = R_1 + R_2 = 528 + 325 = 853 \Omega$

$$\text{Error} = \Delta R_1 + \Delta R_2 = 3 + 5 = 8 \Omega$$

08. Ans: (b)

09. Ans: (a)

10. Ans: (d)

11. Ans: (c)

12. Ans: (b)

13. Ans: (d)

14. Ans: (d)

15. Ans: (d)

$$\text{Sol: } R = \frac{V}{I}; \Delta R = \Delta V + \Delta I$$

$$\Rightarrow \Delta R = 2.4 + 1$$

$$\Delta R = 3.4\%$$

16. Ans: (c)

17. Ans: (d)

18. Ans: (a)

19. Ans: (b)

Sol: $I = I_1 + I_2$

$$\frac{\partial I}{\partial I_1} = \frac{\partial I}{\partial I_2} = 1$$

$$\text{S.D} = \sqrt{\left(\frac{\partial I}{\partial I_1}\right)^2 (\partial I_1)^2 + \left(\frac{\partial I}{\partial I_2}\right)^2 (\partial I_2)^2}$$

$$= \sqrt{(2)^2 + (1)^2} = 2.24$$

$$\therefore I = 400 \pm 2.24 \text{ A}$$

20. Ans: (c)

21. Ans: (d)

22. Ans: (d)

23. Ans: (d)

24. Ans: (a)

Level-2 Key

01. (b) 02. (c) 03. (c) 04. (c) 05. (a)

06. (i) 48V (ii) 60V (iii) 12V

07. (b)

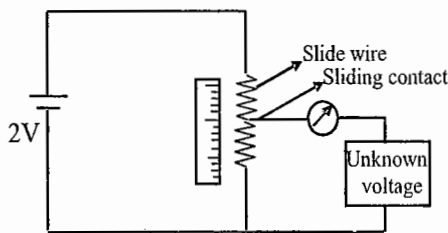
6

Potentiometers & Instrument Transformers

Potentiometer or Null detector

It is an instrument designed to measure an unknown voltage by comparing it with a known voltage.

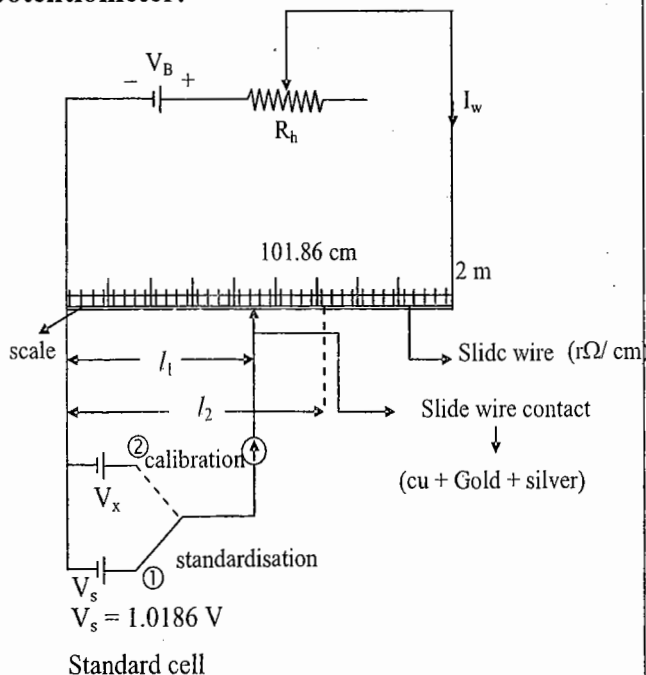
Basic potentiometer circuit:



It measures unknown voltage by comparing it with a known voltage or standard cell voltage.

Used to measure unknown voltages accurately

Circuit diagram of basic slide wire potentiometer:



$\ell \rightarrow$ length of slide wire of potentiometer.

$\ell_1 \rightarrow$ length of slide wire at which the galvanometer is balanced with standard cell voltage (V_s).

$\ell_2 \rightarrow$ length of slide wire at which the galvanometer is balanced with unknown EMF (V_x)

$I_w =$ working current

$$I_w = \frac{V_B}{(R_n + lr)}$$

Standardisation:

The process of adjusting working current (I_w), with the help of rheostat such that voltage drop across portion of slide wire is made equal to standard cell voltage.

Note: Standardisation is process of converting voltage in terms of centimeter scale

Standardisation calibration

$$V_s = I_w \cdot \ell_1 \cdot r$$

$$I_w = \frac{V_s}{\ell_1 \cdot r} \dots (1)$$

$$V_x = I_w \cdot \ell_2 \cdot r$$

$$I_w = \frac{V_x}{\ell_2 \cdot r} \dots (2)$$

By equation 1 and 2

$$V_x = V_s \frac{\ell_2}{\ell_1}$$

Note: Once standardization is done working current is kept constant.

Applications:

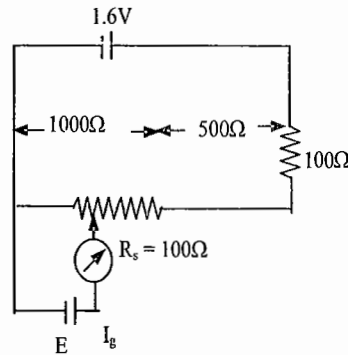
For more accuracy these are used for calibration of ammeter, voltmeter & watt meters.



Example 1:

An unknown voltage source E (with negligible internal resistance) is connected to a potentiometer circuit as shown in the figure below. If the galvanometer current $10\mu\text{A}$ with the direction as indicated, then value of E is

- (a) 0.9910V
- (b) 0.9991V
- (c) 1.0012V
- (d) 1.0048V



Sol: $I_g = 10\mu\text{A}$, $E = ?$

Write KVL for loop 1

$$-1.6 + (100 + 500)I_w + 1000(I_w + I_g) = 0$$

$$1600I_w + 1000I_g = 1.6$$

$$I_w = \frac{1.6 - 1 \times 10^{-3}}{1600} = 9.9375 \times 10^{-4} \text{ A}$$

Write KVL for loop 2

$$-E + 100 \times 10 \times 10^{-6} + 1000(10 \times 10^{-6} + 9.9375 \times 10^{-4}) = 0$$

$$E = 1.00475$$

- Practically in laboratories we are using Crompton's type of potentiometer.

Instrument Transformers

Transformers used in conjunction with measuring instruments for measurement purposes are called "Instrument transformers". The transformer used for measurement of current is called a "current transformer" or simply "CT". Transformers used for voltage measurements are called "voltage transformers" or "potential transformers" or simply "P. T" for short.

Ratio of Instrument Transformers:

1. Transformation Ratio:

$$R = \frac{\text{Primary wdg current}}{\text{secondary wdg current}} \quad \text{for a C. T.}$$

$$= \frac{\text{Primary wdg voltage}}{\text{secondary wdg voltage}} \quad \text{for a P. T.}$$

It is the ratio of the magnitude of the primary phasor to the secondary phasor.

2. Nominal Ratio:

It is the ratio of rated primary winding current (or voltage) to the rated secondary winding current (or voltage).

$$K_n = \frac{\text{rated primary winding current}}{\text{rated secondary winding current}} \quad \text{for a C. T.}$$

$$= \frac{\text{rated primary winding voltage}}{\text{rated secondary winding voltage}} \quad \text{for a P. T.}$$

3. Turns Ratio:

This is defined as

$$n = \frac{\text{number of turns of secondary winding}}{\text{number of turns of primary winding}} \quad \text{for a C. T.}$$

$$= \frac{\text{number of turns of primary winding}}{\text{number of turns of secondary winding}} \quad \text{for a P. T.}$$

The ratio errors are present in instrument transformers

$$(a) \% \text{ ratio error } (\% \sigma) = \frac{k_n - R}{R} \times 100$$

$$(b) \text{ Phase angle error } (\theta)$$

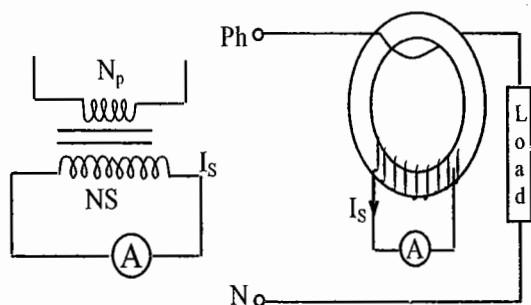
Ratio correction factor:

$$(R.c.f) = \frac{\text{Transformation ratio}(R)}{\text{No min al ratio}(k_n)}$$

The output of CT is called as burden.



Current Transformers:



C.T. primary is connected in series with the load current to be measured. Hence it is called a series T.I.F.

C.T secondary is short circuited by m.I Ammeter.

The ratio mentioned on C.T. name plate is a current ratio i.e. I_p/I_s .

Circular strip wound core is used for the C.T.

Note: C.T secondary should not be open circuited when it's primary is energized it's dangerous.

Reason: Secondary current is zero so secondary EMF is zero but to maintain same current in secondary terminals more magnetizing current is drawn because of this very high voltage is induced then insulation breakdown. It's dangerous to the person operating it.

Transformation ratio $(R) = \frac{I_p}{I_s}$

$$R = \frac{nI_s + I_0 \sin(\alpha + \delta)}{I_s} \quad (\because \delta = 0)$$

$$R = n + \frac{I_0 \sin \alpha}{I_s}$$

$$R = n + \frac{I_e}{I_s}$$

$$R = n \left[1 + \frac{I_e}{I_p} \right]$$

Phase angle error:

The primary and secondary phasor difference is to be 180° in case of ideal Transformer. But in practical this angle having a difference of θ is called phase angle error.

From the above phasor diagram.

$$\tan \theta = \frac{ab}{OC + cb} = \frac{I_0 \cos(\alpha + \delta)}{nI_s + I_0 \sin(\alpha + \delta)}$$

$$[\because I_0 \sin(\alpha + \delta) \ll nI_s]$$

$$\tan \theta \cong \frac{I_0 \cos \alpha}{nI_s} \quad (\because \delta \approx 0)$$

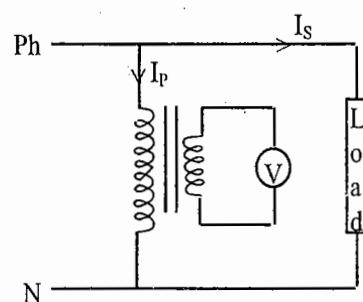
$$\therefore \theta = \frac{I_m}{nI_s} \text{ rad}$$

$$\theta = \frac{I_m}{nI_s} \times \frac{180}{\pi} \text{ degree}$$

Both ratio error and phase angle errors are due to the excitation current ' I_0 '.

Specifically the ratio error is due to the iron loss component ' I_e ' and phase angle error is due to the magnetizing component I_m .

Potential T.I.F:



P.T primary is connected in parallel with the load hence it is called as a shunt transformer.

The standard P.T. secondary voltage rating is 120V.



- P.T. secondary is open circuited by high resistance moving Iron voltmeter.
- It's basically a step down transformer.
- The effect of ratio error and phase error are less in P.T compared to C.T because it is involving with voltage measurement.
- For high voltage appreciation core type is used and for low type voltage, shell type core is used.
- Because cost point of view, potential transformers are not suitable for above 100kV voltage measurement. For this we are using (C.V.T) capacitive voltage transformers are used.

Actual ratio :

$$R = n + \frac{n_I (R_p \cos \Delta + X_p \sin \Delta) + I_e R_p + I_m X_p}{V_s}$$

$$\% \text{RatioError}(\sigma) = \frac{K_n - R}{R} \times 100$$

Phase angle error :

$$\theta = n + \frac{I_s (X_p \cos \Delta - R_p \sin \Delta) + I_e R_p \times I_m X_p}{n V_s} X_p$$

⇒ total resistance total reactance referred to primary side.

Difference between D.T and P.T.

D.T	P.T
1. in case of D.T (distribution transformer) is designed for changes in loads the voltage should be maintained constant i.e It depends voltage regulation.	1. Potential Transformer takes care of voltage ratio Between the primary and secondary.

2. we require η is main criteria in case of distribution Transformer	2. Accuracy is main criteria in case of P.T
3. D.T uses rectangular core	3. T uses circular core
4. Overloading capacity of Distribution Transformer is 20 to 30 % of full load	4. overloading capacity of potential is 2 to 3 % of full load

Example 2:

The constant for 3phase 3element integrating wattmeter is 0.15 revolution of disc per kwh, if the meter is normally used with a potential transformer of ratio 33000/110V and the current transformer of ratio 1000/10A, find the error expressed as a percentage of the correct reading from the following test figures for the instrument only:

Line voltage = 100V; Current 6.25A ; Power factor = 0.9 Time to complete 40 revolutions = 66s

- (a) -59.27% low (b) -29.28% low
(c) -63.32% low (d) -73.69% low

Ans: (a)

Sol: Actual energy consumed during the test period

$$= \sqrt{3} \times \text{ratio of P.T} \times \text{ratio of C.T} \times V_s I_s \cos \phi \times t$$

$$= \sqrt{3} \times \frac{33000}{110} \times \frac{1000}{10} \times 110 \times 6.25 \times \cos 0.9 \times \frac{66}{3600} \times 10^{-3}$$

$$= 654.85 \text{ kWh}$$

Energy recorded by meter during the test period

$$= \frac{40}{0.15} = 266.66 \text{ KWh}$$

$$\therefore \text{Percentage error} = \frac{266.66 - 654.85}{654.85} \times 100$$

$$= -59.27\% \text{ low}$$

Class Room Practice Questions

LEVEL-1 Questions

01. A transfer instrument employed in the standardization of a polar type a.c. potentiometer is
(a) an electrostatic instrument
(b) a thermal instrument
(c) a dynamometer instrument
(d) a moving coil instrument
02. Consider the following statements:
D.C potentiometer is the best means available for measurement of d.c. voltages because
1. The precision in measurement is independent of the type of detector used.
2. It is based on null – balance technique.
3. It is possible to standardize before a measurement is undertaken.
4. It is possible to measure d.c. voltages ranging in value from milli volts to hundreds of volts.
Of these statements:
(a) 2 and 3 are correct
(b) 1 and 4 are correct
(c) 2 and 4 are correct
(d) 3 and 4 are correct
03. Standardization of potentiometer is done in order that, they become
(a) accurate
(b) precise
(c) accurate and direct reading
(d) accurate and precise
04. When a potentiometer is used for measurement of voltage of an unknown source, the power consumed in the circuit of the unknown source under null conditions is
(a) very high (b) high
(c) small (d) ideally zero
05. A potentiometer may be used for...
(a) measurement of resistance
(b) measurement of current
(c) calibration of ammeter
(d) All of the above
06. To measure a resistance with the help of a potentiometer it is
(a) necessary to standardize the potentiometer
(b) not necessary to standardize the potentiometer
(c) necessary to use a volt ratio box in conjunction with the potentiometer
(d) none of the above
07. Basically a potentiometer is a device for
(a) comparing two voltages
(b) measuring a current
(c) comparing two currents
(d) measuring a voltage
(e) none of the above
08. In order to achieve high accuracy, the slide wire of a potentiometer should be
(a) as long as possible
(b) as short as possible
(c) neither too small not too large
(d) very thick
09. To measure an A.C voltage by using of A.C potentiometer, it is desirable that the supply for the potentiometer is taken.
(a) from a source which is not the same as the unknown voltage
(b) from a battery
(c) from the same source as the unknown voltage
(d) any of the above
10. The standardization of a.c. potentiometer is done by



- (a) directly using a.c. standard voltage Sources
 - (b) using d.c. standard sources and transfer instruments
 - (c) using d.c. standard sources and d'Arsonval galvanometer
 - (d) using a.c. standard sources and transfer instruments
11. Precautions are essential for ensuring that the secondary of a CT is not open circuited when the primary circuit carries a current because
- (a) dangerously high voltage might develop across the secondary
 - (b) the ferromagnetic core may develop residual magnetism
 - (c) the reflected impedance may prevent the flow of current in the primary circuit
 - (d) None
12. The primary current in a current transformer is detected dictated by
- (a) the secondary burden
 - (b) the core of the transformer
 - (c) the load current
 - (d) none of these
13. Performance of current transformer is primarily dependent on the core material and the way the core is made up. Ratio and phase angle errors will be less, if the core is of ----- permeability and of ----- form.
- (a) high, rectangular or ring
 - (b) low, rectangular or ring
 - (c) high, triangular or ring
 - (d) none
14. The ratio and phase angle errors in a well designed current transformer (CT) are kept within the specified limits by using
- (a) ferrite core
 - (b) strip wound core
 - (c) some fractional turns

(d) None

15. Usually a C.T has
- (a) poorer overload capacity than a P.T.
 - (b) the same overload capacity as a P.T
 - (c) a higher overload capacity than a P.T
 - (d) no overload capacity
16. It can be stated that
- (a) C.T operates at almost the same flux density as a P.T
 - (b) C. T. operates with a higher flux density than a P.T
 - (c) C. T operates with considerably lower flux density than a P.T
 - (d) No generalization can be made with regard to the flux densities in C. T and P.T
17. The errors in C.T are mainly due to
- (a) leakage flux
 - (b) excitation e.m.f required
 - (c) core loss
 - (d) secondary load

Solutions

LEVEL-1

- | | |
|--------------|--------------|
| 01. Ans: (b) | 02. Ans: (d) |
| 03. Ans: (c) | 04. Ans: (d) |
| 05. Ans: (d) | 06. Ans: (b) |
| 07. Ans: (a) | 08. Ans: (a) |
| 09. Ans: (c) | 10. Ans: (b) |
| 11. Ans: (a) | 12. Ans: (c) |
| 13. Ans: (a) | 14. Ans: (b) |
| 15. Ans: (c) | 16. Ans: (c) |
| 17. Ans: (c) | |

Electronic Measurements

Contents

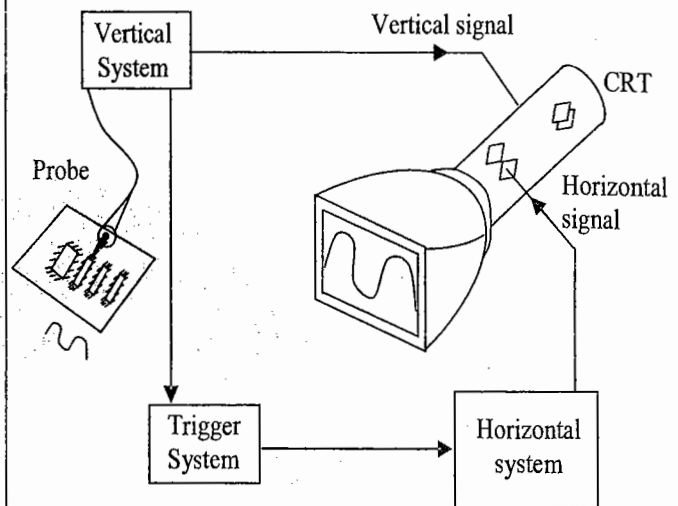
<i>S. No</i>	<i>Chapter Name</i>	<i>Page No.</i>
01	Cathode Ray Oscilloscope	89 - 98
02	Digital Voltmeters	99 - 103
03	Q - Meter	104 - 108

1

Cathode Ray Oscilloscope

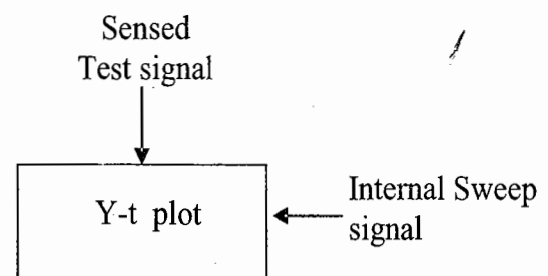
- The cathode ray oscilloscope is an electronic instrument that presents a high-fidelity graphical display of the rapidly changing voltage at its input terminals. The cathode ray oscilloscope is probably the most versatile and useful instrument available for signal measurement.
- Unlike meters, which only allow the user to measure amplitude information, the oscilloscope allows the user to view the instantaneous voltage versus time such displayed plot of the signal can be used for various measurements like peak voltage, frequency, phase, time period, rise time etc. It can also indicate the nature and magnitude of noise that may be corrupting the measurement signal. The more expensive models can measure signals at frequencies up to 500MHz and even the cheapest models can measure signals up to 20 MHz. One particularly strong merit of the oscilloscopes is its high input impedance, typically $1M\Omega$, which means that the instrument has a negligible loading effect in most measurement situations. As a test instrument, it is often required to measure voltages whose frequency and magnitude are totally unknown.
- However, it is not a particularly accurate instrument and is best used where only an approximate measurement is required. Further disadvantages of oscilloscopes include their fragility (being built around a cathode ray tube) and their moderately high cost.
- There are two main classes of oscilloscopes: analog oscilloscopes and digital oscilloscopes.

- A simplified block diagram of an analog oscilloscope is shown in below figure



- The display section of the oscilloscope (i.e., cathode ray tube) has two inputs to it namely vertical(Y) input and Horizontal(X) input. The signals applied to these inputs are driven to corresponding deflection plates and control the position of the electron beam that plots the waveform on the screen. There are two types of plots that can be displayed based on mode of operation of oscilloscope.

1) Sweep mode of operation:

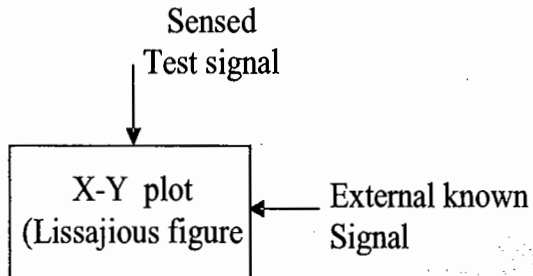


Using
Y-t plot



Various measurements regarding the test signal can be carried out like peak voltage, frequency, phase, time period, resistive etc.

2) X-Y Display mode of operation:



Using Lissajous figure, two measurements can be carried out namely frequency measurement and phase measurement.

• Other important measurements that can be made by oscilloscope are:

- i) Measurement of impedance
- ii) Measurement of power and power factor
- iii) Measurement of degree of modulation
- iv) Measurement of checking of switching times
- v) Tracing of hysteresis loop for magnetic material
- vi) Study of transistors phenomenon
- vii) Fault testing of windings of electrical machines
- viii) Determination of characteristics of thermionic values

Oscilloscope Specifications:

- 1) **Sensitivity:** a series of attenuators and pre-amplifiers exist at the input to the oscilloscope. These condition the measured signal to the optimum magnitude for input to the main amplifier and vertical deflection plates, thus enabling the instrument to measure a very wide range of different signal magnitudes. Selection of the appropriate input amplifier/attenuator is made by setting a

VOLT/DIV control associated with each oscilloscope channel. This defines the magnitude of the input signal that will cause a deflection of one division on the screen.

- 2) **Bandwidth:** One important oscilloscope specification is related to the speed of the waveforms that can be measured. This is determined by the bandwidth of the oscilloscope and it is found that the capability of the oscilloscope to accurately display the waveform falls off with increasing frequency. The Bandwidth is defined as the range of frequencies over which the oscilloscope amplifier gain is within 3dB of its peak value. The oscilloscope specification for bandwidth will typically be quoted in the format: Bandwidth = -3dB at 1500 MHz. The -3 dB specification means that an oscilloscope with a specified inaccuracy of $\pm 2\%$ and bandwidth of 100MHz will have an inaccuracy of $\pm 5\%$ when measuring 30MHz signals, and this inaccuracy will increase still further at higher frequencies. Thus, when applied to signal-amplitude measurement, the oscilloscope is only usable at frequencies up to about 0.3 times its specified bandwidth. In most oscilloscopes, the amplifier is direct coupled, which means that it amplifies d.c. voltages by the same factor as low-frequency a.c. ones. For such instruments, the minimum frequency measurable is zero and the bandwidth can be interpreted as the maximum frequency where the sensitivity (deflection/volt) is within 3 dB of the peak value. In all measurement situations, the oscilloscope chosen for use must be such that the maximum frequency to be measured is well within the bandwidth.

- 3) **Risetime:** The rise time is the transit time between the 10% and 90% levels of the response when a step input is applied to the oscilloscope. The oscilloscope must have a



sufficiently fast rise time to capture the rapid transitions accurately, otherwise important information may not be displayed and the results could be misleading. Oscilloscopes are normally designed such that: $\text{Bandwidth} \times \text{Rise time} = 0.35$

Thus, for a bandwidth of 100 MHz, rise time $= 0.35/100\,000\,000 \Rightarrow 3.5\text{ns}$.

- Types of Oscilloscopes:

- 1) **Analog Oscilloscope:** The analogue oscilloscope is the traditional form of oscilloscope that has been used in laboratories for many years. It relies on analogue techniques and takes in the vertical and sometimes horizontal signals, amplifying them in an analogue format and displaying them on a cathode ray tube.
- 2) **Digital Storage Oscilloscope (DSO):** The digital storage oscilloscope (DSO) is the conventional form of digital oscilloscope. It uses a raster type screen like that used on a computer monitor or television and in this way displays an image that fills the screen and may include other elements in addition to the waveform. These additional items may include text on the screen and the like.
- 3) **Digital Phosphor Oscilloscope (DPO):** The digital phosphor oscilloscope (DPO) is a highly versatile form of oscilloscope that uses

a parallel processing architecture to enable it to capture and display signals under circumstances that may not be possible using a standard DSO. The key element of a DPO is that it uses a dedicated processor to acquire waveform images. In this way it is possible to capture transient events that occur in digital systems more easily. These may include spurious pulses, glitches and transition errors. It also emulates the display attributes of an analogue oscilloscope, displaying the signal in three dimensions: time, amplitude and the distribution of amplitude over time, all in real time.

- 4) **Sampling Oscilloscope:** These oscilloscopes are used for analyzing very high frequency signals. They are used for looking at repetitive signals which are higher than the sample rate of the scope. They collect the samples by assembling samples from several successive waveforms, and by assembling them during the processing, they are able to build up a picture of the waveform. The oscilloscope specifications for these items may detail a frequency capability or bandwidth sometimes as high as 50 GHz. However these scopes are very expensive.

Class Room Practice Questions

LEVEL-1 Questions

01. One cycle of a square wave signal observed on an oscilloscope is found to occupy 6 cm at a scale setting of 30 $\mu\text{s}/\text{cm}$. What is the signal frequency?
(a) 1.8 kHz (b) 5.55 kHz
(c) 18 kHz (d) 55.5 kHz
02. When the horizontal deflection plates of a CRO are kept at the ground potential and a 30 volt dc is applied to the vertical deflecting plates, the bright spot moves 1 cm away from the centre. If with the same setting, a 30 volt ac is applied to the vertical deflecting plates, then the picture observed on the screen would be
(a) a spot approximately 3 cm away from the centre
(b) a vertical line 2 cm long
(c) a vertical line approximately 3 cm long
(d) two spots 2 cm vertically above each other
03. When a sinusoidal signal of 220 V, 50 Hz produces on CRO a vertical deflection of 2 cm at a particular setting of the vertical gain control, what would be the value of the voltage to be applied to produce a deflection of 3 cm for the same vertical gain?
(a) 330 V (b) 110 V
(c) 220 V (d) 55 V
04. The bandwidth of a CRO is from 0 to 20 MHz. The fastest rise time which a square wave can have, in order that it is accurately reproduced by the CRO is
(a) 0.175 μs (b) 17.5 ns
(c) 35 ns (d) 52.5 ns
05. The input impedance of a CRO is nearly
(a) zero
(b) around 10 ohms
(c) around 100 ohms
(d) around one mega ohm.
06. The oscilloscope has an input capacitance of 50 pF and a resistance of 2 $\text{M}\Omega$ and the voltage divider ratio (k) of 10. What are the parameters of a high-impedance probe?
(a) $C_1 = 5.55 \text{ pF}$ and $R_1 = 9 \text{ M}\Omega$
(b) $C_1 = 5.55 \text{ pF}$ and $R_1 = 18 \text{ M}\Omega$
(c) $C_1 = 3.33 \text{ pF}$ and $R_1 = 9 \text{ M}\Omega$
(d) $C_1 = 1.11 \text{ pF}$ and $R_1 = 18 \text{ M}\Omega$
07. The input impedance of a CRO is 1 M ohm in parallel with 10 pF. If the CRO is required to display pulse using a 10:1 attenuator, the attenuator will have to use a
(a) 9 M ohm resistor
(b) 1.11 pF capacitor
(c) Parallel combination of 9 M ohm resistor and 1.11 pF capacitor
(d) Series combination of 9 M ohm resistor and 1.11 pF capacitor.
08. An oscilloscope has an input impedance consisting of 1 $\text{M}\Omega$ and 20 pF in parallel. A high impedance probe connected to the input of this oscilloscope has a 10 $\text{M}\Omega$ series resistance. This 10 $\text{M}\Omega$ resistance
(a) need not be shunted
(b) should be shunted by a 2 pF capacitor
(c) should be shunted by a 20 pF capacitor
(d) should be shunted by a 200 pF capacitor
09. Which one of the following is the correct statement?
Active probe used in a CRO
(a) Is bulkier than passive ones
(b) Cannot measure small signals
(c) Cannot couple high frequency signals
(d) Can attenuate more

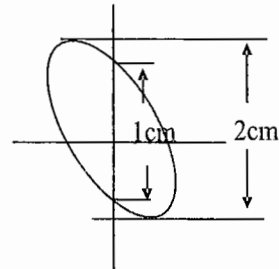


10. Which one of the following measuring devices has minimum loading effect on the quantity under measurement?
 - (a) PMMC
 - (b) CRO
 - (c) Hot wire
 - (d) Electrodynamometer
11. A voltage signal $V(t) = A + B \sin \omega t$ is fed to an Oscilloscope with the coupling mode set to DC, other settings being appropriate. The displayed wave can be expressed as
 - (a) $A + B \sin \omega t$
 - (b) $B \sin \omega t$
 - (c) A
 - (d) $|A + B \sin \omega t|$
12. The sine wave output of a function generator is fed to both the horizontal (X) and vertical (Y) inputs of a CRO. What will be the pattern on the cathode ray screen?
 - (a) A circle
 - (b) An ellipse
 - (c) A straight line with 45° slope
 - (d) Sinusoidal
13. The Lissajous pattern observed on the screen of a CRO is a straight line inclined at an angle of 45° to the x-axis. If the voltage applied to the x-plate is $2 \sin \omega t$, then the voltage applied to the Y-plates will be
 - (a) $2 \sin \omega t$
 - (b) $2 \sin (\omega t + 45^\circ)$
 - (c) $2 \sin (\omega t - 45^\circ)$
 - (d) $2\sqrt{2} \sin (\omega t + 45^\circ)$
14. Two sinusoidal signals having the same amplitude and frequency are applied to the X and Y inputs of a CRO. The observed Lissajous figure is a straight line. The phase shift between the two signals would be.
 - (a) Zero
 - (b) 90 degrees
 - (c) Either zero or 180 degrees
 - (d) Either 90 degrees or 270 degrees

15. An Oscilloscope screen displays a line inclined at 45° . Its Y input is a sine wave of frequency f . Then the X-input should be
 - (a) Sine wave of frequency f and 0° phase shift with the Y- input
 - (b) Sine wave of frequency f and 45° phase shift with the Y- input
 - (c) Sine wave of frequency f and 90° phase shift with the Y- input
 - (d) Sawtooth wave of frequency f

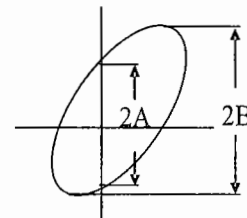
16. An oscilloscope is operated in X-Y mode. The below given ellipse is observed on the screen.

The phase difference between both input signal is



- (a) 330°
- (b) 150°
- (c) 30°
- (d) 210°

17. The elliptical lissajious pattern shown in figure was observed from two sinusoidal signal sources of same frequency on a CRO. The phase angle between the signals is

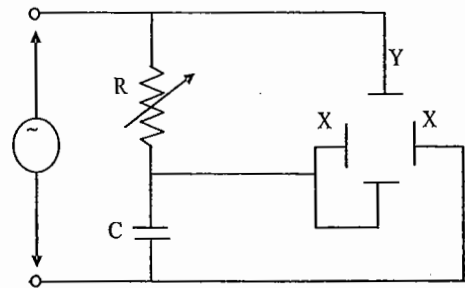


- (a) $\cos^{-1}(A/B)$
- (b) $\tan^{-1}(A/B)$
- (c) $\sin^{-1}(A/B)$
- (d) (A/B)

18. The X – input and Y – input to a CRO are, $5 \cos (\omega t + \phi)$ and $5 \sin (\omega t + \phi)$, respectively. What will be the resulting Lissajous pattern?



- (a) A straight line inclined to an angle ϕ
 (b) An ellipse with an inclined major axis
 (c) A circle
 (d) A horizontal line
19. While measuring the phase difference between the signals $v_1(t) = 10 \sin \omega t$ and $v_2(t) = 10 \sin (\omega t + \phi)$, the Lissajous pattern observed on CRO was a circle. The value of ϕ is
 (a) Zero (b) $\pi/4$
 (c) $\pi/2$ (d) π
20. Voltages $V_y = 100 \sin 1000t$ and $V_x = 50 \sin 1000 t$ are connected to Y and X terminals of a CRO, respectively. What is the shape of figure seen on the CRO?
 (a) A circle (b) A straight line
 (c) An ellipse (d) A parabola
21. A circle is found on the screen of a CRO when 2 time varying signals of same frequency and same magnitude are applied to X and Y plates of the CRO. What is the relative phase difference?
 (a) 0° (b) 90°
 (c) 180° (d) 45°
22. Two equal voltages of same frequency applied to the X and Y plates of a CRO, produce a circle on the screen. The phase difference between the two voltages is
 (a) 30° (b) 60°
 (c) 90° (d) 150°
23. In the CRO plate connections shown in the given figure, the supply frequency is 500 Hz and the capacitance 'C' is $\frac{0.2}{\pi} \mu\text{F}$. The value of resistance 'R' required to obtain a circle on the CRO screen (X and Y plates have equal sensitivities) is



- (a) $2 \text{ k}\Omega$ (b) $5 \text{ k}\Omega$
 (c) $7 \text{ k}\Omega$ (d) $10 \text{ k}\Omega$

24. In an oscilloscope, two Lissajous figure (X) and (Y) are observed. This indicates that ratio of vertical input signal frequency to that horizontal input frequency are

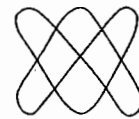


Fig. x



Fig. y

- (a) $5/3$ for X and $3/2$ for Y
 (b) $3/2$ for X and $5/3$ for Y
 (c) $5/3$ for X and $5/3$ for Y
 (d) $3/2$ for X and $3/2$ for Y
25. A signal of 10mV at 75 MHz is to be measured. Which of the following instrument can be used?
 (a) VTVM
 (b) Cathode ray oscilloscope
 (c) moving iron voltmeter
 (d) digital multimeter

LEVEL-2 Questions

01. The screen of a C.R.O has 10 divisions on the horizontal scale. If a voltage signal $5 \sin(314t + 45^\circ) \text{V}$ is examined with a time



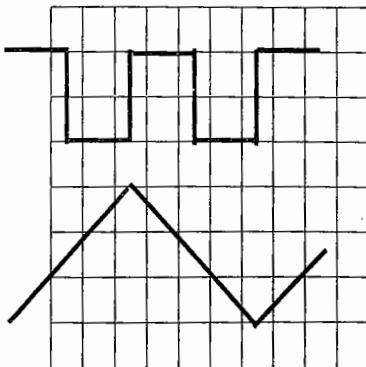
base setting of 5msec per division then the number of cycles of signal displayed on the screen will be

- (a) 4.5 (b) 2
(c) 4 (d) 2.5

02. An oscilloscope is operated with line setting of 0.5ms/div and vertical setting of 100mv/div. Two cycles of sine wave is observed on the screen which occupies 8.8 horizontal divisions and 4.6 vertical divisions. Then, the rms voltage and frequency of sine wave respectively___

- (a) 162.6mV, 227 Hz
(b) 162.6mV, 455 Hz
(c) 325.3v, 227 Hz
(d) 325.3V,455 Hz

03. The time/div and voltage/div axes of an oscilloscope have been erased. A student connects a 1KHz, 5V p-p square wave calibration pulse to channel-1of the scope and observes the screen to be as shown in the upper trace of the fig. An unknown signal is connected to channel-2 (lower trace) of the scope. If the time/div and $\frac{\text{volt}}{\text{div}}$ on both channels are the same, the amplitude (p-p) and period of the unknown signal are respectively **(GATE-EE-06)**



- (a) 5V, 1ms (b) 5V, 2ms
(c) 7.5V, 2ms (d) 10V, 1ms

04. If the bandwidth of an oscilloscope is given as direct current to 10MHz, what is the fastest rise time a sine wave can have to be produced accurately by the oscilloscope?

- (a) 35 n sec (b) 10 n sec
(c) 3.5 n sec (d) 0.035 n sec

05. A dc voltage of 1 V is applied to the X – plates to a CRO and an ac voltage $2 \sin 100 t$ is applied to the Y–plates.The resulting display on the CRO screen will be a

- (a) Vertical straight line
(b) Horizontal straight line
(c) Sine wave
(d) Slant line

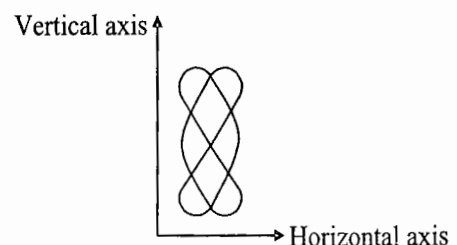
06. Which of the following measurements can be made using Lissajous figures?

1. Frequency
2. phase difference
3. Time interval between pulses
4. pulse width
5. Fundamental and higher harmonic components.

Select the correct answer using the code given below

- (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 4 and 5

07. A screen pattern oscillogram, shown in the given figure is obtained when a sine-wave signal of unknown frequency is connected to the vertical input terminals, and at the same time, a 600 Hz sine-wave voltage is connected to the horizontal input terminals of an oscilloscope. What is the value of unknown frequency?





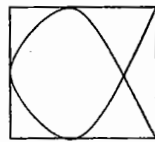
- (a) 300 Hz (b) 400 Hz
(c) 600 Hz (d) 900Hz

08. A Lissajous pattern on an oscilloscope has 5 horizontal tangencies and 2 vertical tangencies. The frequency of the horizontal input is 1000 Hz. What is the frequency of the vertical input?

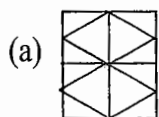
- (a) 400 Hz (b) 2500 Hz
(c) 4000Hz (d) 5000 Hz

09. A lissajous pattern, as shown in figure below, is observed on the screen of a CRO when voltages of frequencies f_x and f_y are applied to the x and y plates respectively. $f_x : f_y$ is then equal to

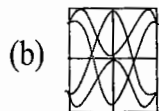
- (a) 3 : 2
(b) 1 : 2
(c) 2 : 3
(d) 2 : 1



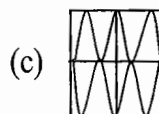
10. In an oscilloscope the input to the horizontal plates is a 100Hz voltage signal. The lissajous patterns (A), (B) and (C) will be generated when different frequency voltage signals are applied to vertical plates. Match each lissajous pattern to the corresponding frequency by



P. $f_y = 50$



Q. $f_y = 66.66$



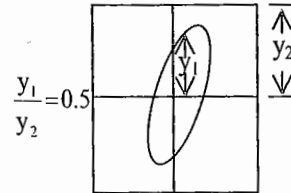
R. $f_y = 125$

(S) $f_y = 150$

(T) $f_y = 200$

(U) $f_y = 300$

11. Two sine waves of the same frequency are impressed on the X and Y plates of a CRO and the Lissajous figure seen is shown in the diagram. The phase difference between the signals is



- (a) 30° or 330° or 150° or 210°
(b) 30° or 330° or 150°
(c) 30° or 330°
(d) 30°

12. Lissajous pattern shown in a double-beam cathode-ray oscilloscope screen for two sinusoidal voltages of equal magnitude and of the same frequency but of phase shift of 30° is

- (a) a circle
(b) a straight line at 45° in the first and third quadrant
(c) an ellipse in the first and third quadrant
(d) an ellipse in the second and fourth quadrant

13. A certain oscilloscope with 4 c.m by 4 c.m screen has its own sweep out put fed to its input. If the x and y sensitivities are same, the oscilloscope will display a.

- (a) triangular wave
(b) diagonal line
(c) sine wave
(d) circle

14. A 1000 Hz sinusoidal voltage is connected to both X and Y inputs of a CRO. Which of the following waveforms is seen on CRO ?

- (a) Sine wave (b) Circle
(c) Ellipse (d) Straight line



15. The X-and Y-inputs of a CRO are respectively $V\sin\omega t$ and $-V\sin\omega t$. The resulting Lissajous pattern will be
 (a) a straight line
 (b) a circle
 (c) an ellipse
 (d) a figure of eight
16. The X and Y inputs to a CRO are respectively $10 \cos(100t + \phi)$ and $10 \sin(100t + \phi)$. The resulting Lissajous pattern is.
 (a) A straight line inclined at an angle ϕ
 (b) A horizontal line
 (c) An ellipse with axis making an angle ϕ
 (d) A circle

Solutions

LEVEL-1

01. Ans: (b)

Sol: $T_{\text{signal}} = N_H \times \frac{\text{time}}{\text{divison}}$
 $= 6 \text{ cm} \times 30 \frac{\mu\text{Sec}}{\text{cm}} = 180 \mu\text{s}$
 $\therefore f_{\text{signal}} = \frac{1}{180 \mu\text{s}} = 5.55 \text{ kHz}$

02. Ans: (b)

Sol: During +ve half cycle, spot moves in up direction by 1c.m.
 During -ve half cycle, spot moves in down direction by 1c.m.
 Therefore a vertical line 2cm long.

03. Ans: (a)

Sol: Voltage gain = $\frac{\text{voltage output}}{\text{deflection}} = \frac{220}{2} = 110 \text{ V/cm}$
 Applied voltage for 3 cm deflection

= voltage gain $\times$ deflection
 $= 110 \times 3$
 $= 330 \text{ V}$

04. Ans: (b)

Sol: We know Rise time = $\frac{0.35}{\text{Band width}}$

$\therefore \text{Fastest Rise time} = \frac{0.35}{20 \times 10^6} = 17.5 \text{ ns}$

05. Ans: (d)

Sol: CRO has very high input impedance in the order of mega ohms.

06. Ans: (b)

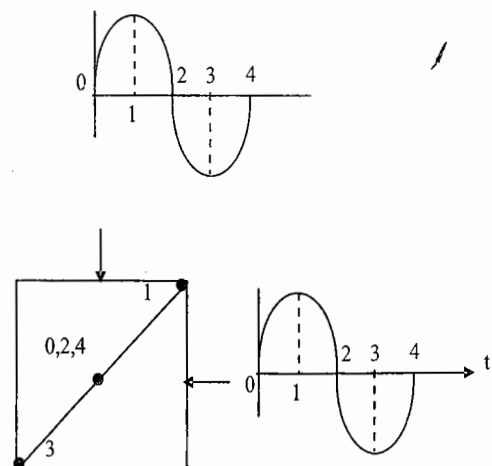
Sol: $R_1 = (10 - 1)R_i$
 $= 9 \times 2 \text{ M}\Omega$
 $= 18 \text{ M}\Omega$
 $C_1 = \frac{C_i}{(10 - 1)} = \frac{50 \text{ Pf}}{9} = 5.55 \text{ Pf.}$

10. Ans: (b)

Sol: Device having highest input impedance will have least loading error.

12. Ans: (c)

Sol: When two identical sinusoidal waves are applied to both the inputs of CRO, pattern obtained on screen is as shown below.





13. Ans: (a)

Sol: When both the inputs to CRO are identical the pattern of Lissajous figure will be straight line making 45° to X-axis

16. Ans: (b)

Sol: For a given ellipse $\Rightarrow$

$$\begin{aligned}\text{Phase difference} &= 180 - \sin^{-1} \left(\frac{Y_{\text{intercept}}}{Y_{\text{maximum}}} \right) \\ &= 180 - \sin^{-1} \left(\frac{1}{2} \right) \\ &= 150^\circ\end{aligned}$$

17. Ans: (c)

Sol: phase angle $= \sin^{-1} \left(\frac{Y_{\text{intercept}}}{Y_{\text{maximum}}} \right)$

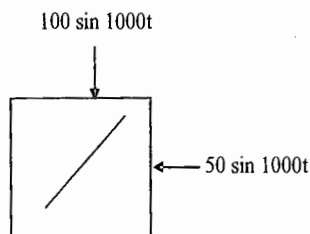
$$= \sin^{-1} \left(\frac{2A}{2B} \right) = \sin^{-1} \left(\frac{A}{B} \right)$$

18. Ans: (c)

Sol: When inputs of CRO have phase difference of 90° , the pattern obtained on screen will be a circle.

20. Ans: (b)

Sol: The shape of figure is as shown below.



23. Ans: (b)

Sol: Given: displayed L.F is circle,

$$C = \frac{0.2}{\pi} \mu\text{F}, f = 500 \text{ Hz}, R = ?$$

ϕ is 90° as circle is displayed on the screen

$$\text{i.e. } V_R = V_C$$

$$R = X_C$$

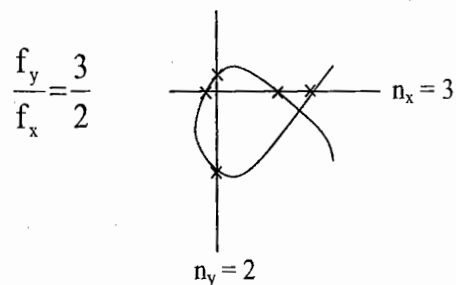
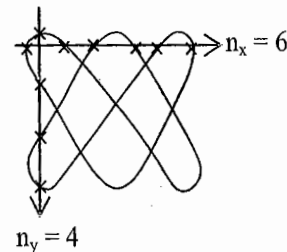
$$R = \frac{1}{2\pi fC}$$

$$\begin{aligned}\Rightarrow R &= \frac{1}{2\pi \times 500 \text{ Hz} \times \frac{0.2}{\pi} \mu\text{F}} \\ &= \frac{10^6}{200} = 5 \text{ k}\Omega\end{aligned}$$

24. Ans: (d)

Sol:

$$\frac{f_y}{f_x} = \frac{6}{4} \Rightarrow \frac{3}{2}$$

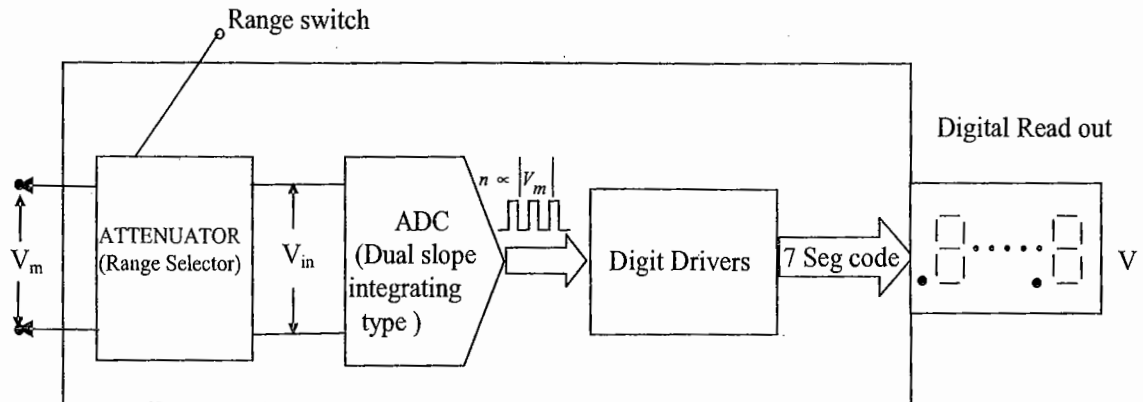


Level-2 Key

01. (c)	02. (d)	03. (c)	04. (a)	05. (a)
06. (a)	07. (b)	08. (b)	09. (c)	10. ()
11. (c)	12. (c)	13. (b)	14. (d)	15. (a)
16. (d)				

Digital Voltmeters

- A Digital Voltmeter is a Voltage measuring unit whose block diagram is shown below:

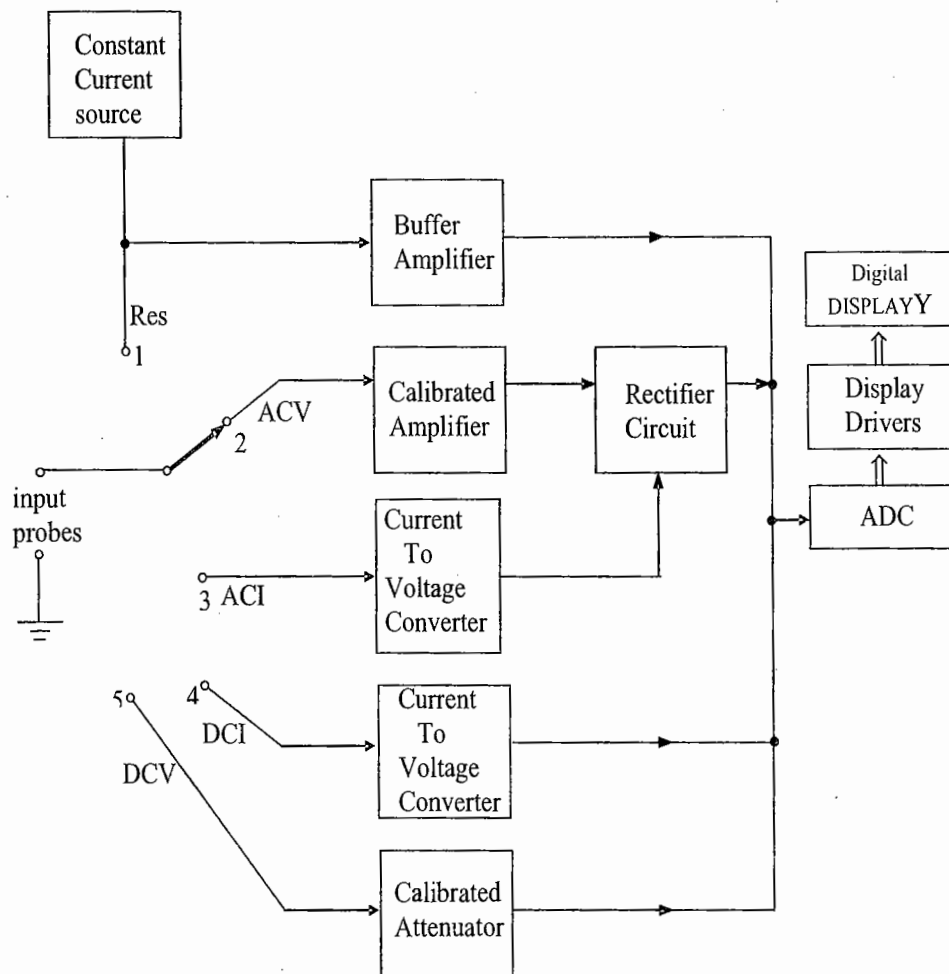


Analog DC voltage to be measured is first converted into pulses by Analog-to-Digital converter. Such Digital output is then driven to a digital display via Digit Drivers.

- The most preferred ADC in the design of Digital meters is Dual Slope Integrating type ADC since it offers High Accuracy, High Noise rejection, High Stability and Less Hardware complexity than other types of ADCs (Non Integrating types)
- Advantages of Digital meters over Analog meters:**
 - High input Impedance
 - Less Loading Error
 - No Observational Errors, No frictional Errors
 - Better Accuracy
 - Superior resolution
 - Fast operating speed
 - Easy readability
 - Programmability
 - Computerized control
 - Auto Ranging and Over Ranging facilities



Block diagram of digital multi meter is shown below:



Class Room Practice Questions

LEVEL-1 Questions

01. A $3\frac{1}{2}$ digit, 2V full scale dual slope DVM is used to measure a time varying voltage $V(t) = (1 + \sin 100\pi t)V$ then, the DVM indicates
(a) 1.999 V (b) 1.414 V
(c) 2.000 V (d) 1.000 V
02. A $3\frac{1}{2}$ digit multi meter has an accuracy specification of $(\pm 0.5\% \text{ of reading} \pm 5 \text{ counts})$. If the meter reads 2.00 mA on a full scale of 20mA, the worst-case error in the reading is
(a) 0.5% (b) 2.5%
(c) 3.0% (d) 5.5%
03. A $3\frac{1}{2}$ digit, 200mV full scale DVM has an accuracy specification of $\pm 0.5\%$ of reading plus 5 counts. When the meter reads 100mV, the voltage being measured is:
(a) Any value between 99.5mV and 100.5mV
(b) Any value between 99.0mV and 101.0mV
(c) Exactly 99.5mV
(d) Exactly 100mV
04. A $3\frac{1}{2}$ digit DVM has a lowest measuring range of 200 mV. Hence it can be concluded that
(a) Its best resolution is 0.1mV
(b) It's poorest resolution is 0.2 mV
(c) Its accuracy is at least 0.05%
(d) The maximum voltage that can be measured in this range is 199.9mV

LEVEL-2 Questions

01. The type of A/D converter normally used in a $3\frac{1}{2}$ digit multi meter is
(a) Dual – Slope integrating type
(b) Voltage to frequency converter type
(c) Flash (or parallel) type
(d) Successive approximation type
02. A $4\frac{1}{2}$ digit DMM has the error specification as 0.2% of reading +10 counts. If a dc voltage of 100 V is read on its 200V full scale. The maximum error that can be expressed in the reading is
(a) $\pm 0.1\%$ (b) $\pm 0.2\%$
(c) $\pm 0.3\%$ (d) $\pm 0.4\%$
03. What is the range for a $3\frac{1}{2}$ digital meter?
(a) 0 to 1999 (b) 0 to 1500
(c) 0 to 999 (d) 0 to 19999
04. A digital voltmeter has $4\frac{1}{2}$ digit display. The 1V range can read up to
(a) 1.0000V (b) 1.1111V
(c) 0.9999V (d) 1.9999V
05. A $3\frac{1}{2}$ digit digital multimeter can display
(a) 999.5 (b) 5.999
(c) 555.5 (d) 1.999
06. A digital voltmeter has a read- out range from 0 to 9999 counts. When full scale reading is 9.999 V, the resolution of the full scale reading is



- (a) 0.001 (b) 1000
(c) 3 digit (d) 1 mv
07. A $3\frac{1}{2}$ digit voltmeter having a resolution of 100 mV can be used to measure maximum voltage of
(a) 100V (b) 200V
(c) 1000 V (d) 5000 V
08. The selected voltage range of $4\frac{1}{2}$ digit digital voltmeter having a resolution of 10mV is
(a) 200V (b) 20V
(c) 40V (d) 400V
09. A 4-digit DVM (digital voltmeter) with a 100mV lowest full-scale range would have a sensitivity of how much value while resolution of this DVM is 0.0001?
(a) 0.1mV (b) 0.01mV
(c) 1.0mV (d) 10mV
10. What is the reading of 0.5245V on 1V range in four and a half digit voltmeter displayed as ?
(a) 0.5245 (b) 00.524
(c) 000.52 (d) 0000.5
11. A $3\frac{1}{2}$ digit digital voltmeter has an internal reference of 200mV. It's full scale count and resolution are
(a) 2000 and 1mV
(b) 4000 and 0.1mV
(c) 2000 and 0.1mV
(d) 4000 and 1mV
12. The resolution of $4\frac{1}{2}$ digit DVM is
(a) 0.001 (b) 0.0001
(c) 0.1 (d) 0.00001
13. The resolution of $4\frac{1}{2}$ digit DVM in 10V range is
(a) 1mV (b) 10mV
(c) 4mV (d) 0.001mV
14. The resolution of $4\frac{3}{4}$ digit DVM in 10V range is
(a) 1mV (b) 10 mV
(c) 4 mV (d) 0.001 mV
15. **Assertion(A):** The resolution of $3\frac{1}{2}$ digit voltmeter is 0.001.
Reason(R): Addition of $\frac{1}{2}$ digit to a digital voltmeter increases the range of the meter
16. In a digital voltmeter, 'over-ranging' implies that
(a) the next four digits are switched on
(b) 1/2 digit is switched off
(c) 1/2 digit is switched on
(d) an over- range indicator starts glowing
17. Which one of the following is not true of digital instruments?
(a) Loading of the circuit under measurement is less
(b) Accuracy is better
(c) Free from observational errors
(d) Can present the reading in overall context of range
18. **Assertion (A):** Digital multimeters are more accurate than analog multimeters.
Reason (R): Digital multimeters are convenient in reading.
19. An average-reading digital multimeter reads 10V when fed with a triangular wave, symmetric about the time axis. For the same input an rms reading meter will read.



- (a) $20/\sqrt{3}$ V (b) $10/\sqrt{3}$ V
(c) $20\sqrt{3}$ V (d) $10\sqrt{3}$ V

Solutions

LEVEL-1

01. Ans: (d)

Sol: DVM measures average value.

The average value of the above input signal is 1.

So due to the $3\frac{1}{2}$ digit DVM it indicates as 1.000 V

02. Ans: (c)

Sol: The DMM has $3\frac{1}{2}$ digit display

It's accuracy specification is $\pm$ (0.5% of reading + 5 counts)

→ 2mA reading on the 20mA full scale.

0 2 . 0 0 mA

→ 1 count on this 20mA full scale is 00.01mA

→ Therefore, error can be calculated as:

Error = $\pm$ (0.5% of reading + 5 counts)

$$= \pm \left(\frac{0.5}{100} \times 02.00\text{mA} + 5 \times 00.01\text{mA} \right)$$

$$= \pm \left(\frac{0.5}{100} \times 2\text{mA} + 5 \times 0.01\text{mA} \right)$$

$$= \pm (0.01\text{mA} + 0.05\text{mA})$$

$$= \pm (0.06\text{mA})$$

→ % error in reading of 2mA is

$$\% \text{error} = \pm \frac{0.06\text{mA}}{2\text{mA}} \times 100 = \pm 3\%$$

03. Ans: (b)

Sol:

→ DVM has $3\frac{1}{2}$ digit display

→ It's Accuracy specification is $\pm$ (0.5% of reading plus 5 counts)

→ 100mV reading on the 200mV full scale
1 0 0 . 0 mV

→ 1 count on this 200mV full scale is 000.1mV

→ Therefore, error can be calculated as Error = $\pm$ (0.5% of reading + 5 counts)

$$= \pm \left[\frac{0.5}{100} \times 100.0\text{mV} + 5 \times 000.1\text{mV} \right]$$

$$= \pm \left[\frac{0.5}{100} \times 100\text{mV} + 5 \times 0.1\text{mV} \right]$$

$$= \pm [0.5\text{mV} + 0.5\text{mV}] = \pm 1\text{mV}$$

The voltage being measured is 100mV $\pm$ 1mV i.e., Any value between 99.0mV and 101.0mV

04. Ans: (d)

Sol:

→ The DVM has $3\frac{1}{2}$ digit display

Therefore, its scale resolution is 0.001

→ Its resolution in 200mV range is 100mV

→ The maximum voltage that can be measured in this 200mV lowest Range: 199.9mV

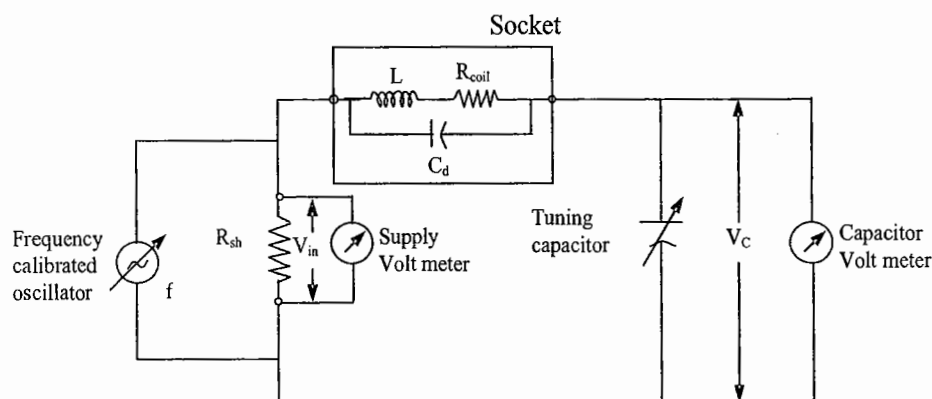
Level-2 Key

01. (a)	02. (d)	03. (a)	04. (c)	05. (d)
06. (d)	07. (b)	08. (a)	09. (b)	10. (a)
11. (a)	12. (b)	13. (a)	14. (a)	15. (b)
16. (c)	17. (d)	18. (b)	19. (d)	

Q - Meter

- The Q-meter is an instrument designed for the measurement of Q-factor of the coil as well as for the measurement of electrical properties of coils like Self inductance, Self capacitance, Resistance. This instrument operates on the principle of series resonance i.e. at resonate condition of an ac series circuit voltage across the capacitor is equal to Q-times of the applied voltage to the circuit. If the voltage applied across the circuit is kept-constant then voltmeter connected across the capacitor can be calibrated to indicate Q directly.
- Circuit diagram of "Direct connection" of a Q-meter is shown in figure. A wide-range os-

cillator with frequency range from 50 kHz to 50 MHz is used as a power supply to the circuit. The output of the oscillator is shorted by a low-value resistance, R_{sh} usually of the order of 0.02 ohm. So it introduces almost no resistance into the oscillatory circuit and represents a voltage source with a very small or of almost negligible internal resistance. The voltage across the low-value shunt resistance R_{sh} , V is measured by a thermocouple meter and the voltage across the capacitor, V_c is measured by an electronic voltmeter.



- For carrying out the measurement, the unknown coil is connected to the test terminals of the instrument, and the circuit is tuned to resonance either by varying the frequency of the oscillator or by varying the resonating capacitor C . Readings of voltages across capacitor C and shunt resistance R_{sh} are obtained and Q-factor of the coil is determined as follows:
- By definition Q-factor of the coil,
 $Q = X_L / R$
 And when the circuit is under resonance condition: $X_L = X_C$
 i.e., $Q = X_C / R$

$$\begin{aligned} V_c &= IX_c \\ &= (V_{in}/Z) X_c \\ &= (V_{in}/R) X_c \quad (\text{because } Z=R \text{ at resonance}) \\ &= (X_c/R) V_{in} \text{ or } (X_L/R) V_{in} \\ &= Q V_{in} \end{aligned}$$

Where Q is Known as Voltage-magnification factor.

$$\begin{aligned} Q &= V_c / V_{in} \\ &= \text{Q-Capacitor Voltmeter / Supply Voltmeter reading} \end{aligned}$$

- This relation ($Q = V_c/V_{in}$) is used to calibrate the voltscale of capacitor voltmeter into Q-scale



- This Q-factor is called the circuit Q because this measurement includes distributed capacitance of coil, shunt resistor R_{sh} . So, the calculated Q-factor of the coil will be somewhat less than the Actual Q-factor.
- There are two more connections of Q meter namely series connection and shunt connection as given below.

At resonance,

$$\omega L = \frac{1}{\omega C_1} \text{ and } Q_1 = \frac{\omega L}{R} = \frac{1}{\omega C_1 R}$$

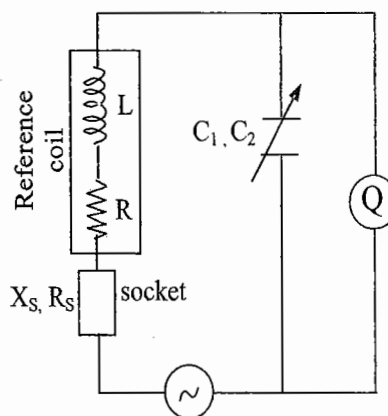
$$X_s = \frac{C_1 - C_2}{\omega C_1 C_2}$$

$$R_s = \frac{Q_1 C_1 - Q_2 C_2}{\omega C_1 C_2 Q_1 Q_2}$$

$$Q_x = \frac{X_s}{R_s} = \frac{(C_1 - C_2) Q_1 Q_2}{Q_1 C_1 - Q_2 C_2}$$

i) Series Connection:

In series connection method the unknown test impedance is connected in series with the Reference coil. This method is employed for measurement of low test impedance i.e., low value resistors, small coils and large capacitors.



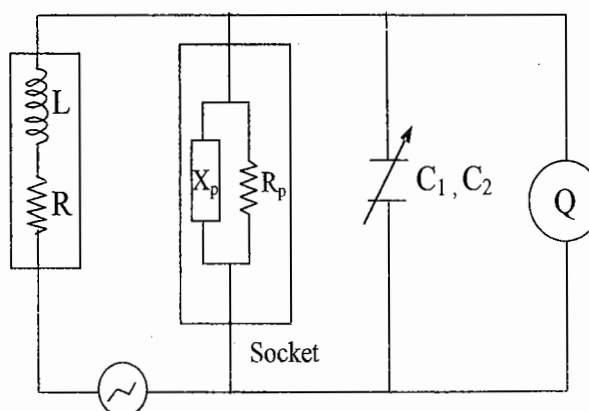
ii) Parallel Connection:

In parallel connection, the unknown test impedance is connected in parallel to the tuning capacitor. This method is employed for measurement of high impedance i.e., high value resistors, certain inductors and small capacitors.

$$\frac{1}{R_p} = \frac{\omega C_1}{Q_2} = \frac{1}{R Q_1}$$

$$X_p = \frac{1}{\omega (C_2 - C_1)}$$

$$Q_x = \frac{R_p}{X_p} = \frac{(C_1 - C_2) Q_1 Q_2}{Q_1 C_1 - Q_2 C_2}$$



The main error in the measurement of Q is due to distributed or stray capacitance of the circuitry. To check for this, the Q is measured at two frequencies f_1 and $2f_1$. It should be same if not,

$$C_d = \frac{C_1 - 4C_2}{3}$$

Class Room Practice Questions

LEVEL-1 Questions

01. In a Q-meter, a small resistance R is added to the series resonance circuit to inject the oscillatory voltage to the circuit. If R_s is the apparent series resistance of the circuit at resonance, then the value of the actual Q will be equal to

(a) observed $Q = \frac{1}{1 + \frac{R}{R_s}}$

(b) observed $Q \left(1 + \frac{R}{R_s} \right)$

(c) observed $Q = \frac{1}{1 + \frac{R_s}{R}}$

(d) Observed $Q \left(1 + \frac{R_s}{R} \right)$

02. If Q_e is the effective Q of the coil, C is the resonance capacitance and C_d is the distributed capacitance, then the true Q in a Q-meter will be

(a) $Q_e [(C + C_d)/C]$

(b) $Q_e [C/(C + C_d)]$

(c) $Q_e [C_d/(C + C_d)]$

(d) $Q_e [(C + C_d)/C_d]$

03. Consider the following statements regarding sources of error in a Q-meter:

1. If a coil with resistance R is connected in direct measurement mode and if the residual resistance of Q-meter is $0.1 R$, then the measure Q of the coil would be 1.1 times the actual Q .

2. If the inductance to be measured is less than 0.1 mH , the error due to presence of residual inductance cannot be neglected.

3. The presence of distributed capacitances in a coil modifies the effective Q of the coil.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 2 only

(c) 2 and 3 only (d) 1 and 3 only

04. The true value of Q of a coil is 245 and the measured value is 244.5. Then, the ratio of resonating capacitance in Q-meter circuit to distributed capacitance of coil is

(a) 0.002 (b) 2.045

(c) 489 (d) 4.80

05. In direct connection of Q-meter, the voltmeter connected across resonating capacitor is known as

(a) multiply Q by meter

(b) coil - Q

(c) Circuit Q meter

(d) Q-voltmeter

06. A very low - loss coil is tested with a Q-meter and the distributed (self) capacitance of the coil is found to be 820 pF . Resonance occurred at an angular frequency of 10^6 rad/sec , with a capacitance of 9.18 nF . The inductance of the coil is

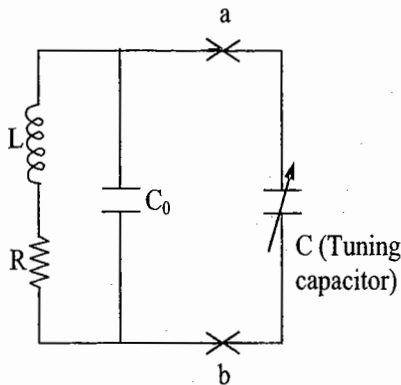
(a) 100 pH (b) $100 \text{ }\mu\text{H}$

(c) 100 nH (d) 10 mH



LEVEL-2 Questions

01. In the given circuit, C_0 is the distributed capacitance of the coil and C is the tuning capacitor. If $C = C_1$ for the fundamental frequency and $C = C_2$ for the second harmonic, then the value of C_0 can be expressed as.



- (a) $C_0 = \frac{C_1 - 2C_2}{3}$ (b) $C_0 = \frac{C_1 - 4C_2}{3}$
(c) $C_0 = \frac{C_1 - C_2}{3}$ (d) $C_0 = C_1 - 2C_2$

02. **Assertion(A):** The Q-meter measures the Q-factor of a coil when the circuit is in resonance.

Reason(R): The Q-meter of a coil depends only on its inductance and not on its resistance

03. A reading of 120 is obtained when a standard inductor was connected in the circuit of a Q-meter and the variable capacitor is adjusted to a value of 300 pF. A lossless capacitor of unknown value C_x is then connected in parallel with variable capacitor and the same reading was obtained when the variable capacitor is readjusted to a value of 200 pF. The value of C_x in pF is

- (a) 100 (b) 200
(c) 300 (d) 500

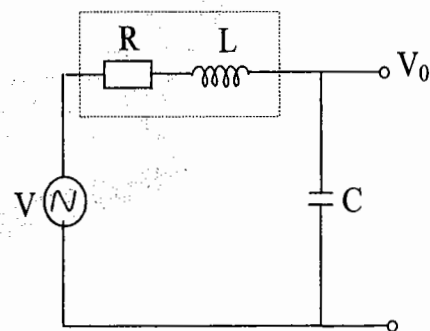
04. A coil with a resistance of 10Ω is connected in direct measurement mode of Q-meter. Resonance occurs when the oscillator frequency is 1MHz and resonating capacitance is set at 65pF. Then, the magnitude of percentage error introduced in measurement of Q by insertion resistance of 0.02Ω is

- (a) 0.5% (b) 0.2%
(c) 0.02% (d) 0.1%

05. The Q-meter works on the principle of

- (a) a mutual inductance
(b) self inductance
(c) series resonance
(d) parallel resonance

06. Figure shows a circuit which has a coil of resistance 'R' and inductance 'L'. At resonance, the Q-factor of the coil is given by



- (a) $\frac{V - V_0}{V}$ (b) $\frac{V_0}{V}$
(c) $\frac{V_0 - V}{V}$ (d) $\frac{V}{V_0}$



07. A coil is turned to resonance of 500 kHz with a resonating capacitor of 36 pF. At 250 kHz, the resonance is obtained with resonating capacitors of 160 pF. What is the self-capacitance of the coil?
 (a) 2.66 pF (b) 5.33 pF
 (c) 8 pF (d) 10.6 pF
08. A Q-meter is supplied with an oscillator having a 500 mV output voltage. While testing an unknown inductor, the voltage across the variable capacitor of the Q-meter, measured by a digital voltmeter, is obtained as 10 V. The Q-factor of the inductor is
 (a) 5 (b) 10
 (c) 20 (d) 0.05

Solutions

LEVEL-1

01. Ans: (b)

Sol: $Q_{\text{true}} = Q_{\text{meas}} \left(1 + \frac{r}{R_{\text{coil}}} \right)$

$$Q_{\text{actual}} = \text{observed-Q} \left[1 + \frac{R}{R_s} \right]$$

02. Ans: (a)

03. Ans: (c)

Sol: $Q_{\text{true}} = Q_{\text{meas}} \left(1 + \frac{0.1R}{R} \right)$

$$Q_{\text{true}} = 1.1 Q_{\text{meas}}$$

As such except 1st statement remaining are correct

04. Ans: (c)

Sol: $1 + \frac{C_d}{C} = \frac{Q_{\text{true}}}{Q_{\text{measured}}}$

$$\Rightarrow \frac{C_d}{C} = \frac{245}{244.5} - 1 = 2.044 \times 10^{-3}$$

$$\Rightarrow \frac{C}{C_d} = 489$$

06. Ans: (d)

07. Ans: (b)

Sol: Given data : $C_d = 820\text{PF}$,

$$\omega = 10^6 \text{ rad/sec} \text{ \& } C = 9.18\text{nF}$$

we know

$$L = \frac{1}{\omega^2 [C + C_d]} = \frac{1}{(10^6)^2 [9.18\text{nF} + 820\text{PF}]}$$

$$= 100\mu\text{H}$$

The inductance of coil tested with a Q-meter is 100μH.

Level-2 Key

01. (b)	02. (c)	03. (a)	04. (b)	05. (c)
06. (b)	07. (b)	08. (c)		/

Transducers

Contents

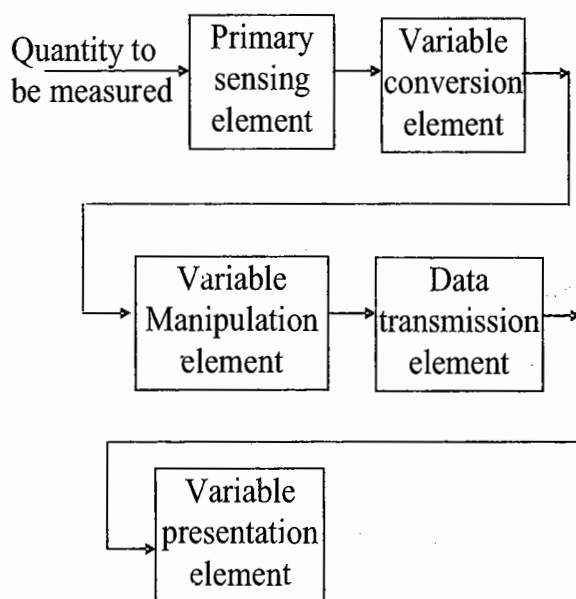
<i>S. No</i>	<i>Chapter Name</i>	<i>Page No.</i>
01	Basics of Transducers	111 - 120
02	Resistive, Inductive & Capacitive Transducers	121 - 137
03	Piezo Electro Transducers	137 - 141
04	Measurement of Temperature	142 - 150
05	Measurement of Flow & Viscosity	151 - 167
06	Intermediate Quantity Measurement	168 - 172

1

Basics of Transducers

It is important to have a systematic organization and analysis of measurement systems. An instrument may be defined as a device or a system which is designed to maintain a functional relationship between prescribed properties of physical variables and must include ways and means of communication to a human observer.

Generalized Measurement system:



Primary sensing element :

The quantity under measurement makes its first contact with the primary sensing element of a measurement system.

Variable conversion element:

The output of the primary sensing element is converted to some other suitable form for the instrument to perform desired function.

Variable manipulation Element :

The function of this element is to manipulate the signal presented to it preserving the original nature of the signal.

Data Transmission Element:

This element transmits the data from one to another.

Data presentation element:

This element conveys the information about the quantity under measurement to the personnel handling the instrument or the system for monitoring, control & analysis purposes.

- The measurand in an instrumentation system makes contact with a primary detection element or input device.
- The measurand or the input signal is called an “information” for measurement system.
- The information may be in the form of a physical phenomenon or it may be an electrical signal.
- The process of detection and conversion of the information into an acceptable form requires energy.
- The ideal situation is, the energy required for above purpose is supplied from outside, not from measurand.
- So that the measurand is not distorted and the analogous output of the detection is a faithful representation of measurand.

Transducers

An electronic instrumentation system consists of a number of components to perform a measurement and record its results.

A generalized measurement system consists of three major components.

- An input device
- A signal conditioning or processing device
- An output device
- The input device receives the measurand or the quantity under measurement and delivers a proportional or analogous electrical signal to the signal conditioning device.



- Here the signal is amplified, attenuated, filtered, modulated or otherwise modified in format acceptable to the output device.

Note: The input quantity for the most instrumentation systems is a “Non – electrical quantity”. In order to use electrical methods and techniques for measurement, manipulation or control, non – electrical quantity is generally converted into an electrical form by a device called a “transducer”.

- We can define transducer as a device which, when accurately transforms energy from one form to another.
- Another name for transducer is “pick – up”.
- The reasons for transforming a physical phenomenon into an electrical form is that the electrical output can be easily used, transmitted and processed for the purpose of measurement. Modern digital computers make the use of these transducers absolutely essential.
- The relationship between the physical parameter and its resulting electrical signal must be linear.
- The operating range of the transducer is called the dynamic range and should be wide.

Transducer mainly consists of two parts

- Sensing element
- Transduction element

Sensing element:

A detector or sensing element is that part of a transducer which responds to a physical phenomenon or a change in a physical phenomenon.

Transduction element:

A transduction element transforms the output of a sensing element to an electrical output.

Classification of Transducers

The transducers can be classified

- Based upon transduction principle
- as primary and secondary transducers
- as passive and active transducers

- as analog and digital transducers
- as transducers and inverse transducers

Based upon transduction principle

The transducers can be classified on the basis of principle of transduction as resistive, inductive, capacitive etc., depending upon how they convert the input quantity into resistance, inductance or capacitance respectively.

Primary and Secondary Transducers:

The first transducer which converts physical phenomenon into displacement, pressure, velocity etc. which is to be accepted by next stage is known as “Primary Transducer”.

The output of the primary transducer is converted subsequently into a usable output by a device called “Secondary Transducer”

Passive and Active Transducers:

Passive transducers: They derive the power required for transduction from an auxiliary power source.

Eg: Resistive, inductive and capacitive transducers.

Active transducers: They do not require an auxiliary power source to produce their output. They are also known as self – generating type since they develop their own voltage or current output.

Eg: piezoelectric, photovoltaic etc.

Analog and digital Transducers:

Analog transducers: These transducers convert the input quantity into an analog output which is a continuous function of time.

Eg: LVDT, thermocouple etc.

Digital Transducers: -These transducers convert the input quantity into an electrical output which is in the form of pulses.

Transducers & Inverse Transducers

Transducer: A transducer can be broadly defined as a device which converts a non – electrical quantity into an electrical quantity.

Inverse transducer: An inverse transducer is defined as a device which converts an electrical quantity into a non – electrical quantity.



Mechanical Devices as Primary Detectors

Type	Operation
A. Contacting spindle, pin or finger	Displacement to displacement
B. Elastic member	
1. Prooving ring	Force to displacement
2. Bourdon tube	Pressure to displacement
3. Bellows	Pressure to displacement
4. Diaphragm	Pressure to displacement
5. Spring	Force to displacement
C. Mass	
1. Seismic mass	Forcing function to displacement
2. Pendulum scale	Force to displacement
3. Manometer	Pressure to displacement
D. Thermal	
1. Thermocouple	Temperature to electric current
2. Bimetallic	Temperature to displacement
3. Temp – stick	Temperature to phase
E. Hydro pneumatic	
1. Static	
(a) Float	Fluid level to displacement
(b) Hydrometer	Specific gravity to displacement
2. Dynamic	
(a) Orifice	Velocity to pressure
(b) Venturi	Velocity to pressure
(c) Pitot tube	Velocity to pressure
(d) Vanes	Velocity to force
(e) Turbines	Linear to angular velocity



Types of Electrical Transducers

Passive Transducers

Electrical parameter And class of transducer	Principle of Operation	Typical applications
Potentiometer device	Positioning of the slider by an external force varies the resistance in a potentiometer or a bridge circuit.	Pressure, displacement
Resistance strain gauge	Resistance of a wire or semiconductor is changed by elongation or compression due to externally applied stress.	Force, torque, displacement
Pirani gauge or hot wire meter	Resistance of a heating element is varied by convection cooling of a stream of gas.	Gas flow, gas pressure
Resistance thermometer	Resistance of pure metal wire with a large positive temperature co-efficient of resistance varies with temperature	Temperature, radiant heat
Thermistor	Resistance of certain metal oxides with negative temperature coefficient of resistance varies with temperature.	Temperature, flow
Resistance Hydrometer	Resistance of a conductive strip changes with moisture content.	Relative humidity
Photoconductive cell	Resistance of the cell as a circuit element varies with incident light.	Photo-sensitive relay.



Capacitance

Variable capacitance pressure gauge	Distance between two parallel plates is varied by an externally applied force.	Displacement, pressure
Capacitor microphone	Sound pressure varies the capacitance between a fixed plate and a movable diaphragm.	Speech, music, noise
Dielectric gauge	Variation in capacitance by changes in the dielectric or dielectric constant.	Liquid level, thickness

Inductance

Magnetic circuit transducer	Self – inductance or mutual inductance of a.c. excited coil is varied by changes in the magnetic circuit.	Pressure, displacement
Reluctance pick up	Reluctance of the magnetic circuits is varied by changing the position of the iron core of coil.	Pressure, displacement, vibrations, position
Differential transformer	The differential voltage of two secondary windings of a transformer is varied by positioning the magnetic core through an externally applied force.	Pressure, force, displacement, position.
Eddy current gauge	Inductance of a coil is varied by the proximity of an eddy current plate	Displacement, thickness
Magnetostriction gauge	Magnetic properties are varied by pressure and stress	Force, pressure, sound.



Voltage and Current

Hall effect pick up	A potential difference is generated across a semiconductor plate (germanium) when magnetic flux interacts with an applied current.	Magnetic flux, current, power
Ionization chamber	Electron flow induced by ionization of gas due to radio – active radiation.	Particle counting, radiation
Photo emissive cell	Electron emission due to incident radiation upon photo emissive surface.	Light and radiation.
Photomultiplier tube	Secondary electron emission due to incident radiation on photosensitive cathode	Light and radiation, Photosensitive relays.

Self – Generating Transducers

Thermocouple and thermopile	An emf is generated across the junction of two dissimilar metals or semiconductors when that junction is heated.	Temperature, heat flow, radiation
Moving coil generator	Motion of coil in a magnetic field generates a voltage.	Velocity, vibrations.
Piezoelectric pickup	An emf is generated when an external force is applied to certain crystalline materials such as quartz.	Sound, vibrations, acceleration, pressure changes.
Photovoltaic	A voltage is generated in a semiconductor junction device when radiant energy stimulates the cell	Light meter, solar cell



Characteristics & Choice of Transducers

When choosing a transducer for any application the input, transfer and output characteristics have to be taken into account.

Input characteristics:

The following are the input characteristics to be taken into consideration.

- Type of input and operating range
- Loading effects

Transfer Characteristics:

The transfer characteristics of transducers require attention of three separate elements.

- Transfer function
- Error
- Response of transducer to environmental influences.

Output Characteristics:

The three conditions in the output characteristics which should be considered are

- Type of electrical output
- Output impedance
- Useful range

Factors influencing the choice of transducers:

The following are the factors influencing the choice of transducer for measurement of a physical quantity:

- Operating principle
- Sensitivity
- Operating range
- Accuracy

- Cross sensitivity
- Errors
- Transient and Frequency response
- Loading effects
- Environmental compatibility
- Insensitivity to unwanted signals
- Usage and Ruggedness
- Electrical aspects
- Stability and reliability
- Static characteristics

Example 01

Which one of the following is an active transducer?

- (a) Photo voltaic cell
- (b) Strain gauge
- (c) Photo emissive cell
- (d) Synchronous

Ans: (a)

Sol: photo voltaic cell is an active transducer

Example 02:

A transducer has an output impedance of $1 \text{ k}\Omega$ and the load resistance is $1 \text{ M}\Omega$, the transducer behaves as

- (a) a constant current source
- (b) a constant voltage source
- (c) a constant power source
- (d) none of the above

Ans: (b)

Class Room Practice Questions

LEVEL-2 Questions

01. Which of the following transducers is classified as an active transducer?

- (a) Metallic strain gauge
- (b) Capacitive microphone
- (c) LVDT
- (d) Piezoelectric transducer

02. Pair of active transducers is

- (a) Thermistor, Solar cell
- (b) Thermocouple, Thermistor
- (c) Thermocouple, Solar cell
- (d) Solar cell, LVDT

03. Match List-I (Parameter) with List-II (Transducer) and select the correct answer:

List - I

- A. Pressure
- B. Temperature
- C. Displacement
- D. Stress

List - II

- 1. Thermistor
- 2. Piezoelectric crystal
- 3. Capacitance transducer
- 4. Resistance strain gauge
- 5. Ultrasonic waves

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 5 | 3 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 1 | 2 | 5 | 4 |
| (d) | 2 | 1 | 3 | 4 |

04. Different types of transducers and signal conditioning circuits are given below, Match the transducer with the appropriate signal conditioning circuit commonly used with it.

List -I

- P. Thermocouple
- Q. Strain gage
- R. Piezoelectric sensor
- S. L. V. D. T.

List -2

- 1. DC Bridge
- 2. Phase sensitive detector
- 3. Charge amplifier
- 4. Cold junction compensation
- 5. Instrumentation amplifier

Codes:

- (a) P-2, Q-3, R-5, S-1
- (b) P-1, Q-5, R-2, S-3
- (c) P-4, Q-1, R-2, S-5
- (d) P-4, Q-1, R-3, S-2

05. In an analog data acquisition unit, what is correct sequence of the blocks starting from the input?

- (a) Transducer – Recorder – Signal conditioner
- (b) Transducer–Signal conditioner Recorder
- (c) Signal conditioner–Transducer Recorder
- (d) Signal conditioner Recorder Transducer

06. Match List - I with List - II and select the correct answer using the code given below the lists:

List - I

- A. Bolometer
- B. Hot - wire Anemometer
- C. C - type Bourdon Tube
- D. Optical pyrometer

List - II

- 1. Measurement of temperature of a furnace
- 2. Measurement of high pressure
- 3. Measurement of flow of air around an aero plane
- 4. Measurement of power at 500 MHz
- 5. Measurement of angular position



Codes:

	A	B	C	D
(a)	2	3	5	1
(b)	4	1	2	3
(c)	2	1	5	3
(d)	4	3	2	1

07. Match the essential amplifier characteristics with the sensing applications given below:

Amplifier Characteristics

P: Charge amplifier with very low bias current and high input impedance

Q: Voltage amplifier with low bias current and very high input impedance

R: Voltage amplifier with very high CMRR

Sensing Applications

L: Strain gauge in unipolar DC Wheatstone bridge

M: Glass electrode pH sensor

N: Piezoelectric sensor for measurement of static force

Codes:

- (a) P – L, Q – M, R – N
(b) P – M, Q – N, R – L
(c) P – N, Q – L, R – M
(d) P – N, Q – M, R – L

08. Match List-I (Instrument) with List-II (Measurand) and select the correct answer:

List – I

- A. McLeod gauge
B. Turbine meter
C. Pyrometer
D. Synchros

List – II

1. Temperature
2. Pressure
3. Flow
4. Displacement

Codes:

	A	B	C	D
(a)	1	4	2	3
(b)	2	3	1	4
(c)	1	3	2	4
(d)	2	4	1	3

09. Match List – I with List – II and select the correct answer using the code given below the lists:

List – I

- A. Variable capacitance device
B. Orifice meter
C. Thermistors

List – II

1. Flow measurement
2. Temperature measurement
3. Pressure transducer
4. Force and torque measurement

Codes:

	A	B	C
(a)	3	1	2
(b)	2	4	3
(c)	2	1	3
(d)	3	4	2

10. Which of the following primary detector type transducers are employed for displacement measurement?

1. Thermistor 2. Diaphragm
3. Thermocouple 4. Pivot Torque

Select the correct answer using the code given below:

- (a) 1 and 2 (b) 2 and 4
(c) 3 and 4 (d) 1 and 4

11. The instrumentation amplifiers are used principally to amplify signals from which of the following?

- (a) Transducers (b) Active fillers
(c) Choppers (d) D/A converters

12. Which displacement transducer is used for accurate and linear measurement?

- (a) LVDT
(b) Strain gauge
(c) Potentiometer
(d) Capacitive displacement transducer



13. The lower limit of useful working range of a transducer is determined by
- (a) Minimum useful input level
 - (b) By transducer error and noise
 - (c) Cross sensitivity
 - (d) Dynamic response.

14. Match the following

P. Radiation Pyrometer	W. Angular velocity measurement
Q. Dall tube	X. Vacuum pressure measurement
R. Pirani gauge	Y. Flow measurement
S. Gyroscope	Z. Temperature measurement.

- (a) P-Z, Q-W, R-X, S-Y
- (b) P-Z, Q-Y, R-X, S-W
- (c) P-W, Q-X, R-Y, S-Z
- (d) P-Z, Q-X, R-W, S-Y

15. Which one of the following is not a self generating type transducer?
- (a) Thermocouple and thermopile
 - (b) Piezoelectric pick – up
 - (c) Photovoltaic cell
 - (d) Magnetostriction gauge

Level – 2 Key

- 01.(d) 02. (c) 03. (d) 04. (d) 05. (b)
06. (d) 07. (d) 08. (b) 09. (a) 10. (b)
11. (a) 12. (a) 13. (c) 14. (b) 15. (d)

2

Resistive, Inductive & Capacitive Transducers

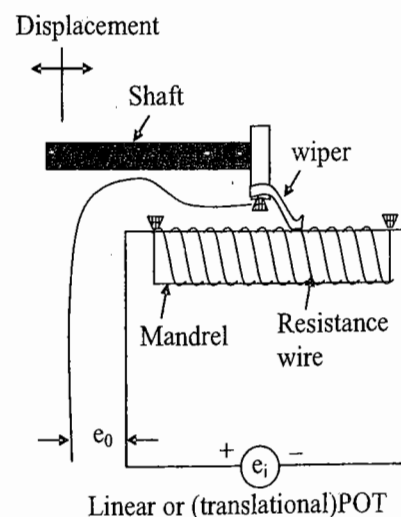
- It is generally seen that methods which involve the measurement of change in resistance are preferred to those employing other principles.
- This is because both alternating as well as direct currents and voltage are suitable for resistance measurements.
- The resistance of a metal conductor is expressed by a simple equation that involves a few quantities.
- The relation is $R = \frac{\rho L}{A}$
- Any method of varying one of the quantities involved in the above relationship can be design basis of an **Electrical Resistance Transducer**.

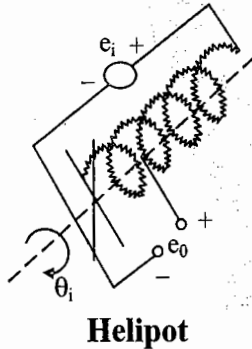
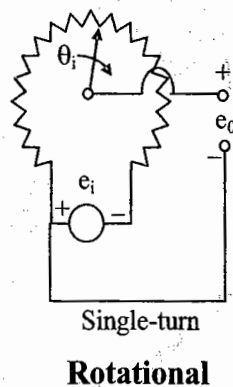
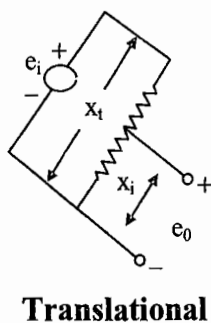
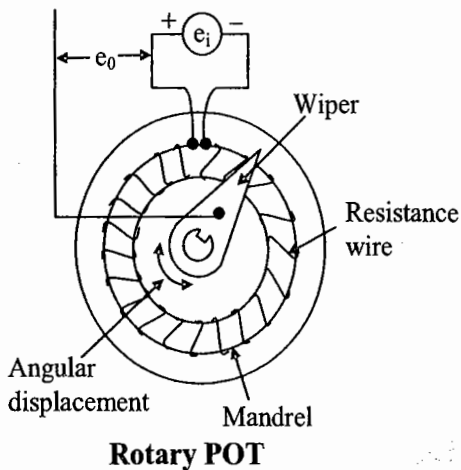
The following table indicates the non-electrical quantities that can be measured by the resistance transducer.

Effect	As a primary transducer	Through the use of secondary transducers
1. Change in length	Linear and angular displacements, thickness	Temperature, pressure, weight, force, torque, level, density, fluid flow rate, viscosity, velocity, acceleration and altitude of vehicles in space.
2. Change in length and area of cross section	High pressure strain	All the above quantities except velocity and altitude
3. Change in resistivity	Temperature	Thermal conductivity, gas composition, fluid flow rate, wind velocity and direction, low pressure and vacuum and infrared radiation.

Potentiometer

- Basically a potentiometer or simply a POT (a resistive potentiometer used for the purpose of voltage division is called a POT) consists of a resistive element provided with a sliding contact.
- This sliding contact is called a wiper.
- The motion of the sliding contact may be translatory or rotational.
- Some POTs use the combination of the two motions, i.e., translational as well as rotational.
- These POTs have their resistive element in the form of a helix and therefore they are called "**Helipots**".





- A full single turn potentiometer may provide accurate measurements upto 357° .
- Multiturn potentiometers may measure upto 3500° of rotation through use of helipots.

Power Rating of Potentiometers

- The potentiometers are designed with a definite power rating which is related directly to their heat dissipating capacity.
- Normally available pots are of single turn with a diameter of 50 mm with a wide range of ohmic values ranging from 100Ω to $10K\Omega$ in steps of 100Ω .

- Power rating is typically 5W at an ambient temperature of 21°C .
- Since power $P = e_i^2 / R_P$, the maximum input excitation voltage that can be used is $(e_i)_{\max} = \sqrt{PR_P}$

Linearity and Sensitivity of Pots:

- Linearity implies resistance of POTs should be low.
- To achieve good sensitivity the output voltage e_o should be high which in turn requires a high input voltage e_i and this calls for a high value of resistance of POT.
- Therefore linearity and sensitivity are two conflicting requirements.

Materials Used for potentiometers:

Based on the materials used POTs may be classified as (i) Wire Wound

(ii) Non – wire wound

- Wire wound potentiometers :
platinum, nickel chromium, nickel copper
- Non-wire wound potentiometers :
cermet, hot moulded carbon, carbon film, thin metal film.

Advantages:

- Inexpensive
- Simple to operate
- Electrical efficiency is high

Disadvantage:

- Linear POT require a large force to move their sliding contacts.

Example 1:

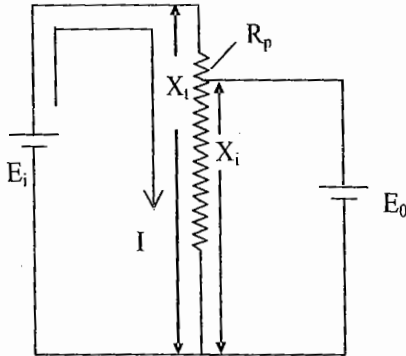
An ideal or unloaded potentiometer used as a displacement transducer has a stroke of 100mm and its resistance over this length is 1000Ω . Make calculation for the power dissipated by the coil. It



may be presumed that the transducer has an overall sensitivity of 0.1 V/mm.

Ans: 0.1 watt

Sol:



Given

$$X_t = 100 \text{ mm}$$

$$R_p = 1000 \Omega$$

$$\text{Sensitivity (S)} = \frac{0.1 \text{ V}}{\text{mm}} = \frac{E_0}{X_i}$$

$$E_0 = 0.1 \text{ V} \rightarrow X_i = 1 \text{ mm}$$

$$\text{When } X_i = X_t = 100 \text{ mm}$$

$$E_0 = E_i = 10 \text{ V}$$

$$I = \frac{E_i}{R_p}$$

$$= \frac{10 \text{ V}}{1000 \Omega} = 0.01 \text{ Amp}$$

So power dissipated by the coil is

$$\begin{aligned} & (I)^2 \times R_p \\ &= (0.01)^2 \times 1000 \Omega \\ &= 0.1 \text{ watt} \end{aligned}$$

Strain Gauge

Strain gauges are used for measurement of strain and associated stress and these can be used as secondary transducers in many applications. They are based on the principle of piezoresistive effect.

Piezoresistive effect:

If a metal conductor is stretched or compressed, its resistance changes on account of the fact that both length and diameter of conductor change. Also there is a change in value

of resistivity of the conductor when it is strained. This property is known as piezoresistive effect.

Theory of Strain Gauges:-

Let us consider a strain gauge made of circular wire.

The wire has the dimensions,

$$\text{Length} = L$$

$$\text{Area} = A$$

Diameter = D before being strained.

The material of the wire has a resistivity ρ .

$\therefore$ Resistance of unstrained gauge, $R = \rho L/A$.

Let a tensile stress s be applied to the wire.

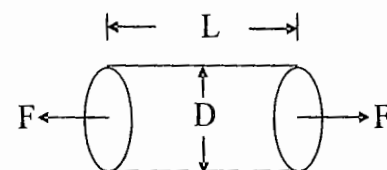
This produces a positive strain causing the length to increase and area to decrease. Thus when the wire is strained there are changes in its dimensions.

Let ΔL = change in length

ΔA = change in area

ΔD = change in diameter

ΔR = change in resistance



Gauge factor:

It is defined as the ratio of per unit change in resistance to per unit change in length.

$$\text{Gauge factor, } G_f = \frac{\Delta R/R}{\Delta L/L}$$

$$\text{or } \frac{\Delta R}{R} = G_f \frac{\Delta L}{L}$$

$$= G_f \times \epsilon$$

$$\epsilon = \text{strain} = \Delta L/L$$

Dividing equation (1) with $\Delta L/L$ we get



$$G_f = 1 + 2\nu + \frac{\Delta\rho/\rho}{\epsilon}$$

$$= 1 + 2\nu + \frac{\Delta\rho/\rho}{\epsilon}$$

Resistance change due to change in length to change in area of piezoresistive effect.

If the change in the value of resistivity of a material when strained is neglected, the gauge factor is $G_f = 1 + 2\nu$

Characteristics of strain gauges:

- Element should be of low resistance temperature coefficient.
- Should have high value of gauge factor
- Should have linear characteristics
- Should have very high resistance.
- Should have very good frequency response.

Types of Strain Gauges:

- Unbonded metal strain gauges
- Bonded metal wire strain gauges
- Bonded metal foil strain gauges
- Semiconductor strain gauges
- Diffused metal strain gauges

Temperature compensation:

The resistive type strain gauges are sensitive to temperature. Therefore it becomes necessary to account for variations in the strain gauge resistance which occurs on account of temperature changes.

Temperature compensation may be provided by

- Use of adjacent arm balancing or compensating gauge.

Use of Dummy gauge

One of the ways in which temperature error can be eliminated by using adjacent arm compensating gauge is to use a dummy gauge in the adjacent arm. This arrangement is shown. Gauge 1 is installed on the test specimen gauge is installed on a like piece of material and is not subjected to any strain.

The gauges installed on the test temperature. A gauge is called dummy gauge in case it is not subjected to any strain. Active gauge is one which is subjected to strain.

Initially when the bridge is balanced,

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

Supposing a change in temperature occurs, the resistance R_1 & R_3 change by an amount ΔR_1 and ΔR_3 respectively.

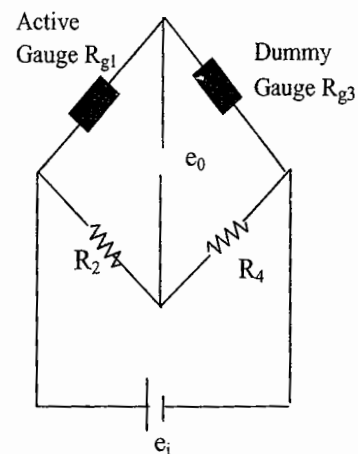
$$\text{Hence for balance, } \frac{R_1 + \Delta R_1}{R_3 + \Delta R_3} = \frac{R_2}{R_4}$$

$$\text{or } \frac{R_4}{R_2} (R_1 + \Delta R_1) = (R_3 + \Delta R_3)$$

$$\text{or } \frac{R_4}{R_2} R_1 + \frac{R_4}{R_2} \Delta R_1 = R_3 + \Delta R_3$$

$$\text{But } \frac{R_4}{R_3} R_1 = R_3 \therefore \frac{R_4}{R_2} \Delta R_1 = \Delta R_3$$

Suppose $R_4 = R_2$ This requires that $\Delta R_1 = \Delta R_3$



It means that for the bridge to remain insensitive to variations in temperature the gauges R_1 and R_2 should have their resistances change by equal amount when subjected to variation in temperature. Therefore the active gauge R_1 and the dummy gauge R_3 should be identical.



The use of dummy gauge for temperature compensation is simple and effective and should be employed whenever possible.

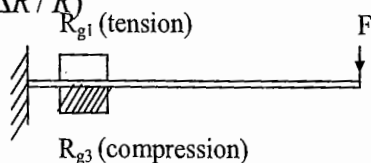
- **Use of two active gauges in adjacent arms**

In certain applications, where equal and opposite strains are known to exist, it is possible to attach two similar gauges in such a way that one gauge experience a positive strain and the other a negative strain. Thus instead of having an arrangement wherein one gauge acts as the active gauge and the other as the dummy gauge, we have now an arrangement wherein both the gauges are active gauges.

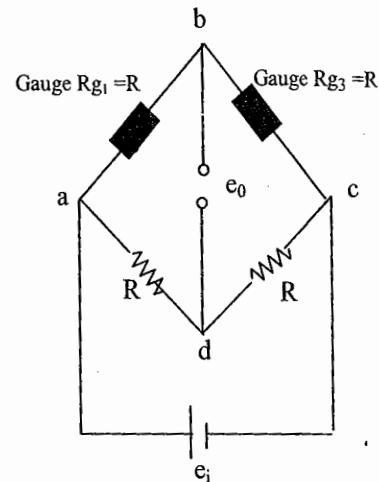
Fig shows the two gauges mounted on a cantilever. The gauge R_{g1} is on top of the cantilever and hence experiences tension or a positive strain. The R_{g3} is at the bottom surface

of the cantilever and hence experiences a compression or a negative strain.

- The bridge arrangement for the two gauges is shown in Fig., There are two active gauges in the 4 arm bridge and hence it is called a Half Bridge.
- The temperature effects are cancelled out by having $R_2 = R_3$ and using two identical gauges in the opposite arms of the bridge.
- When no strain is applied both points b and d are at the same potential, $e_i/2$ and the value of output voltage $e_0=0$.
- When the arrangement shown in Fig is subjected to strain, the resistance of gauge 1 increases and that of gauges R_{g3} decreases. Resistance of gauge R_{g1} when strained is $R(1 + \Delta R/R)$



Two gauges used for measurement of strain



Adjacent arm compensation using two active gauges

Resistance of gauge R_{g3} when strained is $R(1 - \Delta R/R)$

Now $R_2 = R_4 = R \therefore$ Potential of point d is $= e_i/2$

$\therefore$ Potential of point

$$b = \frac{R(1 + \Delta R/R)}{R(1 + \Delta R/R) + R(1 - \Delta R/R)} \times e_i$$

$$= \frac{1 + \Delta R/R}{2} e_i$$

$\therefore$ Change in output voltage when strain applied is,

$$\Delta e_0 = \frac{1 + \Delta R/R}{2} e_i - \frac{e_i}{2} = \frac{\Delta R/R}{2} e_i = \frac{G_f \epsilon}{2} e_i$$

Thus the output voltage from a half bridge is twice that from a quarter bridge and therefore the sensitivity is doubled. In addition, the temperature effects are cancelled. The gauge, sensitivity of a half bridge is

$$S_g = 2kR_g G_f$$

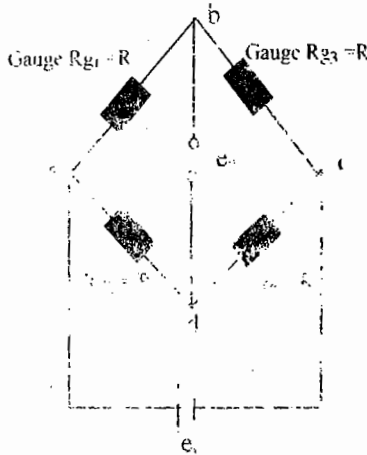
- **Use of four active gauges**

- Fig shows a cantilever using 4 strain gauges for the measurement of strain. All the four gauges are similar and have equal resistance when strained

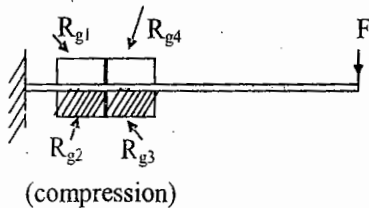
$$R_{g1} = R_{g2} = R_{g3} = R_{g4} = R.$$

- These gauges are connected in the arms of a Wheatstone bridge as shown in fig., since the bridge has 4 active gauges with one gauge in

each of the four arms it is called a **Full Bridge**.



Bridge circuit for measurement of strain four using active gauges



- When no strain is applied the potential of points b and d are both equal to $e_i/2$ and hence the output voltage $e_o = 0$.

When strained, the resistance of various gauges are:

For R_{g1} and R_{g4} : $R(1 + \Delta R/R)$ and for R_{g2} and R_{g3} : $R(1 - \Delta R/R)$

Potential of b when strain is applied

$$\frac{R(1 + \Delta R/R)}{R(1 + \Delta R/R) + R(1 - \Delta R/R)} e_i = \frac{1 + \Delta R/R}{2} e_i$$

Potential of d when strain is applied

$$\begin{aligned} &= \frac{R(1 - \Delta R/R)}{R(1 - \Delta R/R) + R(1 + \Delta R/R)} e_i \\ &= \frac{1 - \Delta R/R}{2} e_i \end{aligned}$$

Therefore change in output voltage

$$\begin{aligned} \Delta e_o &= \frac{1 + \Delta R/R}{2} e_i - \frac{1 - \Delta R/R}{2} e_i = (\Delta R/R) e_i \\ &= G_f \epsilon e_i \end{aligned}$$

Four active-arm bridges are extensively used when strain gauges are used as secondary transducers to give maximum sensitivity combined with full temperature compensation. The effect of increasing the number of active gauges is the same if a low impedance detector is used.

The gauge sensitivity of a full bridge

$$S_g = 4K R_g G_f$$

- Self compensation
- Use of special external control circuitry.

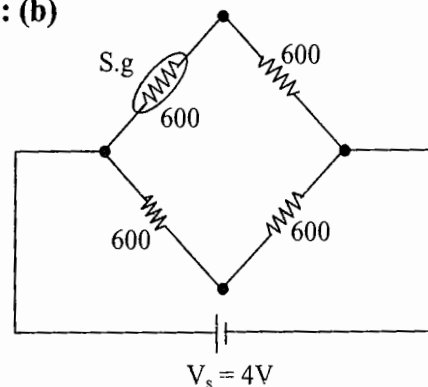
Example 2:

A strain gauge has a nominal resistance of 600Ω and a gauge factor of 2.5. The strain gauge is connected in a dc bridge with three other resistances of 600Ω each. The bridge is excited by a 4 V battery. If the strain gauge is subjected to a strain of $100 \mu\text{m/m}$, the magnitude of the bridge output will be

- (a) 0 V
- (b) $250 \mu\text{V}$
- (c) $500 \mu\text{V}$
- (d) $750 \mu\text{V}$

Ans: (b)

Sol:



Given

$R = 600$ (strain gauge resistance)

$G.F = 2.5$

$$\frac{\Delta L}{L} = 100 \mu\text{m}$$

$$= 100 \times 10^{-6}$$

$$\text{So, } \Delta R = G.F \times \frac{\Delta L}{L} \times R$$



$$\begin{aligned}
 &= 2.5 \times 100 \times 10^{-6} \times 600 \\
 &= 0.15 \\
 \text{So, } V_B &= \frac{V_s}{4} \times \frac{\Delta R}{R + 2\Delta R} \\
 &= \frac{4}{4} \times \frac{0.15}{600 + 2(0.15)} \\
 &= 2.4987 \times 10^{-4} \\
 &\approx 250 \mu V
 \end{aligned}$$

Variable inductance transducer

The variable inductance transducers work upon one of the following three principles :

- Change of self inductance
- Change of mutual inductance.
- production of eddy currents

Based on change of self inductance:

- The self inductance of a coil,

$$L = N^2 / R$$

Where N = number of turns
R = reluctance of the magnetic circuit
- The reluctance of the magnetic circuit $R = l / \mu A$

$$\therefore \text{Inductance } L = N^2 \mu (A/l)$$

$$= N^2 \mu G \text{ -----(1)}$$

where μ = effective permeability of the medium in and around the coil

$G = A / l$ = geometric form factor

A = area of cross-section of coil

l = length of the coil

- It is clear from equation (1) that the variation in inductance may be caused by :
- Change in number of turns, N
- Change in geometric configurations, G.
- Change in permeability, μ

Based on change of Mutual Inductance :

- An inductive transducer working on the principle of variation of mutual inductance uses multiple coils.

- The mutual inductance between the two coils is $M = K \sqrt{L_1 L_2}$
 L_1 and L_2 = self-inductance of two coils
K = coefficient of coupling
- Thus mutual inductance between the coils can be varied by variation of self-inductances or the coefficient of coupling
- The mutual inductance can be converted into a self-inductance by connecting the coils in series
- The self-inductance of such an arrangement varies between $L_1 + L_2 - 2M$ to $L_1 + L_2 + 2M$ with one of the coils being stationary while the other is movable.

Air cored coils:

They can be operated at a higher carrier frequency because of absence of eddy current losses in air cores

Iron cored coils:

Advantages:

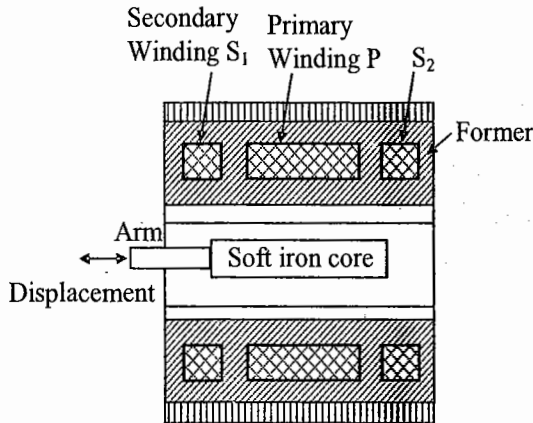
- Their size is much smaller than that air cored transducers on account of high permeability of iron cores.
- They are less likely to cause external magnetic fields and are less affected by stray magnetic fields

Based on production of Eddy currents:

- These inductive transducers work on the principle that if a conducting plate is placed near a coil carrying alternating current, eddy currents are produced in the conducting plate.
- The conducting plate acts as a short-circuited secondary winding of a transformer.
- The eddy currents flowing in the plate produce a magnetic field of their own which acts against the magnetic field production by the coil.
- This results in reduction of flux and thus the inductance of the coil is reduced.
- The nearer is the plate to the coil, the higher are the eddy currents and thus higher is the reduction in the inductance of the coil.

Linear Variable Differential Transformer (LVDT)

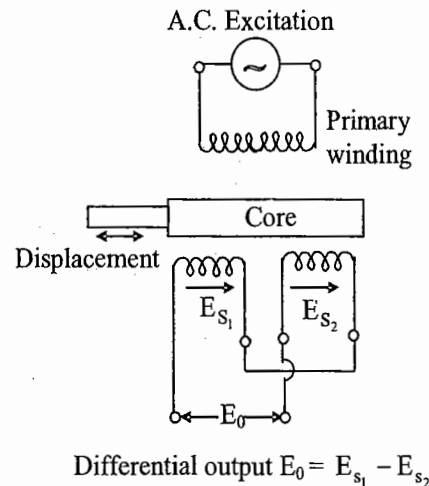
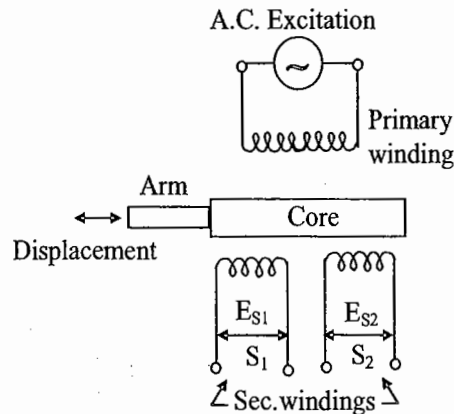
- The most widely used inductive transducer to translate the linear motion into electrical signals is the linear variable differential transformer (LVDT).
- The basic construction of LVDT is shown in figure



- The transformer consists of a single primary winding p and two secondary windings S_1 and S_2 wound on a cylindrical former.
- The secondary windings have equal number of turns and are placed identically on either side of the primary winding.
- The primary winding is connected to an alternating current source.
- A movable soft iron core is placed inside the former.
- The displacement to be measured is applied to the arm attached to the soft iron core.
- Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating current voltages in the two secondary windings.
- The output voltage of secondary S_1 is E_{S1} and that of secondary S_2 is E_{S2} .
- In order to convert the outputs from S_1 and S_2 into a single voltage signal, the two secondaries S_1 and S_2 are connected in series opposition.

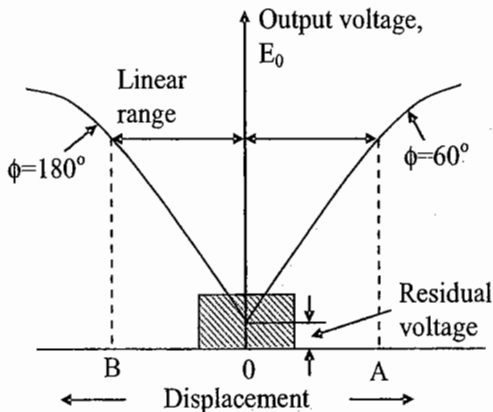
Thus the output of the transducer is the difference of the two voltages.

Differential output voltage, $E_o = E_{S1} - E_{S2}$



- When the core is at NULL position, the flux linking with both the secondary windings is equal and hence equal emfs are induced in them.
- Thus at NULL position, $E_{S1} = E_{S2}$ and hence the output voltage $E_o = 0$.
- If the core is moved to the left of the NULL position, more flux links with winding S_1 and less with winding S_2 .
- The output voltage E_{S1} of the secondary winding S_1 is more than E_{S2} , the output voltage of secondary winding S_2 .
- The magnitude of output voltage $E_o = E_{S1} - E_{S2}$ and is in phase with the primary voltage.

- If the core is moved to the right of the null position, the flux linking with winding S_2 becomes larger than that with winding S_1 .
- Thus $E_o = E_{S2} - E_{S1}$ which is 180° out of phase with primary voltage.
- The following figure shows the variations of output voltage against displacement for various positions of the core.



Advantages:

- High range
- Friction and Electrical Isolation
- Immunity from external effects
- High input and high sensitivity
- Ruggedness
- Low hysteresis
- Low power consumption

Disadvantages:

- Relatively large displacements are required for appreciable differential output.
- Sensitive to stray magnetic fields
- Temperature affects the performance of transducer.
- Dynamic response is limited mechanically by the mass of the core and electrically by the frequency response of the applied voltage.

Example 3:

The output voltage of a LVDT is 1.5V at maximum displacement. At a load of $0.5 \text{ M}\Omega$, the

deviation from linearity is maximum and it is $\pm 0.003 \text{ V}$ from a straight line through origin. the linearity at the given load is _____.

Ans: $\pm 0.2\%$

Sol: We know for LVDT

The percentage linearity

$$= \frac{\pm 0.003}{1.5} \times 100$$

$$= \pm 0.2\%$$

Example 4:

The output of the LVDT with input mechanical motion of 10 Hz and excitation frequency 400 Hz will contain frequencies.

- 10 Hz and 400 Hz
- 400 Hz only
- 10 Hz only
- 390 Hz and 410 Hz

Ans(d)

Sol: For LVDT

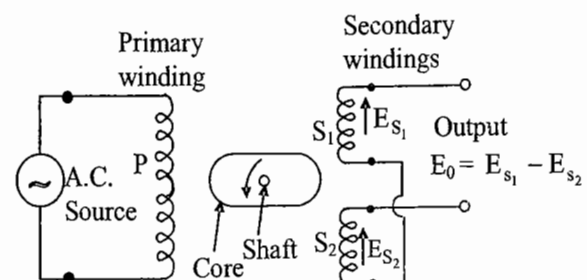
$f_{ex} \pm f_x$ is the modulated wave frequency
so modulated wave contains frequency

$$400 \text{ Hz} \pm 10 \text{ Hz}$$

$$= 390 \text{ Hz to } 410 \text{ Hz}$$

Rotary Variable Differential Transformer (RVDT)

- A variation of linear variable differential transformer (LVDT) which can be used to sense angular displacement is Rotary Variable Differential Transformer (RVDT).
- The circuit of RVDT is shown in the following figure.



- It is similar to LVDT except that its core is cam shaped and may be rotated between the windings by means of a shaft.
- The operation is similar to LVDT.
- Clockwise rotation produces an increasing voltage of a secondary winding of one phase while counter-clockwise rotation produces an increasing voltage of opposite phase.
- Hence, the amount of angular displacement and its direction may be ascertained from the magnitude and phase of the output voltage of the transducer.

Synchros

A synchros is an electromagnetic transducer which is commonly used to convert the angular position of the shaft into an electric signal.

- There are two types of synchro systems.
 - (i) Control or error detecting type
 - (ii) Torque transmission type

Resolvers

- Resolvers are used for conversion of angular position of a shaft into Cartesian coordinates.
- The output of the transducer is in the form of two signals, one proportional to the sine of the angle and the other proportional to cosine of the angle.
- Resolvers are classified into two groups
 - Computing resolvers
 - Synchro resolvers
- Computing resolvers are used for generating sine, cosine and tangent functions as well as for solving geometric relationships.
- Synchro resolvers are used for data transmission.

Applications:

- Vector resolution
- Vector composition
- Vector angle and component resolution
- Pulse amplitude control and pulse resolution

- Phase shifting

Capacitive transducer

The principle of operation of capacitive transducers is based upon the familiar equation for capacitance of a parallel plate capacitor.

$$\text{Capacitance, } C = \epsilon A/d$$

$$= \epsilon_r \epsilon_0 A/d$$

Where A = overlapping area of plates

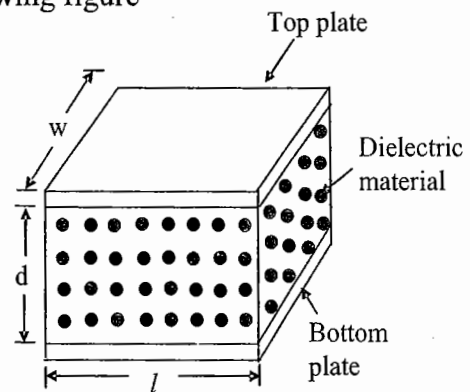
D = distance between two plates

$\epsilon = \epsilon_r \epsilon_0$ = permittivity of medium

ϵ_r = relative permittivity

ϵ_0 = permittivity of free space

A parallel plate capacitor is shown in the following figure



The capacitive transducer works on the principle of change of capacitance which may be caused by:

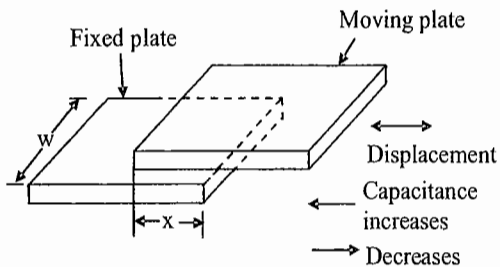
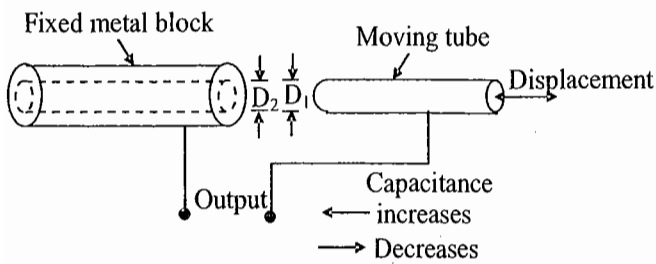
- Change in overlapping area A.
- Change in distance d between the plates.
- Change in dielectric constant

Change in Area of plates :

The equation for capacitance,

$$C = \epsilon A/d = \epsilon_r \epsilon_0 A/d$$

- The capacitance is directly proportional to the area A of the plates
- Thus the capacitance changes linearly with change in area of the plates.
- The elementary diagram of two types of capacitive transducers are shown in the following figures.



- For a parallel plate capacitor, the capacitance is

$$C = \epsilon A / d = \epsilon x w / d$$

Where x = length of overlapping part of plates

w = width of overlapping part of plates

$$\text{Sensitivity } S = \frac{\partial C}{\partial x} = \epsilon \frac{w}{d} \quad \text{F/m}$$

- This is suitable for measurement of linear displacements ranging from 1 mm to 10 mm
- Accuracy is as high as 0.005 %
- For a cylindrical capacitor, the capacitance is:

$$C = \frac{2\pi\epsilon x}{\log_e(D_2/D_1)} \quad \text{F}$$

Where x = length of overlapping part of cylinders

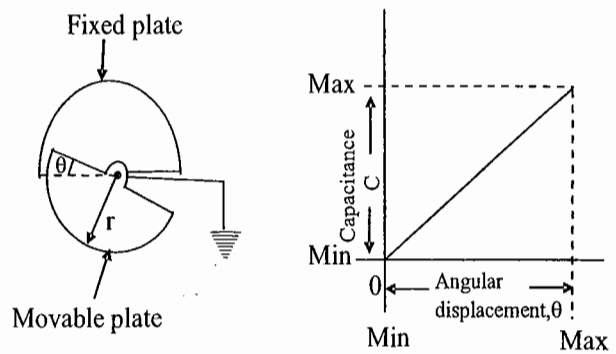
D_2 = inner diameter of outer cylindrical electrode

D_1 = outer diameter of inner cylindrical electrode

- Sensitivity $S = \frac{\partial C}{\partial x} = \frac{2\pi\epsilon}{\log_e(D_2/D_1)} \quad \text{F/m}$

Measurement of angular displacement:

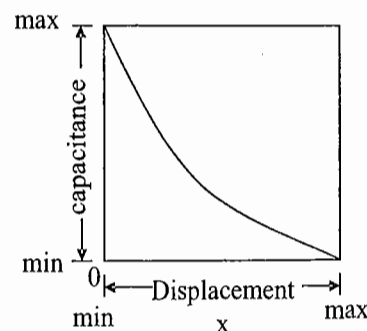
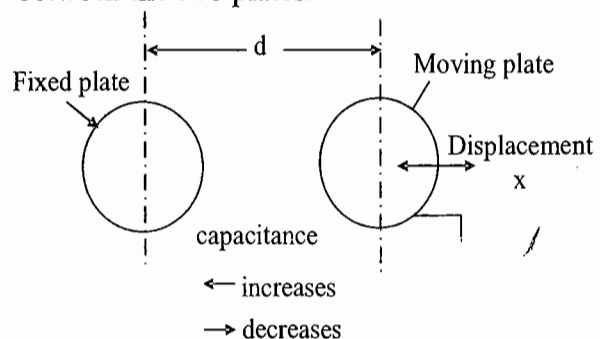
The following figure shows a two plate capacitor used for measurement of angular displacement.



- One plate is fixed and other is movable.
- The angular displacement to be measured is applied to the movable plate which changes the effective area between the plates and thus changes the capacitance.
- Maximum value of capacitance
Capacitance at an angle θ is
Where θ = angular displacement in radian.
- Sensitivity $S = \frac{\partial C}{\partial \theta} = \frac{\epsilon r^2}{2d}$
- The variation of capacitance with angular displacement is linear.

Change in distance between the plates:

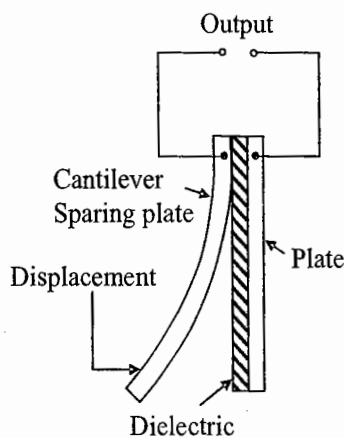
- The following figure shows the basic form of a capacitive transducer utilizing the effect of change of capacitance with change in distance between the two plates.



- One is fixed plate and the displacement to be measured is applied to the other plate which is movable.
- The capacitance C , varies inversely as the distance d between the plates, the response of the transducer is non-linear.

Capacitive transducer using cantilever spring plate:

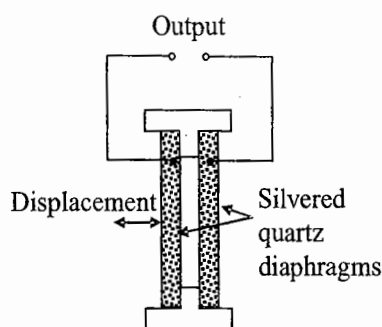
The following figure shows a capacitive transducer using cantilever spring plate



- The displacement when applied to the cantilever type spring plate moves it towards the second plate decreasing the distance.
- This increases the capacitance of the capacitor.

Capacitive transducer using Quartz diaphragms:

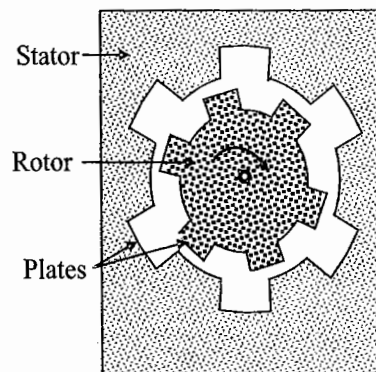
- The following figure shows a capacitive transducer using Quartz diaphragms.



- Two silvered quartz diaphragms form a capacitor
- The displacement of these diaphragms varies the capacitance of this capacitive transducer.

Rotational displacement measurement :

- The following figure shows an arrangement for rotational displacement measurement.



- As the rotor plates of the capacitor are displaced in the counter clockwise direction the capacitance increase
- The change in capacitance is a measure of the angular displacement.
- Effectively used for the measurement of torque.

Advantages:

- They require extremely small forces to operate them and hence are very useful for use in small systems.
- They are extremely sensitive
- They have a good frequency response. This response is as high as 50 kHz and hence they are very useful for dynamic studies.
- They have a high input impedance and therefore the loading effects are minimum.
- A resolution of the order of 2.5×10^{-3} mm can be obtained with these transducers.
- The capacitive transducers can be used for applications where stray magnetic fields render the inductive transducers useless.



- The force requirements of capacitive transducers is very small and therefore they require small power to operate them.

Disadvantages:

- The metallic parts of the capacitive transducers must be insulated from each other. In order to reduce the effects of stray capacitances, the frames must be earthed.
- The capacitive transducers show non-linear behaviour many a times on account of edge effects. Therefore guard rings must be used to eliminate this effect. Guard rings are also a must in order to eliminate the effect of stray electric fields, especially when the transducers have a low value of capacitance of the order of pF.
- The cable connecting the transducer to the measuring point is also a source of error. The cable may be source of loading resulting loss of sensitivity.

Applications:

- Used for the measurement of both linear and angular displacements.
- Used for the measurement of force and pressure. The force and pressure to be measured are first converted to displacement which causes a change of capacitance.
- Used directly as pressure transducers in all those cases where the dielectric constant of a medium changes with pressure.
- Used for measurement of humidity in gases since the dielectric constant of gases changes with change in humidity thereby producing a change in capacitance.
- Commonly used in conjunction with mechanical modifiers for measurement of volume, density, liquid, level, weight etc.

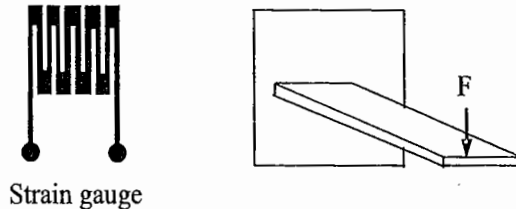
Class Room Practice Questions

LEVEL-1 Questions

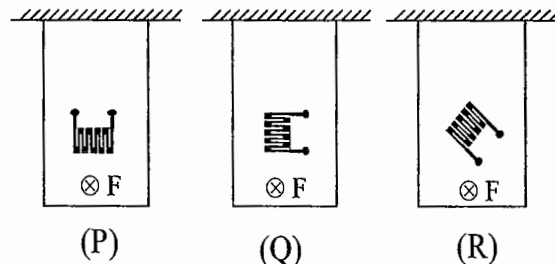
01. A resistance potentiometer is a
 - (a) first order instrument
 - (b) zero order instrument
 - (c) second order instrument
 - (d) none of the above
02. In a resistance potentiometer, the non linearity
 - (a) increases with increase of load to potentiometer resistance
 - (b) decreases with increase of load to potentiometer resistance
 - (c) is not dependent upon load to potentiometer resistance
 - (d) none of the above
03. In a resistance potentiometer high value of resistance of POT leads to
 - (a) high value of sensitivity
 - (b) low value of sensitivity
 - (c) low value of non-linearity
 - (d) low value of error.
04. For a semi – conductor strain gauge which is true statement?
 - (a) has a much higher gauge factor than that of a metal wire gauge
 - (b) employs piezoelectric property of undoped silicon
 - (c) does not require temperature compensation
 - (d) exhibits very little gauge factor variation as compared to that of metal wire gauges
05. A unity ratio quarter bridge strain measuring circuit produces an output of 1 mV for a strain of 500 microstrain when the bridge excitation is 4 volts. The gauge factor of the element is then

- (a) 1 (b) 2 (c) 3 (d) 4

06. The figure below shows various configurations of bonding a strain gauge to a cantilever subjected to a bending force F.



Top views of possible configuration



Which configuration gives the maximum change in resistance for this force?

- (a) P (b) Q (c) R
 - (d) All have equal change in resistance
07. Poisson's ratio for a metal is 0.35. Neglecting piezo-resistance effect, the gauge factor of a strain gauge made of this metal is
 - (a) 0.65 (b) 1 (c) 1.35 (d) 1.70
 8. A strain gauge with a resistance of 250 ohms undergoes a change of 0.150 ohm during a test. The strain is 1.5×10^{-4} . then the gauge factor is
 - (a) 2.0 (b) 3.0 (c) 4.0 (d) 100
 9. Which one of the following gives Gauge factor of a strain gauge?

(a) $\frac{\Delta L}{L} / \frac{\Delta R}{R}$ (b) $\frac{\Delta R}{R} / \frac{\Delta L}{L}$



- (c) $\frac{\Delta R}{R} / \frac{\Delta D}{D}$ (d) $\frac{\Delta R}{R} / \frac{\Delta \rho}{\rho}$
10. In strain gauge torque transducers, the strain gauges are mounted at which one of the following?
 (a) 0° to the shaft axis
 (b) 45° to the shaft axis
 (c) 90° to the shaft axis
 (d) At any angle with the shaft axis
11. Why is strain gauge bridge sometimes excited with ac?
 (a) It has a stable performance with ac
 (b) Its sensitivity is more with ac
 (c) Power frequency pick-up can be avoided with ac
 (d) Ac output can be easily amplified
12. Semiconductor strain gages typically have much higher gage factors than those of metallic strain gages, primarily due to:
 (a) Higher temperature sensitivity
 (b) Higher Poisson's ratio
 (c) Higher piezoresistive coefficient
 (d) Higher magnetostrictive coefficient
13. Linear variable differential transformer has
 (a) two primary coils connected in phase and a secondary coil
 (b) two primary coils connected in opposition and a secondary coil
 (c) one primary coil and two secondary coils connected in phase
 (d) one primary coil and two secondary coils connected in opposition
14. A linear variable differential transformer (LVDT) is
 (a) a displacement transducer
 (b) an impedance matching transformer
 (c) a differential temperature sensor
 (d) an auto transformer

15. The dynamic characteristics of capacitive transducer are similar to those of
 (a) low pass filters
 (b) high pass filters
 (c) notch filters
 (d) band stop filters.
16. What is the order of minimum displacement that can be measured with capacitive transducers?
 (a) 1 cm (b) 1 mm
 (c) 1 μ m (d) 1×10^{-12} m.

LEVEL-2 Questions

01. An LVDT is used to measure displacement. The output of the LVDT is connected to a voltmeter of range 0 to 5 V through an amplifier having a gain of 250. For a displacement of 0.5 mm, the output of the LVDT is 2 mV. The sensitivity of the instrument would be:
 (a) 0.1 V/mm (b) 0.5 V/mm
 (c) 1 V/mm (d) 5 V/mm
02. A resistance wire strain gauge with a gauge factor of 2 is bonded to a steel structural member subjected to a tensile stress of 100 Mega Newtons/metre². The modulus of elasticity of steel is 200 Giga Newtons per square metre. The percentage change in gauge resistance due to the applied stress is
 (a) 0.1 (b) 0.2
 (c) 0.4 (d) 0.5
03. The gauge factor of the material of strain gauge is such that the resistance changes from 1000 ohms to 1009 ohms when subjected to a strain of 0.0015. The Poisson's ratio for the material of the gauge wire is



- (a) 1.75 (b) 2
(c) 2.5 (d) 6
04. An ac LVDT is given 6.3 V input and produces 5.2 V for a range of ± 0.25 inch. When the core is -0.25 inch from the centre, what is the output produced?
(a) -2.0 V (b) $+2.0$ V
(c) -2.6 V (d) $+2.6$ V
05. A strain gauge with a nominal resistance of 120Ω and gauge factor of 2 undergoes a strain of 10^{-5} . What is the change in resistance in response to the strain?
(a) 240Ω (b) 24Ω
(c) $2.4 \times 10^{-3}\Omega$ (d) $2.4 \times 10^{-2}\Omega$
06. A capacitive transducer is made of two concentric cylindrical electrodes. The outer diameter of the inner cylindrical electrode is 4 mm and dielectric medium is air. The inner diameter of the outer electrode is 4.1 mm. When 150 V is applied across the electrodes, what is the electric field intensity?
(a) 1500 V/mm (b) 3000 V/mm
(c) 4500 V/mm (d) 6000 V/mm
07. A resistance strain gauge of gauge factor 2 is used as a transducer element. Neglecting piezoresistive effects, what is the value of Poisson's ratio?
(a) 0.5 (b) 1
(c) 1.6 (d) 2
08. A resistance strain gauge with gauge factor of 3 is subjected to a stress of 3000 kg/cm^2 when fastened to a steel rod. The modulus of elasticity of steel is $2.1 \times 10^6 \text{ kg/cm}^2$. The percentage change in resistance of the strain gauge element is
(a) 0.1428% (b) 24.84%
(c) 0.4284% (d) 4.3%
09. The wire in a metallic strain gauge is 0.1 m long and has an initial resistance of 120 ohm. On application of an external force, the wire length increases by 0.1mm and the resistance increases by 0.21 ohm. The gauge factor of the strain gauge will be
(a) 3.00 (b) 2.00
(c) 1.75 (d) 2.85
10. An LVDT has the following specifications: Input = 6.3 V, Output = 5.2 V, range ± 1.25 cm
Then the output voltages produced due to core movement from $+1.1$ cm to -0.4 cm will be respectively
(a) $+4.576$ V and -1.664 V
(b) $+2.288$ V and -0.832 V
(c) $+4.0$ V and -1.0 V
(d) $+2.0$ V and -1.0 V
11. A 1 m length wire has a resistance of 150Ω . When it is subjected to strain, its length becomes 1.01 m. The measurement is conducted by a strain gauge whose gauge factor is 2. The change in resistance of the wire is
(a) 0.5Ω (b) 1.0Ω
(c) 2.0Ω (d) 3.0Ω
12. A 100Ω resistive potentiometer is used with an input supply voltage of 10 V. The temperature of the POT is if the thermal resistance 30°C/W and the ambient temperature is 40°C .
(a) 60°C (b) 80°C
(c) 70°C (d) 100°C



Level - 1 Key

01.(b) 02. (b) 03. (a) 04. (a) 05. (b)
06. (a) 07. (d) 08. (c) 09. (b) 10. (b)
11. (c) 12. (c) 13. (d) 14. (a) 15. (b)
16. (c)

Level - 2 Key

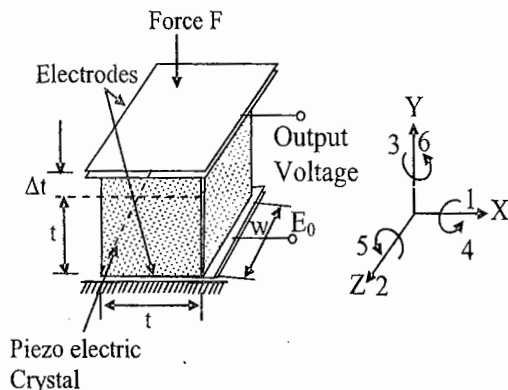
01.(c) 02. (a) 03. (c) 04. (c) 05. (c)
06. (a) 07. (a) 08. (c) 09. (c) 10. (a)
11. (d) 12. (c)



Piezo Electric Transducers

Piezo-electric transducers are active transducers. They work on the principle of piezo-electric effect. A piezo-electric material is one in which an electric potential appears across certain surfaces of a crystal if the dimensions of the crystal are changed by the application of a mechanical force.

- Each piezo-electric crystal has three sets of axes :
 - An optical axes
 - Three electrical axes (X) at 120° with each other and
 - Three mechanical axes (Y) at 120° with each other and at right angle to the respective electric axes.
- If an electric stress is applied in the direction of a electrical axis (X-axis), a mechanical strain is produced in the direction of the mechanical axis (Y), which is perpendicular to the relevant X-axis, conversely if a mechanical stress is applied along a Y-axis, electric charges appear on the faces of the crystal perpendicular to the relevant Y-axis.
- This effect is known as piezo-electric effect.



- The materials that exhibit a significant and useful piezo-electric effect are divided into two categories:
 - (i) Natural group

(ii) Synthetic group

- **Natural Group:**

Quartz and Rochelle salt

- **Synthetic Group :**

Lithium sulphate, ethylene diamine tartarate, Ammonium dihydrogen phosphate, dipotassium tartarate, potassium dihydrogen phosphate.

- Synthetic or artificial crystals, they themselves do not possess the piezo-electric property. By applying high voltages across the electrodes heating to higher temperature and slow cooling induces the property.
- With synthetic crystals, if the temperature is more than a threshold value called curie-point, the property is lost.
- Properties of crystal depends on
 - cutting of the crystal
 - Thickness of the crystal
 - Dimensions of the plate and
 - Means of mounting
- The piezo-electric effect can be made to respond to mechanical deformations of the material in many different modes.
- The modes can be:
 - Thickness expansion
 - Transverse expansion
 - Thickness shear and
 - Face shear
- The mode of motion effected depends on the shape of the body relative to the crystal axis and location of the electrodes.
- Mechanical deformation generates a charge and this charge appears as a voltage across the electrode.

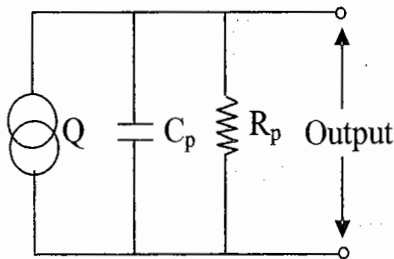


The voltage is $E = Q / C$

- The piezo-electric effect is direction sensitive. A tensile force produces a voltage of one polarity while a compressive force produces a voltage of opposite polarity.
- The magnitude and polarity of the induced surface charges are proportional to the magnitude and direction of the applied force F . The polarity of induced charges depends upon the direction of applied force.

Equivalent circuit of piezo-electric transducers:

The basic equivalent circuit of a piezo-electric transducer is shown in the following figure

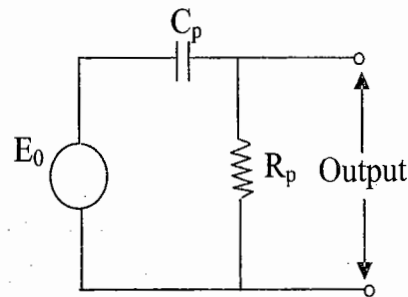


- The source is charge generator.
- The value of the charge is $Q = dF$
- The charge generated is across the capacitance C_p of the crystal and its leakage resistance R_p . The charge generator can be

replaced by an equivalent voltage source having a voltage of $E_o = \frac{Q}{C_p} = \frac{dF}{C_p}$

in series with a capacitance, C_p and resistance R_p .

The circuit is shown in the following figure.



Properties of Piezo-electric crystal

- Stability
- High output
- Insensitivity to temperature variations
- Insensitivity to variations in humidity
- Ability to be formed into a useable component

Most commonly used piezo-electric crystals:

- Quartz is the most stable piezo-electric material. However, its output is quite small.
- Rochelle salt provides highest output but it can be warm over limited temperature and humidity range.

Class Room Practice Questions

LEVEL-1 Questions

01. A piezoelectric type 50 MPa full-scale pressure sensor with built-in electronics has a sensitivity of 100 mV per MPa. If this sensor is subjected to a static pressure of 10 MPa then its output will be

- (a) 0 mV (b) 100 mV
(c) $1000/\sqrt{2}$ mV (d) 1000 mV

02. For Signal conditioning of piezoelectric type transducer we require

- (a) a charge amplifier
(b) a differential amplifier
(c) an instrumentation amplifier
(d) a transconductance amplifier

03. Consider the following statements piezoelectric materials serve as

1. A source of ultrasonic waves.
2. When electric field is applied, the mechanical dimensions of the substances are not at all altered.
3. Converts electrical energy to mechanical and vice versa
4. Converts thermal energy to electrical energy.

Which of these statements is/are correct?

- (a) 1 Only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2, 3 and 4

04. Consider the following statements: Piezoelectric materials

1. Crystal can be shown as electrical equivalent circuit similar to an inductor and a capacitor (Tank circuit)
2. Quartz, Rochelle salt, tourmaline.
3. Used in voltage stabilizers.
4. This exhibits the reverse effect electrostriction.

Which of these statements are correct?

- (a) 1, 2 and 4 only (b) 1 and 2 only
(c) 2 and 4 only (d) 1, 2, 3 and 4

LEVEL-2 Questions

1. A piezoelectric transducer is a voltage source of 10 V with an internal impedance of 10 MΩ. It is connected to a digital oscilloscope with an input impedance of 10MΩ directly. What is the voltage measured by the oscilloscope?

- (a) 20 V (b) 10 V
(c) 5 V (d) 2.5 V

2. Which one of the following materials is not used as a piezoelectric transducer?

- (a) Rochelle salt
(b) Lithium sulphate
(c) Barium titanate
(d) Tungsten oxide

3. In a piezoelectric crystal oscillator, if x represents the mass of the crystal, then the oscillation or tuning frequency is linearly proportional to

- (a) the mass of the crystal
(b) the square root of the mass of the crystal
(c) the square of the mass of the crystal
(d) the inverse of the square root of the mass of the crystal

4. Which one of the following quantities can be measured with the help of the piezoelectric crystal?

- (a) Acceleration (b) Flow
(c) Temperature (d) Velocity

5. A piezoelectric crystal having a thickness of 2mm and a voltage sensitivity of



0.02 V-m/N is subjected to a pressure of 20×10^3 pa. What is the output voltage?

- (a) 0.775V (b) 0.80 V
(c) 0.002×10^{-6} V (d) 0.2×10^{-6} V

Level - 1 Key

- 01.(a) 02. (a) 03. (b) 04. (a)

Level - 2 Key

- 01.(c) 02. (d) 03. (d) 04. (a) 05. (b)



Measurement of Temperature

The most important of the measurand variables encountered in industrial processes is Temperature. Because there are changes in the physical chemical state of most substances when they are heated or cooled. The measurement of temperature is very common in the field of instrumentation.

There are lot of devices available for the measurement of temperature such as

- Resistance thermometers
- Thermocouples
- Thermopiles
- Bimetallic thermometers
- Radiation pyrometers etc.
- Thermistors

The above mentioned devices can be classified into two groups.

- Contact Type
- Non-contact type (Radiation Type)

Contact Type:

In this category measuring device indicator is placed in direct contact with the heat source.

Ex:

- Resistance thermometers
- Thermistors
- Thermocouples
- Bimetallic Thermometers
- Thermopiles

Non-contact type (Radiation Type):

- In this category temperature measuring devices operate either on total radiation i.e. heat and light from the hot body or on the visible radiation.
- The devices operating on total radiation from the hot body are called the Radiation type

devices while those operating on visible radiation are called optical type devices.

- These temperature measuring devices are particularly suitable for the measurement of very high temperatures.

Resistance Thermometers (RTD)

- Resistance thermometer often referred to as resistance-temperature detector (RTD) operates upon the fact that almost all pure metals have the property of varying their resistance with temperature and change in resistance is almost directly proportional to the change in temperature.
- The range of temperature over which this is valid is decided by the temperature coefficient of resistance, chemical inertness and its crystal structure which should not undergo permanent changes with in this range.
- Electrical resistance with temperature, for most metallic materials can be represented by an expression.

$$R_t = R_0 [1 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \dots \alpha_n t^n]$$

Where

$R_0 \rightarrow$ Resistance in Ω at reference temperature (usually at ice point, 0°C)

$R_t \rightarrow$ Resistance in Ω at temperature t

$\alpha_1 \rightarrow$ Temperature coefficient of resistance in $\Omega/\Omega/^\circ\text{C}$

$\alpha_2, \alpha_3, \dots, \alpha_n \rightarrow$ Coefficients determined on the basis of two or more known resistance – temperature (calibration) points.

- The number of terms necessary depends on the material, the desired accuracy, and the range of operation.



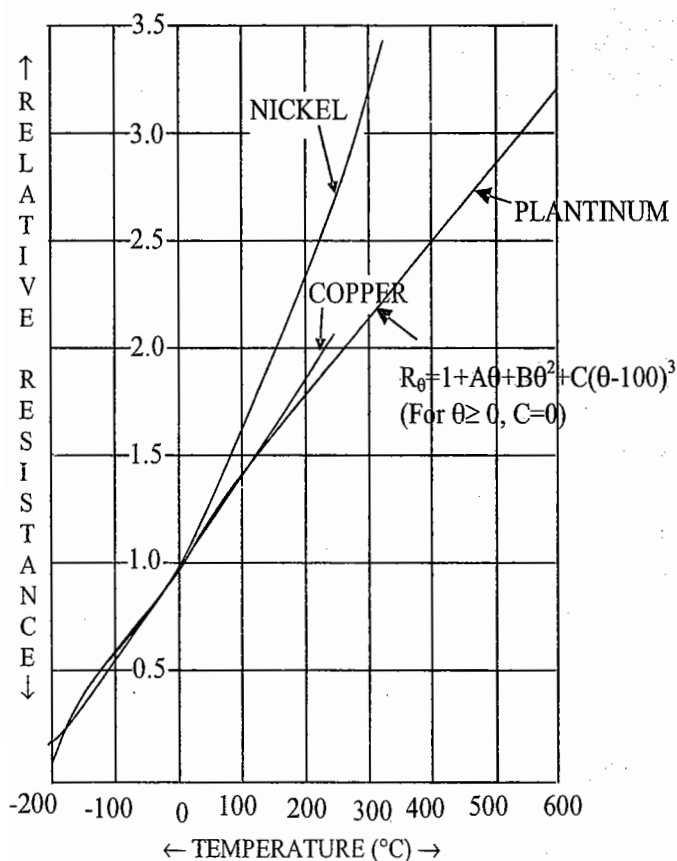
- In the narrow range of operation α_2 and higher order coefficients are negligibly small.
 $\therefore$ The expression becomes
 $R_t = R_0[1 + \alpha_1 t]$
- The RTD is applicable for measurements of small temperature difference as well as for wide ranges of temperature.
- The RTD doesn't generate its own voltage, so a voltage source is required to be incorporated into the measuring circuit.
- The main draw-back of an RTD lies in its large size and sophisticated instrumentation.
- Wheat-stone Bridges are usually employed for measurement of variations in resistance, owing to changes in temperature, arising in RTDs.

Requirements for the Resistance Materials used in RTDs:

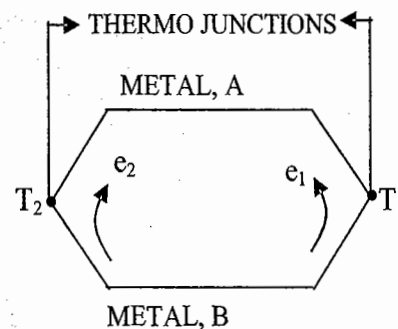
- High temperature coefficient of resistance (in order to give substantial change in resistance for a relatively small change in temperature i.e. larger sensitivity.)
- High Resistivity (to ensure small wire length for high resistance or a larger resistance for the same wire length and so increased output).
- Linearity of relation between resistance and temperature (for convenience in measurement)
- Stability of the electrical characteristics of the material and resistance to contamination for good repeatability.
- Sufficient mechanical strength (so that resistance wire could be drawn very fine for reducing response time and at the same time
- adequate ruggedness in construction is obtained).
- Platinum, Nickel, and copper are the most commonly used resistance materials.
- Gold and silver are rarely used owing to their low resistivities.
- Tungsten has relatively high resistivity, but its use is limited for high temperature applications, as it is extremely brittle and difficult to work.
- Electrolytic copper of the highest purity is readily available in the market, having a highly consistent temperature coefficient of resistance which is slightly higher than that of platinum. But its use is limited to low-range industrial applications because of low resistivity and tendency toward oxidation at higher temperatures.
- In the range below 125°C, a gold-silver alloy having the same characteristics as platinum can be employed.
- For low temperature measurements certain Phosphor-Bronze alloys are found satisfactory.
- RTDs made out of Nickel wire can be employed with good stability and repeatability in temperature range of -70°C to 150°C .
- Nickel is less expansive than platinum and has a somewhat higher temperature coefficient and higher tensile strength but it is more non-linear than copper and above 300°C , it undergoes transformations which make its resistance-temperature curve irregular.
- Platinum is the most suitable material for RTDs most laboratory work and for industrial measurements of high accuracy over a wide temperature range.
- Platinum is commercially available in a pure form is relatively stable under different environmental conditions, and its resistance temperature curve is simple and holds over a wide range of temperatures (-263°C to $+545^\circ\text{C}$) with high precision.
- Its resistivity tends to increase less rapidly at higher temperatures than for the other metals (unique quality).
- It can withstand high temperatures while maintaining excellent stability.

Materials used in RTDs:

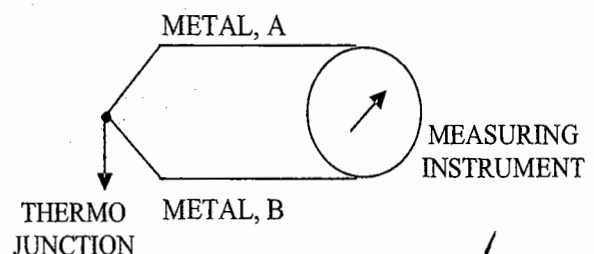
Name of the material	Temperature coefficient of resistance/ °C	Temperature range		Accuracy	Melting point
		Min	Max		
Platinum	0.00391	-250°C	900°C	± 0.5°C	1775.5
Copper	0.00425	-200°C	150°C	± 0.2°C	1083
Nickel	0.0066	-70°C	150°C	± 0.2°C	1455
Tungsten	0.0045	-200°C	1000°C	-	3370


Thermo Couples

Thermocouples are perhaps the most simple and most widely used devices for measurement of temperature. It essentially consists of two dissimilar metal wires A and B, insulated from each other but welded or brazed together at their ends forming two junctions.



When one end of each wire is connected to a measuring instrument as shown in figure below it becomes an accurate and sensitive temperature measuring device.



The operation of a thermocouple is based on seeback effect.

Seeback effect:

If two wires of different metals are joined together at each end and form a complete electric circuit then current flows in the circuit when the two junctions are kept at different temperatures.



Thermo electric emf:

It is the emf which is responsible for current flow between two dissimilar metals in a thermocouple this emf is generated when junction temperatures difference should greater than zero.

The relation between the thermo-emf setup and temperature difference of hot and cold (or reference) junction is given by

$$E = \alpha (T - T_0) + \beta (T^2 - T_0^2)$$

Where,

E – thermo electric emf (volts)

T – absolute temperature of hot junction

T_0 – absolute temperature of cold junction

α, β - constants (depends upon the metals used)

∴ Thermo-electric emf is a function of temperature difference.

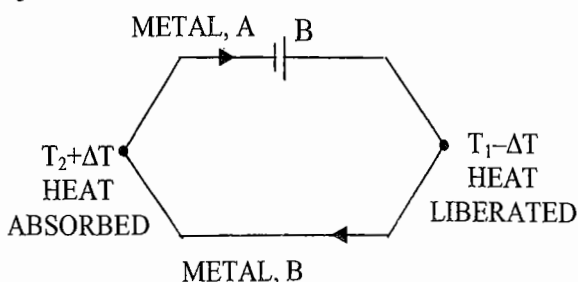
The emf developed by thermocouple depends only on material used and temperature of hot and cold junction. Not depends on shape, size length and area of material used and also method of joining them.

Reverse of seeback effect is also true it is explained Peltier effect.

Peltier effect:

Peltier observed that when an electric current flows across a junction of two dissimilar metals heat is either generated (liberated) or absorbed, depending upon the direction of flow of current.

When the electric current flows in the same direction as the seeback current, then that is absorbed at the hot Junction and liberated at the cold junction.



Another Reversible heat – flow effect is the Thomson effect.

Laws Governing thermocouples:

- Thermocouple circuits are normally required to be broken for introducing a galvanometer type instrument to measure the current or a known emf to balance the generated thermo-electric emf.
- In doing this additional metals and junctions are introduced, which may be at various temperatures therefore the following laws summarize the effects of these.

Law of Homogenous Metals:

A thermoelectric current cannot be sustained in a circuit of a single homogeneous material, however varying in cross section, by the application of heat alone. A consequence of this law is that two different materials are required for any thermocouple circuit.

Law of intermediate metals:

This law states that a third homogeneous material can always be added to a circuit with no effect on the net emf of the circuit as long as its extremities are at the same temperature.

This law makes it possible to use extension wires of a metal different from the thermocouple materials.

Law of intermediate temperatures:

If two dissimilar homogenous metals produce a thermo electric emf of E_1 when the junctions are at temperatures T_1 and T_2 , and a thermoelectric emf of E_2 when the junctions are at T_2 and T_3 , the emf generated with the junctions at temperatures T_1 and T_3 will be $E_1 + E_2$.

The application of this law permits a thermocouple calibrated for a given reference temperature to be employed with any other reference temperature through the use of a suitable correction.



Properties of thermocouples:

Type (ISA Standard)	Metal Alloys Used		Sensitivity in $\mu\text{V}/^\circ\text{C}$	Accuracy in $^\circ\text{C}$	Temperature Range in $^\circ\text{C}$	Spot Maximum in $^\circ\text{C}$
	Positive Wire	Negative Wire				
Type E	Chromel; 90% Nickel and 10% Chromium	Constantan; 57% Chromium and 43% Nickel	40 to 55	$\pm 1.8^\circ$ $\pm 0.5\%$	-200 to 850	1100
Type K	Chromel; 90% Nickel and 10% Chromium	Alumel 94% Ni, 3% Mn; 2% Al and 1% Si	40 to 55	$\pm 2.2^\circ$ $\pm 0.5\%$	-200 to +1100	1300
	Iron	Constantan; 57% Chromium and 43% Nickel	45 to 57	$\pm 2.2^\circ$ $\pm 0.5\%$	-200 to +850	1100
Type T	Copper	Constantan; 57% Chromium and 43% Nickel	15 to 60	$\pm 1.8^\circ$ $\pm 0.5\%$	-200 to +350	500
Type R	Platinum Rhodium; 87% Platinum and 13% Rhodium	Platinum	5 to 12	$\pm 0.25^\circ$	0 to 1000	1200
Type S	Platinum Rhodium 90% Platinum and 10% Rhodium	Platinum	5 to 12	$\pm 0.25^\circ$	450 to 1500	1650
	Rhodium – Iridium 60% Rhodium and 40% Iridium	Iridium	5	$\pm 1.0^\circ$	1200 to 2000	

Merits:

- Thermocouples are cheaper than resistance thermometers.
- The temperature range of thermocouples is about 1400°C . Cooled thermocouples can operate upto 4000°C .
- Except for the refractory group (tungsten-tungsten-rhenium) thermocouples are rugged in construction and can withstand high shock
- Thermocouples can be made in very small sizes.
- Thermocouples can be made from wire pairs as 0.013mm in diameter for millisecond

response. Special high speed thermocouples have time constant as small as $10\mu\text{s}$.

- Thermocouples can be made from heavy-gauge wire or even round stock to withstand the most severe applications.

Demerits:

- Thermocouples have less accuracy than RTDs (± 0.25 to $\pm 1\%$ reading depending on type and temperature range)
- Thermocouples need periodical checking when employed at elevated temperatures.



- Thermocouples need compensating leads as these are placed remote from measuring devices.
- Instrumentation must resolve terms of micro volts per degree of temperature.
- Reference junction compensation is required in thermocouples.

Applications:

- Thermocouple pyrometers, like resistance pyrometers can be applied to measurements at a number of points.
- Thermocouple, because of their small heat capacity and very small bulk are particularly suitable for indication of rapidly changing temperature and for use in localized and in otherwise accessible positions.
- Thermocouple particular application is in the measurement of surface temperature.

Thermistor

- Thermistors are also called the thermal resistors and the name is derived from thermally sensitive resistors, as the resistance of a thermistor varies as a function of temperature.
- Thermistors are essentially semiconductor devices, that behave as resistors with high negative temperature coefficient (usually - 0.04% at room temperature of 25°C) and are at least 10 times as sensitive as the platinum resistance thermometer.
- Thermistors are both positive temperature coefficient and negative temperature coefficient types, but most thermistors are negative temperatures coefficient types.
- Thermistors can detect very small changes in temperature which could not be observed with an RTD or a thermocouple.
- Sensitivity → Thermistor > RTD > Thermocouple

- Thermistors are widely used in applications which involve measurements in the range of - 60°C to 15°C.
- The resistance of thermistors ranges from 0.5Ω to 0.75Ω.
- Thermistors exhibits a highly non-linear characteristic of resistance versus temperature.
- The resistance R of a thermistor at a temperature T can be given by expression.

$$R = \alpha e^{\beta/T}$$

- Where α , β are constants depending upon the material and manufacturing techniques.
- The approximate relationship between resistance and temperature applying to most thermistors can be obtained by re-writing above equation (i) for temperatures T_1 and T_0 .

$$R_1 = R_0 e^{\beta(1/T_1 - 1/T_0)}$$

$\beta \rightarrow$ Thermistor constant, expressed in Kelvin and is of the order of 4000.

- $R_1, R_0 \rightarrow$ Resistance at absolute temperatures T_1 and T_0 .
- The assumption that β is constant gives a relation for the resistance temperature coefficient which is given as Temperature coefficient resistance

$$\alpha = \frac{dR_1}{dT} / R_1 = \frac{-\beta}{T^2}$$

- **Note:** From above equation we know that α is not constant. It actually drops with rise in temperature.

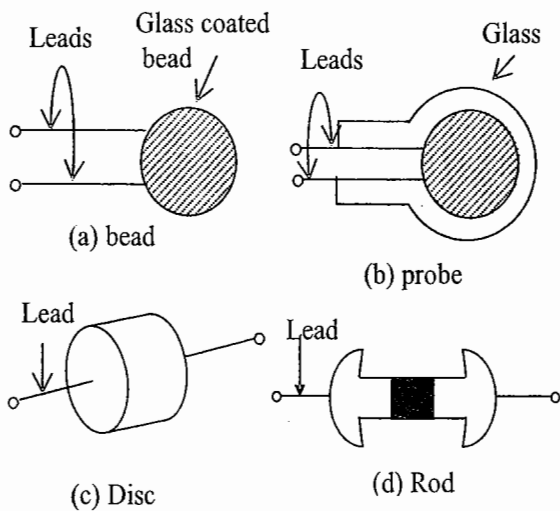
Ex: If $\beta = 4000$ and $T = 25^\circ\text{C}$ or 298 K then temperature coefficient of resistance

$$\alpha = \frac{-4000}{(298)^2} = -0.045 \Omega/\Omega/^\circ\text{C}.$$

Materials used for manufacturing thermistors:

Modern thermistors are manufactured from the oxides of metals like manganese, nickel, cobalt, copper, iron, zinc, aluminium titanium, magnesium and uranium.

Because of their high resistance they can be manufactured in very small sizes in the form of beads, discs and rods.



Note:

- If the thermistor is instead put into a wheat stone bridge the measuring system can detect temperature variations as small as $\pm 0.005^\circ\text{C}$.
- High resistance temperature coefficient characteristic of them results in inherent high sensitivity and high level of output, eliminating the need for extremely sensitive read out devices and lead wire matching techniques respectively.
- The high degree of non-linearity in the resistance-temperature function usually limits the range of the readout instrumentation.
- Calibration, stability and limitations an inter changeability are inherent to thermistors.
- Parallel and series resistors are employed for reducing non-linearity of thermistors and make thermistors easier to use over reasonably wide temperature range.
- The time constant varies directly with the thermal capacity of the thermistor and inversely with dissipation factor.

Features of thermistors:

- They are compact, rugged, and inexpensive.
- When properly aged they have good stability.
- The response time of thermistors can vary from a fraction of second to minutes depending on the size of the detecting mass and thermal capacity of the thermistor.
- The upper operating limit of temperature for thermistor is dependent on physical changes in the material or solder used in attaching the electrical connections and is usually 400°C or less.
- Thermistors can be installed at a distance from their associated measuring circuits if elements of high resistance are used such that the resistance of leads does not affect the readings and hence errors on this count are negligible.
- The average power dissipation can be effectively reduced and the highest sensitivity retained by energizing the thermistor with pulses of measuring power.

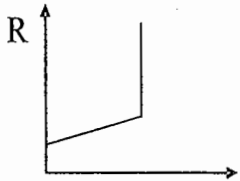
Applications:

Thermistors have a wide range of applications with the most common being direct temperature sensing and surge suppression. The characteristics of NTC and PTC thermistors lend themselves to applications including.

- Liquid level indicators
- Temperature compensation
- Flow measurement.
- Vacuum gages
- Thermal protection
- Amplifier-gain control
- Time delay circuits.
- Thermal switches etc....
- Measurement of gas composition.
- Measurement of difference of two temperatures.

Class Room Practice Questions

LEVEL-2 Questions

01. Which of the following metals is to be preferred in a resistance-thermometer employed for measuring temperature in the range -200°C to 1000°C ?
- (a) Platinum (b) Copper
(c) Nickel (d) Tungsten
02. The RTDs use the principle of change of resistance with temperature. The properties of a conductor material to be used as an element of an RTD should possess the following properties
- (a) The change in resistance per unit change in temperature should be as small as possible
(b) The resistance of the materials should not have continuous and stable relationship with temperature
(c) The change of resistance with temperature should not be a linear function
(d) None of the above
03. Platinum is the commonly used metal for RTDs. This is because
- (a) Platinum has a constant value of resistance temperature co-efficient of $0.004/^{\circ}\text{C}$ for a temperature range between 0 to 100°C .
(b) The resistivity of platinum tends to increase less rapidly at higher temperatures
(c) Platinum is available in pure form for commercial applications and has a stability over higher range of temperature
(d) All of the above
04. The resistance temperature characteristics of a temperature transducer is shown in Fig. The transducer is,
- (a) Nickel RTD
(b) NTC thermistor
(c) Platinum RTD
(d) PTC thermistor
- 
05. Thermistors are essentially semi-conductors
- (a) Well suited to precision measurement of temperature
(b) Widely used in the lower temperature range of -100°C to 300°C
(c) Which behave as resistors with a high negative temperature coefficient of resistance
(d) All of the above
06. A thermistor exhibits
- (a) only a negative change of resistance with increase in temperature
(b) only a positive change of resistance with increase in temperature
(c) can exhibit either a negative or positive change of resistance with increase of temperature depending upon the type of material used
(d) none of the above
07. Thermistors are:
- (a) bi-metallic strips used in thermometers
(b) semi-conductors which have negative coefficient of resistance
(c) metals having small positive coefficient of resistance
(d) transducers used for measuring the thermal properties of materials
08. Thermocouples
- (a) are most commonly used temperature transducers



- (b) require reference junction compensation
(c) have a low output voltage level
(d) all of the above
09. Which one of the following thermocouples has the highest temperature measuring range?
(a) Copper – Constantan
(b) Iron – Constantan
(c) Alumel – Chromel
(d) Platinum Rhodium-Platinum
10. A thermo-couple arrangement is to be used to measure temperature in the range of 700-800°C. Which one of the following pairs would be most suitable?

- (a) Copper-constantan
(b) Iron-constantan
(c) Chromel-alumel
(d) Platinum – Platinum rhodium

Level – 2 Key

01. (a) 02. (d) 03. (d) 04. (d)
05. (d) 06. (c) 07. (b) 08. (d)
09. (a) 10. (b)

5

Measurement of Flow & Viscosity

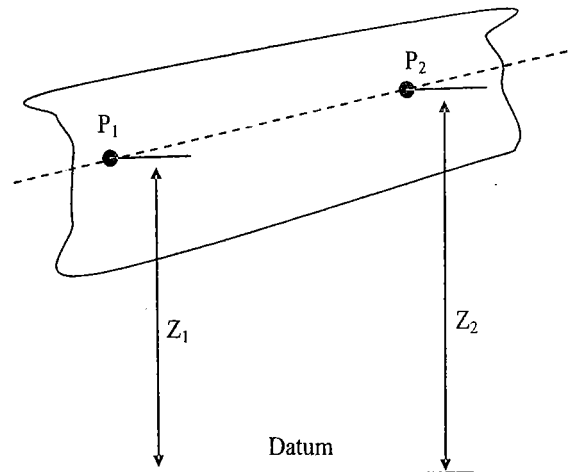
Theory of Variable Head Meters

- The head meter in the form of venturimeter, orifice and flow nozzle is by far the most common flow meter for closed conduits (i.e., pipes). When the fluid flows through these obstruction meters, the fluid accelerates and a reduction in pressure occurs. The difference in pressure before and after the obstruction is measured by means of a differential pressure sensor and is related to the flow rate.
- The equations convenient for practical use and providing the basis of operation of these meters can be derived by following the under given three steps:
 - Write the general energy equation and then modify it for the assumed conditions.
 - Solve the modified equation for down stream velocity.
 - Express the flow rate by the product of velocity and area, i.e.,

$$Q = AV$$
- The equations thus developed will apply equally to any variable head meter. However the individual factors and coefficients differ.
- The energy at any section comprises the internal energy (u), flow work (pv or p/ρ), kinetic energy ($v^2/2$) and gravitational potential energy (gz). Further during the fluid flow between the two chosen sections, heat (q) and mechanical or shaft work (w_s) may also cross the boundaries of the system. Since energy can neither be created nor destroyed, in symbolic form we can write:

$$m_1 \left(u_1 + \frac{P_1}{\rho_1} + \frac{V_1^2}{2} + gz_1 + q \right) = m_2 \left(u_2 + \frac{P_2}{\rho_2} + gz_2 + w_s \right)$$

- Each term in the brackets is expressed per unit mass.
- The symbols used in equation have the following meanings and units:
 - m = fluid mass flow in certain time,
 - u = internal energy,
 - p = static pressure
 - v = specific volume,
 - V = average flow velocity,
 - Z = elevation of centre of fluid above datum,
 - g = local acceleration due to a gravity
 - q = heat entering the system
 - W_s = work output from the system



Flow through a converging section

- Subscript 1 refers to the pipe and the fluid at the upstream pressure connection, subscript 2 refers to the pipe and the fluid at the section of minimum area as indicated in Fig., In actual use, it may not be possible to measure area, pressure, etc., at the smallest section of the stream. In that case subscript 2 refers to the smallest measurable area and to the conditions of fluid at the actual pressure connections. The corrections are then applied



by using a factor called coefficient of discharge.

- We shall make the following assumptions for ease of analysis and for getting simplified results:

- The fluid flow process is adiabatic, i.e., there is no transfer of heat across the system boundaries ($q = 0$)
- no mechanical work is done on or by the fluid ($w_s = 0$)
- the process is frictionless, i.e., $u_1 = u_2$
- the measuring device is horizontal, i.e., the points 1 and 2 are at the same horizontal level ($z_1 = z_2$) and gravity force can be neglected
- there is steady flow of fluid, i.e. the mass flow rate at section 1 is equal to that at section 2, ($m_1 = m_2$)

With these assumptions, the steady flow energy equation can be written as:

$$\frac{P_1}{\rho_1} + \frac{V_1^2}{2} = \frac{P_2}{\rho_2} + \frac{V_2^2}{2} \dots\dots\dots(1)$$

(or)

$$\frac{V_2^2 - V_1^2}{2} = \frac{P_1}{\rho_1} - \frac{P_2}{\rho_2} \dots\dots\dots(2)$$

- The above equation forms the basis of operation of variable head flow measuring devices and is valid both for liquids and ideal gases.
- Further, replacing the density ρ by specific weight by using the relation $w = \rho g$, we get

$$\frac{P_1}{w_1} - \frac{P_2}{w_2} = \frac{V_2^2 - V_1^2}{2g} \dots\dots\dots(3)$$

- This is well-known Bernoulli's equation, each term has units of length m.

Variable head-meters for incompressible fluids

The continuity relation for the flow situation is

$$m = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Where $\rho_1 = \rho_2 = \rho$ for incompressible fluid flow from section 1 to section 2. This yields

$$V_1 = (A_2/A_1)V_2 \dots\dots\dots(4)$$

Solution of equations (1) and (4) gives out flow velocity

$$V_2 = \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}} \dots\dots\dots(5)$$

- The downstream fluid velocity V_2 as given by equation shown above has been derived without considering any losses and so it is the ideal or hypothetical velocity at the minimum section. The actual velocity can be obtained by multiplying the theoretical velocity by a factor C_v called the coefficient of velocity. The coefficient of velocity is the ratio of actual mean velocity to the ideal mean velocity which would occur without any friction loss.

$$C_v = \frac{\text{actual mean velocity}}{\text{ideal mean velocity}}$$

$$\therefore V_2(\text{actual}) = C_v \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

- For obtaining the volume flow rate, we apply the continuity equation.

Discharge = area $\times$ velocity

$$\therefore Q = A_2 C_v \frac{1}{\sqrt{1 - (A_2/A_1)^2}} \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

- During flow through an orifice meter, the fluid jet on leaving the orifice contracts to minimum area at a section called vena-contracta. Area of fluid jet at vena contracta is less than area of the orifice and the two are related by the equation:



- Area of jet at vena contracta = $C_c \times$ orifice area
- Where C_c is the co-efficient of contraction. Thus if the orifice area is A_2 , then the area at minimum section which controls the flow rate and where the path of particles becomes parallel again would be $C_c A_2$.

$\therefore$ Actual discharge

$$Q = (C_c A_2) C_v \frac{1}{\sqrt{1 - (A_2 / A_1)^2}} \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

- For venturi meter and flow nozzle, there is almost no formation of vena contracta and the coefficient of contraction can be taken as unity.
- Combining C_v and C_c into a single factor C_d called the coefficient of discharge, the volumetric flow rate through the meter can be written as

$$Q = C_d \frac{A_2}{\sqrt{1 - (A_2 / A_1)^2}} \sqrt{\frac{2(p_1 - p_2)}{\rho}} \dots \dots \dots (6)$$

- The discharge coefficient C_d is not constant; it depends primarily on the flow Reynolds number and the channel geometry.
- For given meter, A_1 and A_2 have definite values and so the quantity $\sqrt{1 - (A_2 / A_1)^2}$
- Is constant for the meter. This parameter allows for the additional kinetic head of the approaching fluid and is called velocity of approach factor ; its value is always greater than unity. For upstream velocity V_1 equal to zero, the velocity of approach factor $M=1.0$.

$$Q = C_d M A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

- For venture meter, the factors C_d and M are applied separately but for nozzles and orifice plates these factors are combined into a single

factor known as the flow coefficient and designated as K .

$$\therefore Q(\text{actual}) = C_d M A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

For venturi meter

$$= K A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

For nozzles and orifice plates

- The area A_2 is usually determined from measurements obtained at room temperature but in use this area is influenced by the temperature of the fluid flowing through the meter. The actual discharge is essentially worked out from the relation

$$Q(\text{actual}) = C_d M A_2 E \sqrt{\frac{2(p_1 - p_2)}{\rho}} \text{ for}$$

venturi meter.

$$= K A_2 E \sqrt{\frac{2(p_1 - p_2)}{\rho}}$$

for nozzles and orifice plates.

- Where E is thermal expansion correction factor. If the density is replaced by specific weight w , equation can be written as:

$$Q(\text{actual}) = C_d M A_2 E \sqrt{\frac{2g}{w} (p_1 - p_2)}$$

for venturi meter

$$= M A_2 E \sqrt{\frac{2g}{w} (p_1 - p_2)}$$

for nozzles and orifice plate.

- The pressure differential $\frac{(p_1 - p_2)}{w}$ called pressure head or piezometer head, is measured by a differential manometer, pressure gauge or any other pressure measuring device. The gauge reading is same for a given discharge irrespective of the inclination of the pipe line. This can be shown by considering the measuring device laid in horizontal, inclined or vertical position.



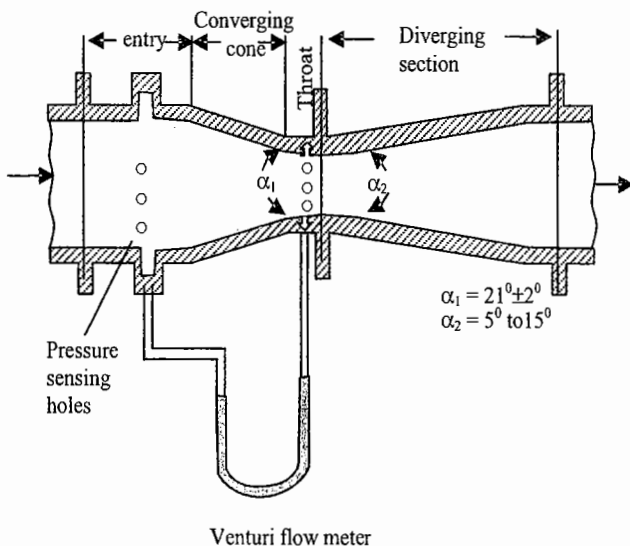
Venturi flow meter

- This simple and reliable device finds an extensive use for water flow measurement, particularly in large size pipes and for large flow rates.

The important constructional features of a venturi meter are shown in the meter consists of

Cylindrical entrance section:

- This section has the size of the pipe to which it is attached. For satisfactory operation, the venturi meter should be preceded by a straight pipe of not less than 5 to 10 pipe diameters and free from fittings, misalignment and other sources of large scale turbulence. If these conditions cannot be met, straightening vanes should be placed upstream from the meter for reduction of rotational motion in flow.



Venturi flow meter

Converging conical section

- The converging takes place at an angle of $21^\circ \pm 2^\circ$. The velocity of fluid increases as it passes through the converging section and correspondingly the static pressure falls.

Throat:

- This is a cylindrical section of minimum area. At this section the velocity is maximum and the pressure is minimum. The throat diameter is usually between $\frac{1}{2}$ to $\frac{1}{4}$ of the inlet diameter. The length of the throat equals its diameter

Diverging section:

- in which there is a change of stream area back to the entrance area. The recovery of kinetic energy by its conversion to pressure is nearly complete and so the overall pressure loss is small. To accomplish a maximum recovery to kinetic energy, the diffuser section is made with an included angle of 5° to 7° . The angle has to be kept less so that the flowing fluid has least tendency to separate out from the boundary of the section. However, with small angles, the length and hence the cost of the meter would increase. So where pressure recovery is not of much importance, the angle of the diverging cone may be kept as high as 14° .
- The pressure recovery is not of much importance, the angle of the diverging cone may be kept as high as 14° .
- The pressure taps are made at the throat and at entrance where the venturi has a diameter equal to that of pipe. The pressure taps may be made from a single hole at piezometer rings thereby giving average pressure values at the two sections.
- The small sized venturi meters, suitable for pipe lines less than 5cm in diameter, are usually made of brass or bronze. The inside surface is smoothly finished to reduce friction. Large venturies are usually made of cast iron; the throat is however lined with brass or bronze and machined to a smooth finish. Very large venturies upto 6m pipe



diameter have been made of smooth surface concrete, the throat only being of machined bronze.

Advantages and limitations:

- High pressure recovery is attainable, i.e., loss of head due to its installation in the pipe lines is small.
- Because of smooth surface, the meter is not much affected by wear and abrasion.
- Well established characteristics; years of applicational experience.
- Less likelihood of becoming clogged with sediments.
- Due to low value of losses, the co-efficient of discharge is high and approaches unity under favourable conditions
- Long laying length; space requirements are more
- Quite expensive in installation and replacement
- Difficult to construct particular so as to provide reproducibility.

Venturi meters are ideally suited for measurement of :

- Large flow of water
- Process fluids, i.e., fluids used in industrial processes
- Industrial wastes
- Gases
- Suspended particles, slurries and dirty liquids
- Venturimeter are not standardized yet to an extent that permits discharge coefficient obtained conditions, the discharge coefficient for a venturimeter is found to be function of Reynolds number and meter size. These meters are not generally useful below 7.5cm pipe diameter.

Example 1:

A horizontal venturimeter with 15cm inlet and 7.5cm throat is used for measuring the flow of cooling water in a manufacturing process. The differential pressure between the inlet and throat is 17.5cm when measured using a mercury U-tube manometer. Make calculation for the water flow rate if the discharge coefficient for the venturimeter is 0.97. (Density of water = 1000kg/m^3 and specific gravity of mercury = 13.6).

Ans: = $0.02915\text{ m}^3/\text{sec}$

Sol: given

$$D_1 = 15\text{ cm}$$

$$A_1 = 0.017\text{ m}^2$$

$$D_2 = 7.5\text{ cm}$$

$$A_2 = 0.0044\text{ m}^2$$

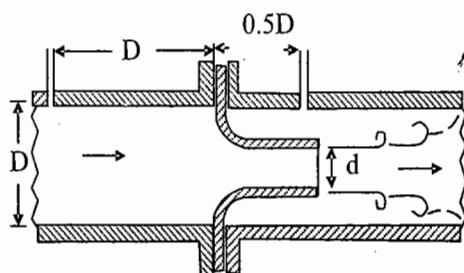
$$C_d = 0.97$$

We know

$$\begin{aligned} Q &= C_d \cdot M \cdot E \sqrt{2g(sg-1) \times h_m} \\ &= 0.97 \times \frac{0.017 \times 0.0044}{\sqrt{(0.017)^2 - (0.0044)^2}} \\ &\quad \times \sqrt{2 \times 9.8 \times 12.3 \times 17.5 \times 10^{-12}} \\ &= 0.02915\text{ m}^3/\text{sec} \end{aligned}$$

Flow Nozzle

- Nozzles are used in engineering practice for the creation of jets



Flow nozzle

and streams for all purposes as well as for fluid metering. When placed in or at the end of a pipe line as metering devices, they are called flow nozzles.



- The flow nozzles comprises a smooth, gradual contraction to throat followed by a free, uncontrolled expansion back to the original pipe flow area. Because the device contains no provision an orderly transformation of velocity pressure into static pressure, the nozzle has a pressure loss of from 80 to 90% of the differential pressure obtained

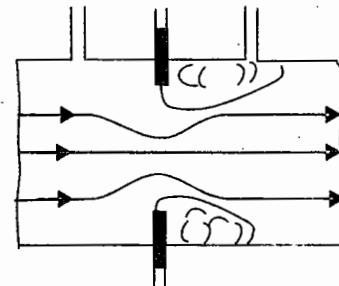
Advantages and Limitations:

- Cheaper than a standard venturi meter and can be installed in an existing main without great difficulty.
- Increased coefficient of discharge when compared with orifice.
- Physical length is less compared to venturi meter.
- Widely accepted for high-pressure/temperature steam flow.
- Good for fluid containing solids that settle.
- Discharge of liquids into the atmosphere can be metered. The down stream pressure p_2 in that case is atmospheric pressure and we need
- to measure the difference between upstream pressure p_1 and atmospheric pressure.
- Pressure recovery is poor and so cannot be used where available pressure head is small or where pressure recovery is a must. The pressure loss is, however, less than that of an orifice plate.
- Compared to orifice meter, it is expensive and also difficult to install.
- Limited to moderate pipe size ; not available above 120cm.

Orifice flow Meter

- The orifice meter consists of a thin, circular metal plate with a hole in it. The plate is held in the pipe line between two flanges called orifice flanges. The flow characteristics of the

orifice differ from those of a nozzle in that minimum section of the stream tube occurs not within the orifice but downstream from the orifice edge. This minimum area is called vena contracta and it is at this area where minimum pressure is obtained. It is not feasible to locate the down stream pressure connection at the vena contracta because the location of vena-contracta depends on Reynolds number, area



Orifice plate

- Ratio between orifice and pipe, the roughness of the pipe and the compressibility of the fluid. It is also sensitive to the upstream velocity profiles and to the sharpness of the upstream edge of the orifice plate. Accordingly downstream pressure tapping is taken at a fixed proportion of the pipe diameter and correction for vena -contracta is made.

Advantages and limitations

- Low initial ease of installation and replacement.
- Characteristics are well known/ and predictable ; years of applicable experience.
- Requires less space as compared with venturi meter
- Can be used in wide range of pipe size (1.25cm to 150cm)
- Pressure recovery is poor. The overall pressure loss varies from 40 to 90% of the differential pressure.
- Necessity of providing straightening vanes upstream.



- Coefficient of discharge has a low value.
- Success of providing straightening vanes upstream.
- Tends to clog; not suitable for slurries or entrained particles.
- More likely to be damaged by pressure transient because of its lower physical strength
- The wide application of the variable head meters is due to following reasons:-
- Possibility to meter all types of liquids
- No moving parts immersed in the liquid being metered; mechanical wear and tear is practically absent.
- Possibility to build the meter to suit any shape of the conduct which need not be horizontal or vertical
- No limit to the maximum flow that can be metered.
- Possibility of indicating, diagrammatic record and integration of the flow rate.
- Possibility to transmit the measured signal over long distance.

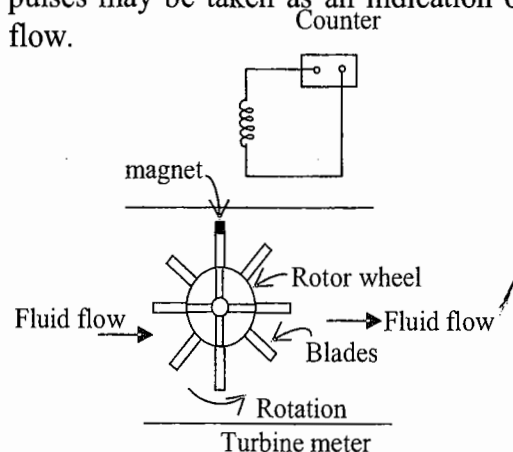
Pitot tubes, anemometers current meters:

- Flow velocity constitutes an important parameter in kinematics and dynamics of fluid flow and, therefore, its measurements is quite essential. Velocity measurement may be made with a view to :
- Determine the volumetric flow rate.
- Forecast the weather from wind velocity measurement.
- Locate the separation points from a knowledge of velocity distribution.
- Flow velocity measurements may be classified as:
 - Instruments like the pitot-tube and hot wire anemometers which measure the local velocity at a point in the channel through which the fluid is flowing

- Instruments like the cup and vane anemometers, the current and turbine meters which measure the average velocity of flow.

Turbine flow Meter

- It consists of a freely rotating wheel (rotor or propeller) with multiple blades. The rotor is supported by ball or sleeve bearings, and is located centrally in the pipe along which the flow occurs. The flowing fluid impinging on the turbine blades imparts a force on the blade surfaces and sets the rotor in motion with an angular speed proportional to the fluid velocity. The rotor speed is measured with a mechanical counter or with an electro-magnetic pick up and associated counter. The electro-magnetic pick up could be a small permanent magnet mounted at the tip of one of the rotor blades with a coil being placed just outside the tube. As the magnet moves past the coil, an e.m.f. is induced. The voltage pulse is picked up for each rotation of the wheel. Faster the fluid flow, the greater the count per second. Each pulse represents a definite flow quantity, and the total number of pulses may be taken as an indication of total flow.



$$Q = \frac{f}{K}$$

- Where f is the pulse frequency, Q is the volume flow rate and K is the flow coefficient

that depends on the flow rate and the viscosity. By minimizing the bearing friction and other losses, the device can be designed to give a linear output.

Advantages and limitations:

- Good accuracy, excellent repeatability and wide range commencing from 0.5 lpm to 15,000 lpm for liquids and 3 lpm to 500,000 lpm for air
- Low pressure drop, good temperature and pressure ratings
- Easy to install and maintain
- Adaptability to flow totalizing and to digital blending systems
- Good transient response
- Relatively high cost and limited use for slurry applications
- Requires in-line mounting and a straight run of pipe (15 diameter) ahead of the meter.

Rota Meter

The Rota meter consists of a tapered metering glass tube, inside which is located the rotor or active element (float) of the rotor. This tapering tube is provided with suitable inlet and outlet connections. The float or bob material has specific gravity higher than that of the fluid to be metered.

- With increase in the flow rate, the float rises in the tube and there occurs an increase in the annular area between the float and the tube. The float adjusts its position in relation to discharge through the passage i.e., the float rises higher or lower depending on the flow rate. The discharge equation for flow through a Rota meter is given by

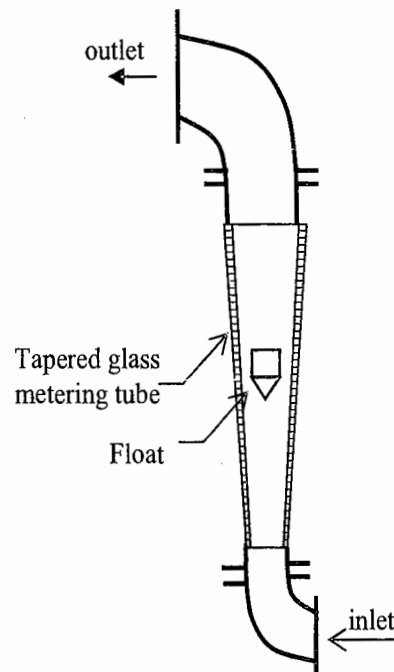
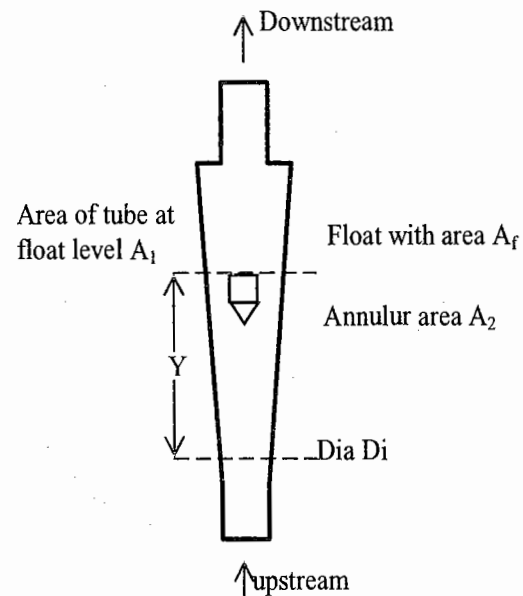
$$Q = C_d A [2gV_f (\rho_f - \rho) / A_f \rho]^{\frac{1}{2}}$$

- Where Q is the volume flow rate, C_d is the discharge coefficient, V_f is the volume of

float, ρ_f is the density of float material, ρ is the density of fluid flowing, A_f is the cross-sectional area of the float and A is the annular area between the float and tube.

- The relationship between discharge and the float position can be worked out by carrying out the following analysis.

Variable-area flow meters





Weight of float = $\rho_f V_f g$

Buoyant force on the float = $\rho V_f g$

- Drag force on the float = $A_f (p_1 - p_u)$ where p_1 and p_u are the pressure on the lower (upstream) and upper (downstream) face of the Rota meter.
- For static equilibrium of float at any position, there is balance between the drag force, buoyant force and weight of float.

That is $A_f (p_1 - p_u) + \rho V_f g = \rho_f V_f g$

That gives $(p_1 - p_u) = \frac{V_f}{A_f} g(\rho_f - \rho)$

- The two surfaces of the float form a kind of constriction (obstruction) and accordingly

$$Q_{act} = C_d \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} \sqrt{2g p_h}$$

Where p_h = Piezometric head

$$= \sqrt{\frac{p_1 - p_u}{\rho g}} = \sqrt{\frac{V_f (\rho_f - \rho)}{A_f \rho}}$$

$A_1 = A_t$ = area of tube at float level

$A_2 = A_t - A_f$ = minimum annular area between the tube and float

C_d = coefficient of discharge

Therefore

$$Q_{act} = C_d \frac{A_t (A_t - A_f)}{\sqrt{A_t^2 - (A_t - A_f)^2}} \sqrt{2g} \times \sqrt{\frac{V_f (\rho_f - \rho)}{A_f \rho}}$$

- If the discharge coefficient is assumed to remain constant with the position of float and further if

$(A_t - A_f)/A_t \ll 1$, then $Q_{act} = K(A_t - A_f)$

Where

$$K = 2g \times \sqrt{\frac{V_f}{A_f} \left(\frac{\rho_f - \rho}{\rho} \right)}$$

is the Rota meter constant.

Let θ be the taper of the Rota meter tube, then

$$A_t = \frac{\pi}{4} (D_i + y \tan \theta)^2 = \frac{\pi}{4} D_i^2 + \frac{\pi}{2} y D_i \tan \theta$$

- Where D_i is the diameter at inlet and y is the float position with respect to inlet

$$\begin{aligned} Q_{act} &= K \left[\frac{\pi}{4} D_i^2 + \frac{\pi}{2} y D_i \tan \theta - A_f \right] \\ &= K \frac{\pi}{4} y D_i \tan \theta + K \left[\frac{\pi}{4} D_i^2 - A_f \right] \\ &= K_1 y + K_2 \dots \dots \dots (13) \end{aligned}$$

- The constants K_1 and K_2 depend upon the shape of Rota meter and the float. Equation(13) shows that there exists a liner relationship between the flow rate and displacement of float in case of a Rota meter..
- The glass tube is often made of high strength borosilicate glass and the flow rate scale is engraved on the tube corresponding to a particular float material. However, where greater strength is required, the metallic tubes are used and the position of float is detected magnetically. Further, for remote indication the position of float can be monitored electrically by using a suitable displacement transducer.
- The Rotameter is equally suitable for measuring both gas and liquid flows. When metering gas flows a small sphere is used in a narrow tube and no guidance needs to be provided to its motion. While metering liquid fluids, spherical slots are cut no a part of the float and these slots cause it to rotate slowly about the axis of tube and keep it central. This spinning helps also to prevent accumulation of any sediment on the top and sides of the float. Alternatively, stability of the bob is

ensured by employing a guide along which the float would slide.

- The float or bob material has a specific gravity higher than that of the fluid to be metered. The density difference ($\rho_f - \rho$) required for metering a particular liquid or gas can be obtained by selecting different materials for the float. An examination of the discharge equation would reveal that when $\rho_f \gg \rho$, the volume flow rate becomes independent of the density ρ of the flowing fluid.
- Rota meters are widely used for metering purge flows, pump-seal fluids and coolants and lubricants for operating machinery. In these applications, flows are relatively small and accuracy requirement are not rigid.

Advantages and limitations

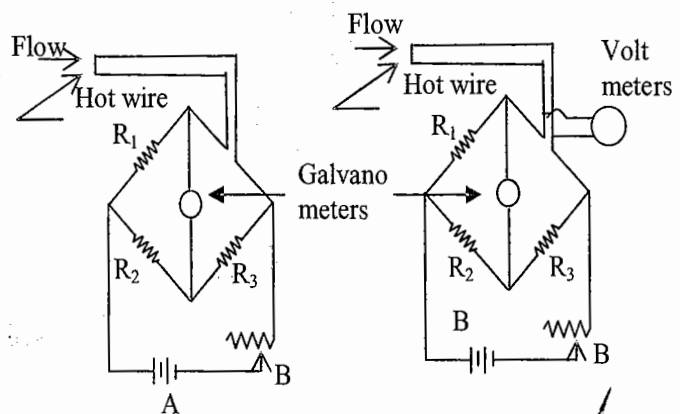
- Simplicity of operation, ease of reading and installation
- Relatively low cost
- Handles wide variety of corrosive fluids
- Easily equipped with data transmission, indication and recording devices
- Possibility of convenient and visible flow comparisons by mounting several Rota meters side by side
- Glass tube subject to breakage
- Limited to small pipe sizes and capacities
- Less accurate compared to venturi and orifice meters
- Must be mounted vertically
- Subject to oscillations in pulsating flows

Hot wire anemometer

- Hot-wire anemometry has been considered as satisfactory approach to the measurement of the mean and fluctuating flow components in a flow field. The sensor is a 5 micron

diameter platinum-tungsten wire welded between the two prongs of the probe and heated electrically as part of a wheat-stone bridge circuit,. When the probe is introduced into the flowing fluid, it tends to be cooled by the instantaneous velocity and consequently there is a tendency for the electrical resistance to diminish. The rate of cooling of wire depends upon the

- Dimensions and physically of wire,
- Difference of the temperature between the wire and the fluid,
- Physically properties of the fluid, and
- Stream velocity under measurement. For a simple hot wire anemometer the first three conditions are effectively constant and the instrument. For a simple hot wire anemometer the first three conditions are effectively constant and the instrument response is then a direct measurement of the velocity.



Hot wire bridge circuits

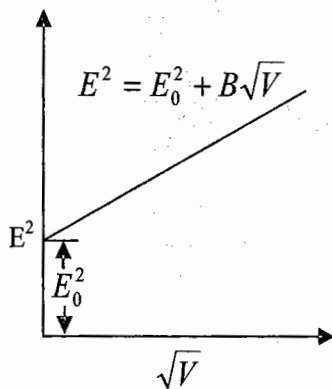
- Depending on the associated electronic equipment, hot wire may be operated in two ways. **Constant-current mode** in this mode the heating current, i.e., voltage across the bridge is kept constant. Initially the circuit is adjusted such that the galvanometer reads zero when the heated wire lies in stationary air. When the air flows, the hot wire cools; the



resistance changes and the galvanometer deflects. The galvanometer deflections are amplified, measured and correlated with air velocity by previous calibration.

Constant-temperature mode:

- In this mode the operating resistance of the wire and hence its temperature is maintained constant. In the event of the tendency of the hot-wire to cool by the flowing fluid, the external bridge voltage is applied to the wire to maintain a sensibly constant temperature. The bridge voltage is varied so as to bring the galvanometer needle to zero ; the reading on the voltmeter is recorded and correlated with air velocity.
- The response of a hot wire placed normal to the flow velocity V is known to be described by :



Response of hot wire

$$E^2 = E_0^2 + BV^n$$

- Where E is the instantaneous value of the bridge voltage, E_0 and B are the constants depending on the cold resistance of the wire and on the properties of the fluid medium. The values of these constants for a given wire are found by calibrating the wire in a uniform flow field against a standard pitot-static probe. Generally with exponent $n = 0.5$, the response of the hot wire in the form of a plot between E^2 and V^n is a straight line.

Electro Magnetic Flow Meter

- The operation of electromagnetic flow-meters is based on Faraday's law of electromagnetic induction, which states that an emf of e volts is induced in a conductor of length l metres when it moves with a transverse velocity of v m/s across a magnetic field of strength B wb/m² and the induced emf in the conductor is given by the expression $e = B/v$ volts
- Since the voltage induced depends on the rate at which the conductor moves through the magnetic field, the magnitude of voltage induced can be used as an indication of the flow-rate of liquid.
- The flowing fluid through the pipe should be conductive and should have a conductivity at least about 10^{-3} mho per meter.
- As the conductive fluid flows through the insulated tube through the magnetic field, a voltage is induced across the electrodes.
- The voltage induced across the electrodes varies directly in proportion to
 - the magnetic field strength, B
 - the distance between the electrodes, d
 - the average velocity of the conductive fluid flowing through the non-conductive tube.
- If the magnetic field strength B and the distance between the electrodes d are kept constant the voltage induced will be directly proportional to the flow-rate of the flowing fluid.

Advantages:

- The main advantages of the electromagnetic flow-meter are that it does not obstruct the flow in any way that may cause pressure drops.
- It has no moving part that may cause friction
- It is insensitive to viscosity, density, and flow disturbances (as long as the velocity profile is symmetrical).

- It has a wide linear range of 10 : 1 with good accuracy and reliability.
- It is simple and rugged in construction.
- Its response is fast and is independent of the properties of the fluid except for the electrical conductivity.
- The only condition is that the pipe should always be full.
- The electromagnetic flow-meter is ideal for bidirectional measurements and is well suited for slurries, and corrosive and solid contaminated liquids.

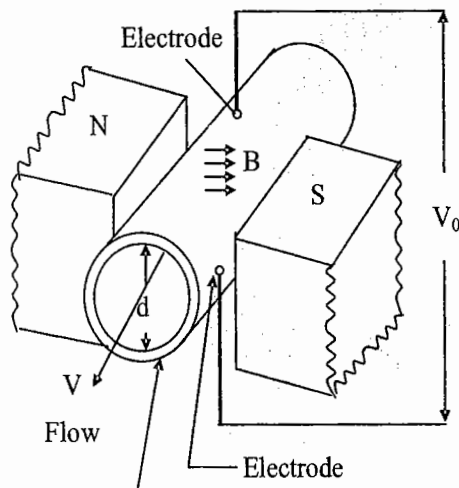


Fig. Electromagnetic flow-meter

Example 2:

Calculate the generated emf (rms value) by the magnetic flow meter if the ordinary tap water flows at the rate of 400 liters per minute through its tube of 8 cm diameter. Assume the maximum flux density of the alternating magnetic field established in the flow meter to be 400 lines per cm^2 (gauss).

Ans: 6.0 mv

Sol: Given, $Q_v = \frac{400}{60} = 6.667 \text{ litre/sec}$

$l = 8\text{cm} = 0.08\text{m}, B = 400 \text{ gauss}$
 $= 400 \times 10^{-4} \text{ tesla}$

We know,

$$E = \frac{4B}{\pi l} Q_v \times 10^{-3} \text{ V} \approx 4.24 \text{ mv}$$

$$E_{\text{rms}} = \frac{2 \times 4.24}{\sqrt{2}} \text{ mv} = 6.0 \text{ mv}$$

Ultrasonic flow meter

- Basically an ultrasonic or acoustic flow-meter consists of two piezoelectric crystals in the liquid or gas separated by a distance. One of the crystal acts as a transmitter and the other as a receiver.
- These flow meters are of two types depending upon the principle on which they operate.
- The first one operates on the basis of measurement of phase shift between oscillations directed downstream and upstream alternately while the second one operates on the basis of measurement of separation of frequency of oscillations directed downstream and upstream simultaneously.
- The schematic diagram of an ultrasonic flow meter operating on first principle, called the "Doppler effect", is shown in Fig. below

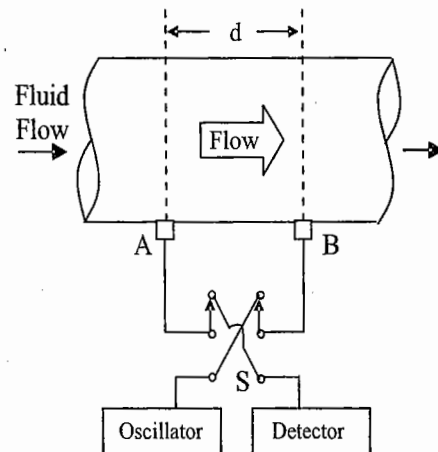


Fig. Schematic Diagram of Ultrasonic Flowmeter Operating

- Two piezo crystals A and B working both as transmitter and receiver of ultrasonic signals

alternately are mounted so that the ultrasonic signals are transmitted between them, as well as through the liquid.

- An oscillator supplies alternately A or B through the switch S when the detector is connected simultaneously to B or A respectively.
- The detector is a phase sensitive device which measures the transit time from upstream to down stream and vice-versa.
- If C is the velocity of sound propagation in medium in m/s, v is the linear velocity of fluid flow in m/s and d is the distance in metres between the two piezo crystals A and B. The transit time in the direction of flow (i.e., from upstream to down stream)

$$\Delta t_1 = \frac{d}{C+v} \dots\dots\dots (i)$$

- The transit time when the ultrasonic signal is traveling in the opposite direction of fluid flow.

$$\Delta t_2 = \frac{d}{C-v} \dots\dots\dots (ii)$$

- Similarly a sinusoidal signal of frequency f Hz traveling in the direction of fluid flow has a phase shift of

$$\Delta \phi_1 = \frac{2\pi f d}{C+v} \text{ radians } \dots\dots(iii)$$

and that traveling against the direction of fluid flow has a phase shift of

$$\Delta \phi_2 = \frac{2\pi f d}{C-v} \text{ radians } \dots\dots(iv)$$

- The difference between two phase shifts is proportional to the flow-rate of the fluid as obvious from the following expression.

$$\begin{aligned} \Delta \phi_2 - \Delta \phi_1 &= \frac{2\pi f d}{C-v} - \frac{2\pi f d}{C+v} \\ &= \frac{4\pi f d v}{C^2 - v^2} = \frac{4\pi f d v}{C^2} = Kv \dots (v) \end{aligned}$$

As the fluid flow velocity $\ll C$, the acoustic velocity.

- The linearity is good – within 2% and linear flow velocity upto about 1mm/s in the low side can be measured easily.
- The dynamic response is limited by the switching frequency. However the uncertainty of the value of C may be a source of an error.
- Flow channel walls should be such as not to permit acoustic transmission. Plastic materials may be used for this purpose.
- The fundamental frequency is determined by the thickness of the piezo crystal. Probes of different frequencies are employed for different groups of fluids, the normal value being in the range of 200 to 5000 kHz.
- The schematic diagram of an ultrasonic flow-meter operating on second principle is illustrated in Fig. Such a flow-meter is called the transit time flow-meter or leading-edge meter.
- In this arrangement two sets of piezo crystals (a transmitter and a receiver as a set) are mounted as shown in the fig. and wave trains are sent at an angle θ with the direction of fluid flow in either way.

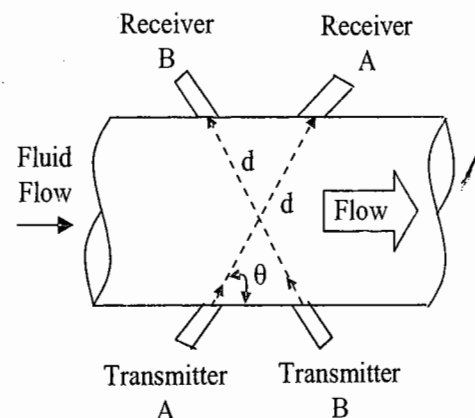


Fig. Schematic Diagram of Transit Time Ultrasonic Flowmeter



- The velocity of the ultrasonic signal from the transmitter A to the receiver A is increased while that from the transmitter B to receiver B is decreased because of fluid flow velocity in the direction or in opposite direction of sound path.
- If C is the sound propagation velocity in medium in m/s, v is the fluid flow velocity in m/s, d is the distance between the transmitter and receiver in meters and θ is the angle between the sound path and the pipe wall, the repetition frequencies in the upstream side and downstream side will be

$$f_a = \frac{C + v \cos \theta}{d} \text{ and } f_b = \frac{C - v \cos \theta}{d}$$

respectively.

The difference in frequency is given by

$$\Delta f = f_a - f_b = \frac{2 v \cos \theta}{d} \dots\dots\dots (vi)$$

- By measuring the difference in the repetition frequency Δf , and knowing the values of θ and d , the velocity of the fluid flow can be determined. Alternatively, the flow velocity can be determined by measuring the time difference between the two pulse trains in either direction as $\Delta t = \frac{1}{\Delta f}$

Advantages:

- Expression shows that output is independent of C , therefore the effects of pressure and temperature are avoided.
- Also the output is linearly proportional to the velocity of fluid flow.

- The measurement is insensitive to viscosity, pressure and temperature variations.
- The other advantages are no obstruction to flow, bidirectional measuring capabilities, good accuracy, fast response, wide frequency range and its versatility in that it can be employed for any pipe size.

Disadvantages:

- However the complexity and relatively high cost limit its use for individual applications.

Example 3:

In an ultrasound flow meter the transducers are separated by a distance $l = 25 \text{ mm}$ with the line joining the transducers making an angle of 60° with the direction of flow. The transit time difference between upstream and downstream measurements is 10 ns with the sound velocity in the medium being 1000 m/s . Assuming that the size of the transducers is very small as compared to the diameter of the pipe, what is the flow velocity?

Ans: 0.4 m/sec

Sol: Given $\Delta t = 10 \times 10^{-9} \text{ sec}$

$$V_s = 1000 \text{ m/sec}$$

$$d = 25 \text{ mm}$$

$$\theta = 60^\circ$$

$$\begin{aligned} \text{So, } V_f &= \frac{\Delta t \times V_s^2}{2d \cos \theta} \\ &= \frac{10 \times 10^{-9} \times (1000)^2}{2 \times 25 \times 10^{-3} \times \cos(60^\circ)} \\ &= 0.4 \text{ m/sec} \end{aligned}$$

Class Room Practice Questions

LEVEL-1 Questions

01. An example of a variable area device for measuring flow is
(a) Flow nozzle (b) Orifice meter
(c) Venturimeter (d) Rotameter
02. A pitot static tube is used to measure the airflow rate in a square tube. If P_d is the differential pressure, the flow rate is proportional to
(a) $\sqrt{P_d}$ (b) P_d (c) $\frac{1}{\sqrt{P_d}}$ (d) P_d^2
03. Which of the following restrictors has the highest discharge co-efficient?
(a) Orifice plate (b) Flow nozzle
(c) Venturi tube (d) None
04. An example of a positive displacement flow meter is
(a) orifice meter
(b) rotary vane type meter
(c) turbine type meter
(d) ultrasonic flow meters
05. Which of the following flow meters has the lowest pressure drop for a given range of flow
(a) Orifice meter (b) Venturi meter
(c) Flow Nozzle (d) Rotameter
06. "Vena Contracta" is the cross section where the flow area is a minimum for a restriction type flowmeter. For an orifice meter, if d is the diameter of the orifice opening, then the area of the vena contracta is approximately.
(a) $\frac{\pi d^2}{4}$ (b) $\frac{0.99 \pi d^2}{4}$
- (c) $\frac{0.8 \pi d^2}{4}$ (d) $\frac{0.6 \pi d^2}{4}$
07. The pressure and velocity at the throat of a venturi tube, measuring the flow of a liquid, are related to the upstream pressure and velocity, respectively, as follows:
(a) Pressure is lower but velocity is higher
(b) Pressure is higher but velocity is lower
(c) Both pressure and velocity are lower
(d) Pressure and velocity are identical
08. For a rotameter, which one of the following statements is TRUE?
(a) The weight of the float is balanced by the buoyancy and the drag force acting on the float
(b) The velocity of the fluid remains constant for all positions of the float
(c) The measurement of volume flow rate of gas is not possible
(d) The volume flow rate is insensitive to changes in density of the fluid
09. The fluid density does not affect the working of
(a) Orifice plate
(b) Rotameter
(c) Electro-magnetic flow meter
(d) Pitot-static probe traverse
10. The hot wire anemometer used for measuring gas velocities is a variable
(a) Resistance transducer
(b) Inductance transducer
(c) Capacitance
(d) Frequency transducer
11. Which of the following flow meters gives good results when the fluid being metered is exceptionally clean?



- (a) Hot wire anemometer
(b) Turbine flow meter
(c) Ultrasonic flow meter
(d) Electro-magnetic flow meter
12. Consider the four following statements in the context of flow meters:
- (a) Hot wire anemometer is preferred to measure highly fluctuating velocities.
(b) In an electromagnetic flow meter the induced voltage is proportional to flow rate.
(c) The obstruction type flow meters have a linear relationship between the pressure difference and the rate of fluid flow
(d) Rota meter is a variable area flow meter and it should always be installed in a vertical pipeline.

LEVEL-2 Questions

1. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Name of the transducer)

- A. Bourdon tube
B. Hot wire anemometer
C. Hydrometer
D. Pitot tube

List-II (Operation)

1. Fluid flow to resistance change
2. Velocity to pressure
3. Pressure to displacement
4. Specific gravity to displacement

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 3 | 2 | 4 |
| (b) | 3 | 1 | 2 | 4 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 4 | 2 | 3 | 1 |

2. Pirani gauge is used to measure
(a) very low pressures
(b) high pressures
(c) pressures in the region of 1 atm
(d) fluid flow

3. Match List I (Transducers) with List II (Measured Quantities) and select the correct answer using the codes given below the Lists: (IES-05)

List I

- A. Hall effect pick up
B. Piezoelectric pick up
C. Hot-wire anemometer
D. Rotameter

List II

1. Pressure
2. Velocity of fluid
3. Flow rate
4. Current

Codes :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 3 | 2 | 1 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 2 | 3 | 4 | 1 |

4. Which one of the following transducers cannot measure flow in non-conducting medium?
(a) Orifice meter
(b) Electromagnetic flow meter /
(c) Turbine meter
(d) Rotameter
5. A hot-wire anemometer is a device used to measure
(a) Gas velocities
(b) Pressure in gases
(c) Liquid discharge
(d) Temperature



Level - 1 Key

01. (d) 02. (a) 03. (c) 04. (b) 05. (b)
06. (d) 07. (a) 08. (a) 9. (c) 10 (c)
11. (a) 12. (b)

Level - 2 Key

- 01.(c) 02. (a) 03. (c) 04. (b) 05. (a)

6

Intermediate Quantity Measurements

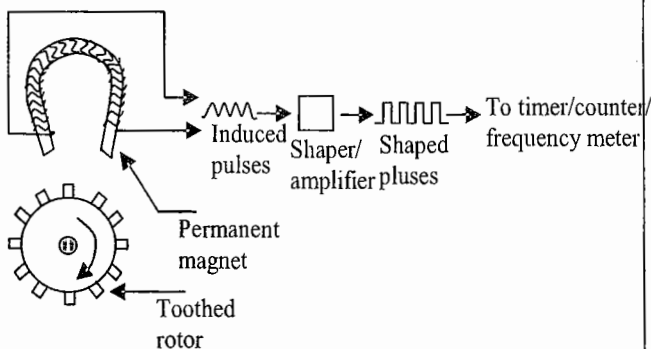
Measurements of Rotational Velocity

Contact less electrical tachometers:

- Tachometers of this type produce pulse from a rotating shaft without any physical contact between the speed transducer and the shaft. This aspect has the distinct advantage in that no load is applied to the machine.

Inductive pick-up tachometer

- The unit consists of a small permanent magnet with a coil round it. This magnetic pick up is placed near a metallic toothed rotor whose speed is to be measured. As the shaft rotates, the teeth pass in front of the pick-up and produce a change in the reluctance of the magnetic circuit. The field expands or collapses and a voltage is induced in the coil. The frequency of the pulses depends upon the number of teeth on the wheel and its speed of rotation. Since the number of teeth is known, the
- Speed of rotation can be determined by measuring the pulse frequency. To accomplish this task, pulse is amplified and squared, and fed into a counter of frequency measuring unit.



Inductive pick-up tachometer

- Let T be the number of teeth on the rotor, N be the revolutions per second and P be the number of pulses per second, then

$$\text{Speed } N = \frac{\text{pulses per second}}{\text{number of teeth}}$$

$$N = \frac{P}{T} \text{ rps} = \frac{P}{T} \times 60 \text{ rpm}$$

- If the rotor has 60 teeth, and if the counter counts the pulses in one second, then the counter will directly display the speed in revolutions per minute.

Example1:

A metallic toothed-rotor having 60 teeth was mounted on a shaft whose speed was to be determined. The pulses induced in the inductive-type magnetic pick-up were registered to be 3600 per second. Estimate the shaft speed. If the frequency meter gives the counts within $\pm 5\text{Hz}$, calculate the range within which the shaft speed can lie.

Ans: $(3600 \pm 5) \text{ rpm}$

Sol: Shaft speed is related to the number of teeth and the number of pulses by the relation.

$$\text{Speed} = \frac{\text{pulses per second}}{\text{number of teeth}}$$

$$= \frac{3600}{60} = 60 \text{ rps} = 3600 \text{ rpm}$$

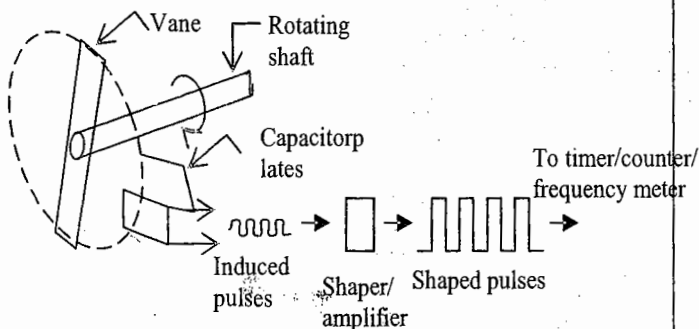
$$\text{Accuracy of measurement} = \pm 5\text{Hz}$$

$$= \pm \frac{5}{60} \times 60 = \pm 5 \text{ rpm}$$

$$\therefore \text{Range of shaft speed} = (3600 \pm 5) \text{ rpm}$$

Capacitive type pick-up tachometer

The device consists of a vane attached to one end of the rotating **machine** shaft. When the shaft rotates between the fixed capacitive plates, there occurs a change in the capacitance. The capacitor forms a part of an oscillator tank so that number of frequency changes per unit of time is a measure of the shaft speed. The pulses thus produced are amplified, and squared, and may then be fed to frequency measuring unit or to a digital counter so as to provide a digital analog of the shaft rotation.



Capacitive pick-up tachometer

Measurement of Acceleration

There are two types of accelerometers generally used for measurement of acceleration:

- Piezo-electric type,
- Seismic type.

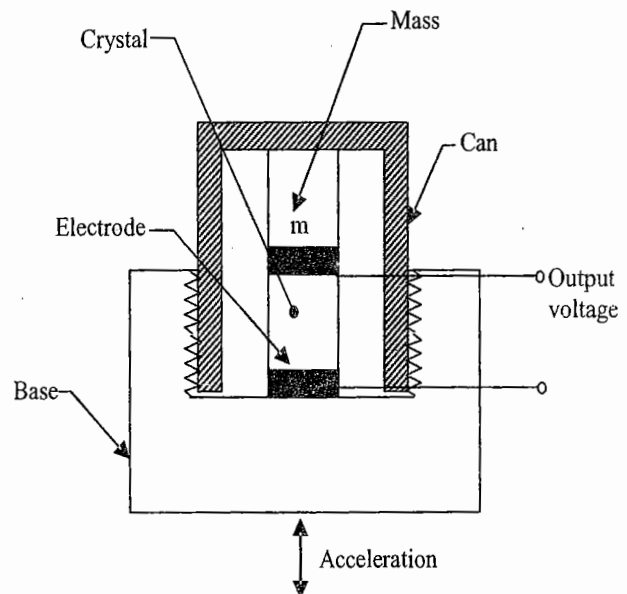
Piezo-electric accelerometer:

The unit is perhaps the simplest and most commonly used transducer employed for measuring acceleration. The sensor consists of a piezo-electric crystal sandwiched between two electrodes and has a mass placed on it. The unit is fastened to the base whose acceleration characteristics are to be obtained. The can threaded to the base acts as a spring and squeezes the mass against the crystal. Mass exerts a force

on the crystal and a certain output voltage is generated. If the base is now acceleration downward, inertial reaction force on the base acts upward against the top of the can. This relieves stress on the crystal. From Newton's second law

$$\text{Force} = \text{mass} \times \text{acceleration}$$

And since the mass is a fixed quantity, the decrease in force is in proportion to the acceleration. Likewise, an acceleration in the upward direction would increase the force on the crystal in proportion to the acceleration. The resulting change in the output voltage is recorded and correlated to the acceleration imposed on the base.



Piezo-electric accelerometer

Advantages and Limitations:

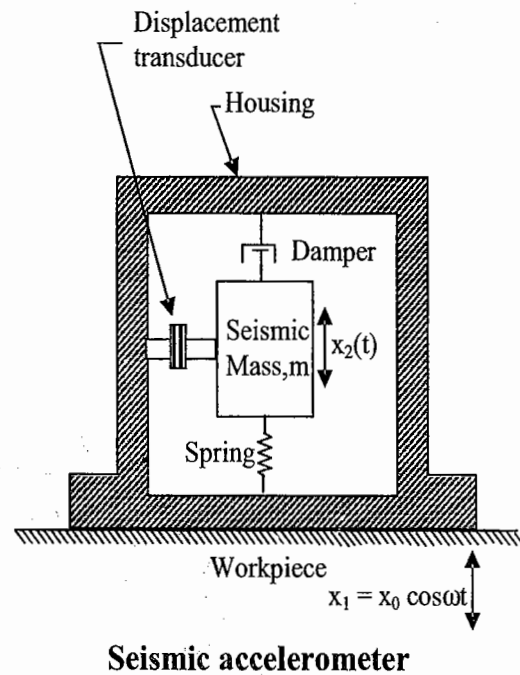
- Rugged and inexpensive device
- High output impedance
- High frequency response from 10 Hz to 50 kHz
- High sensitivity varies from 10 to 100 mV/g where $g = 9.807 \text{ m/s}^2$
- Capability to measure acceleration from a fraction of g to thousand of g



- Somewhat sensitive to changes in temperature
- Subject to hysteresis errors.

Displacement sensing (seismic) accelerometer:

In a seismic accelerometer, the displacement of a mass resulting from an applied force is measured and correlated to the acceleration. Fig. shows the schematics of a common spring-mass-damper system which accomplishes this task. The mass is supported by a spring and damper is connected to the housing frame. The frame is rigidly attached to the machine whose acceleration characteristics are to be determined. When an acceleration is imparted by the machine to the housing frame, the mass moves relative to the frame, and this relative displacement between the mass and frame is sensed and indicated by an electrical displacement transducer.



Class Room Practice Questions

LEVEL-1 Questions

01. A seismic vibration sensor, having natural frequency ω_n and damping ratio ζ , used for measuring amplitude of vibration with frequency ω , should have
 - (a) $\omega_n \ll \omega$ and ζ much greater than 1
 - (b) $\omega_n = \omega$ and ζ slightly less than 1
 - (c) $\omega_n \gg \omega$ and ζ slightly less than 1
 - (d) $\omega_n \ll \omega$ and ζ slightly less than 1
02. In a seismic acceleration sensor, with a displacement transducer as the secondary element, decreasing the mass while maintaining all other parameters of the sensor unchanged will
 - (a) reduce both natural frequency and steady – state sensitivity
 - (b) increase both natural frequency and steady – state sensitivity
 - (c) increase natural frequency but reduce steady – state sensitivity
 - (d) increase natural frequency without affecting steady – state sensitivity
03. In d.c. tachogenerators used for measurement of speed of a shaft, frequent calibration has to be done because
 - (a) the contacts wear off
 - (b) the strength of permanent magnet decrease with age
 - (c) the armature current produces heating effects
 - (d) all of the above.
04. In a drag cup type a.c. tachogenerator, the output voltage is:
 - (a) sinusoidal

- (b) in the form of pulses
- (c) modulated waveform
- (d) constant d.c. because rectifiers are used.

LEVEL-2 Questions

01. A variable reluctance type tachometer has 60 rotor teeth, the records 3600 counts/minutes. The device speed is
 - (a) 60 r.p.s.
 - (b) 1800 r.p.m
 - (c) 3600 r.p.s.
 - (d) 7200 r.p.m.
02. An accelerometer has a seismic mass of 0.05 kg and a spring constant of 3×10^3 N/m. The maximum permissible displacement of the mass is ± 1 mm. The maximum measurable acceleration is
 - (a) 30 m/s^2
 - (b) 60 m/s^2
 - (c) 90 m/s^2
 - (d) 120 m/s^2
03. A variable reluctance type tachometer has 150 teeth on the rotor. The counter records 13,500 pulses per second. The rotational speed is
 - (a) 4800 rpm
 - (b) 5400 rpm
 - (c) 6000 rpm
 - (d) 7200 rpm
04. A variable reluctance type tachometer has 180 teeth on rotor. The speed of the shaft on which it is mounted is 1200 rpm. The frequency of the output pulses is
 - (a) 1800 per second
 - (b) 3600 per second
 - (c) 4800 per second
 - (d) 5600 per second
05. Consider the following devices:
 1. Anemometer
 2. Stroboscope
 3. Accelerometer



The correct sequence of these device to measure the rotational speed, vibration and airflow, respectively is

- (a) 2, 3 and 1 (b) 2, 1 and 3
(c) 1, 3 and 2 (d) 3, 2 and 1

06. A reflectometer consists of
(a) two directional couplers
(b) one directional coupler and an isolator
(c) one directional coupler and a circulator
(d) two directional couplers and a circulator
07. A capacitive transducer is made of two concentric cylindrical electrodes. The outer diameter of the inner cylindrical electrode is 4 mm and dielectric medium is air. The inner diameter of the outer electrode is 4.1 mm. When 150 V is applied across the electrodes, what is the electric field intensity?
(a) 1500 V/mm (b) 3000 V/mm
(c) 4500 V/mm (d) 6000 V/mm

Level - 2 Key

01.(d) 02. (d) 03. (b) 04. (c)

Level - 2 Key

01.(c) 02. (b) 03. (b) 04. (b) 05. (a)
06. (a) 07. (a)